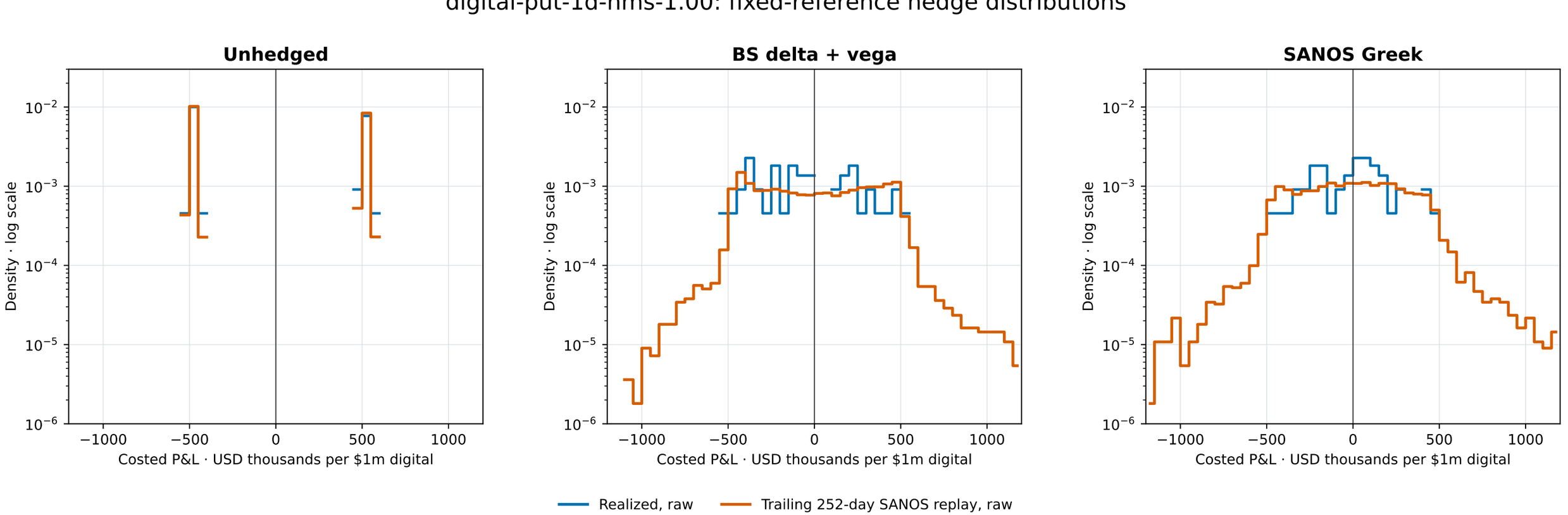
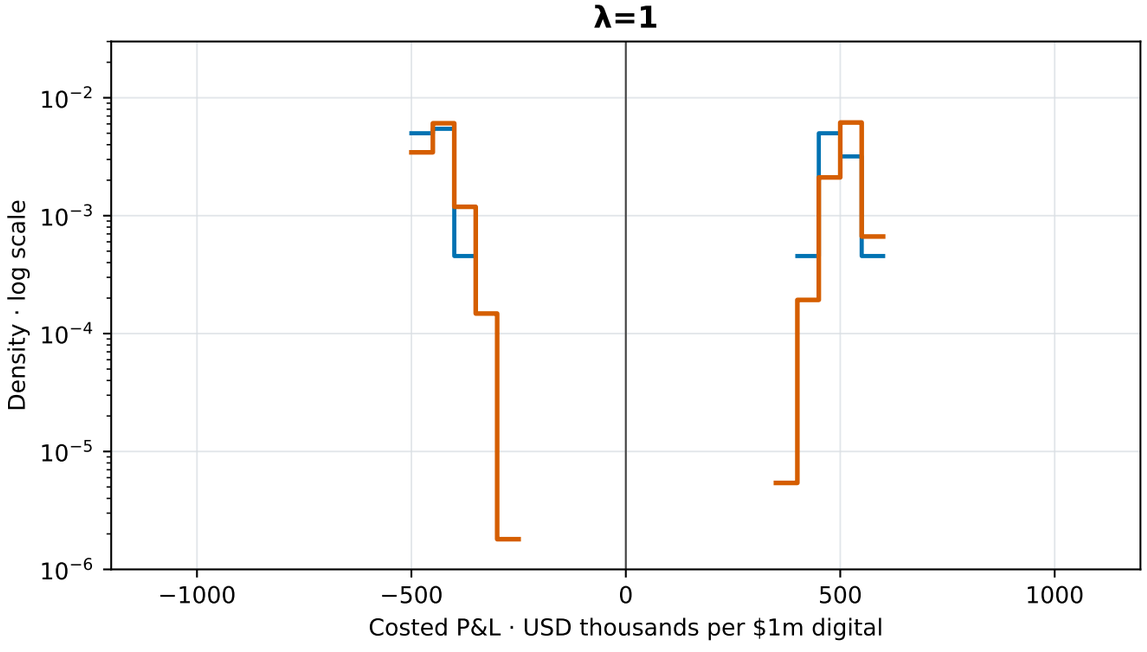
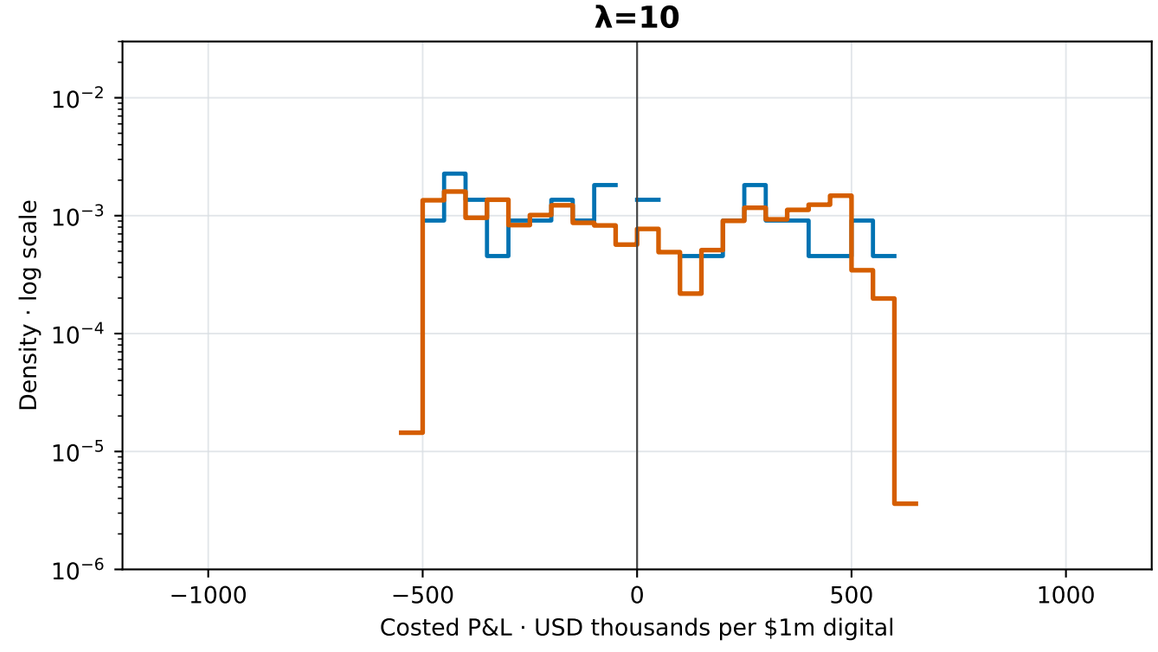
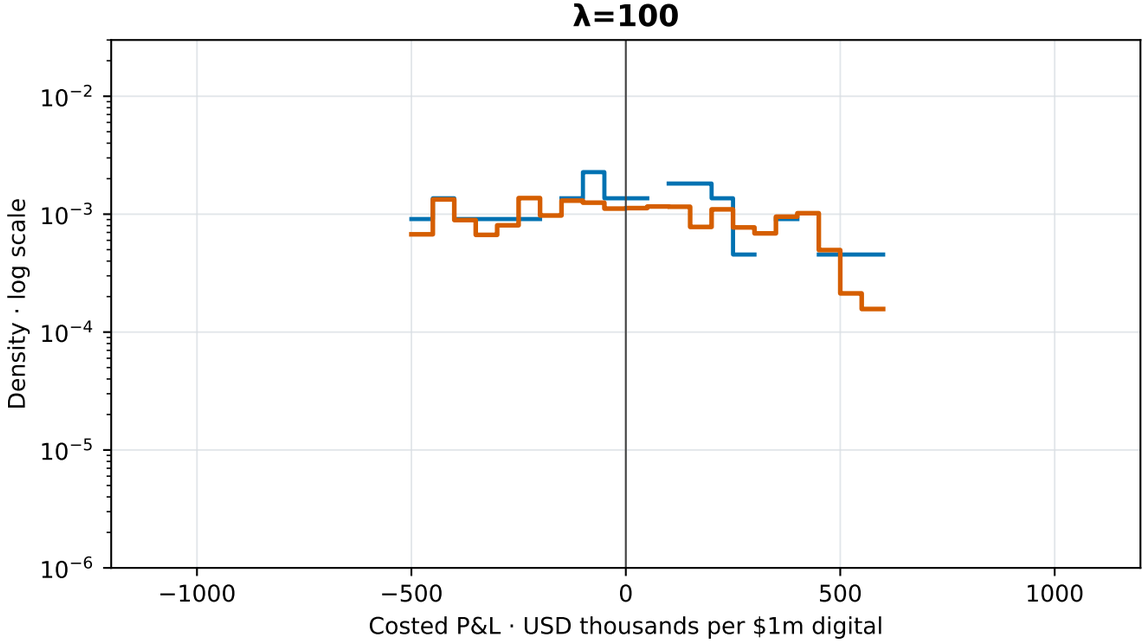
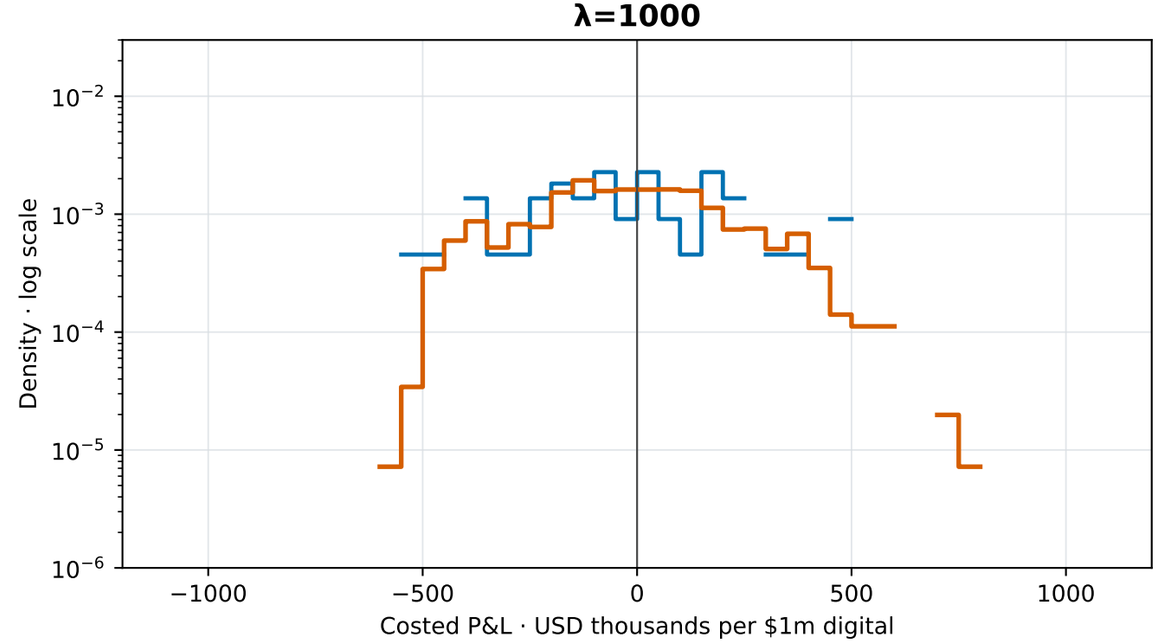

digital-put-1d-nms-1.00: fixed-reference hedge distributions
Unhedged
BS delta + vega
SANOS Greek
Density · log scale
Costed P&L · USD thousands per $1m digital
Realized, raw
Trailing 252-day SANOS replay, raw
No mean correction: 44 realized next-day outcomes versus preceding 252 raw returns. References have no fitted hedge-return law.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-1.00: SANOS historical · Mean-variance





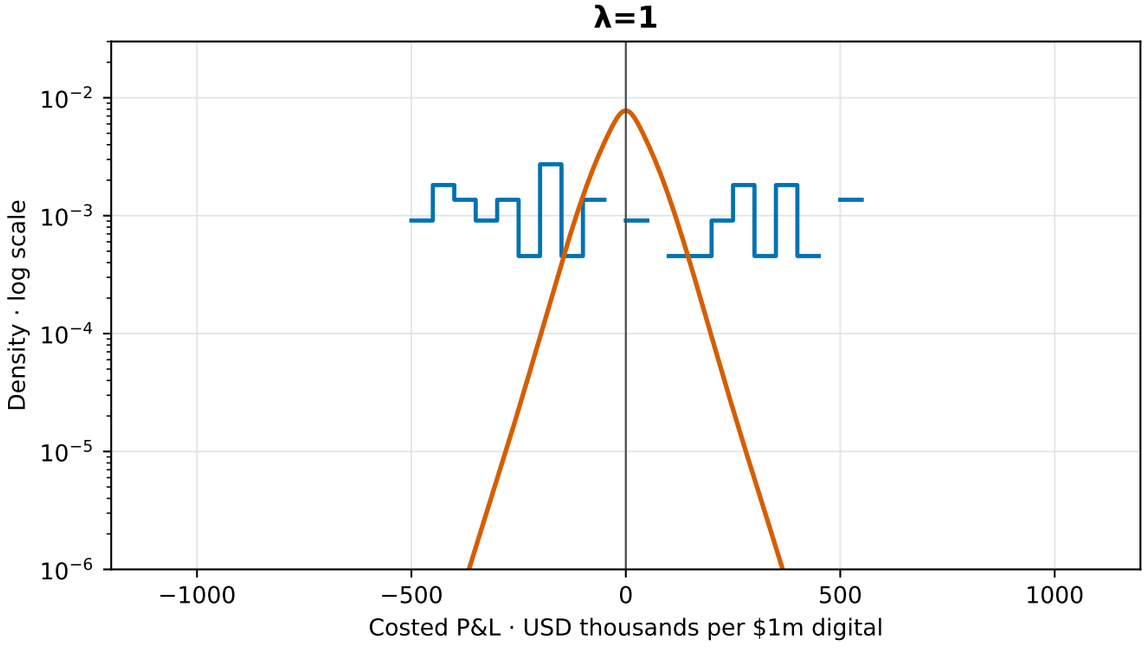


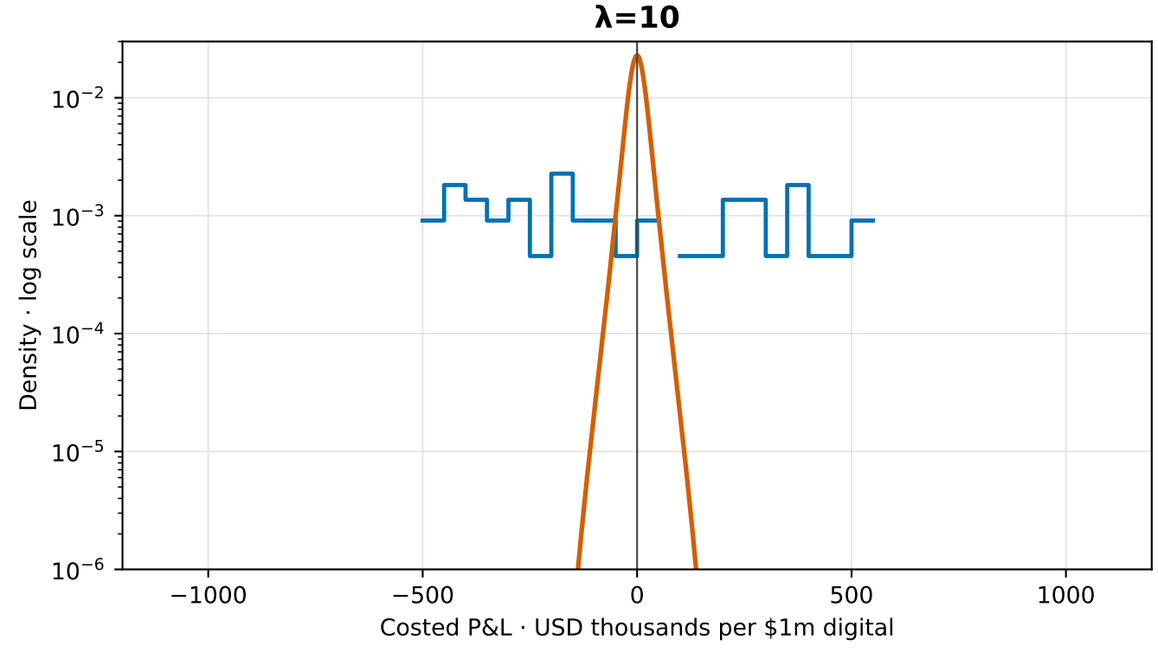


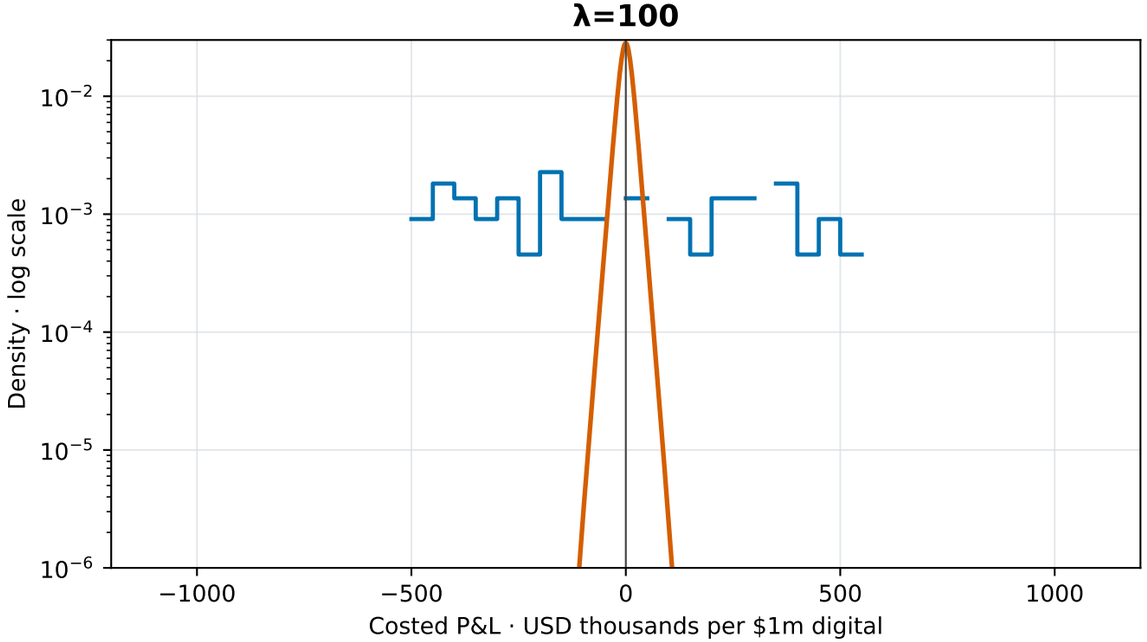
Realized, raw Fitted 252-day demeaned SANOS returns

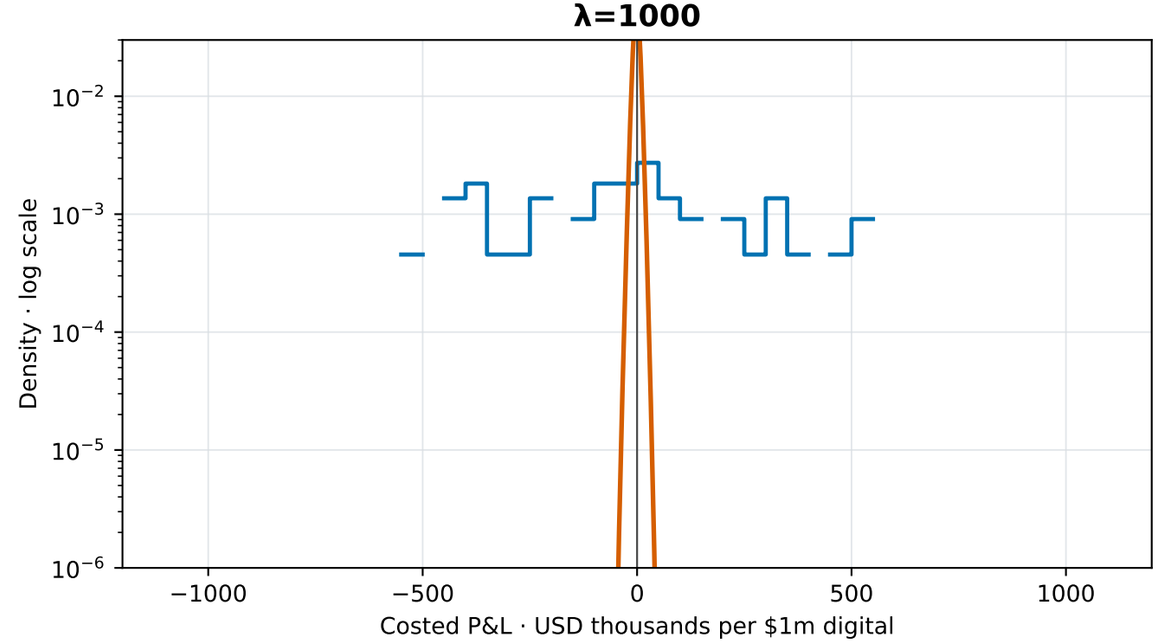
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-1.00: SANOS parametric · Mean-variance





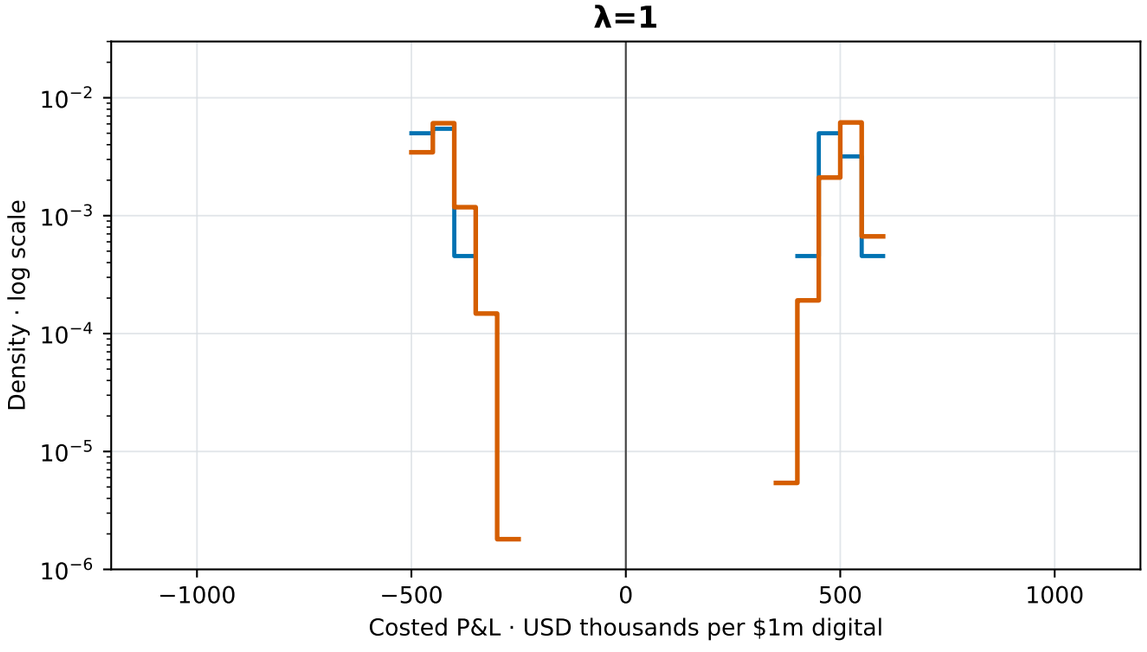


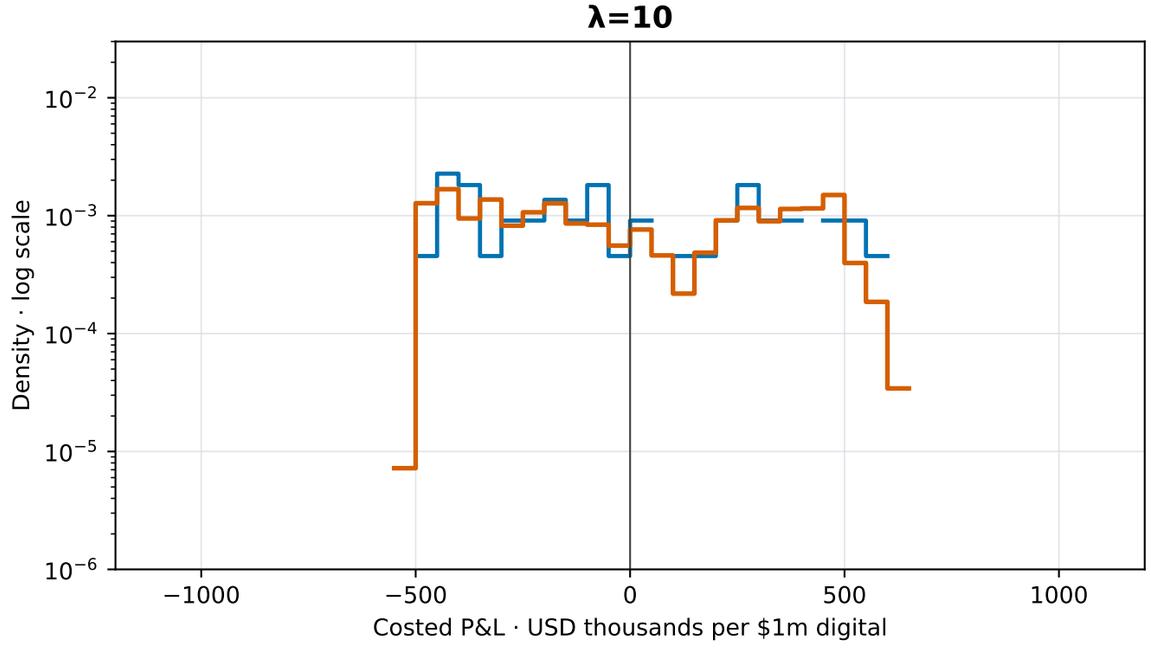


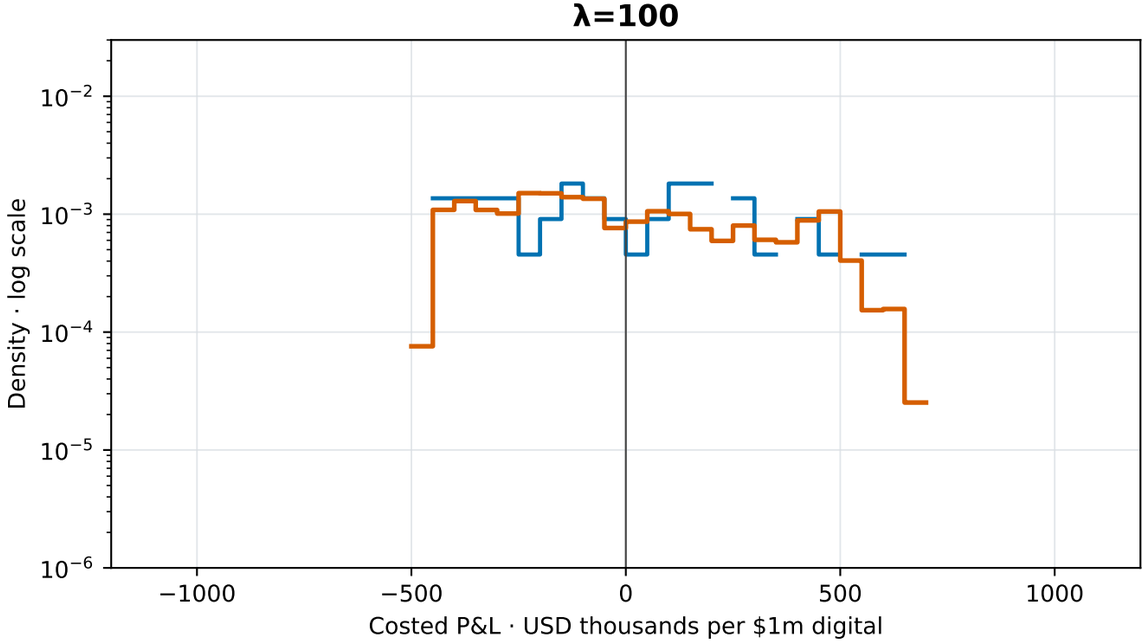
Realized, raw Fitted centered factor Gaussian

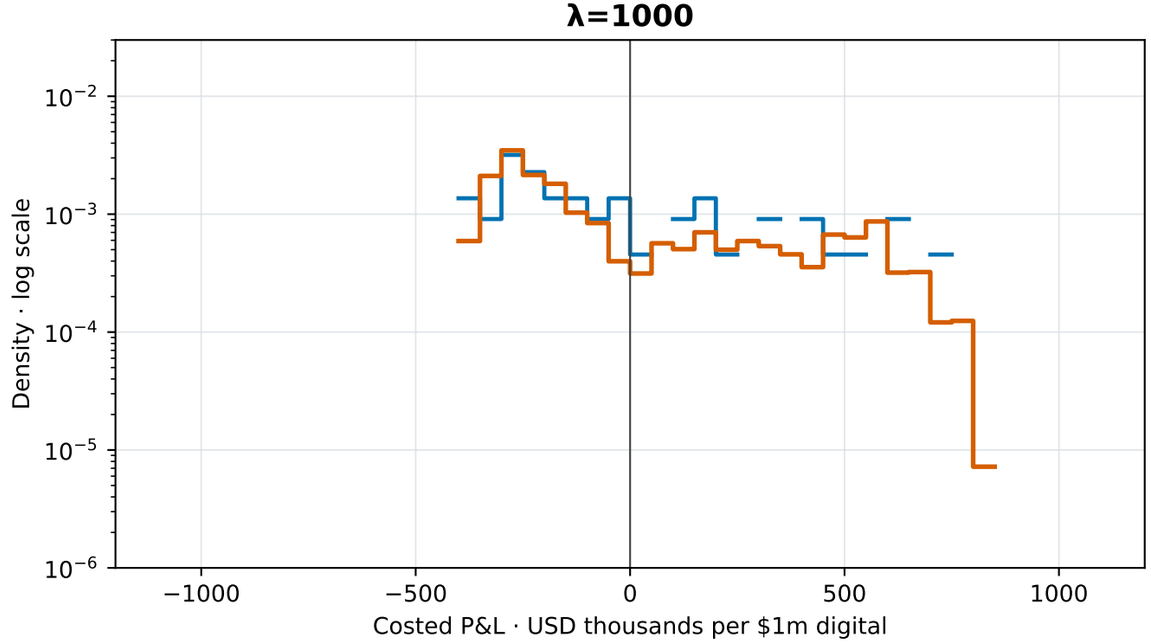
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-1.00: SANOS historical · Entropy





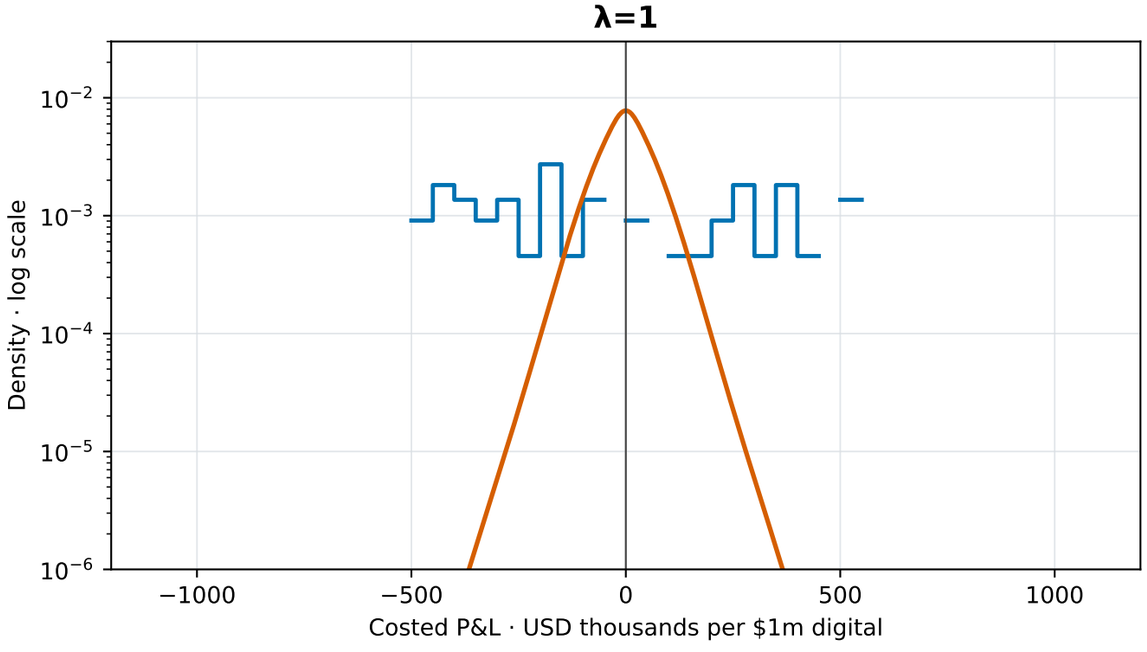


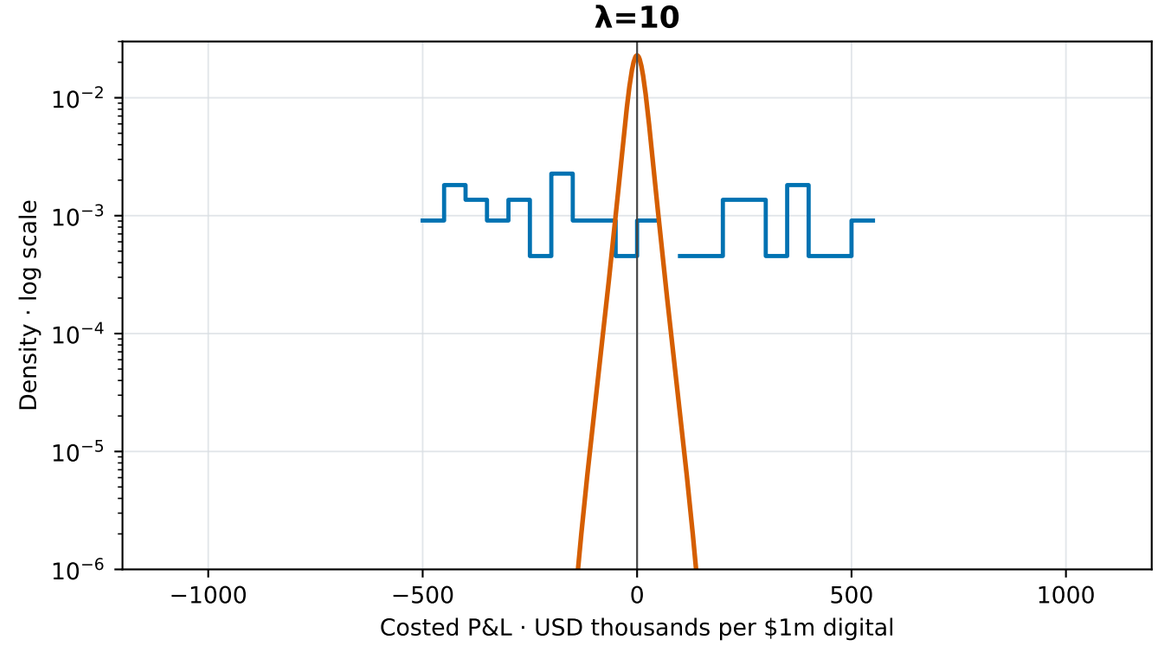


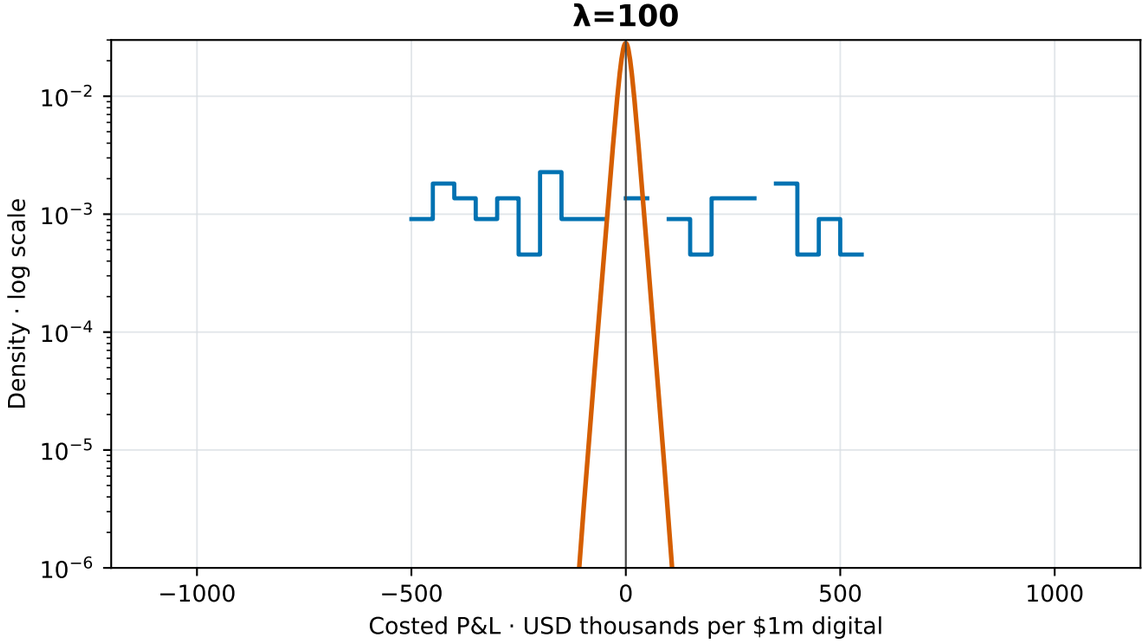
Realized, raw Fitted 252-day demeaned SANOS returns

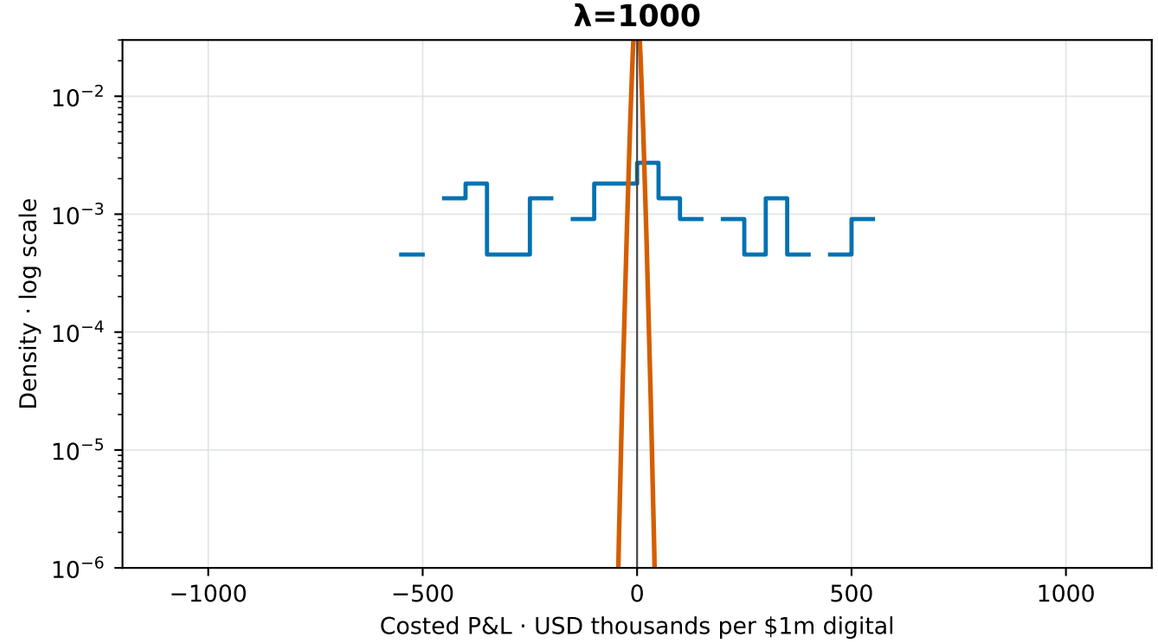
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-1.00: SANOS parametric · Entropy





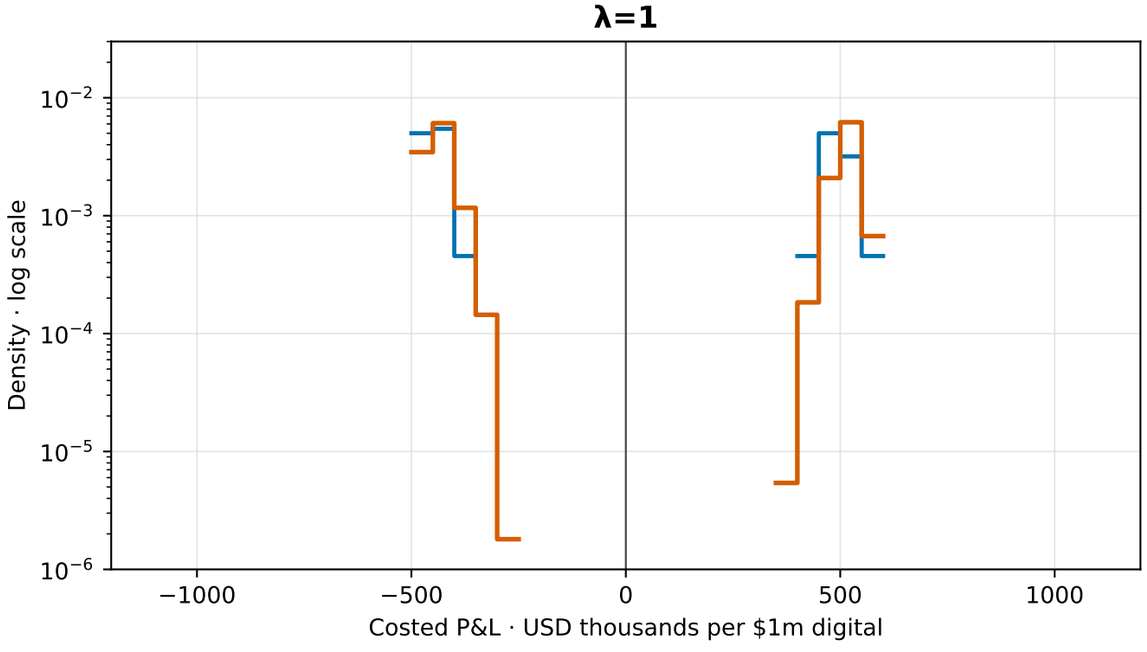


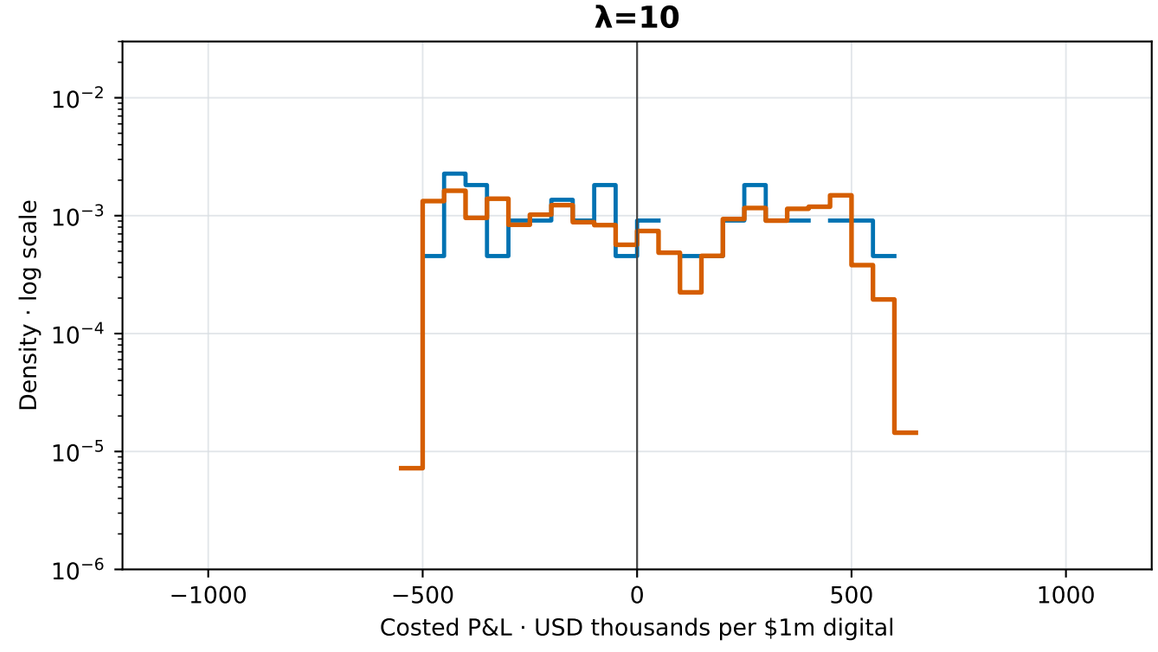


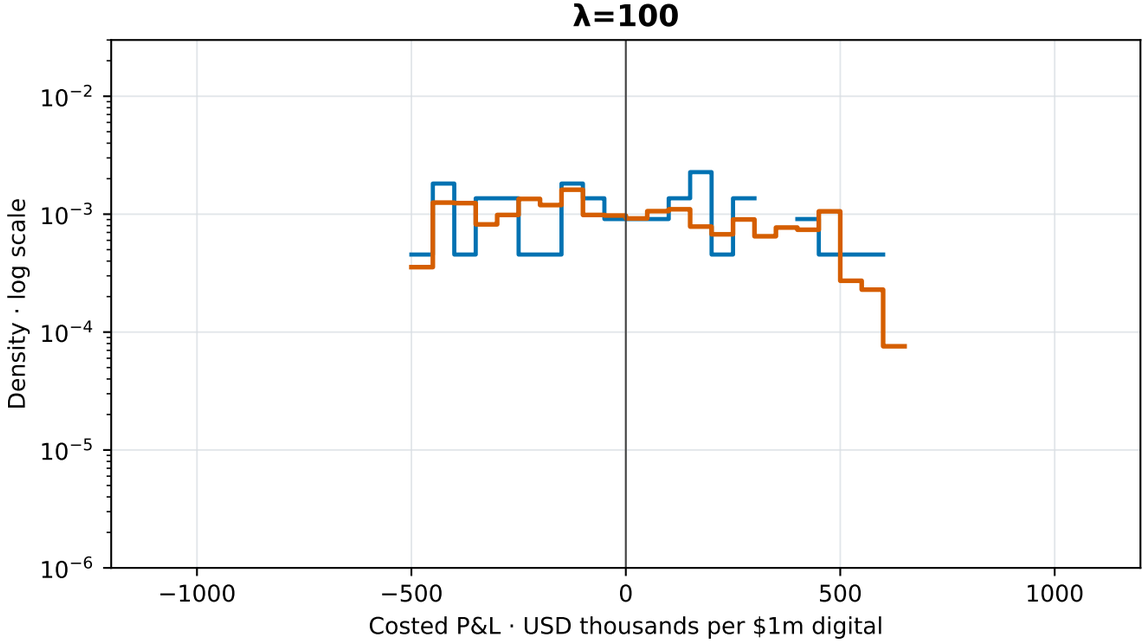
Realized, raw Fitted centered factor Gaussian

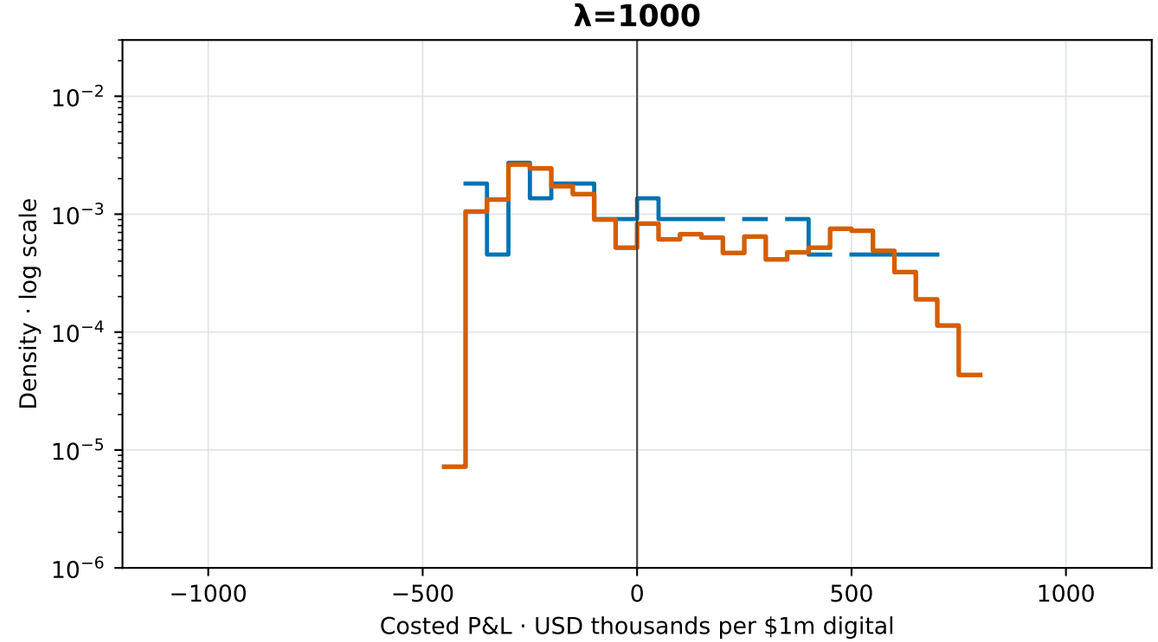
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-1.00: SANOS historical · Power-exponential





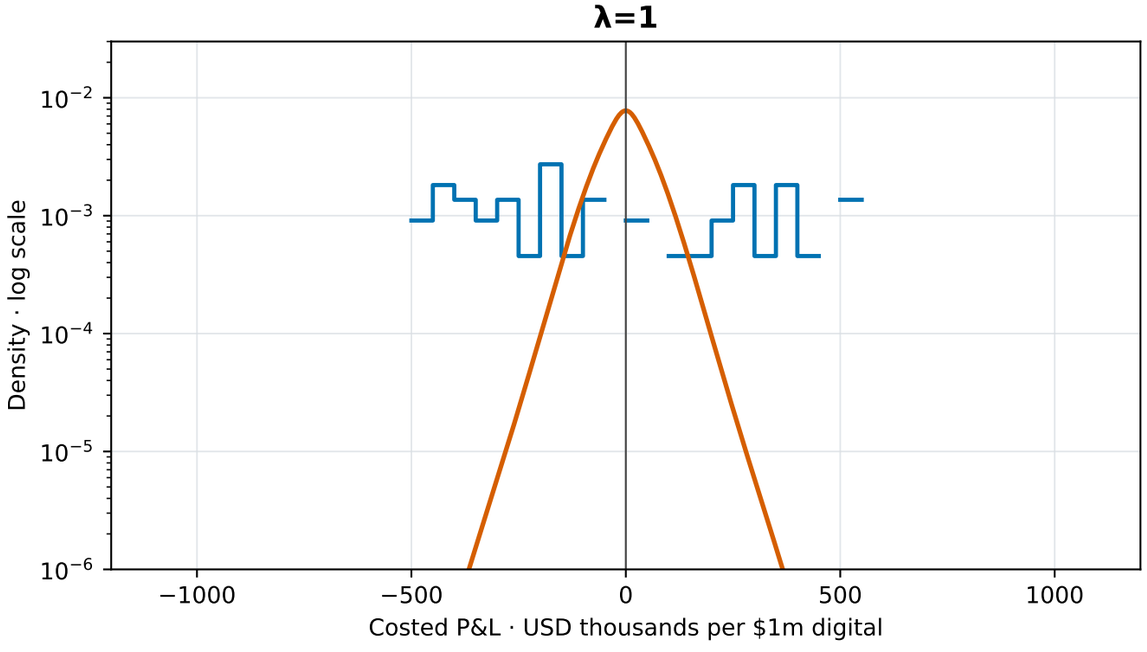


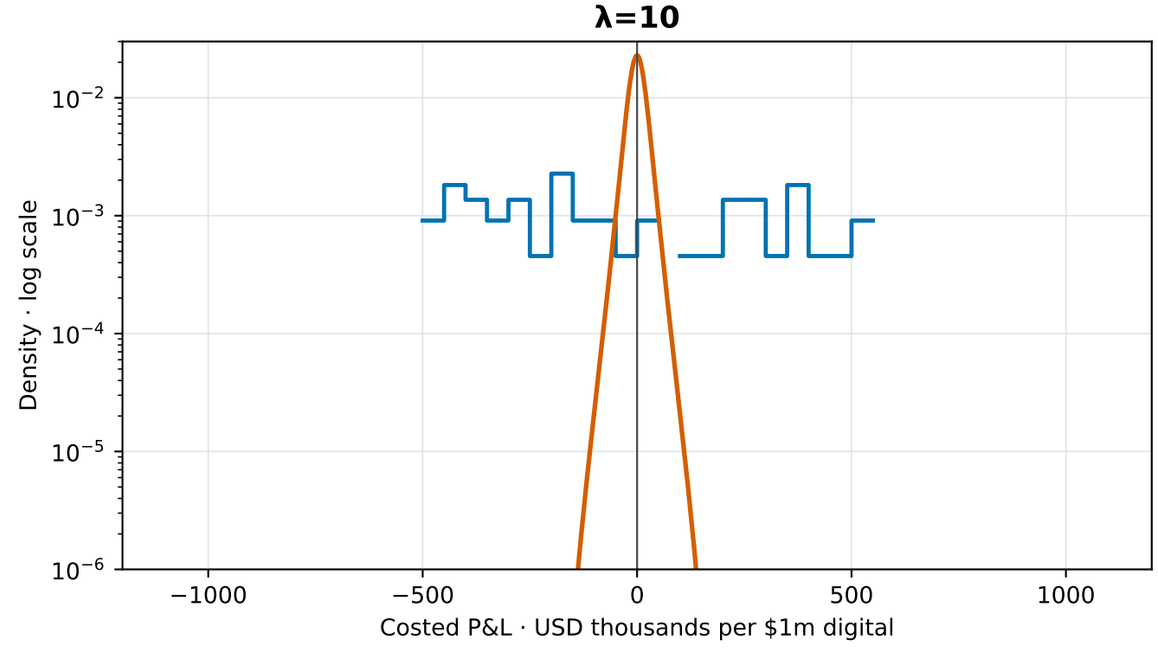


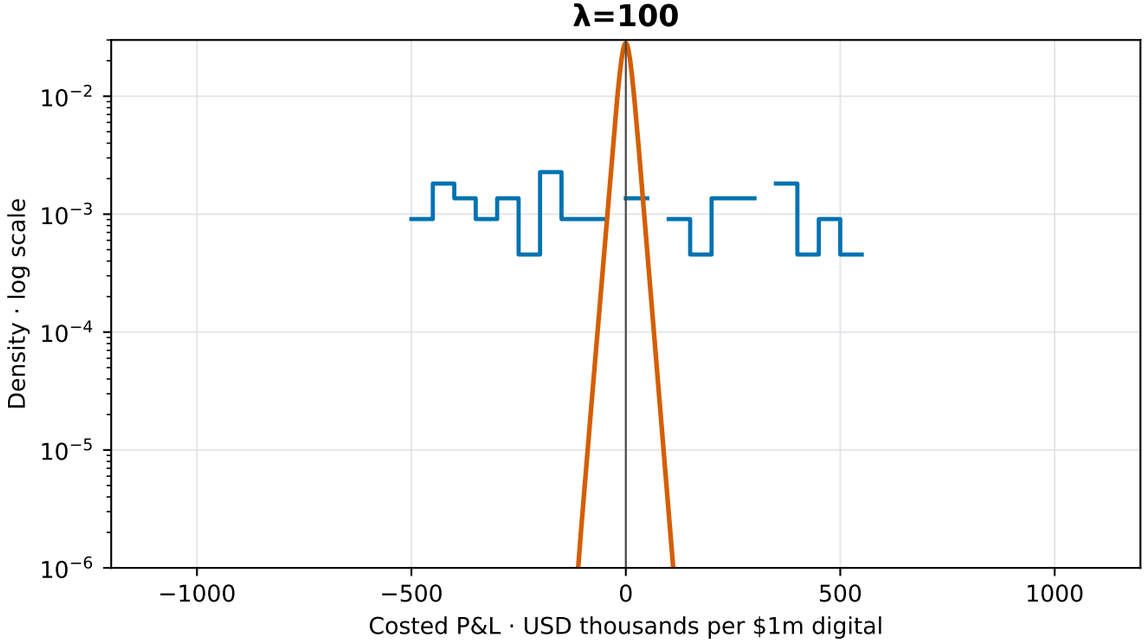
Realized, raw Fitted 252-day demeaned SANOS returns

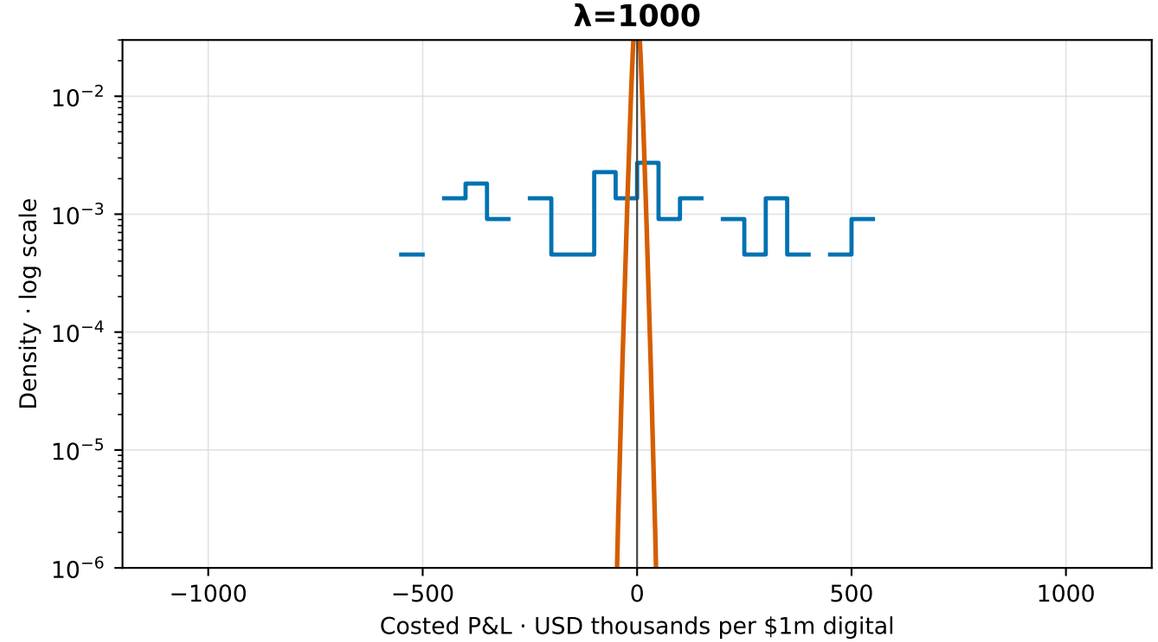
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-1.00: SANOS parametric · Power-exponential



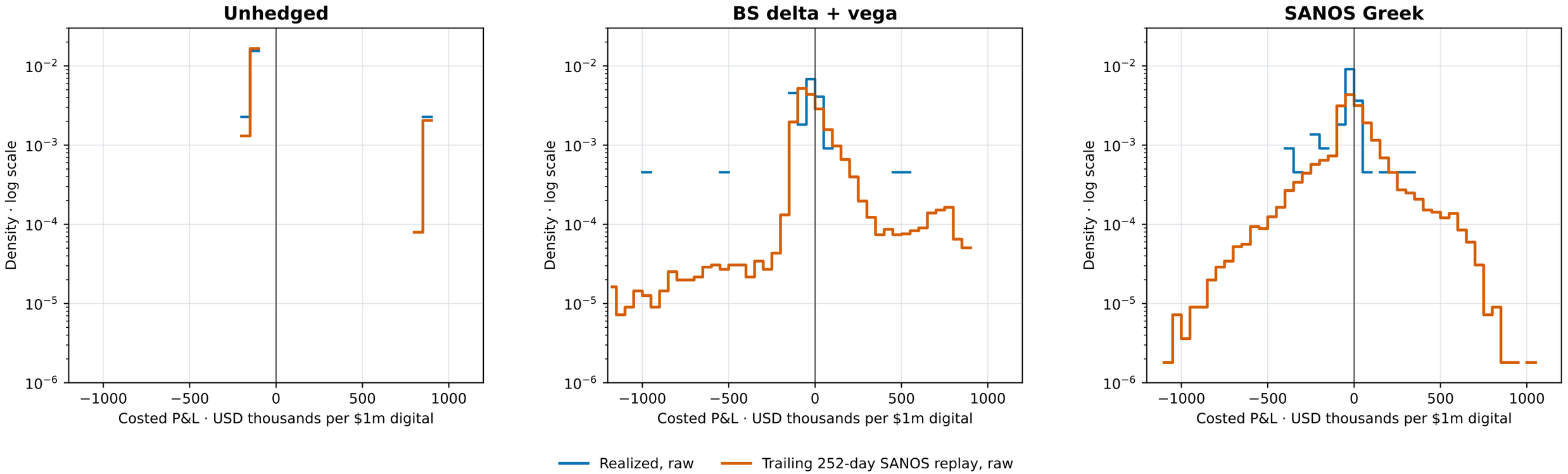




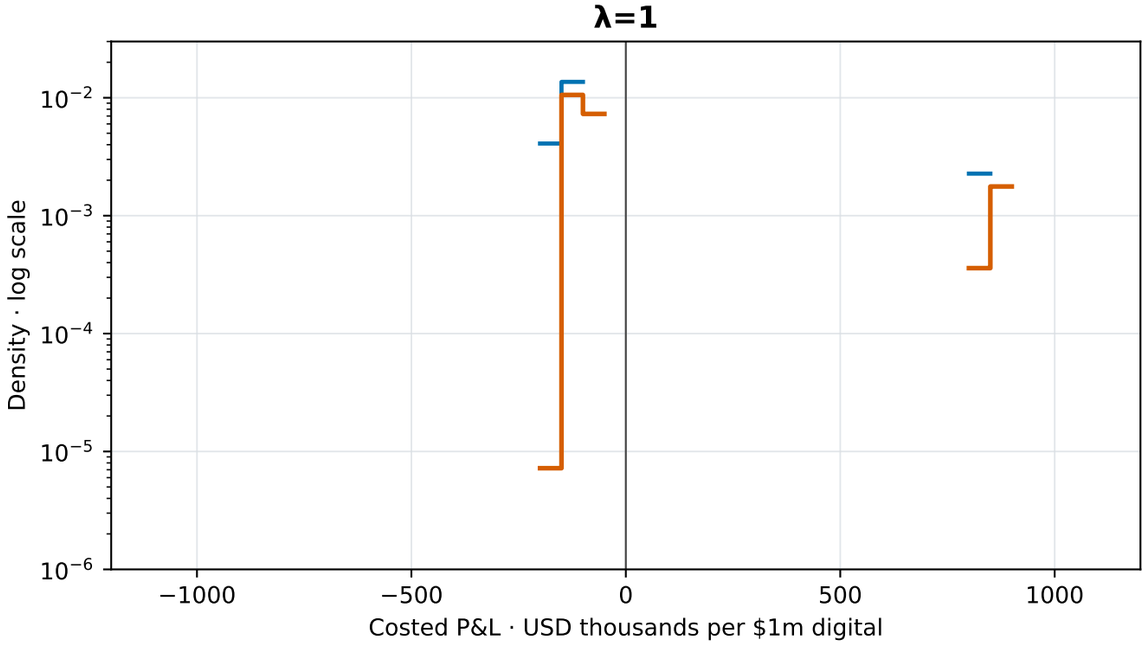


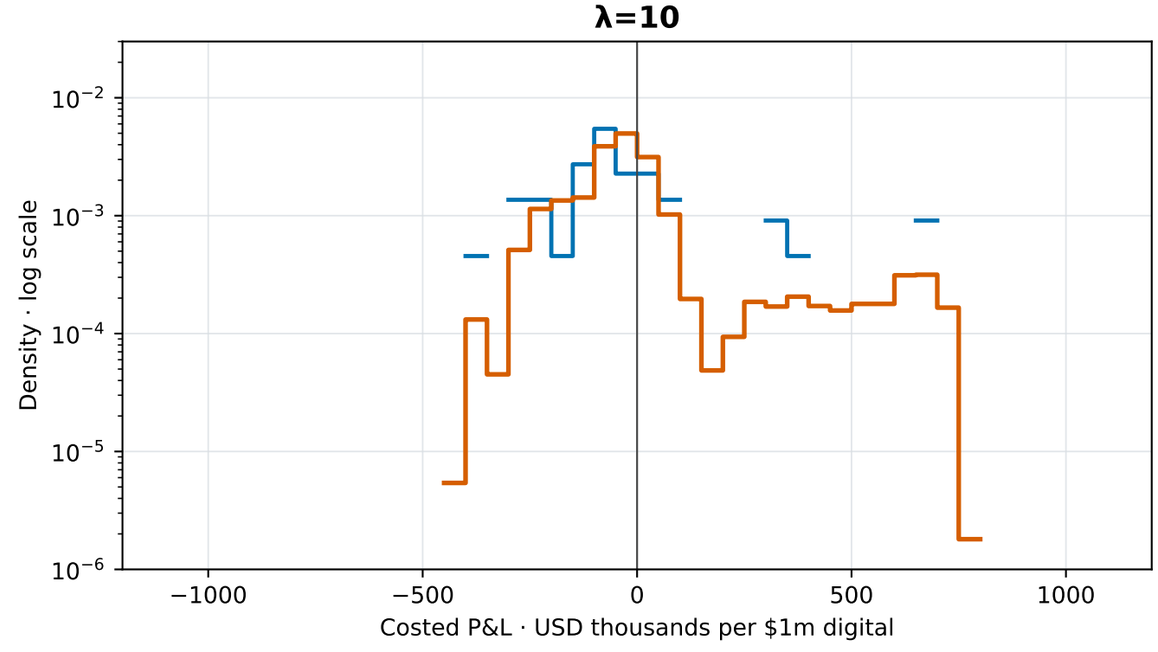
Realized, raw Fitted centered factor Gaussian

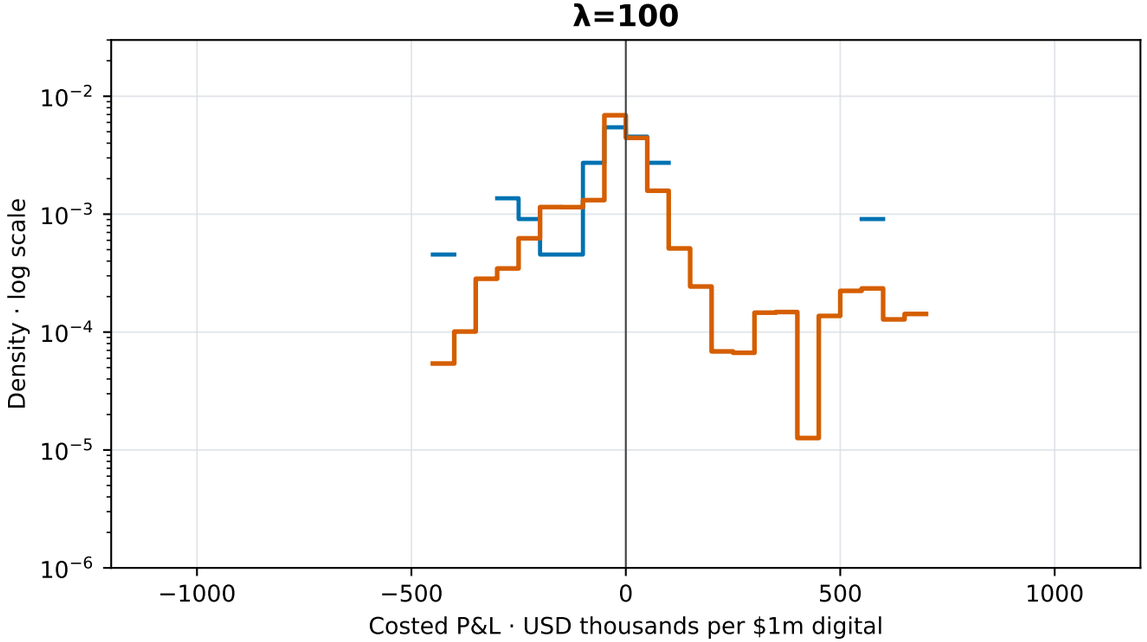
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

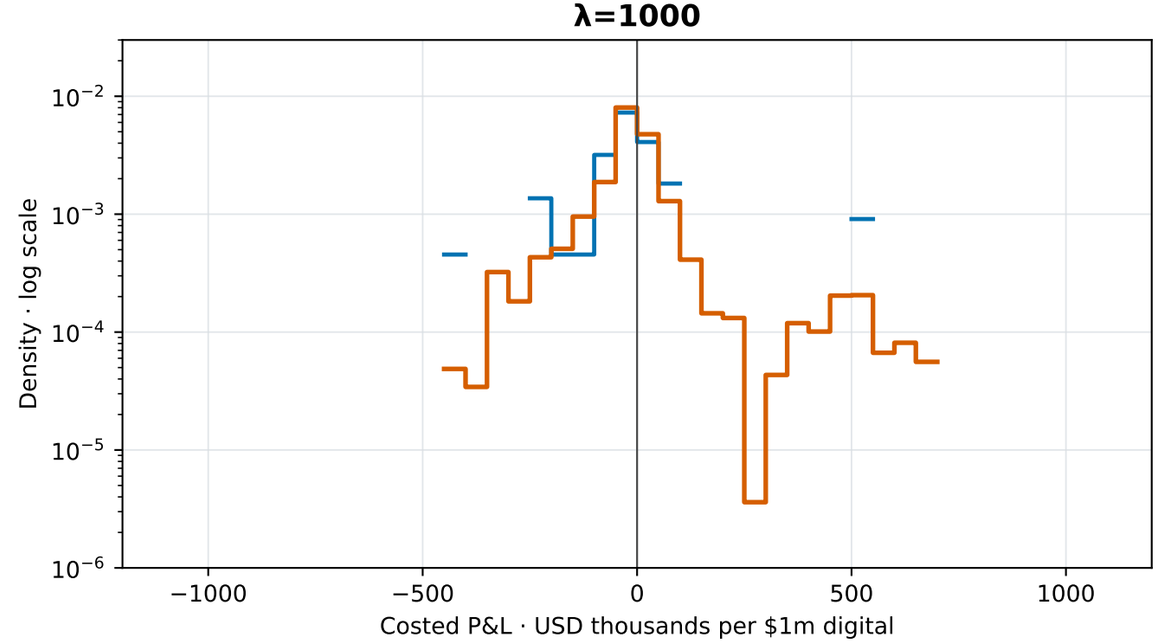

digital-put-1d-nms-0.75: fixed-reference hedge distributions
Unhedged
BS delta + vega
SANOS Greek
Density · log scale
Costed P&L · USD thousands per $1m digital
Realized, raw
Trailing 252-day SANOS replay, raw
No mean correction: 44 realized next-day outcomes versus preceding 252 raw returns. References have no fitted hedge-return law.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.75: SANOS historical · Mean-variance





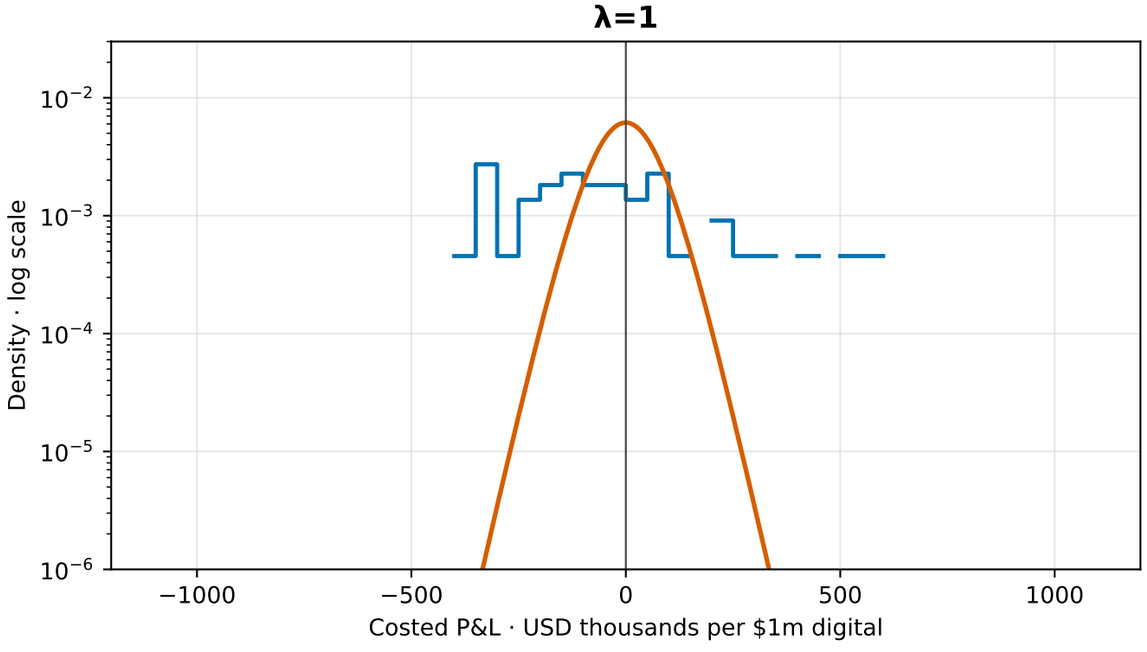


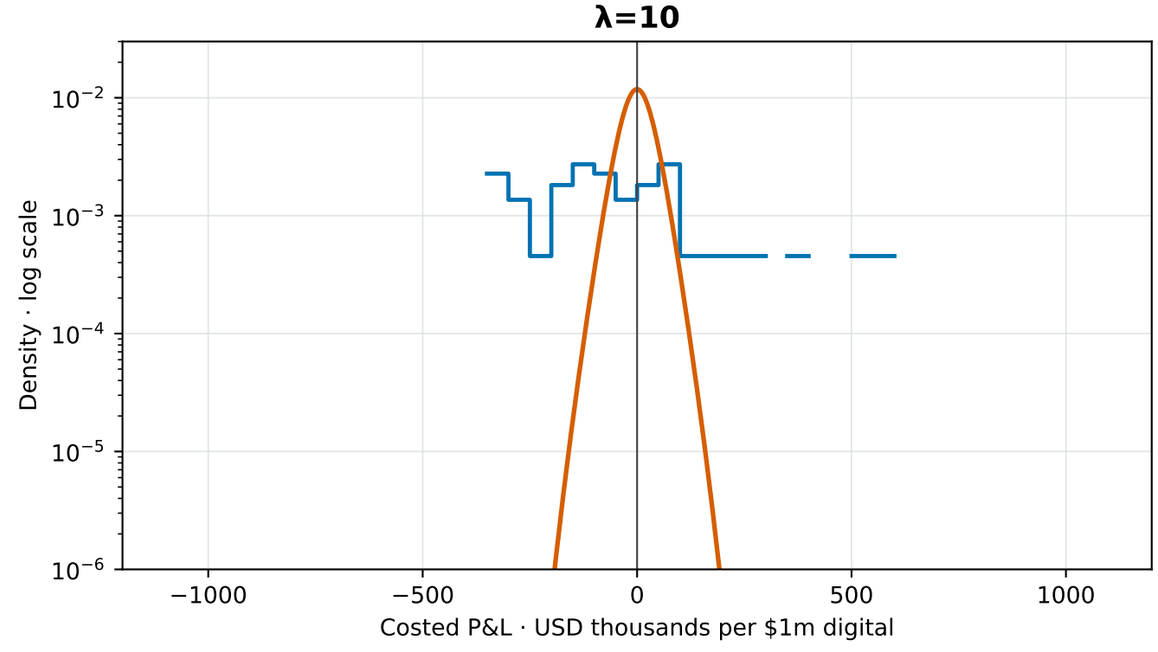


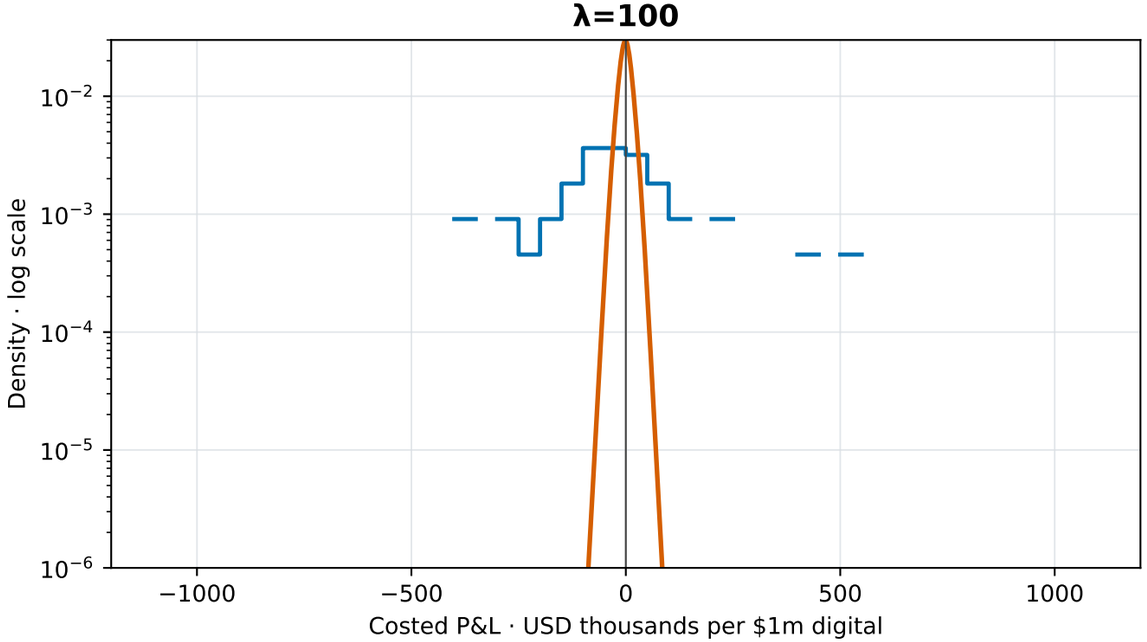
Realized, raw Fitted 252-day demeaned SANOS returns

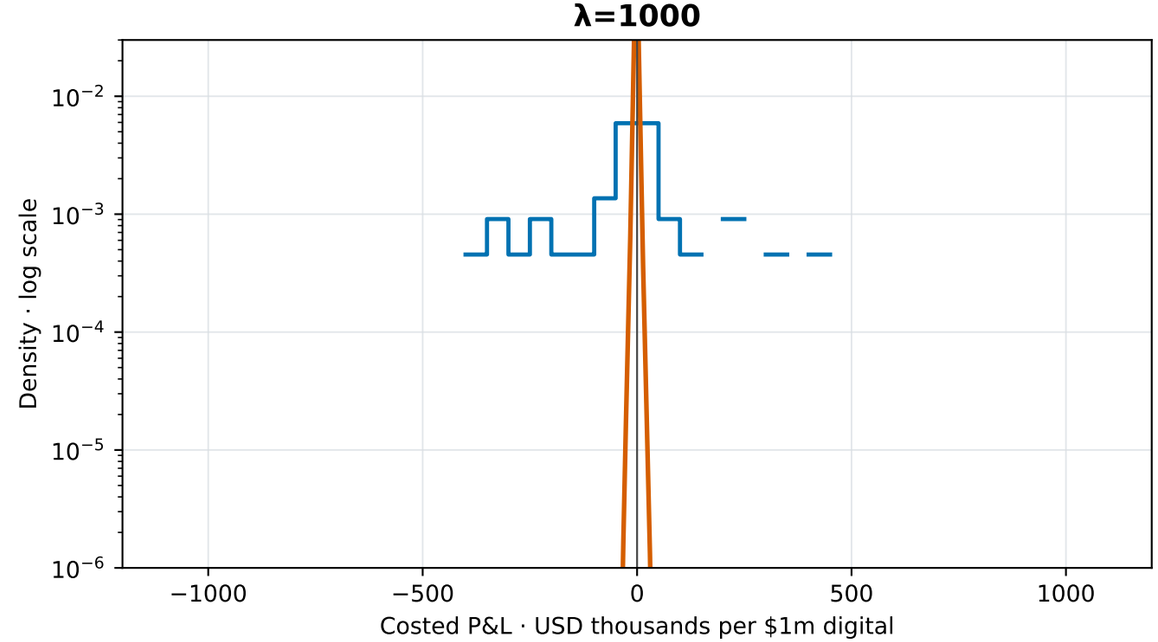
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.75: SANOS parametric · Mean-variance





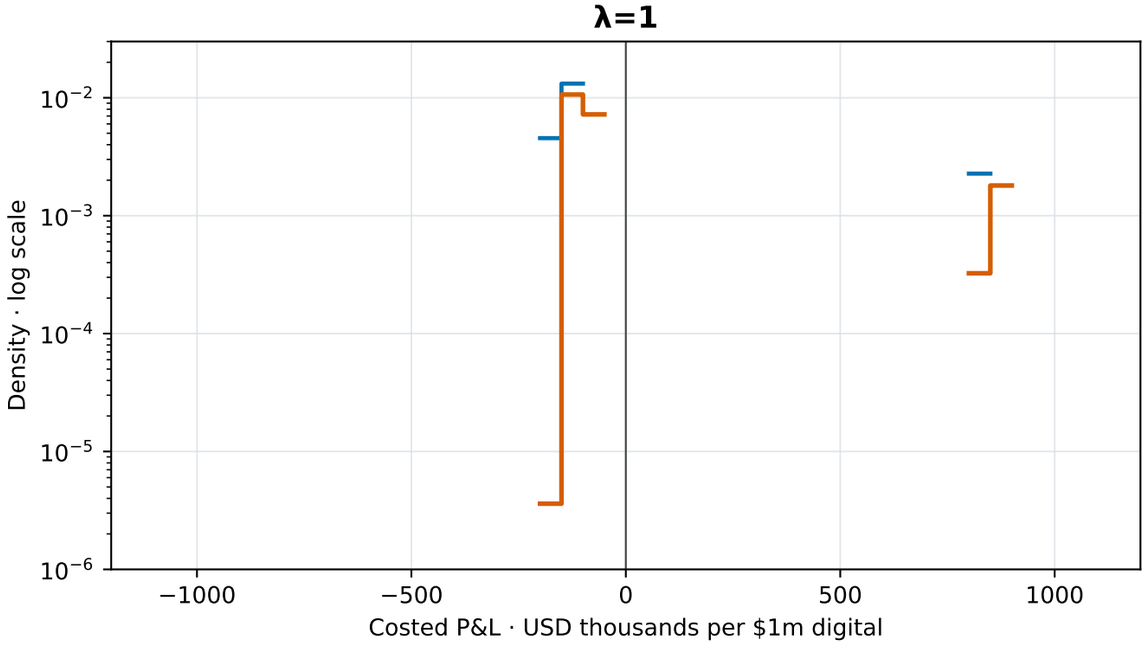


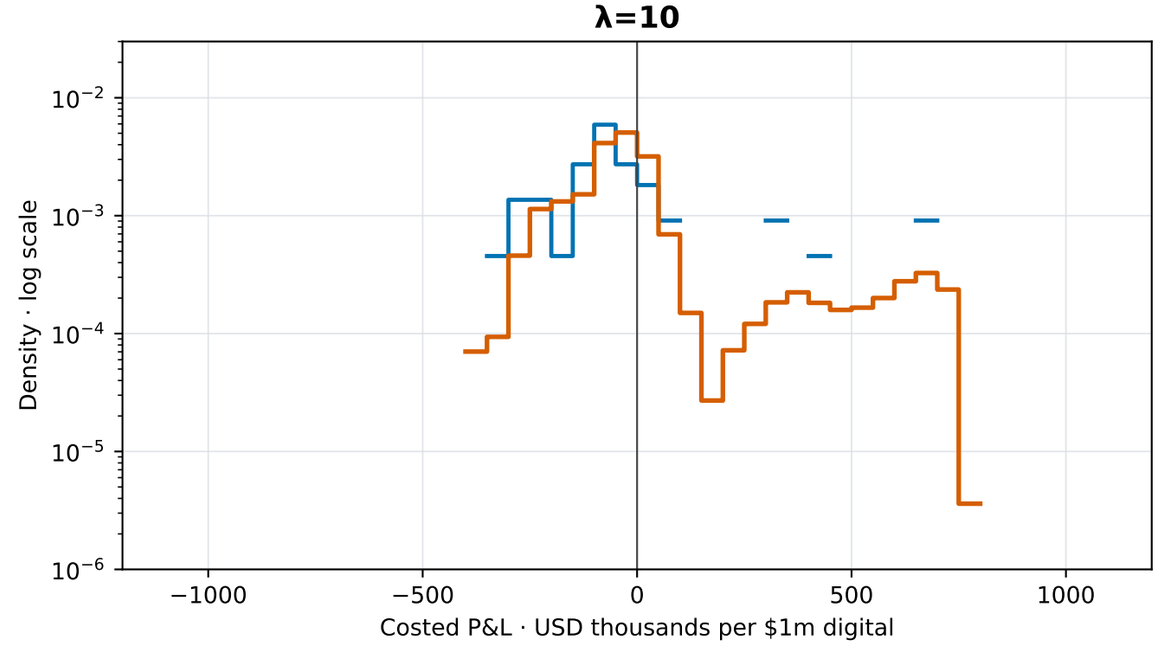


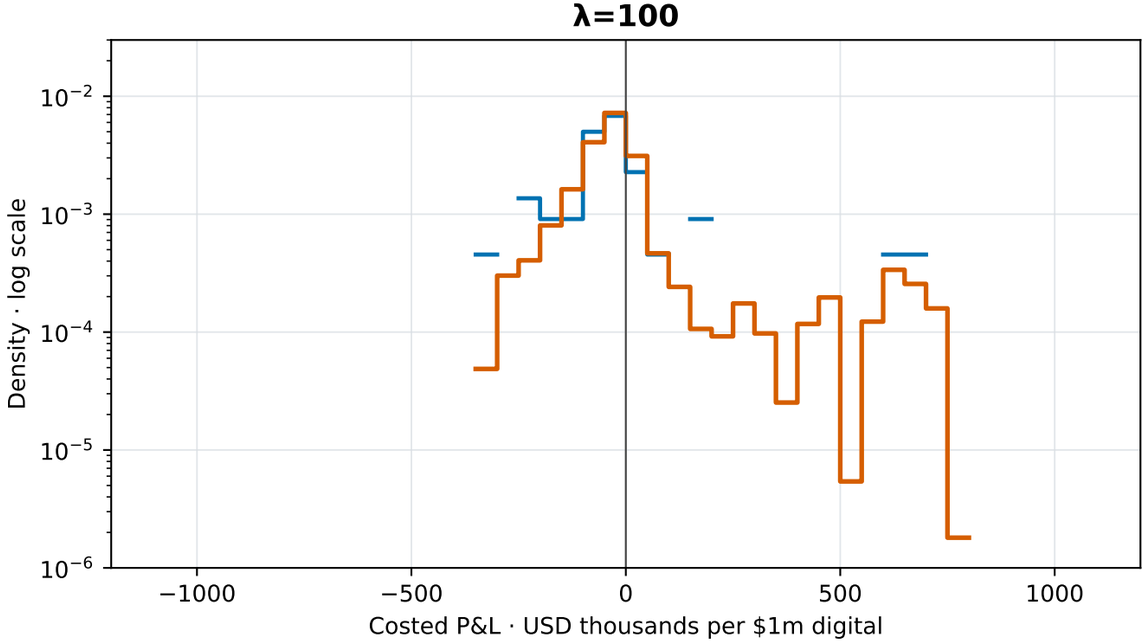
Realized, raw Fitted centered factor Gaussian

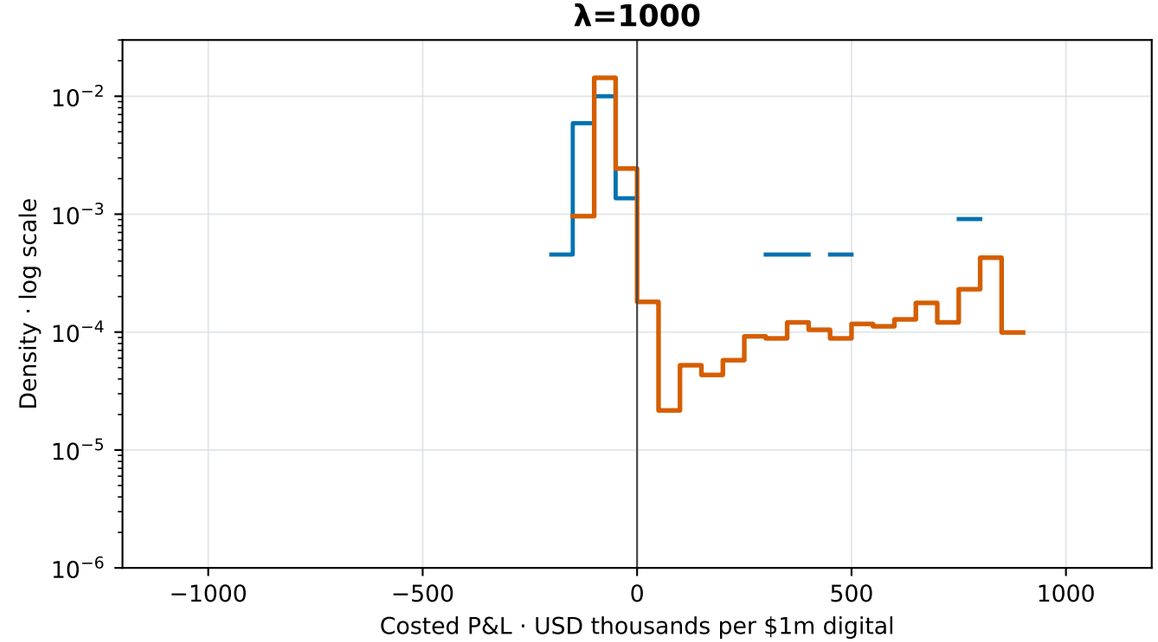
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.75: SANOS historical · Entropy





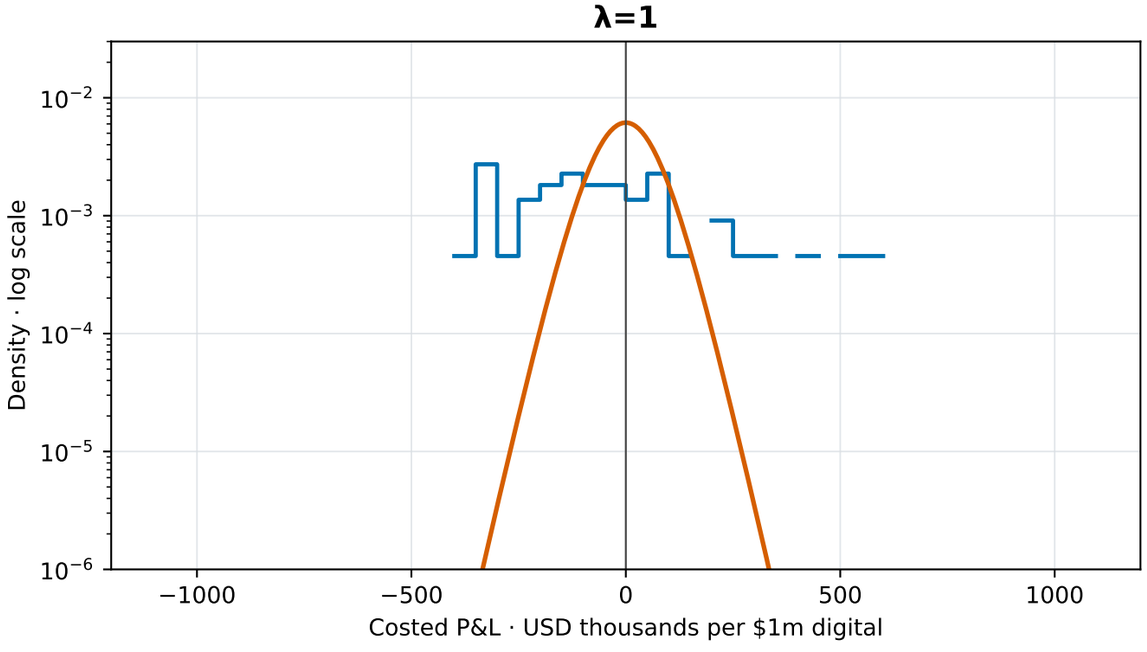


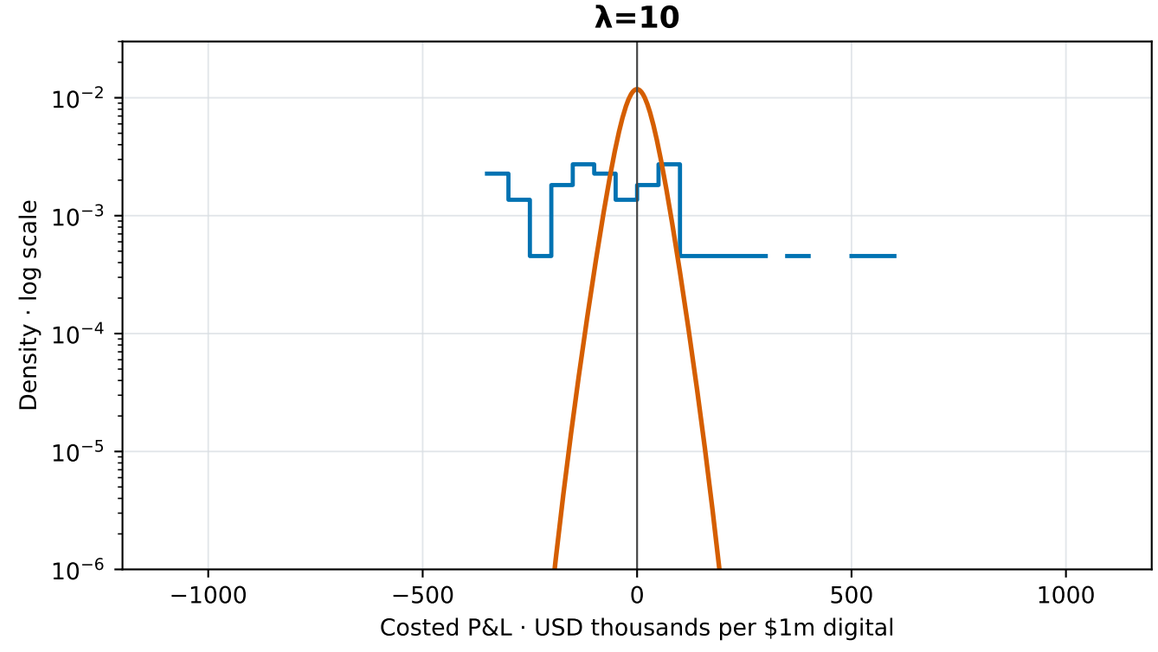


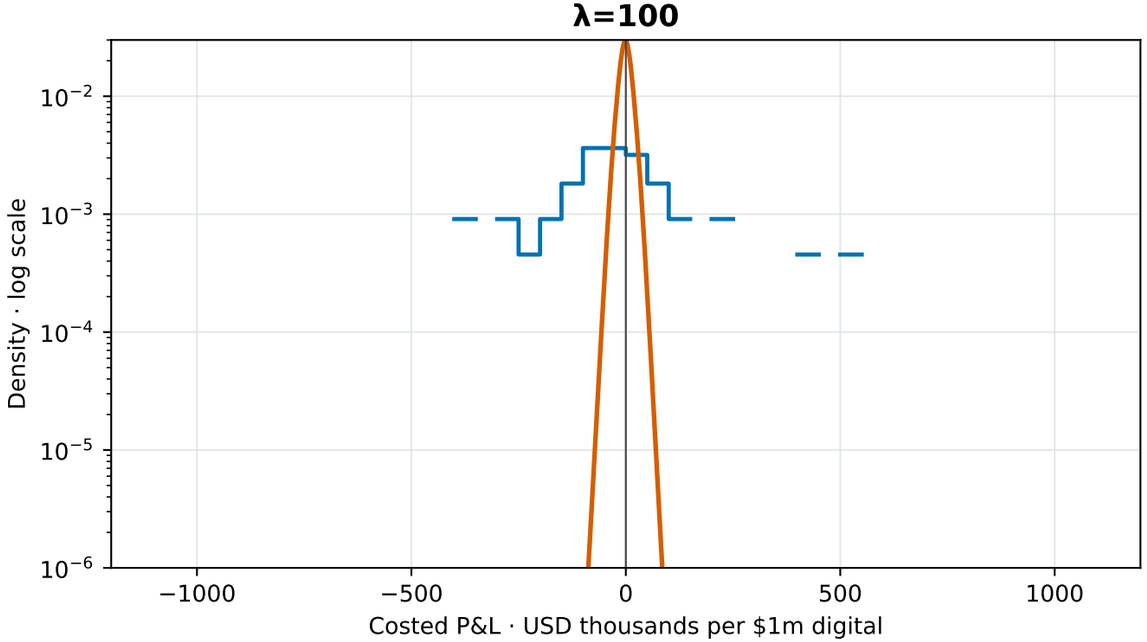
Realized, raw Fitted 252-day demeaned SANOS returns

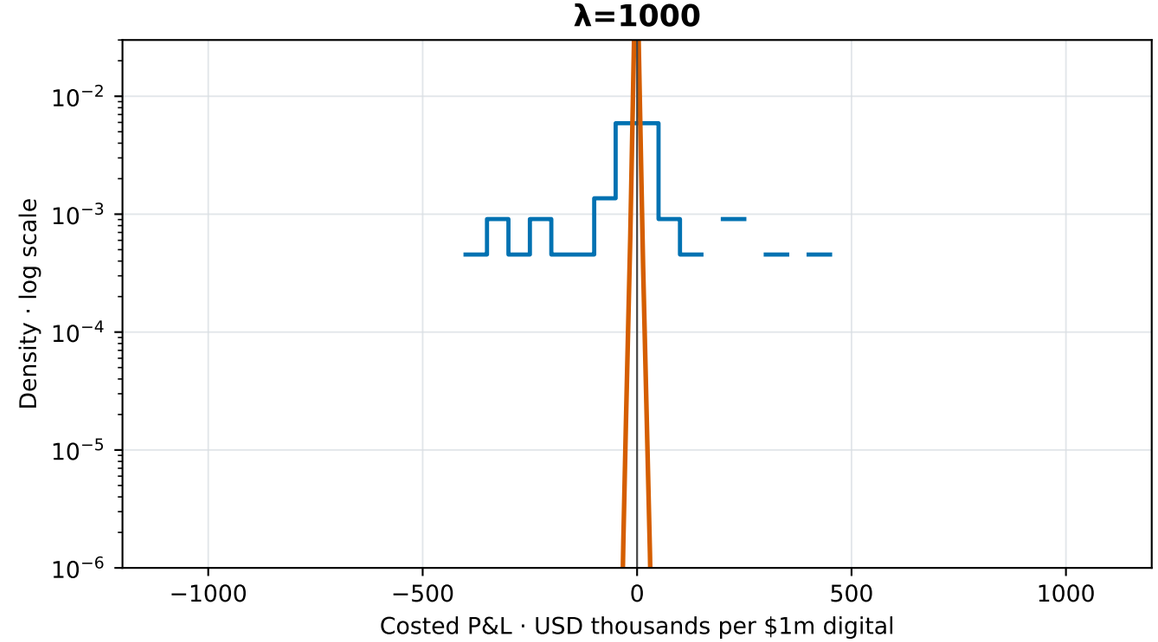
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.75: SANOS parametric · Entropy





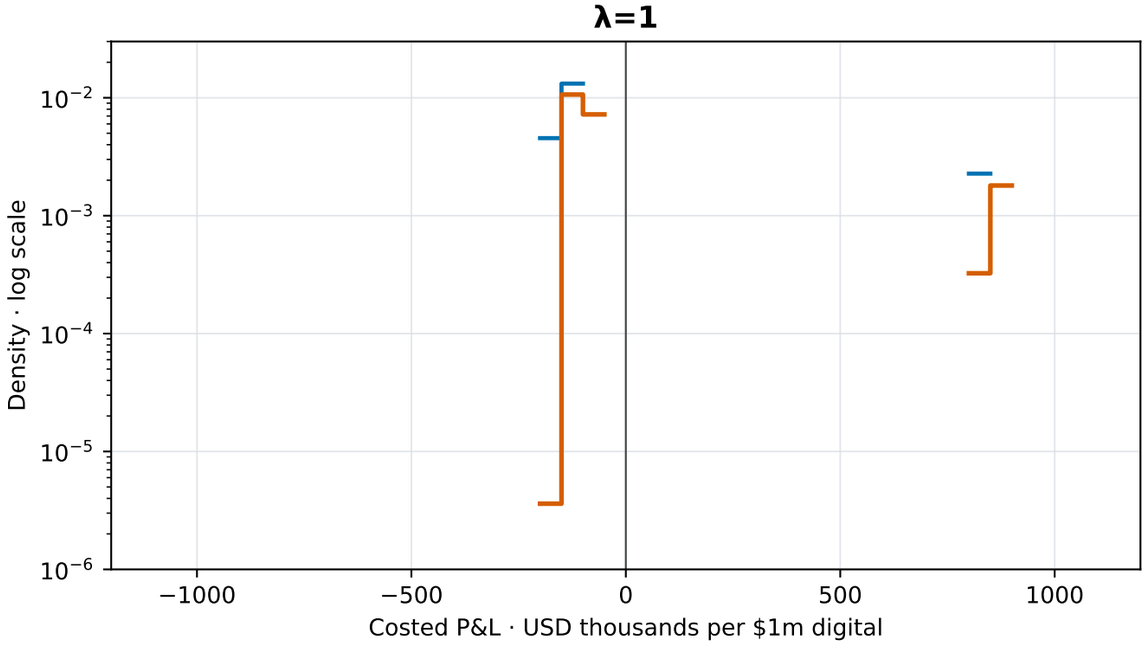


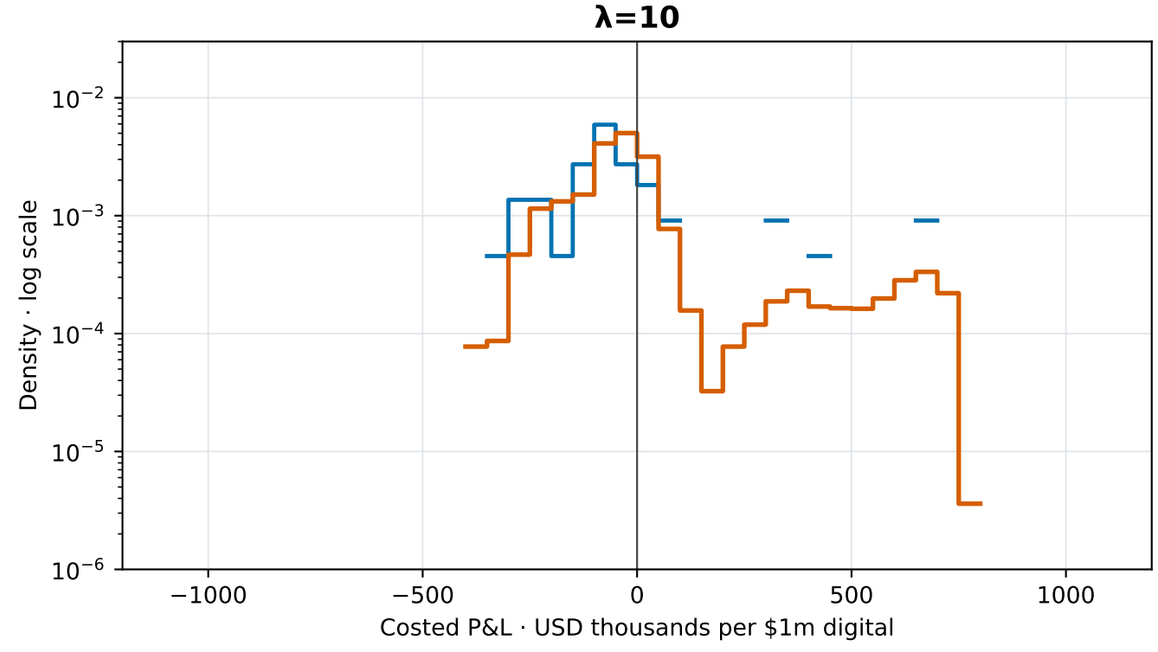


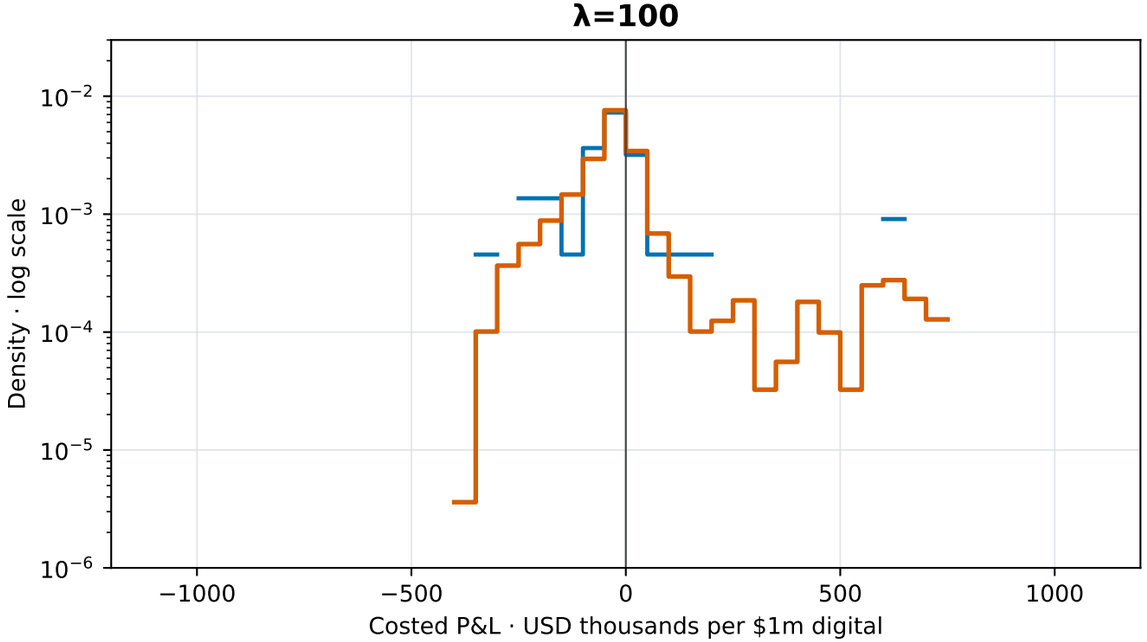
Realized, raw Fitted centered factor Gaussian

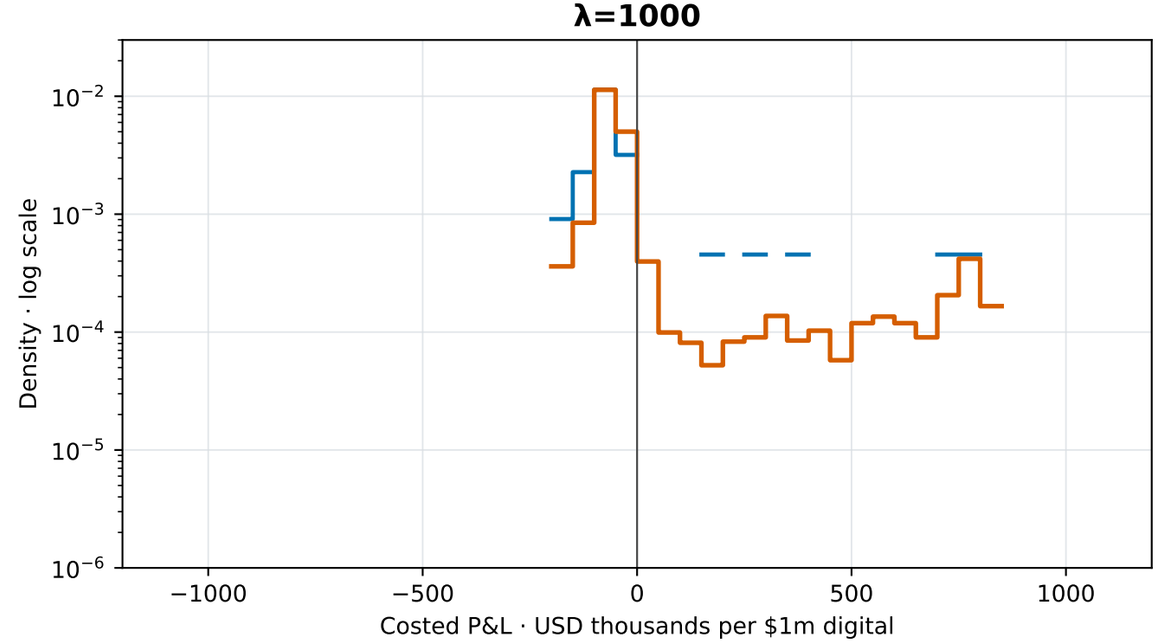
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.75: SANOS historical · Power-exponential





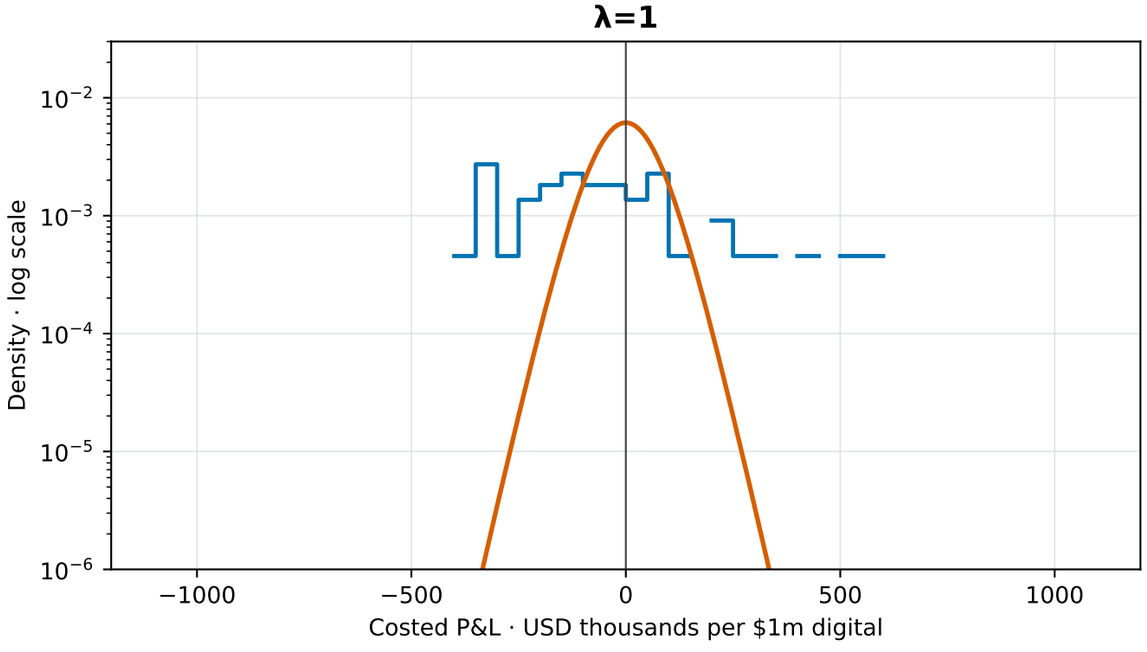


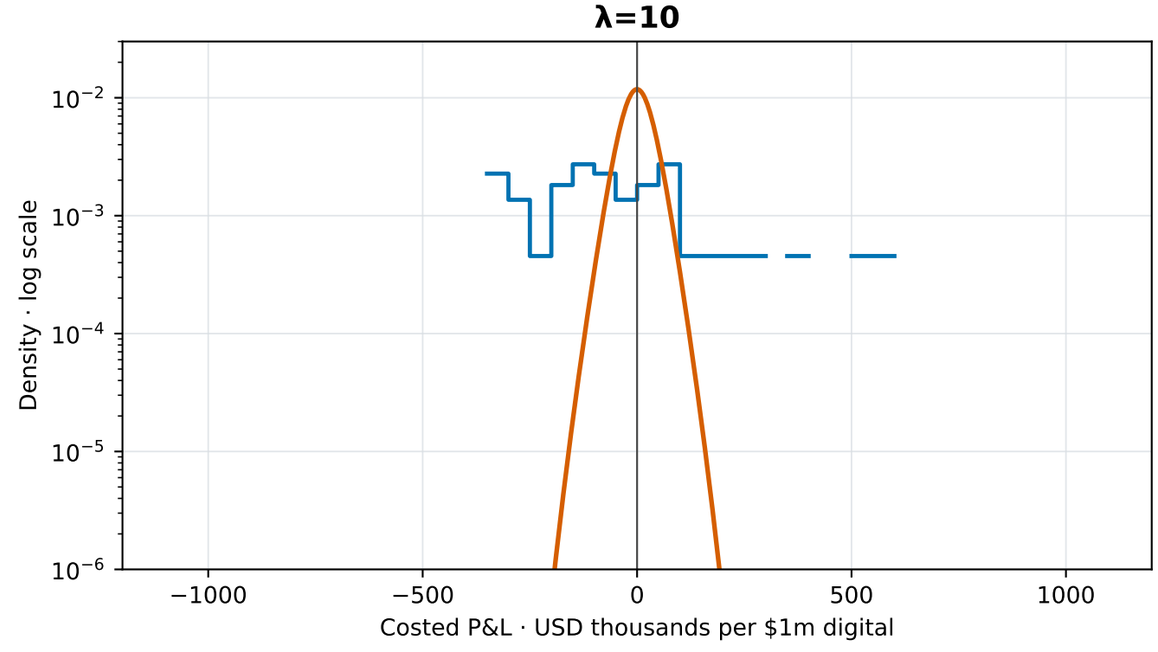


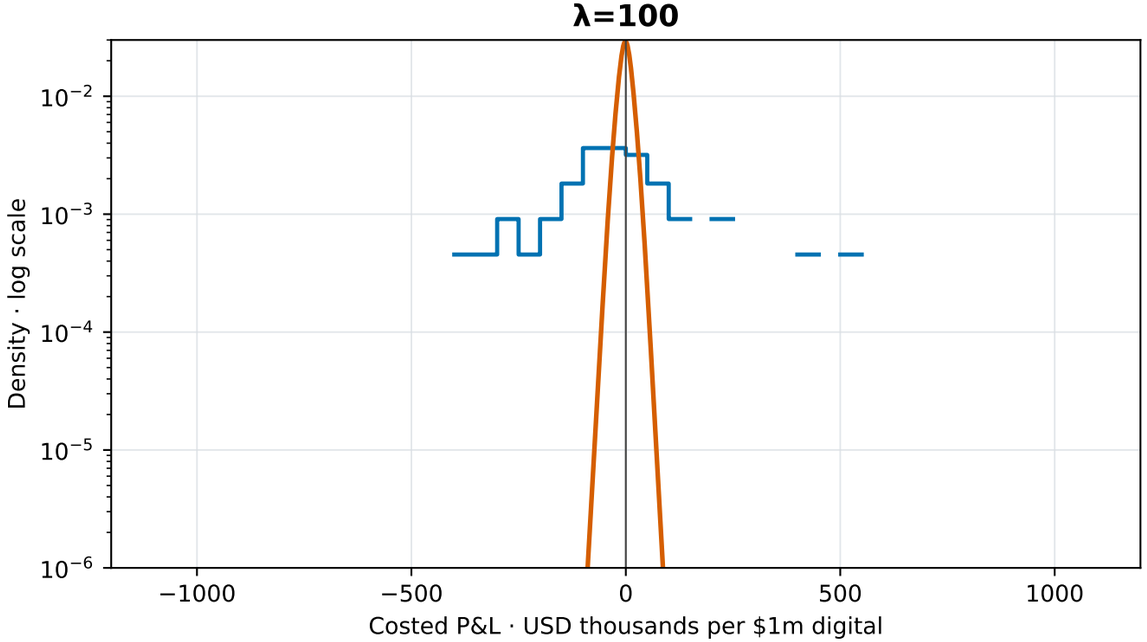
Realized, raw Fitted 252-day demeaned SANOS returns

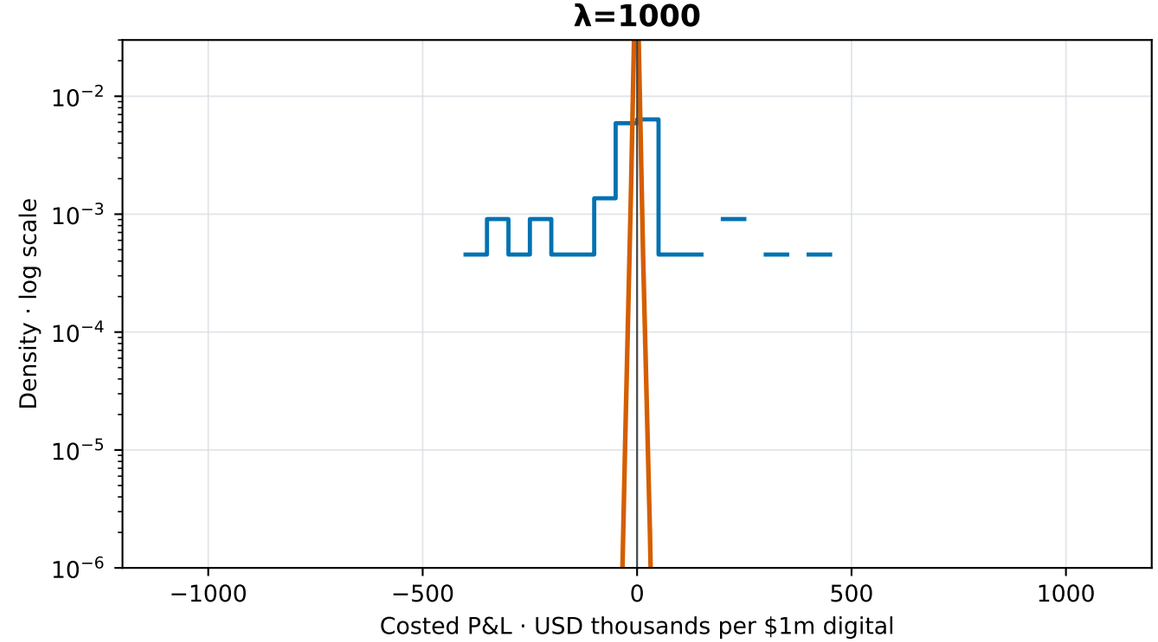
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.75: SANOS parametric · Power-exponential



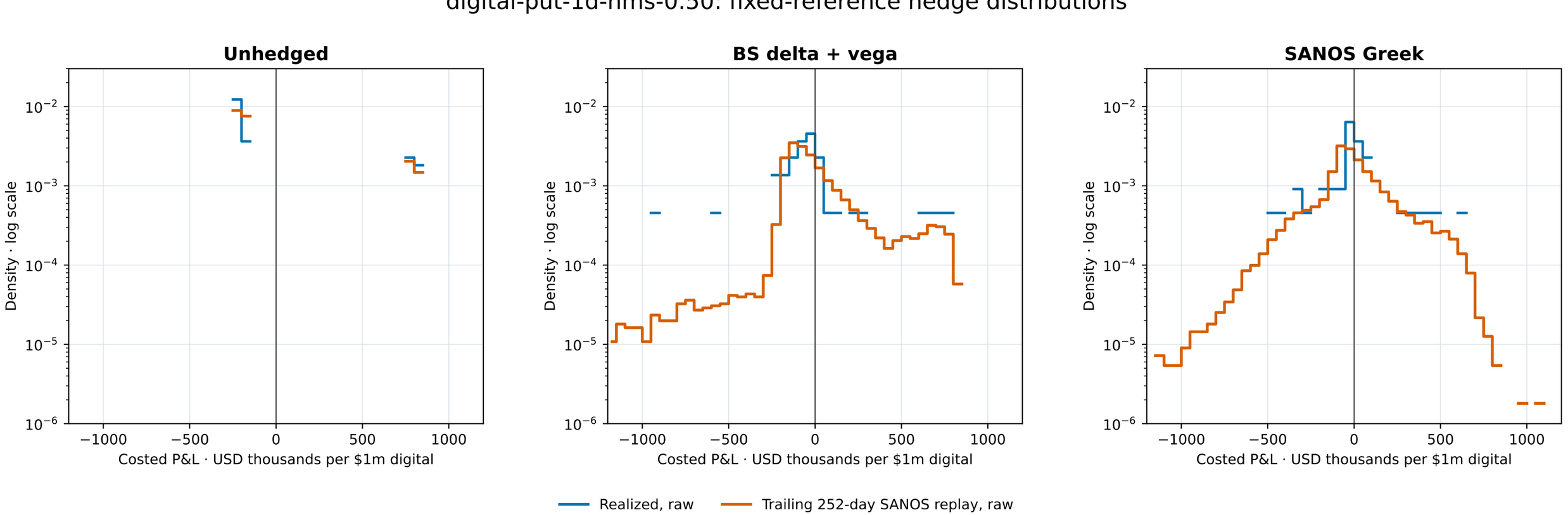




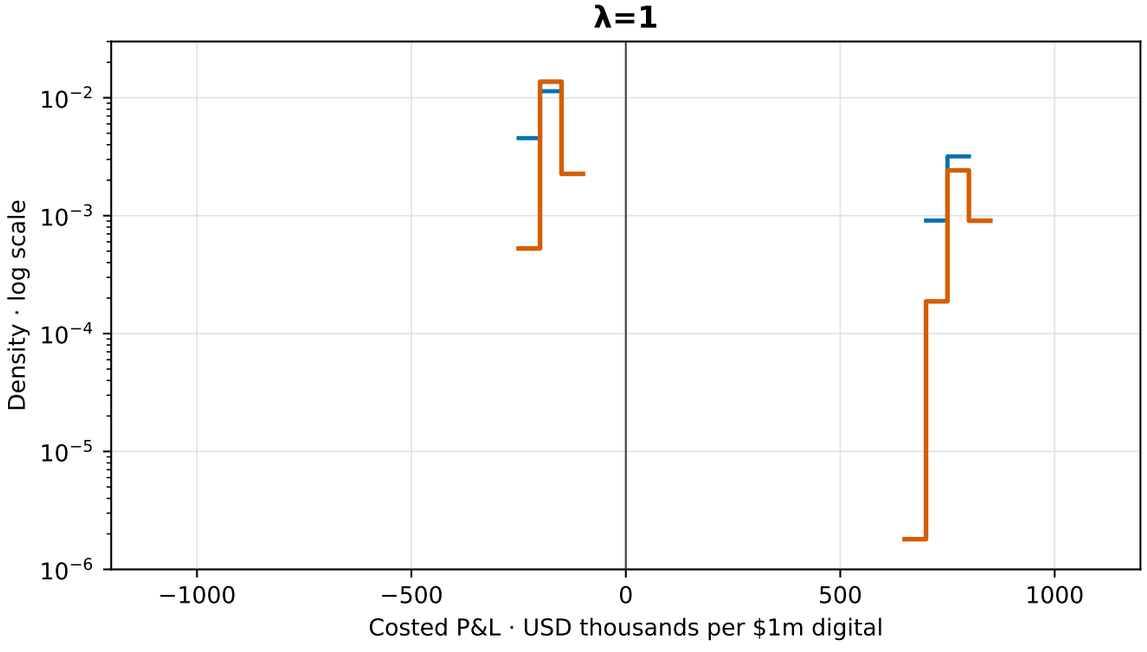


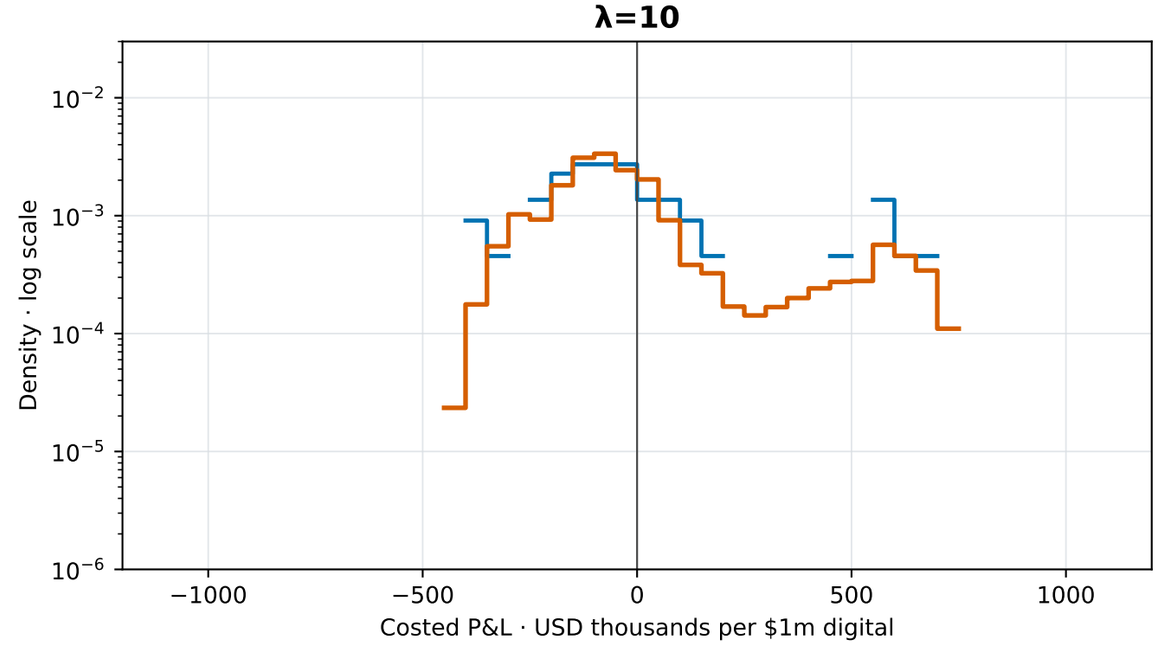
Realized, raw Fitted centered factor Gaussian

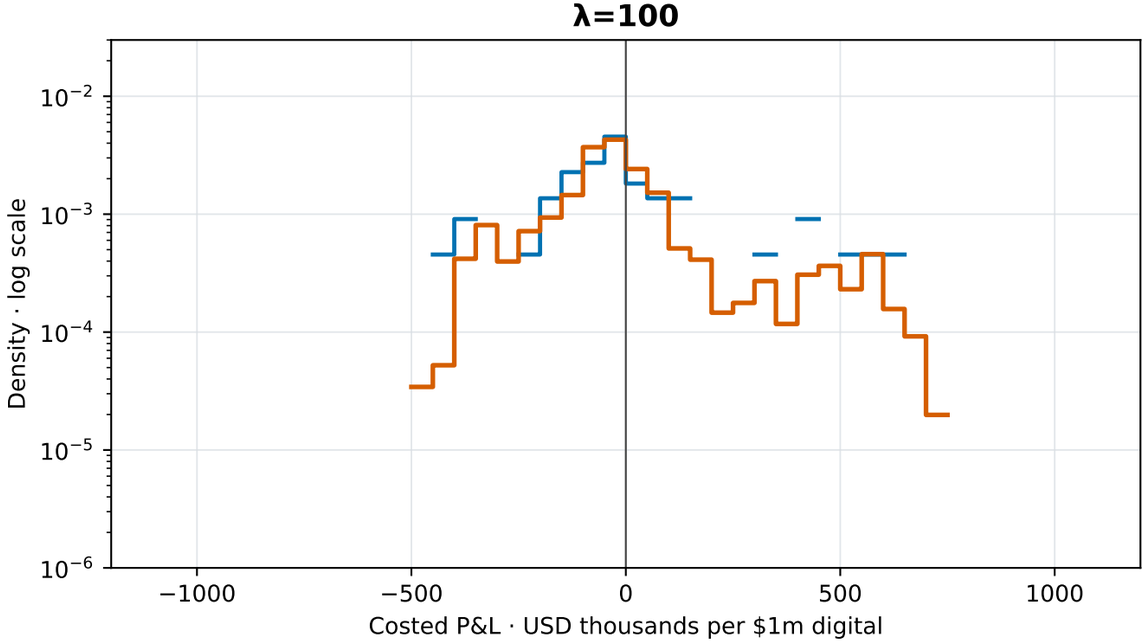
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

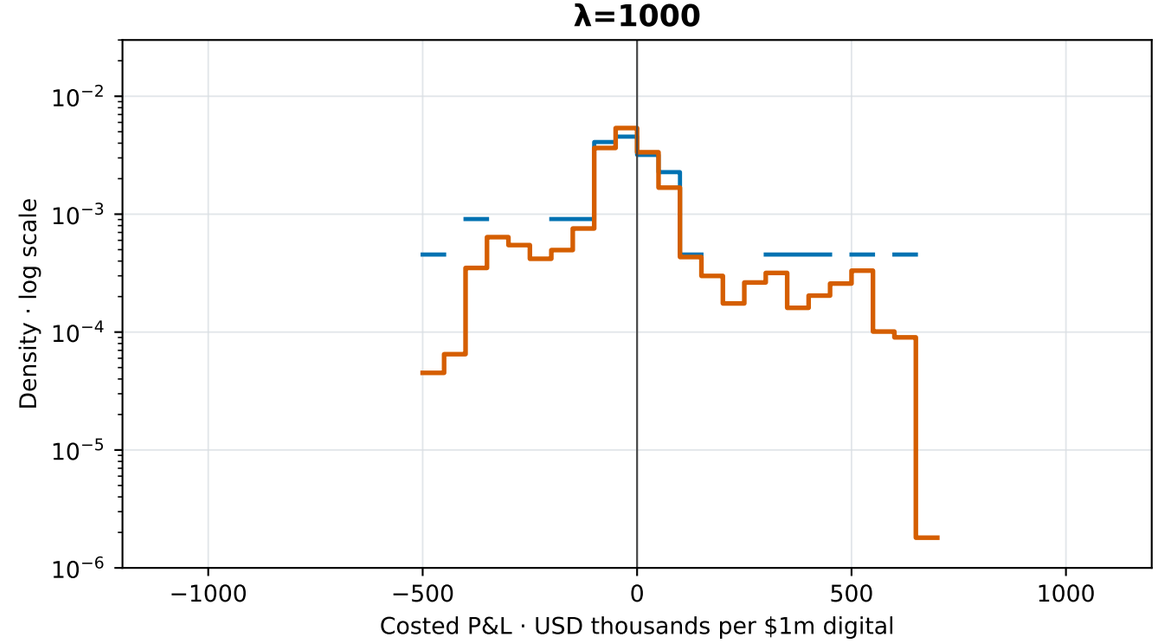

digital-put-1d-nms-0.50: fixed-reference hedge distributions
Unhedged
BS delta + vega
SANOS Greek
Density · log scale
Costed P&L · USD thousands per $1m digital
Realized, raw
Trailing 252-day SANOS replay, raw
No mean correction: 44 realized next-day outcomes versus preceding 252 raw returns. References have no fitted hedge-return law.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.50: SANOS historical · Mean-variance





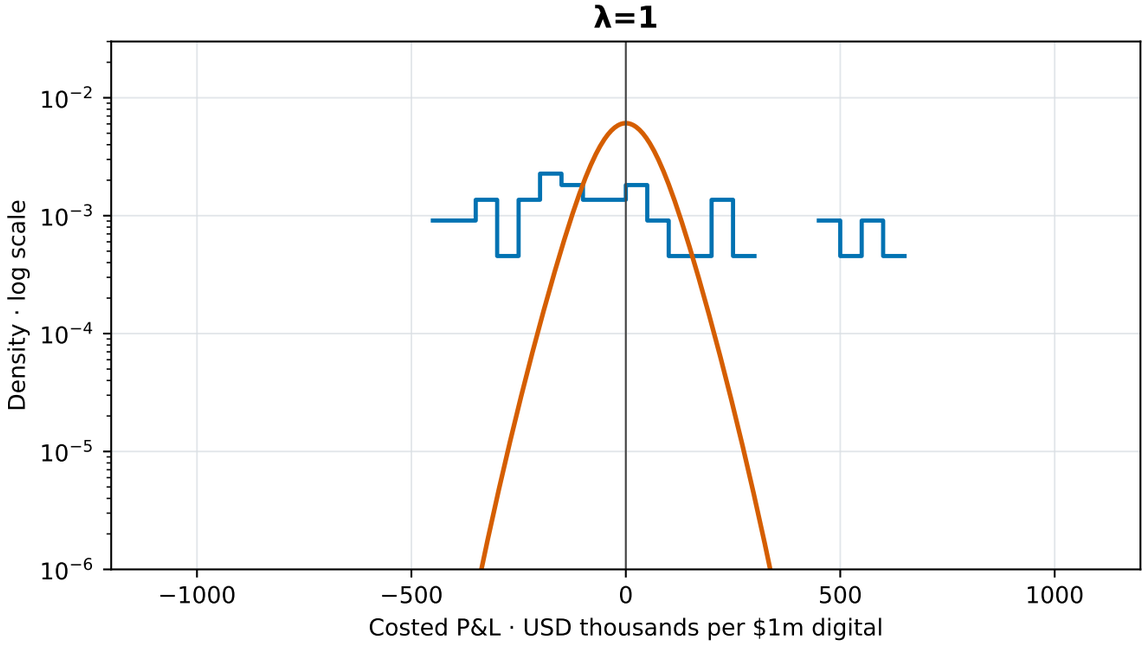


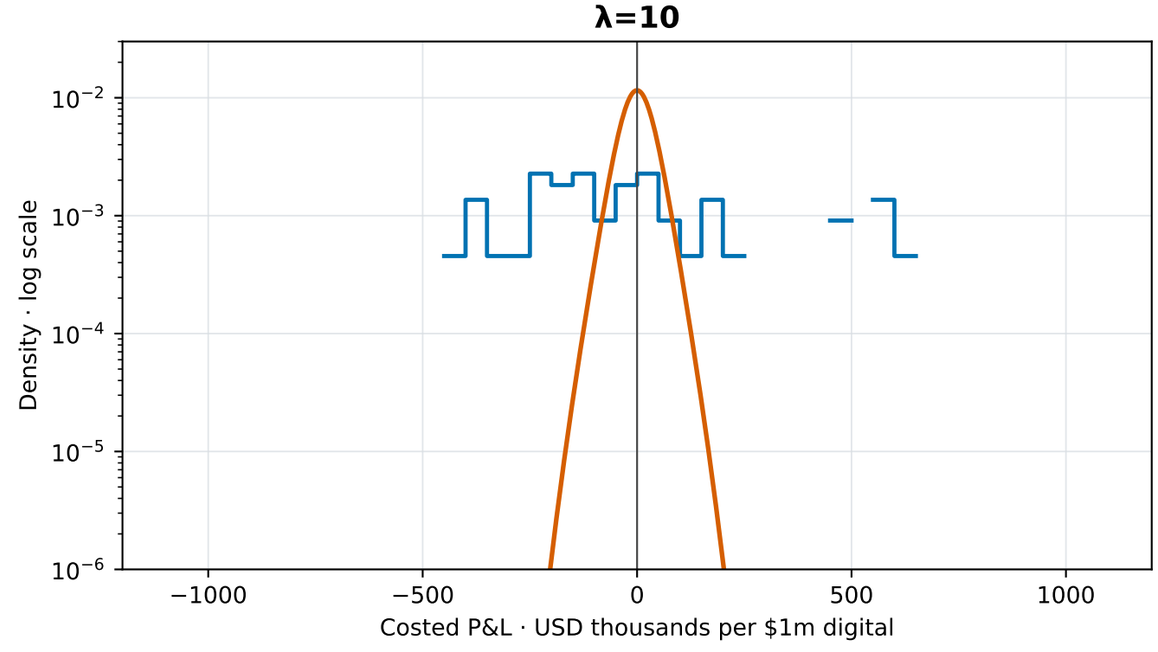


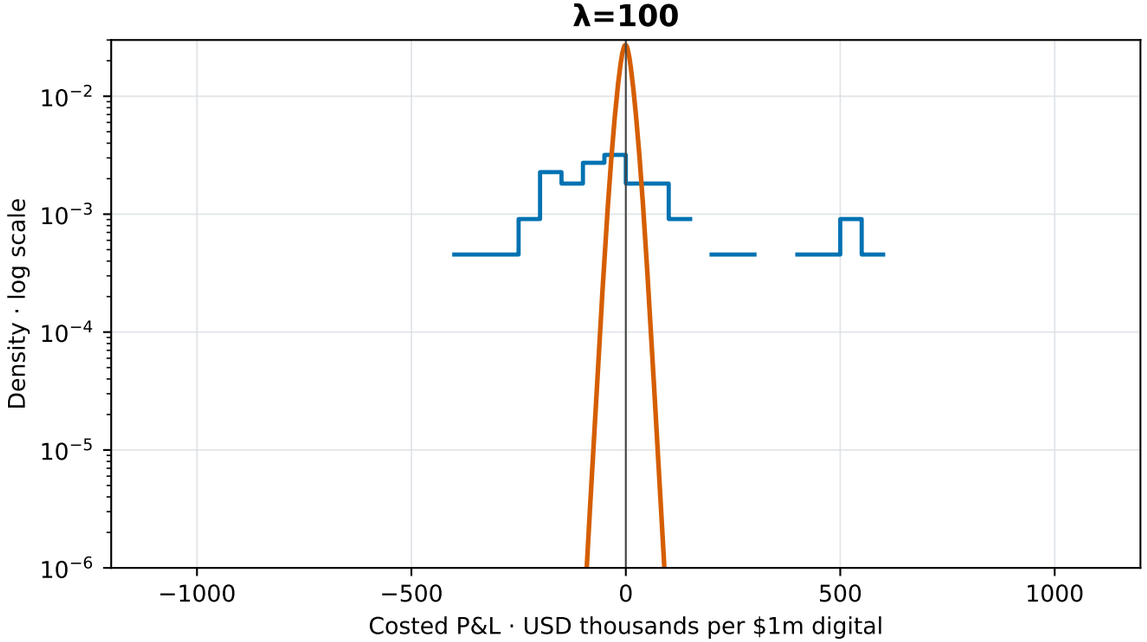
Realized, raw Fitted 252-day demeaned SANOS returns

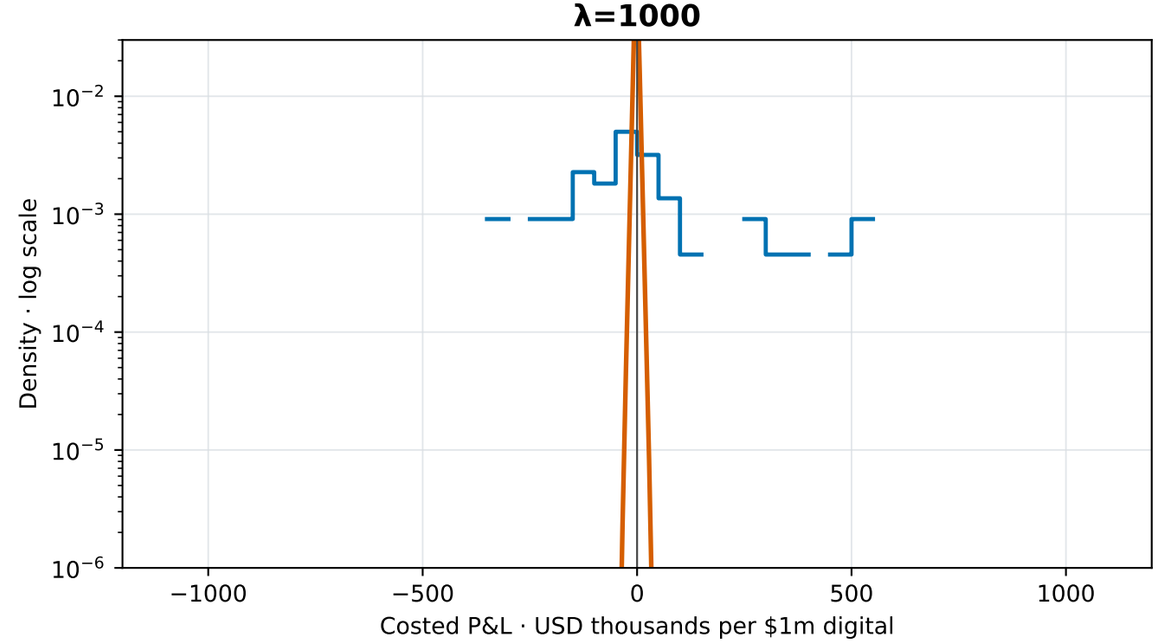
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.50: SANOS parametric · Mean-variance





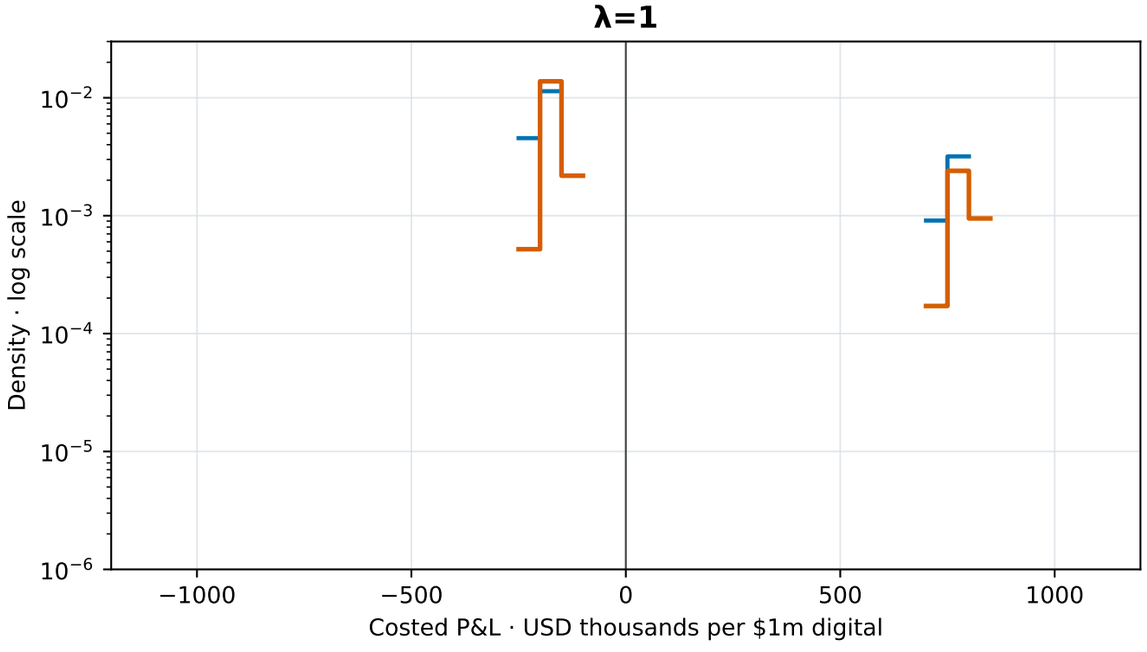


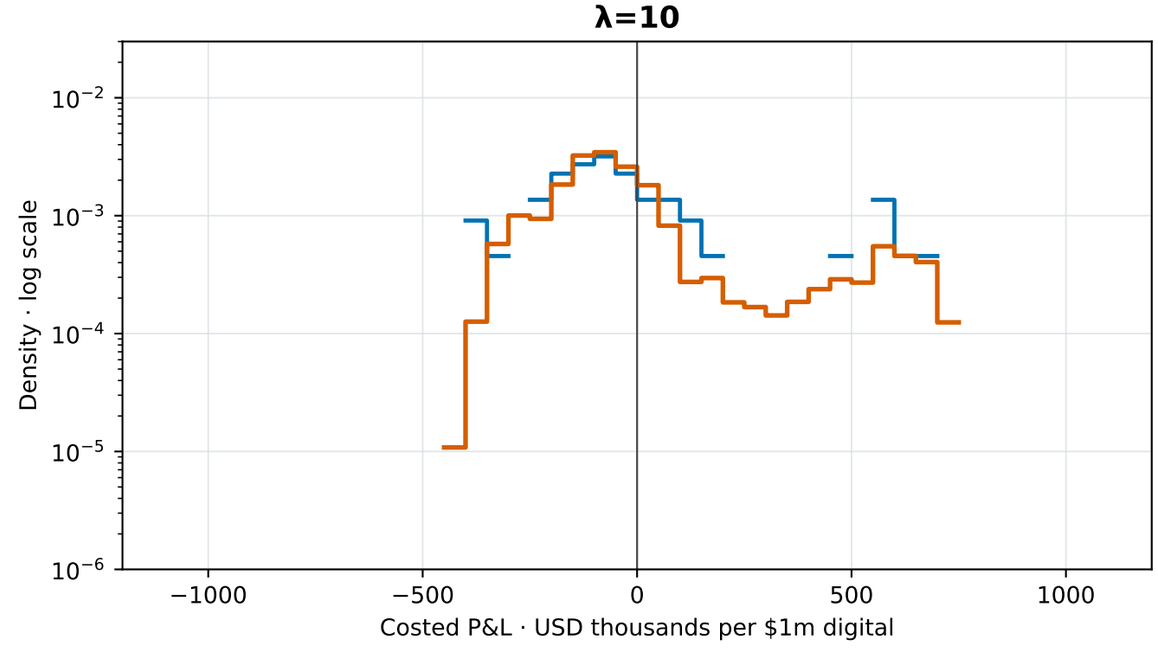


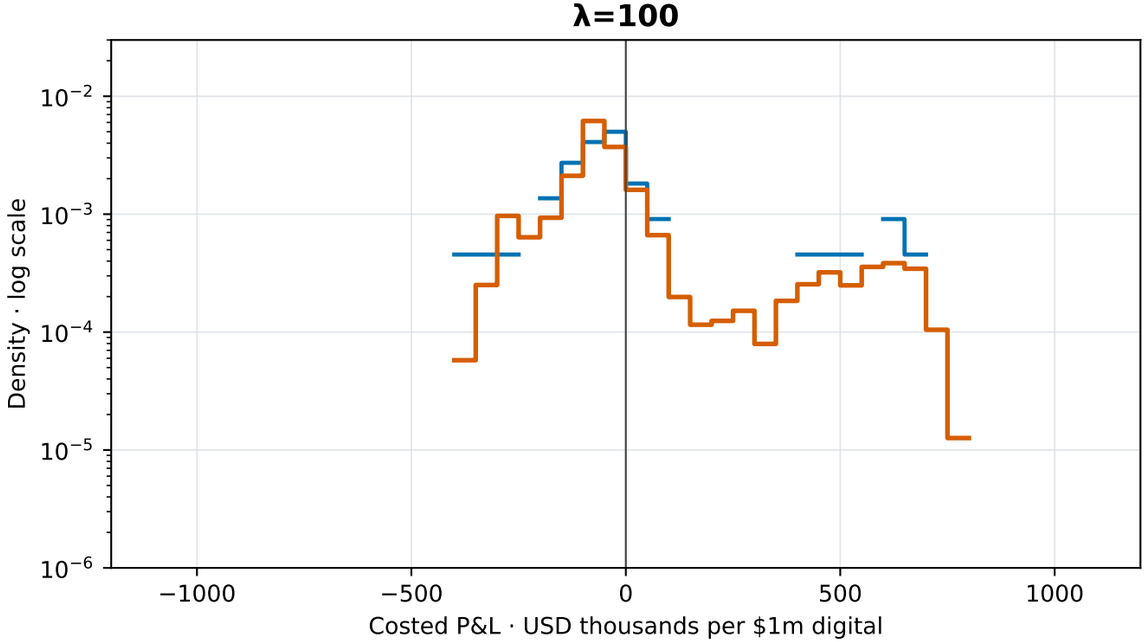
Realized, raw Fitted centered factor Gaussian

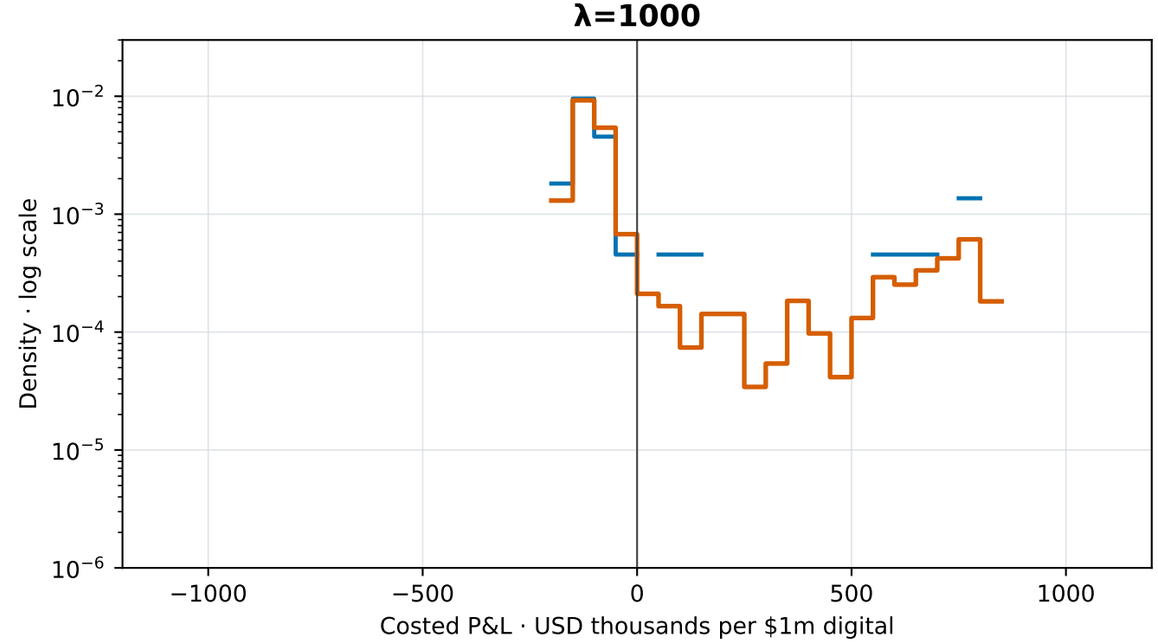
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.50: SANOS historical · Entropy





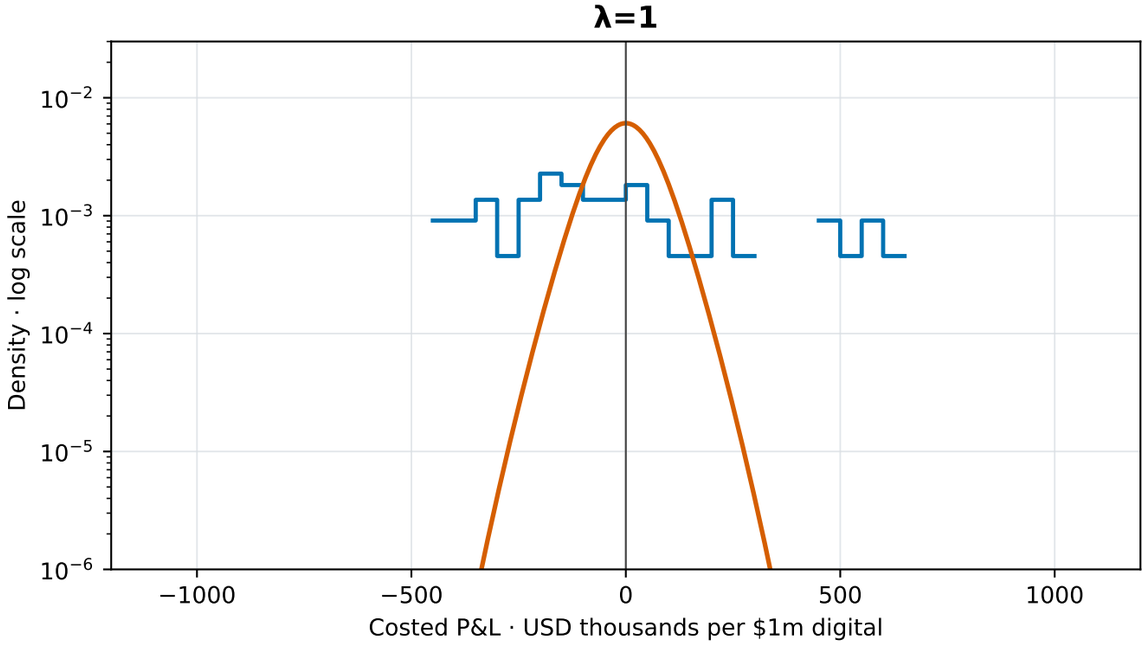


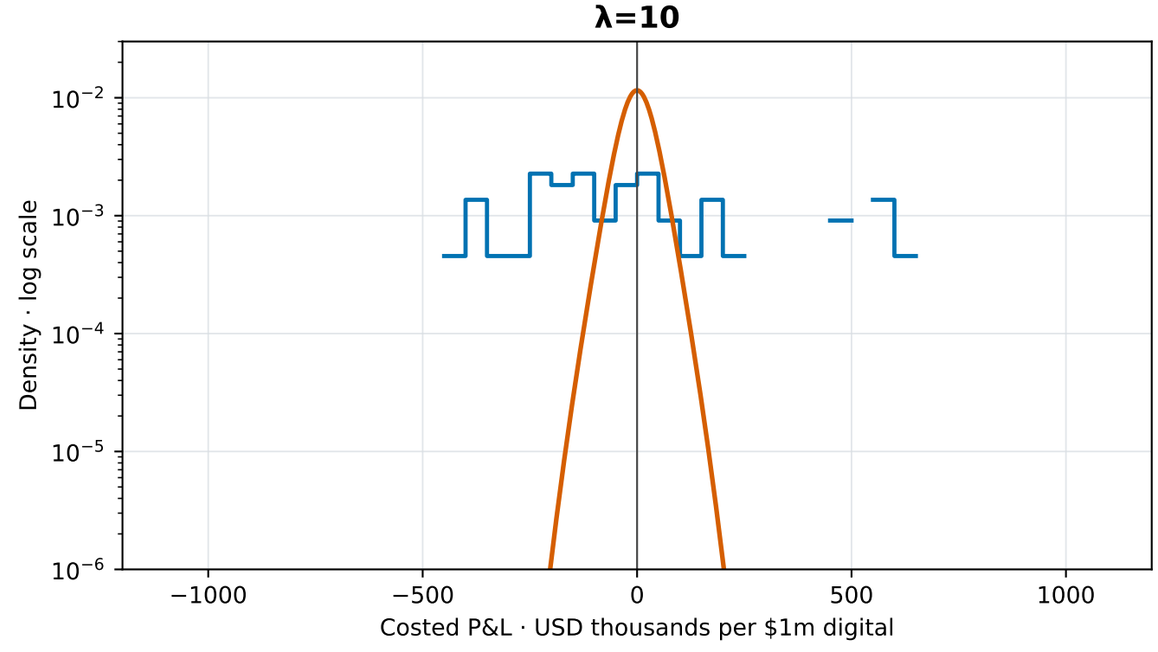


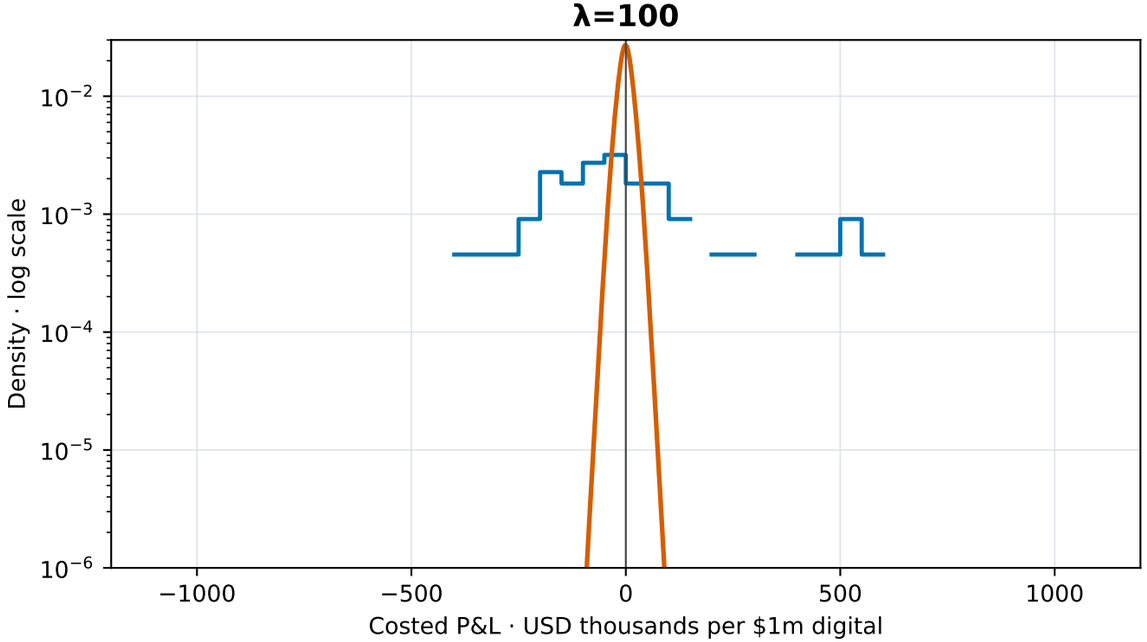
Realized, raw Fitted 252-day demeaned SANOS returns

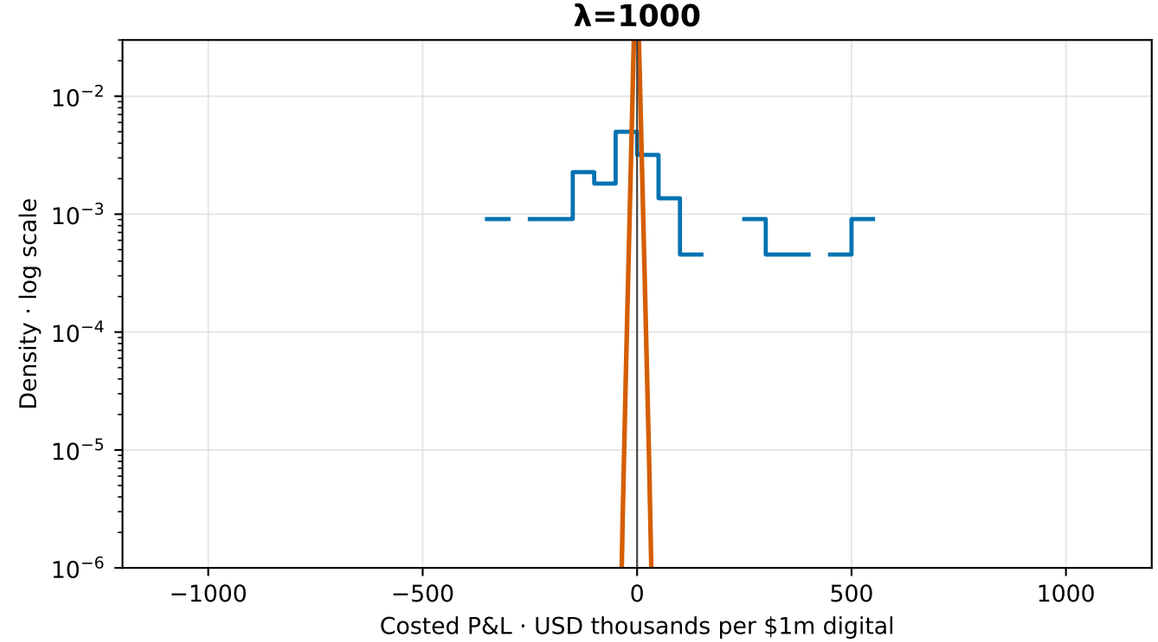
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.50: SANOS parametric · Entropy





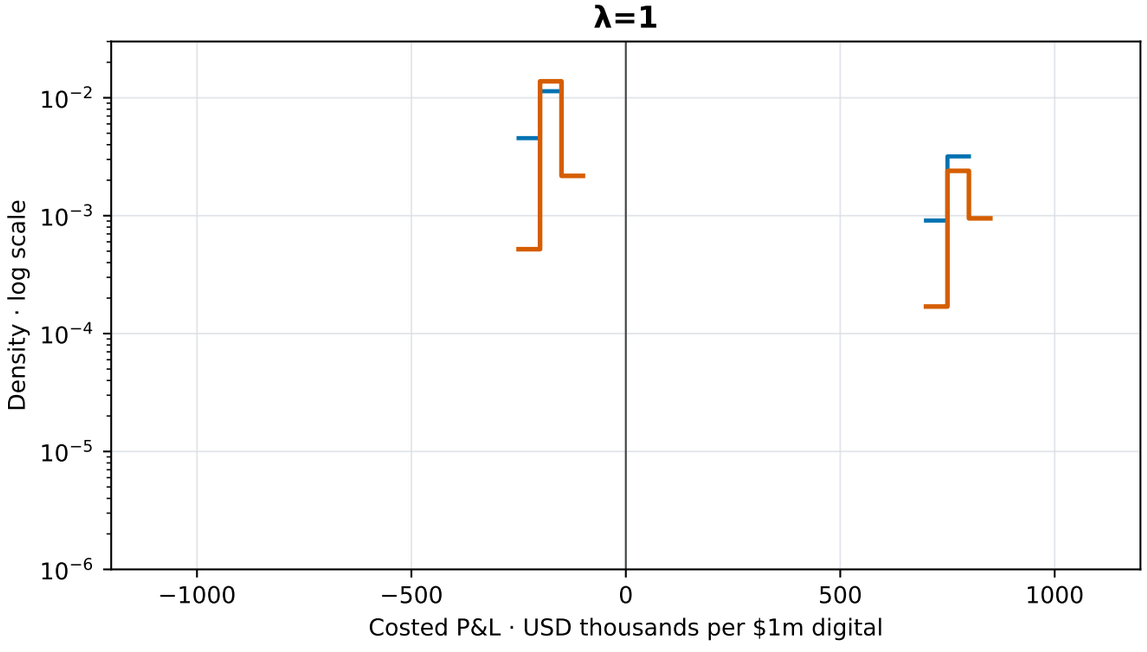


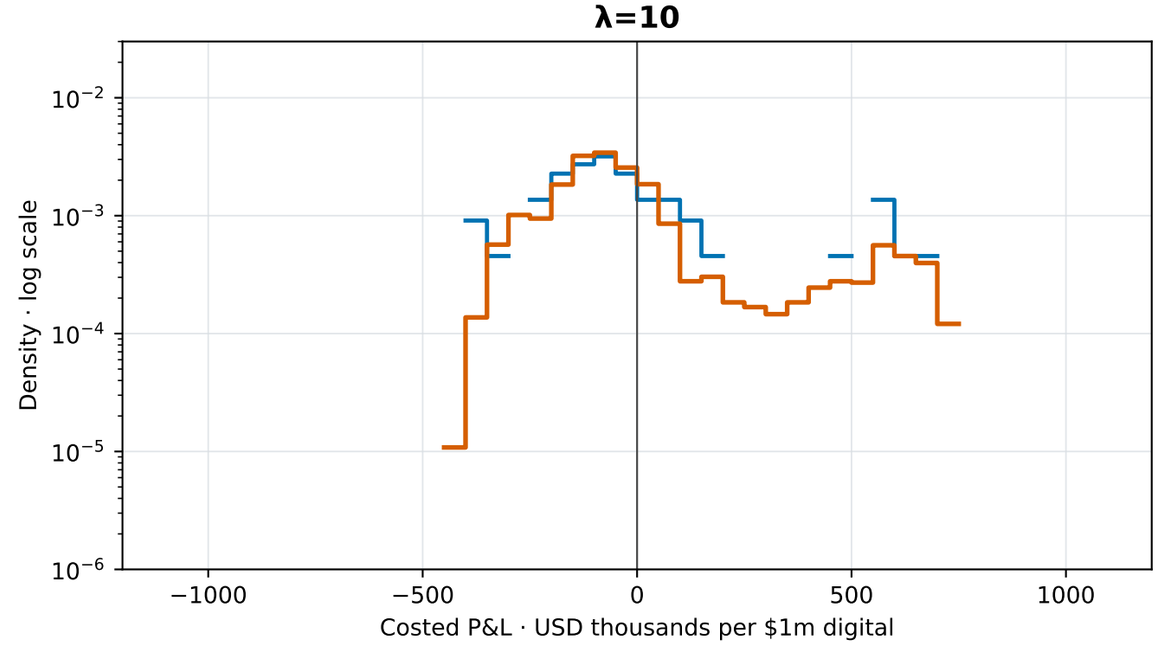


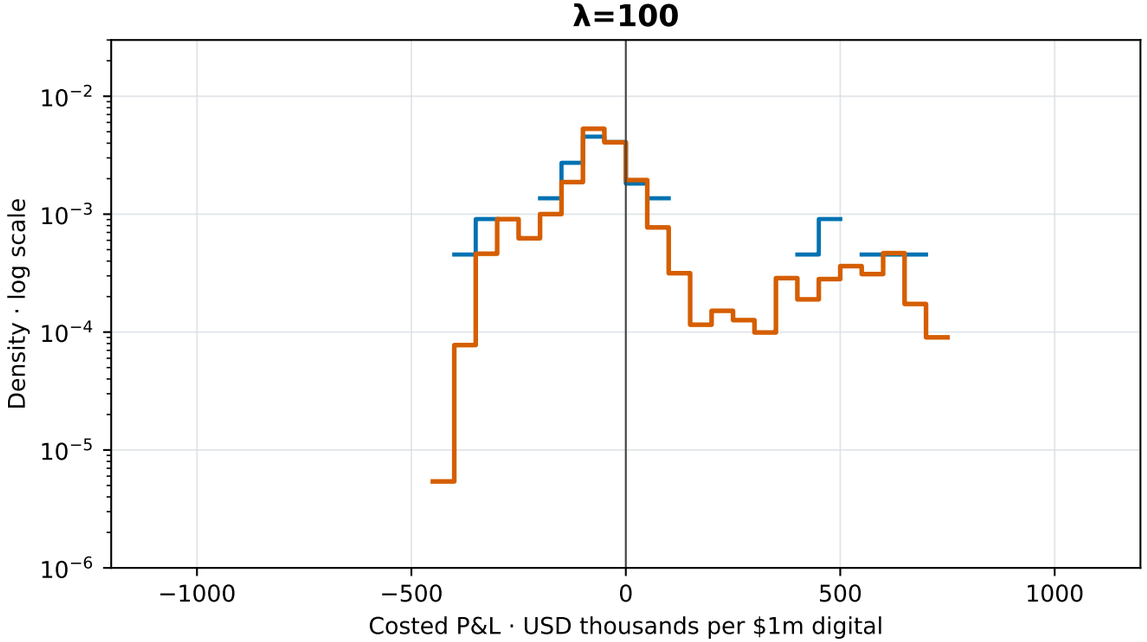
Realized, raw Fitted centered factor Gaussian

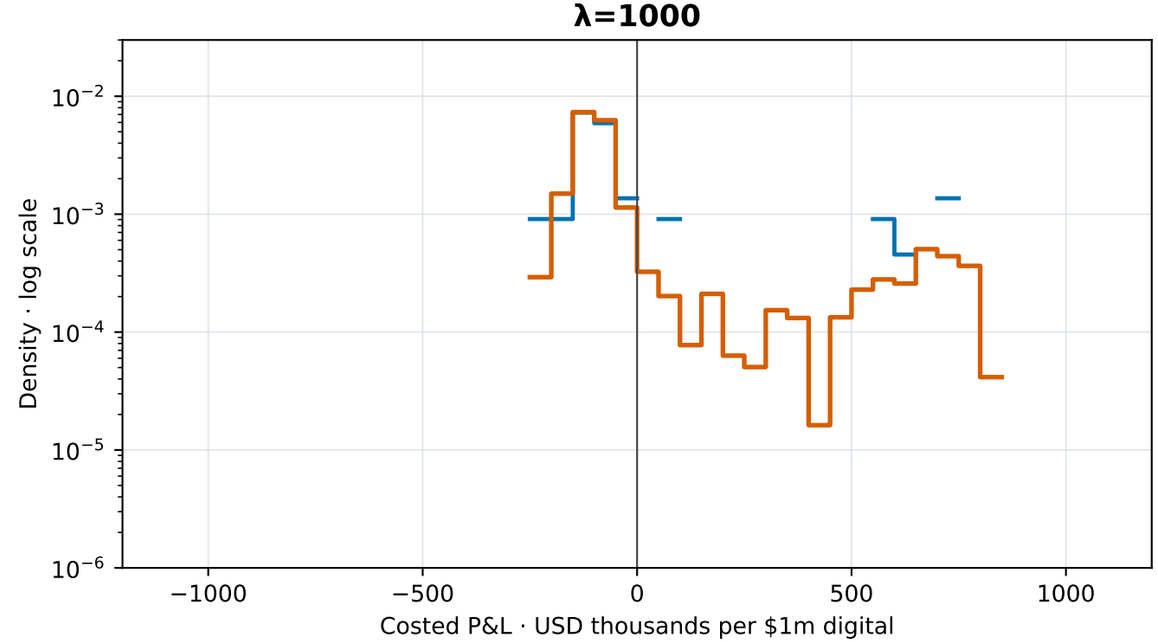
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.50: SANOS historical · Power-exponential





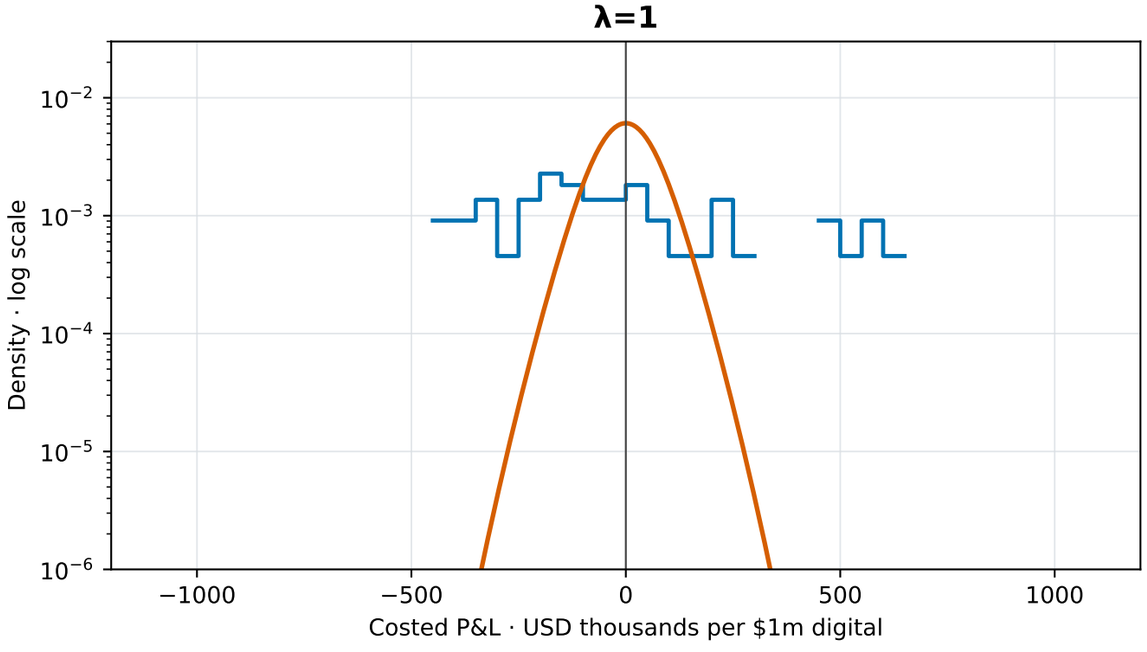


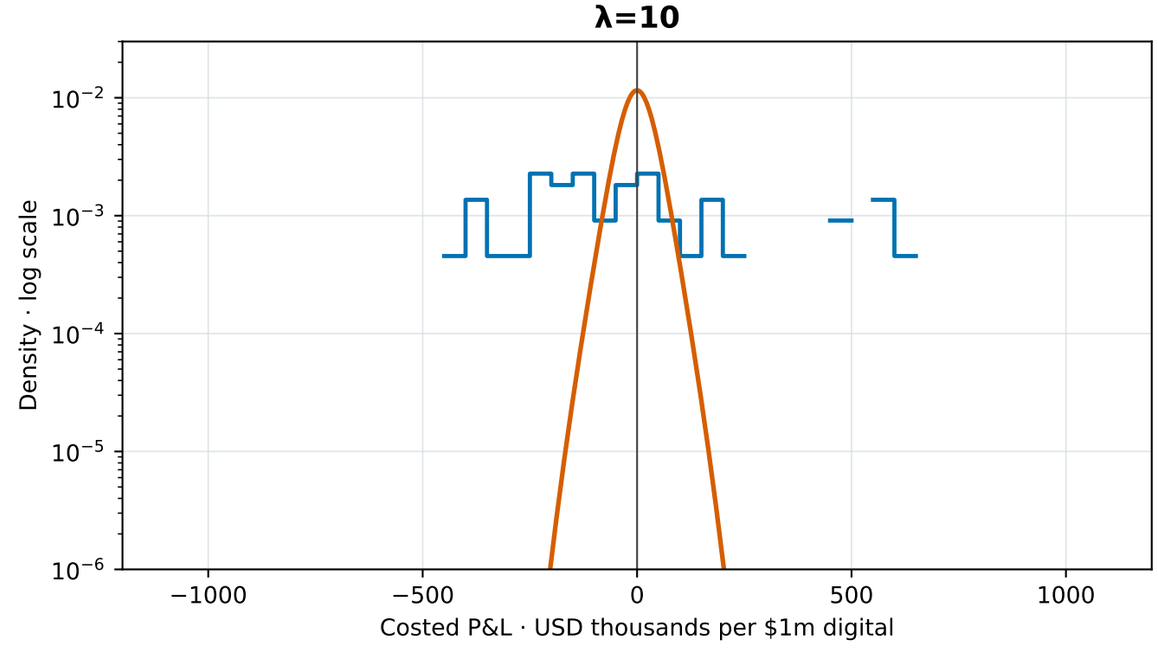


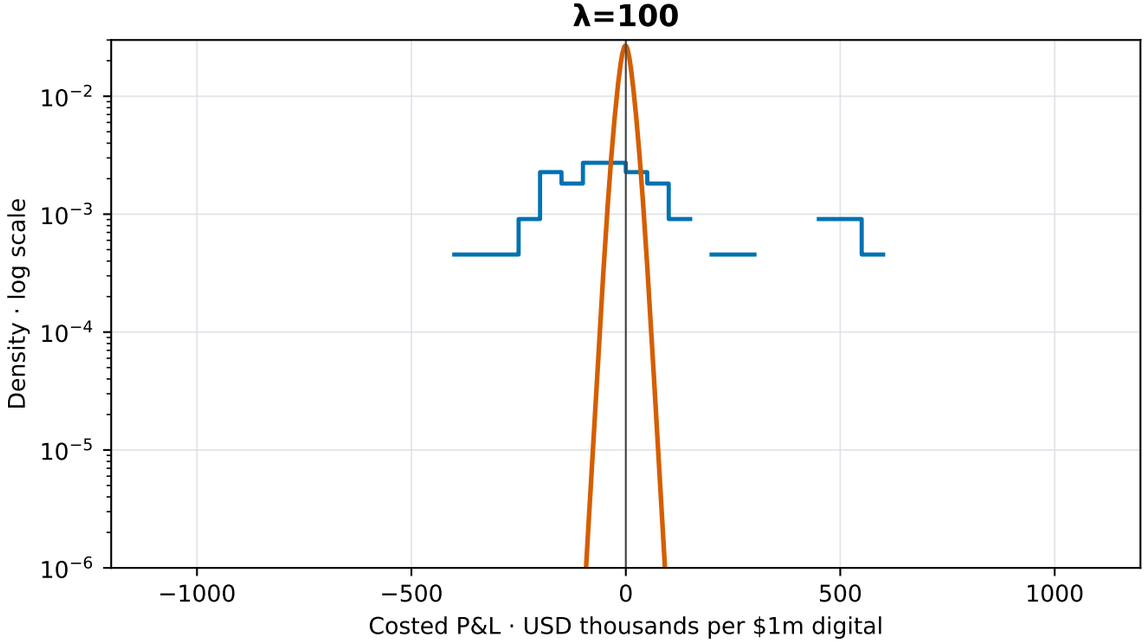
Realized, raw Fitted 252-day demeaned SANOS returns

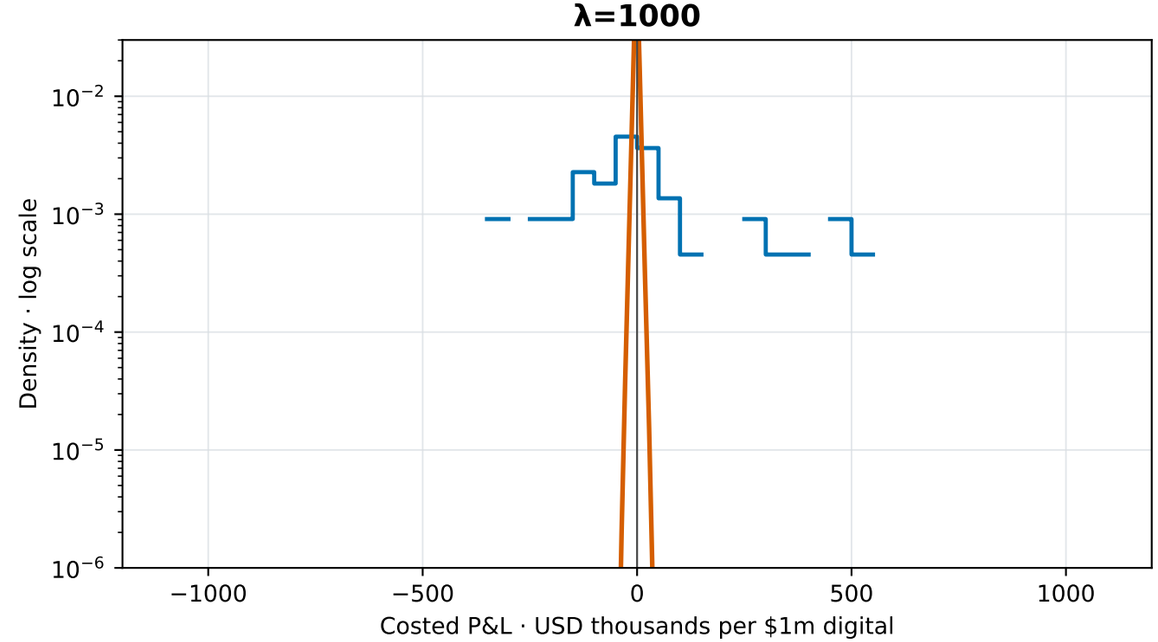
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.50: SANOS parametric · Power-exponential



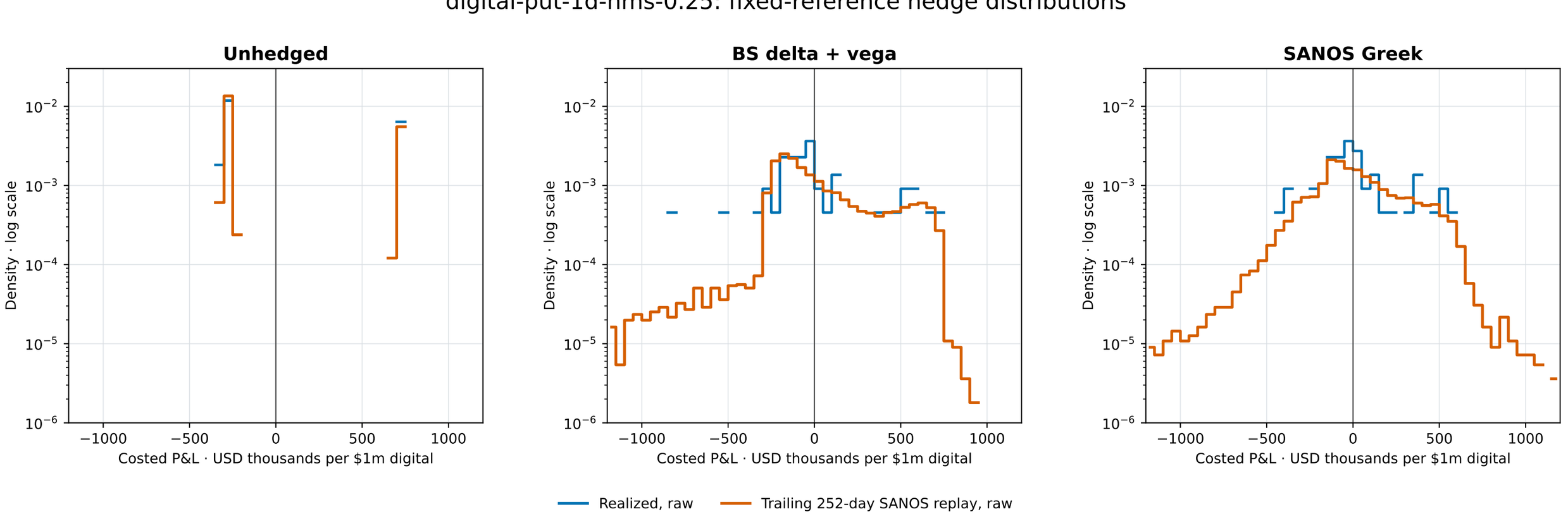




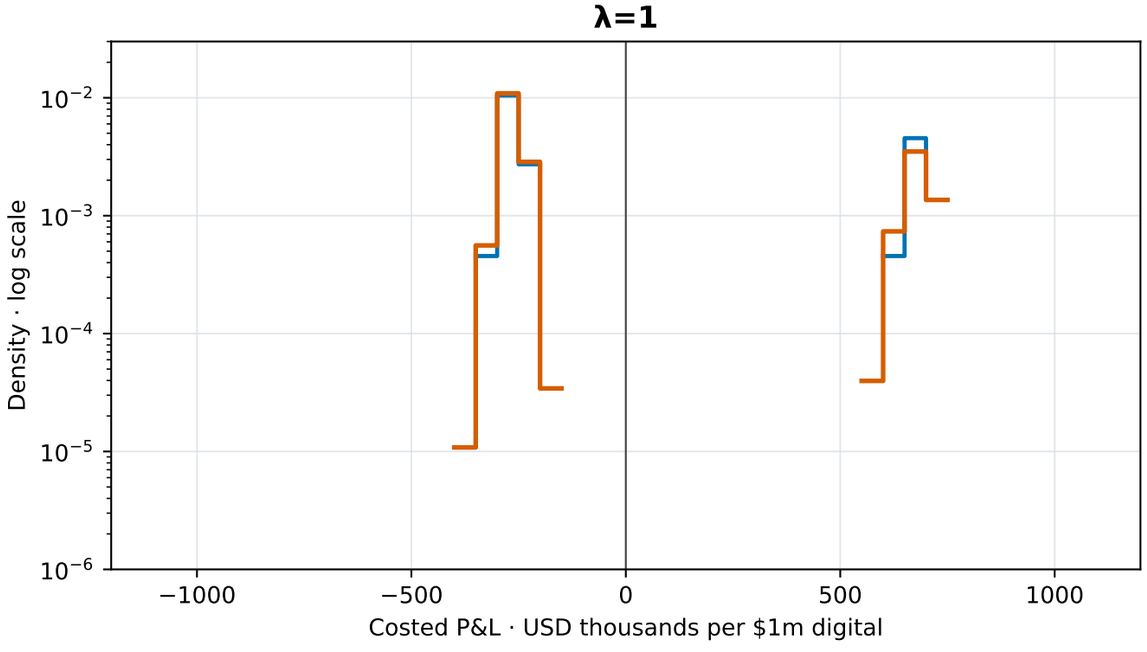


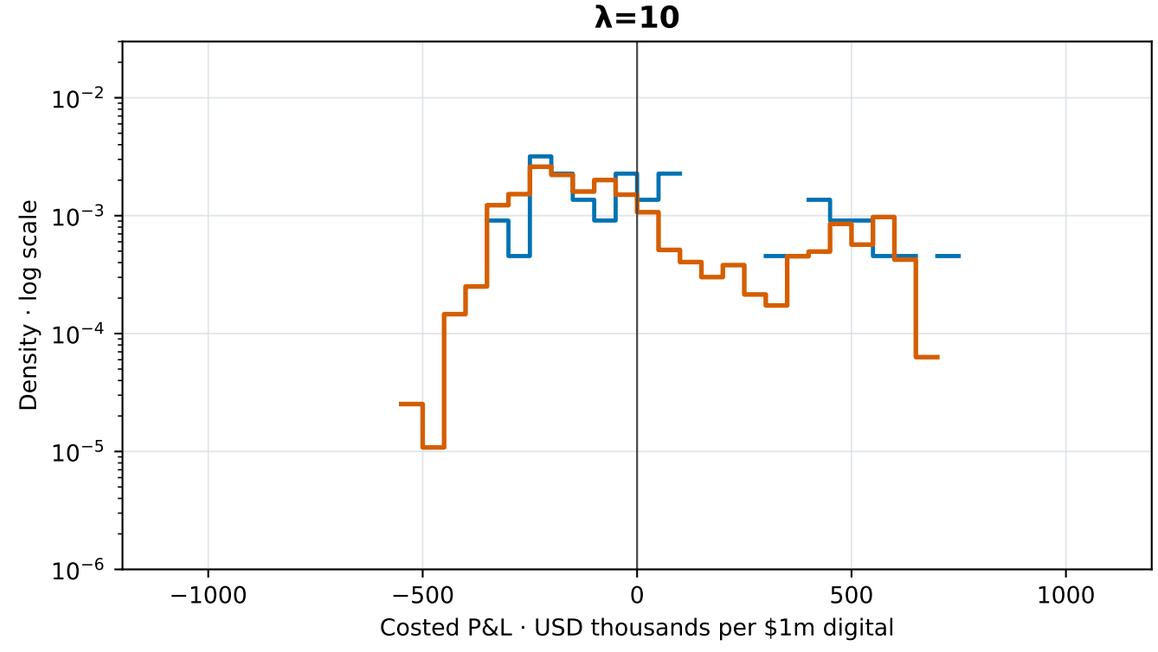
Realized, raw Fitted centered factor Gaussian

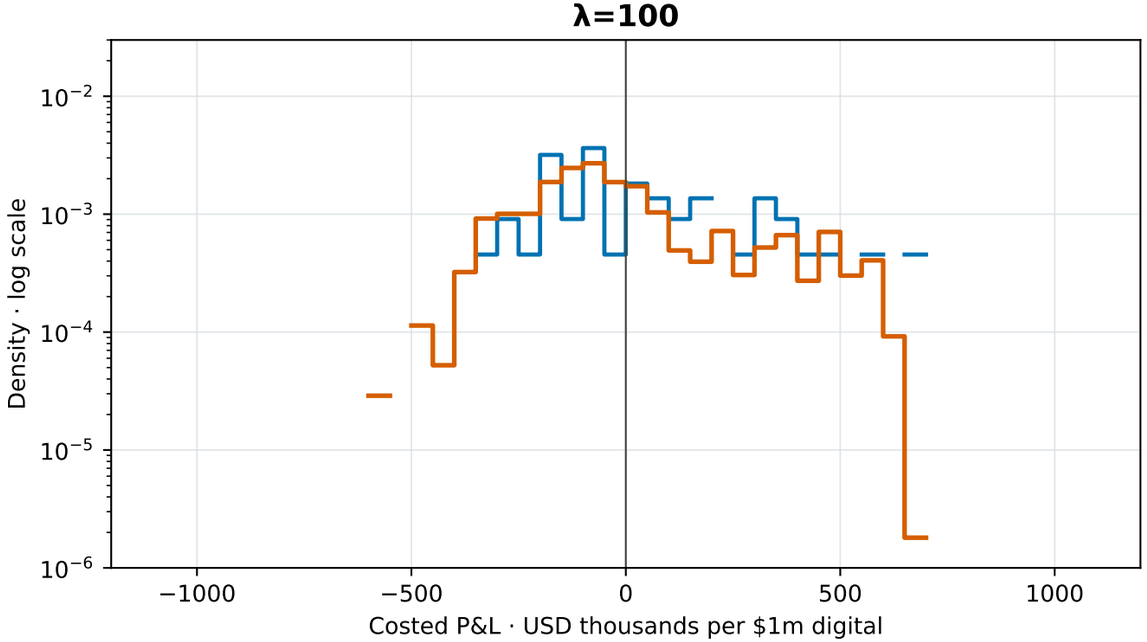
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

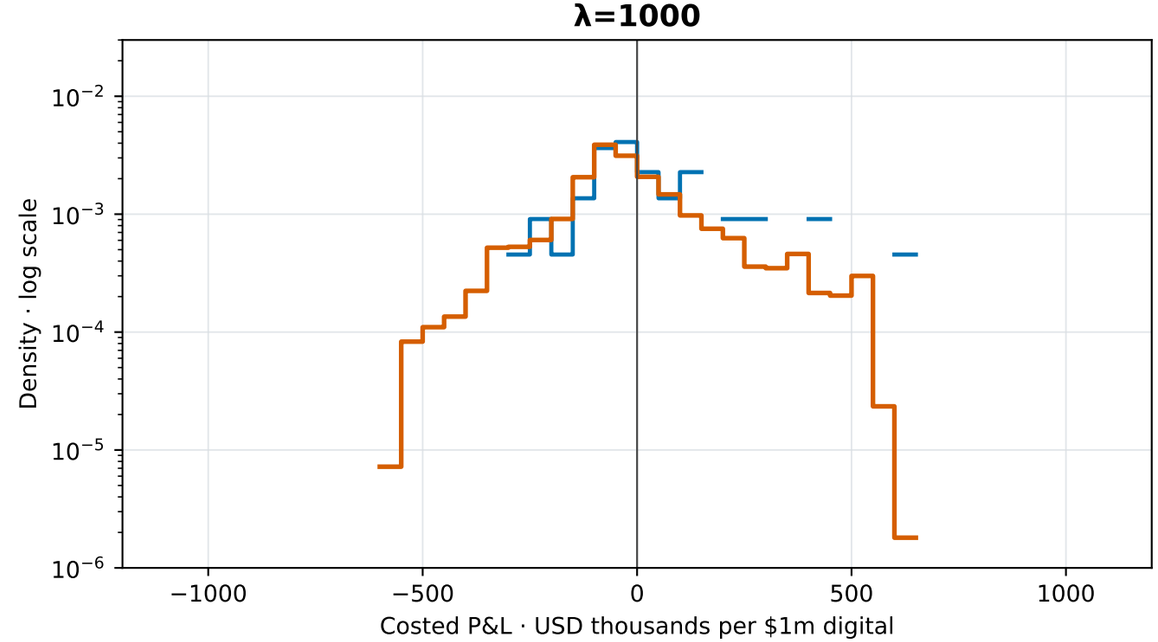

digital-put-1d-nms-0.25: fixed-reference hedge distributions
Unhedged
BS delta + vega
SANOS Greek
Density · log scale
Costed P&L · USD thousands per $1m digital
Realized, raw
Trailing 252-day SANOS replay, raw
No mean correction: 44 realized next-day outcomes versus preceding 252 raw returns. References have no fitted hedge-return law.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.25: SANOS historical · Mean-variance





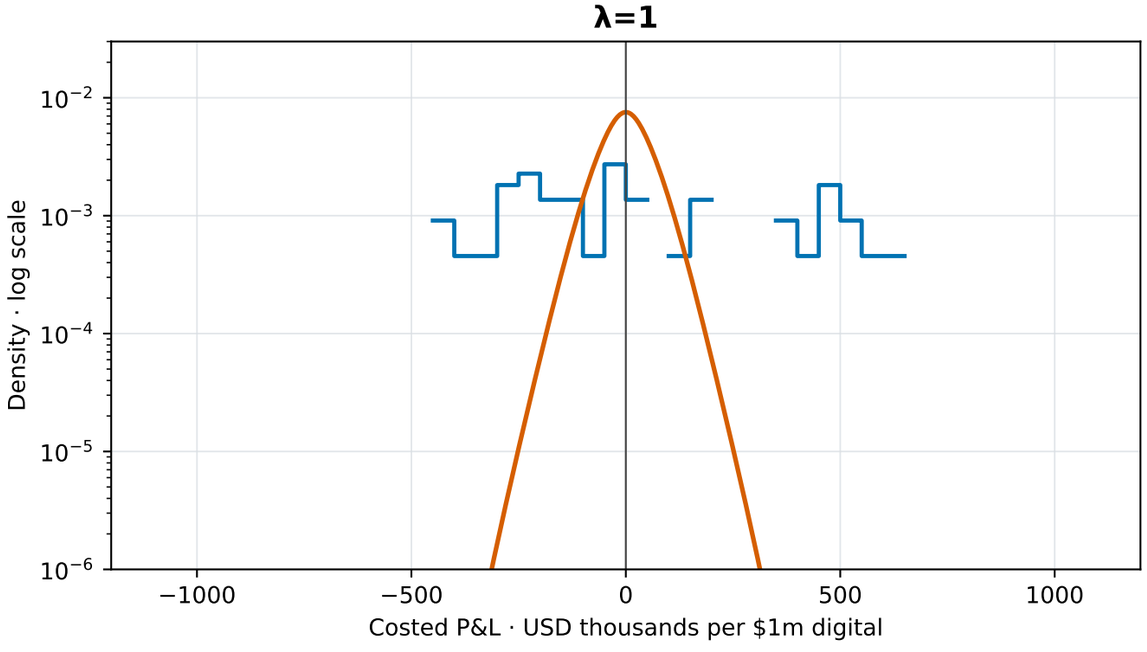


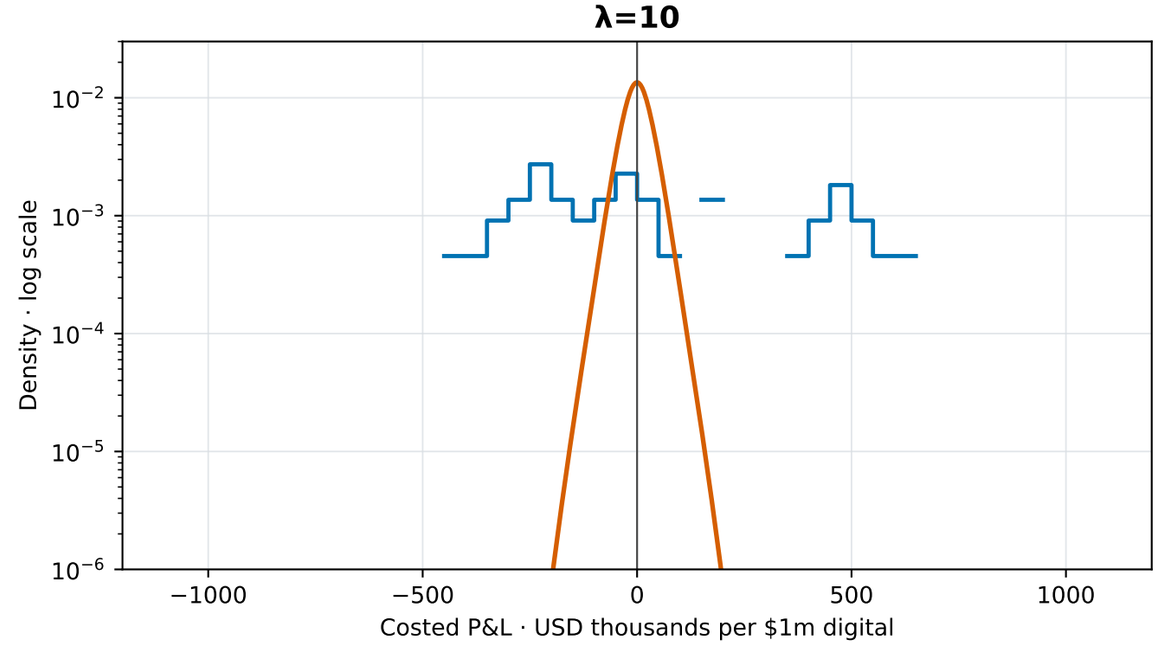


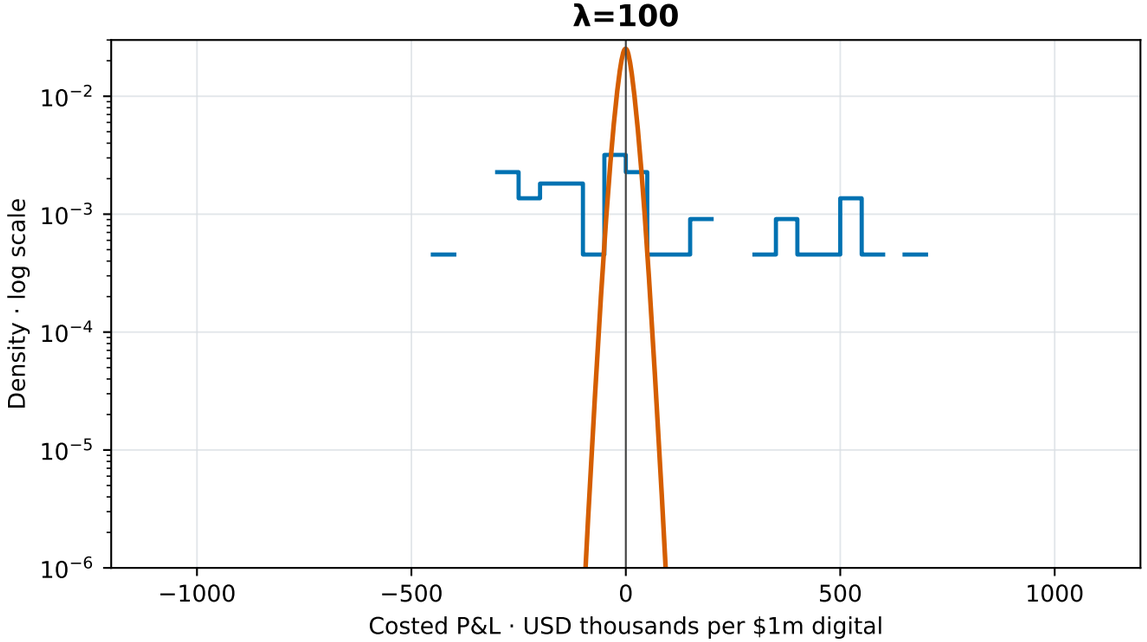
Realized, raw Fitted 252-day demeaned SANOS returns

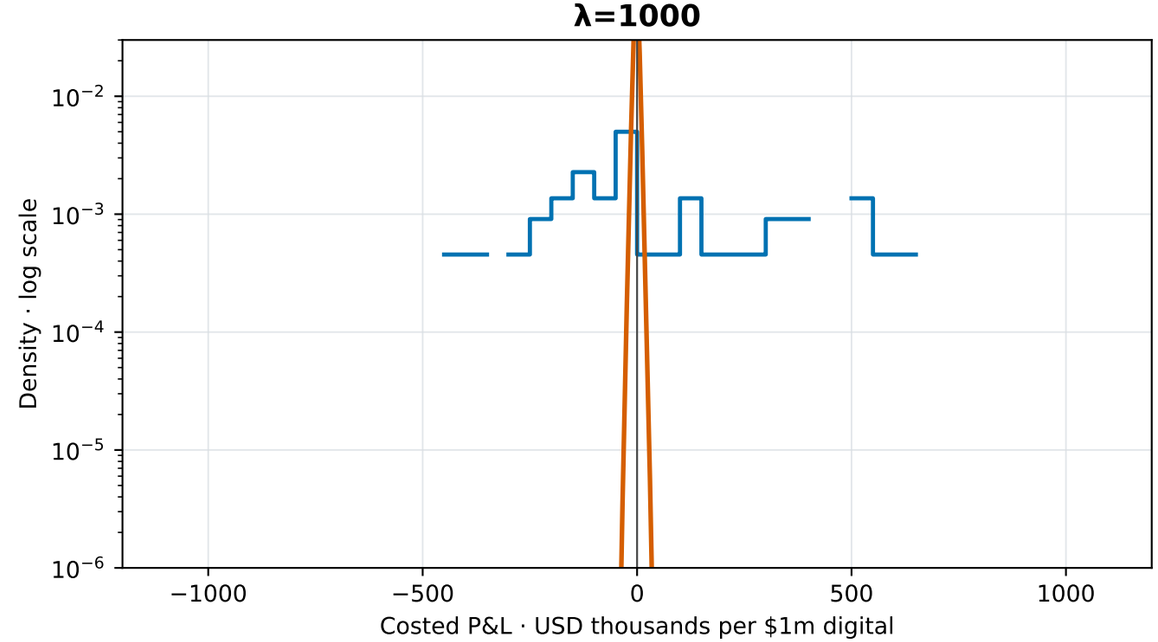
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.25: SANOS parametric · Mean-variance





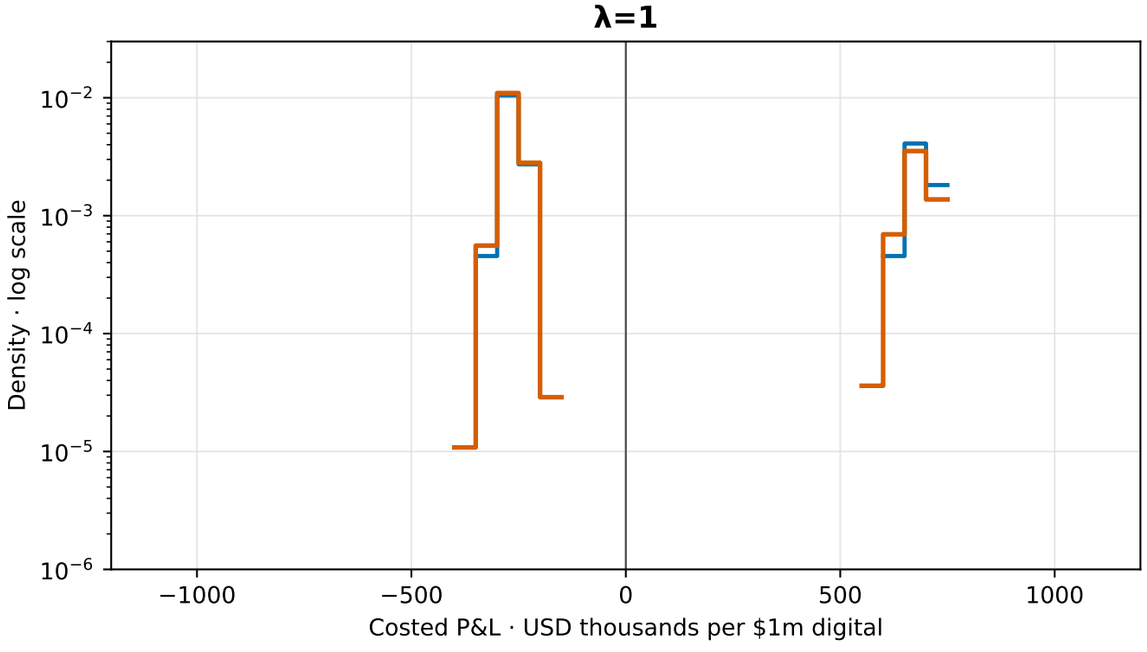


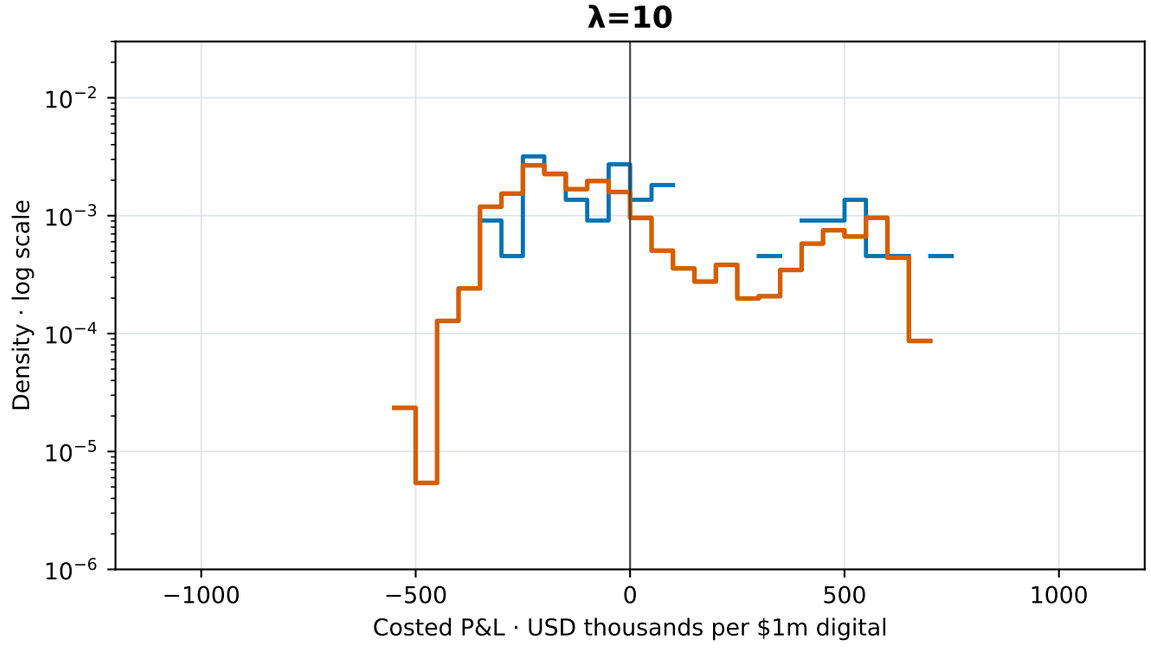


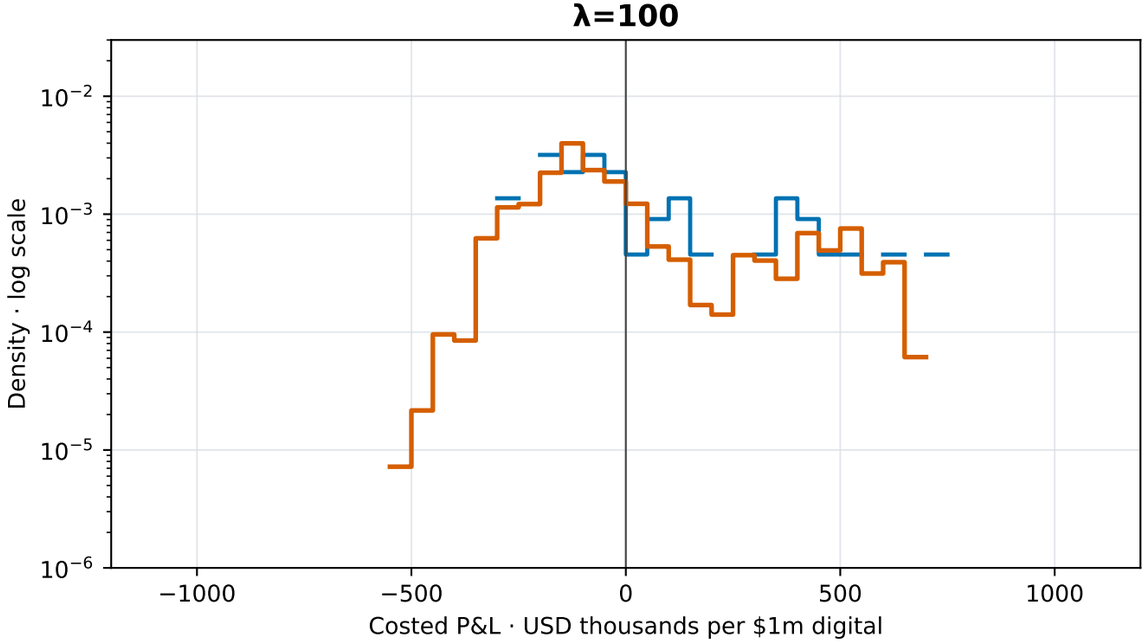
Realized, raw Fitted centered factor Gaussian

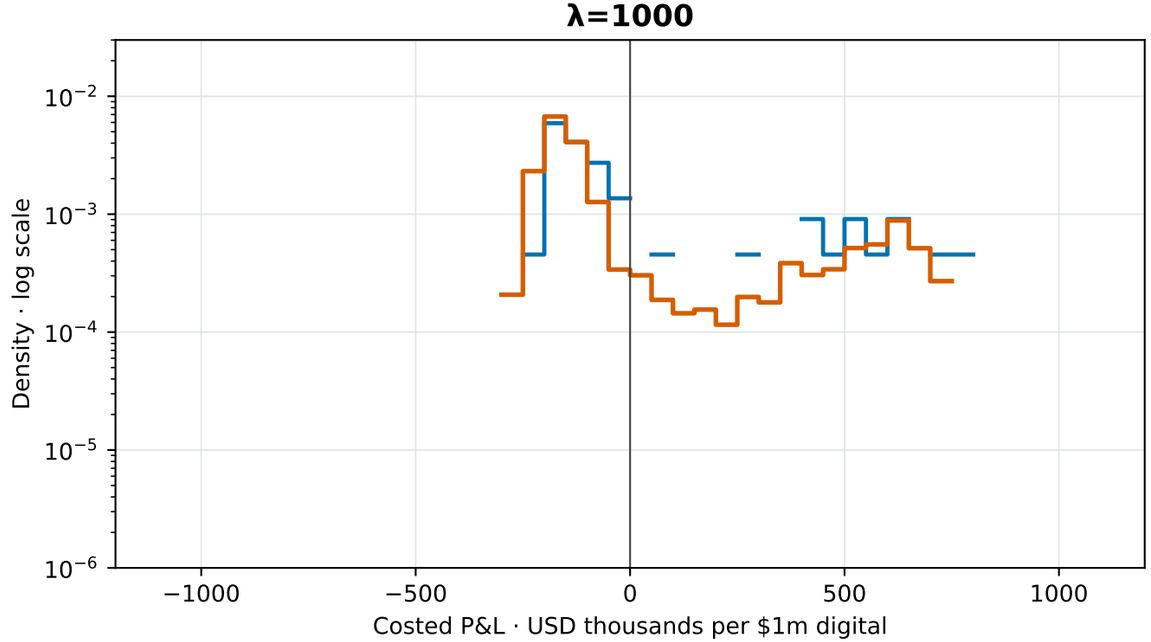
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.25: SANOS historical · Entropy





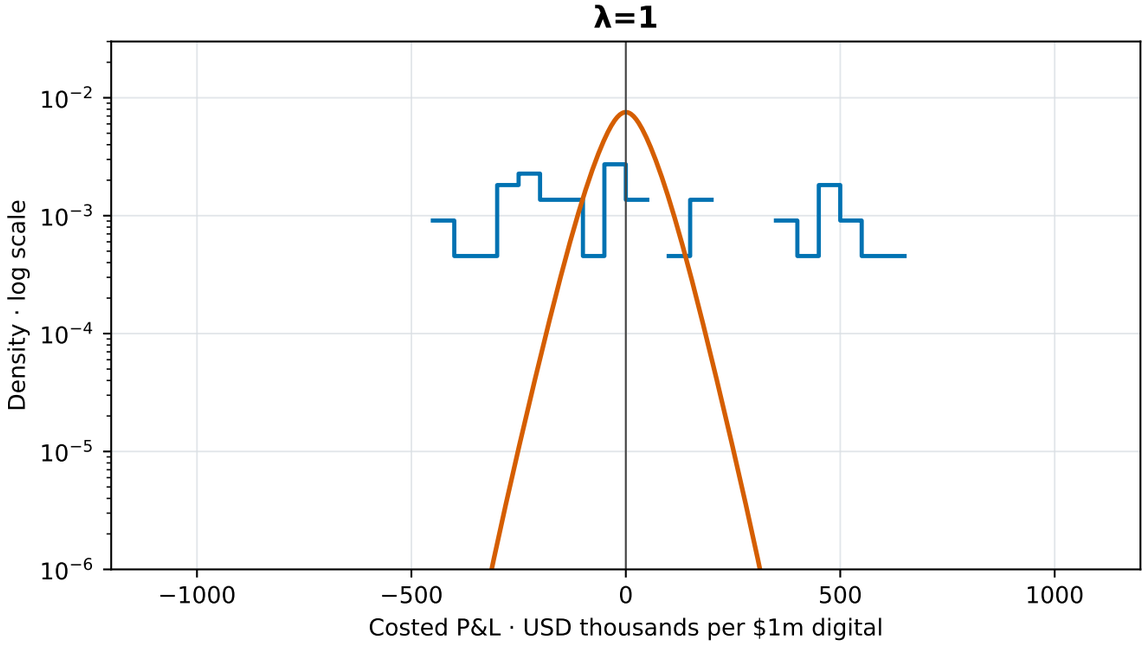


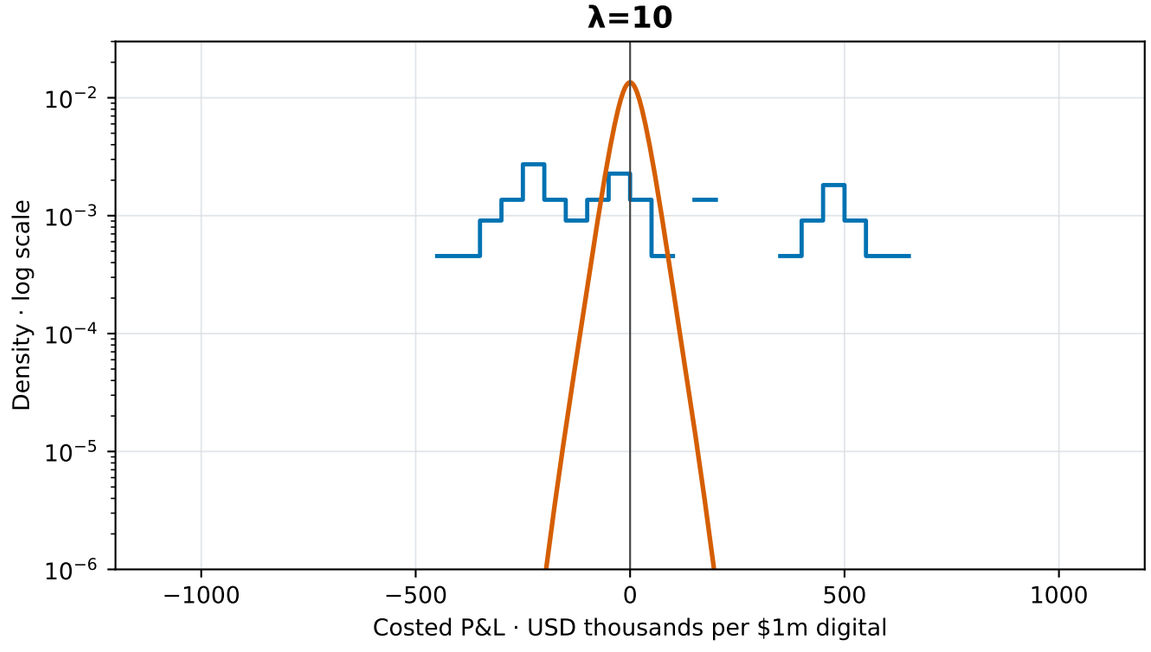


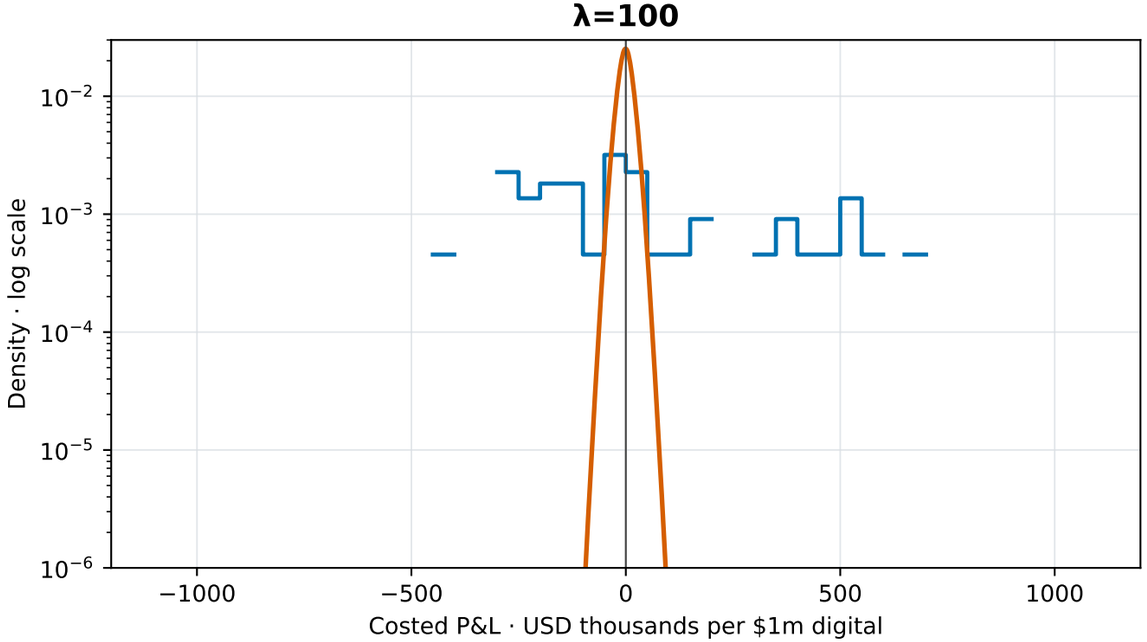
Realized, raw Fitted 252-day demeaned SANOS returns

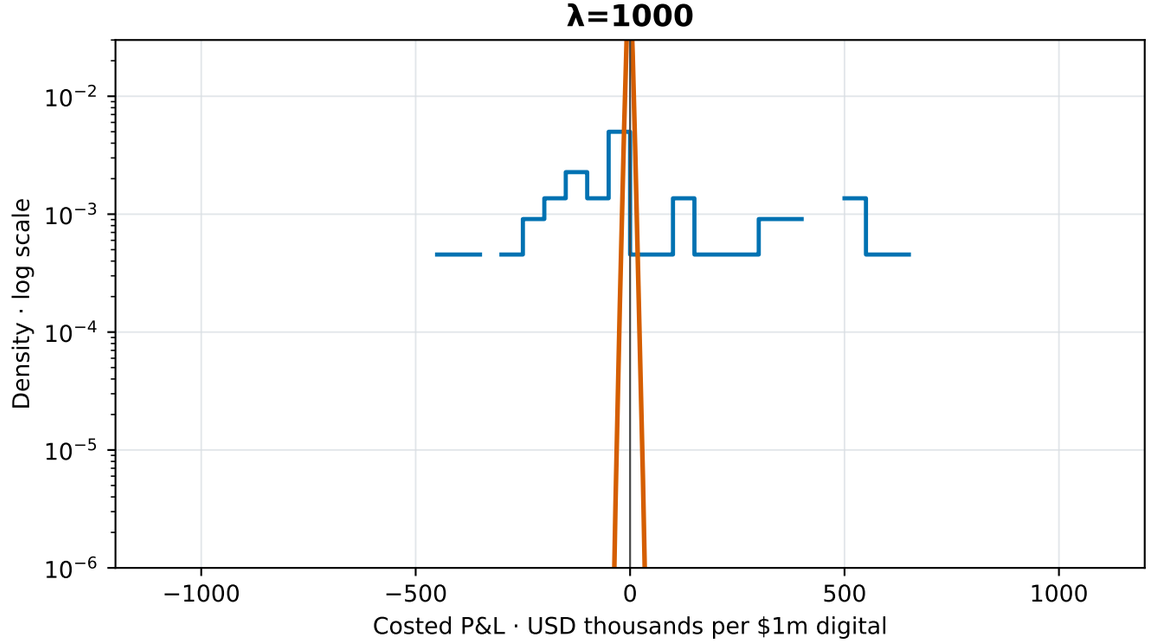
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.25: SANOS parametric · Entropy





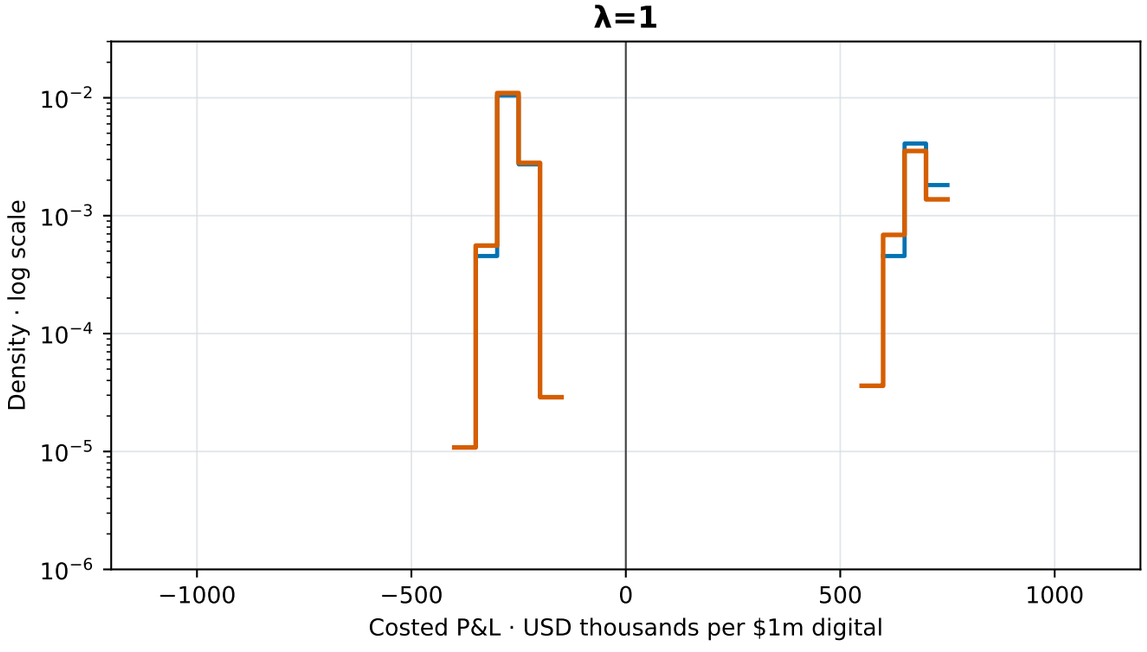


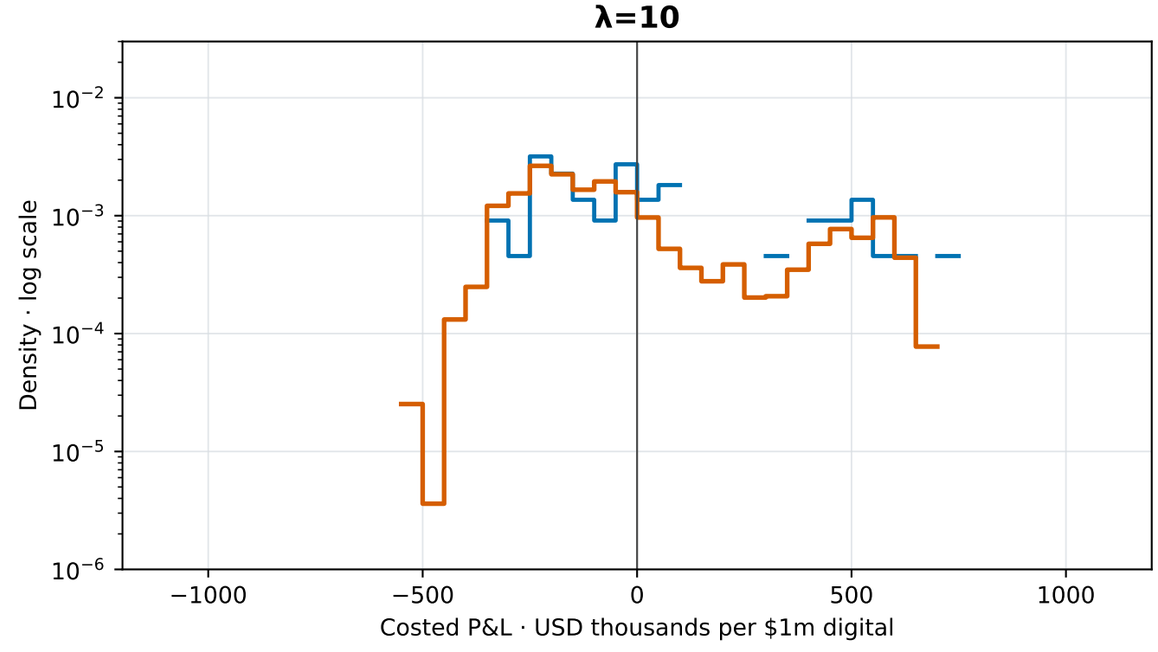


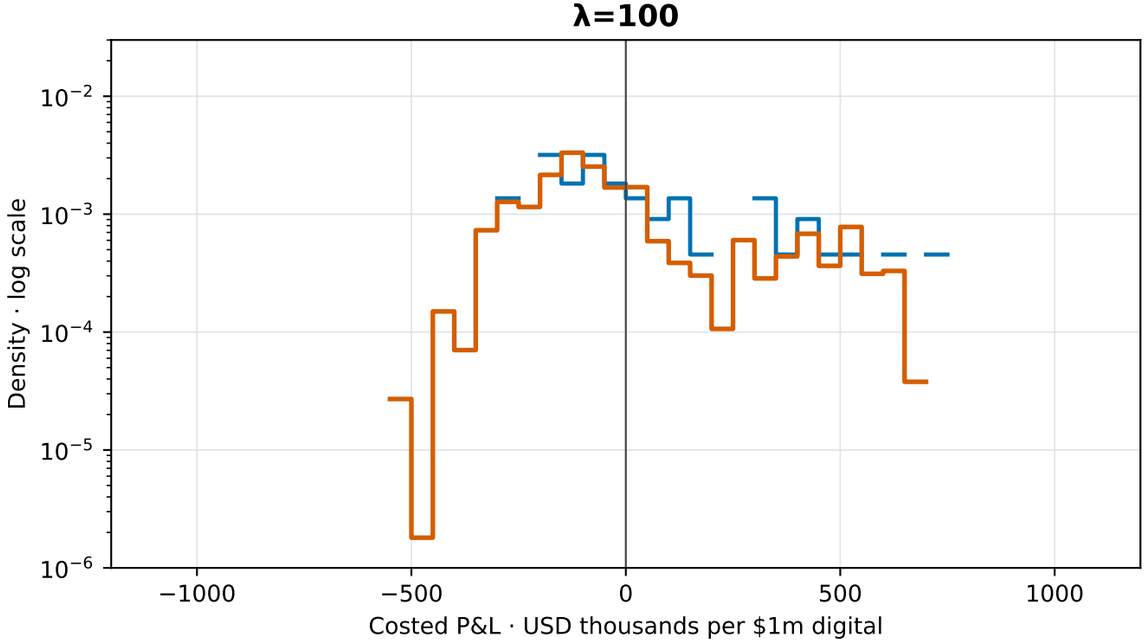
Realized, raw Fitted centered factor Gaussian

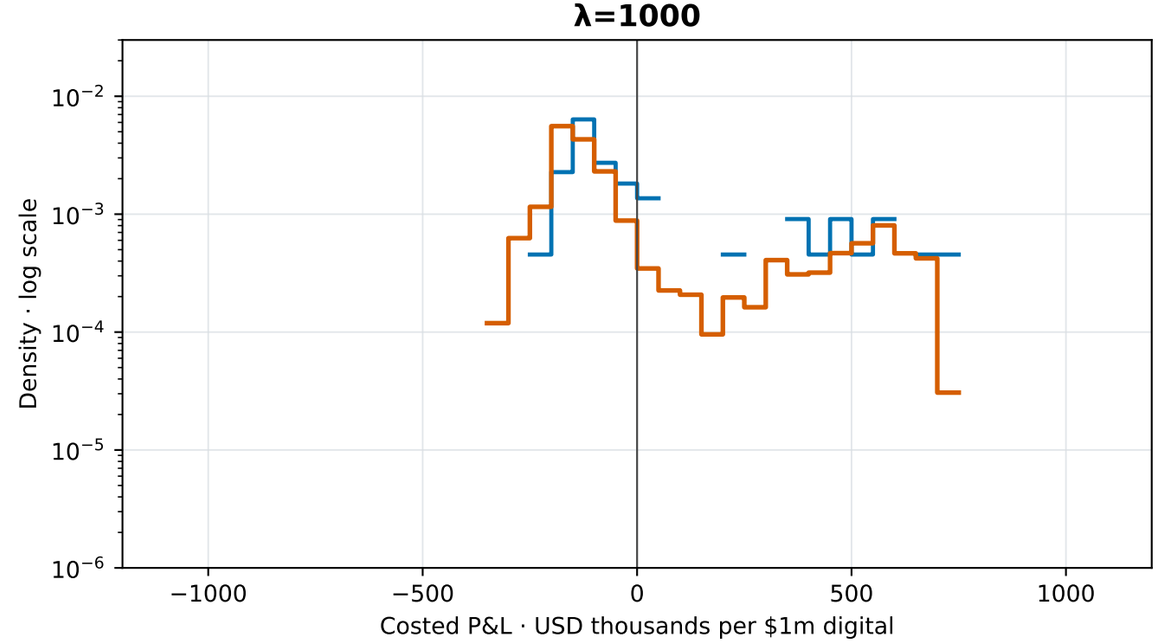
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.25: SANOS historical · Power-exponential





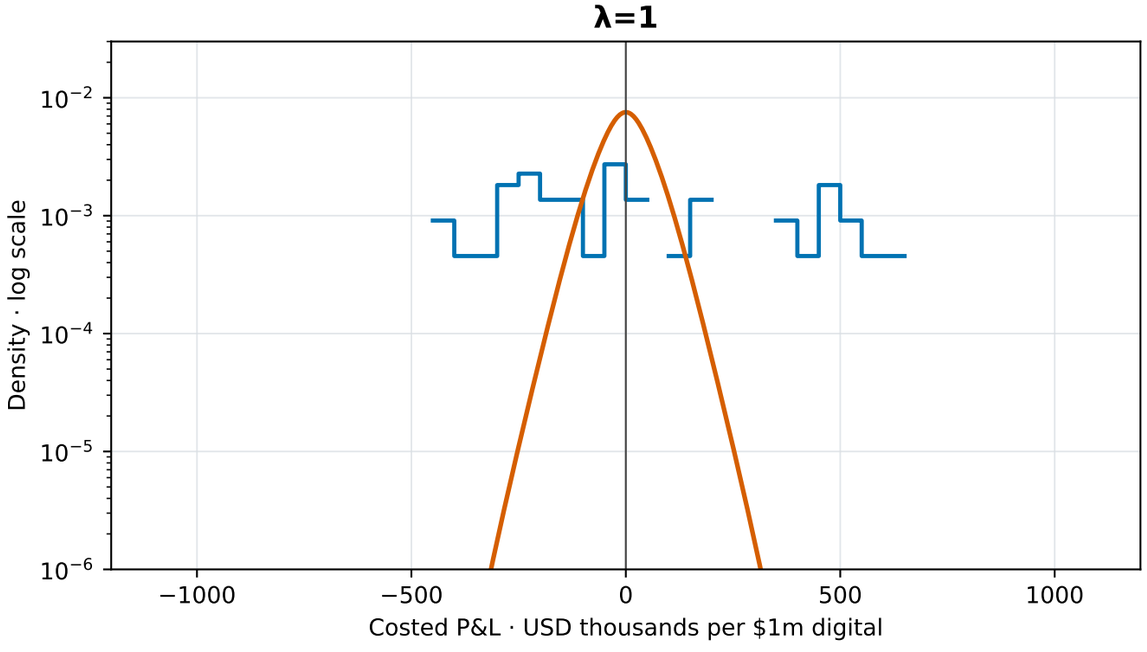


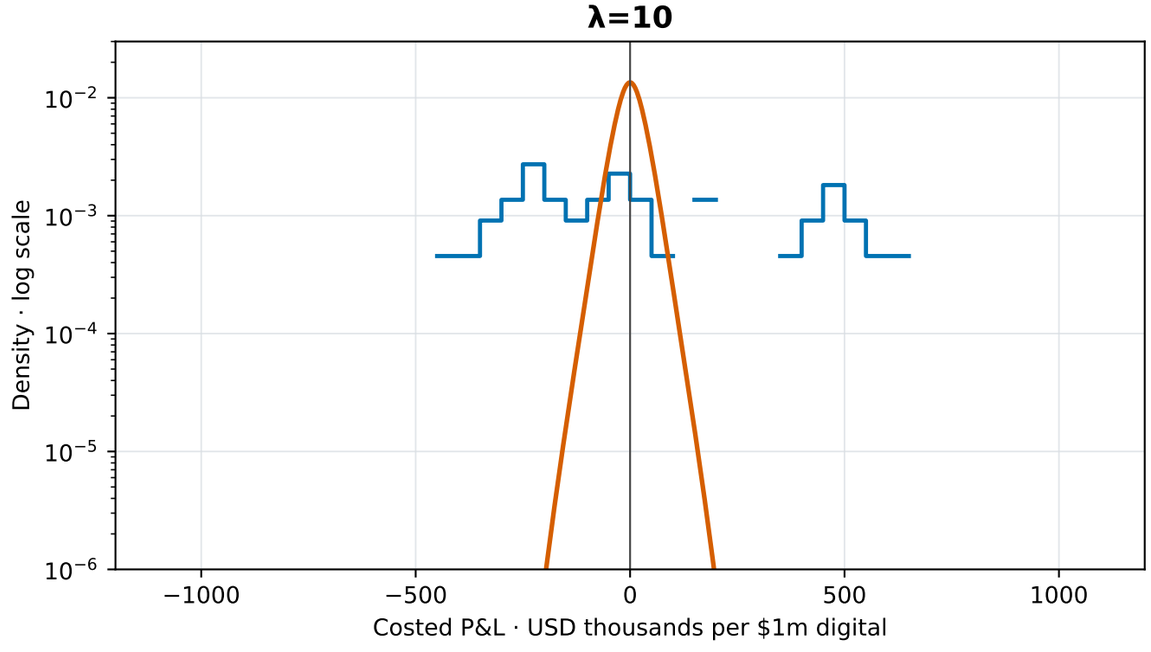


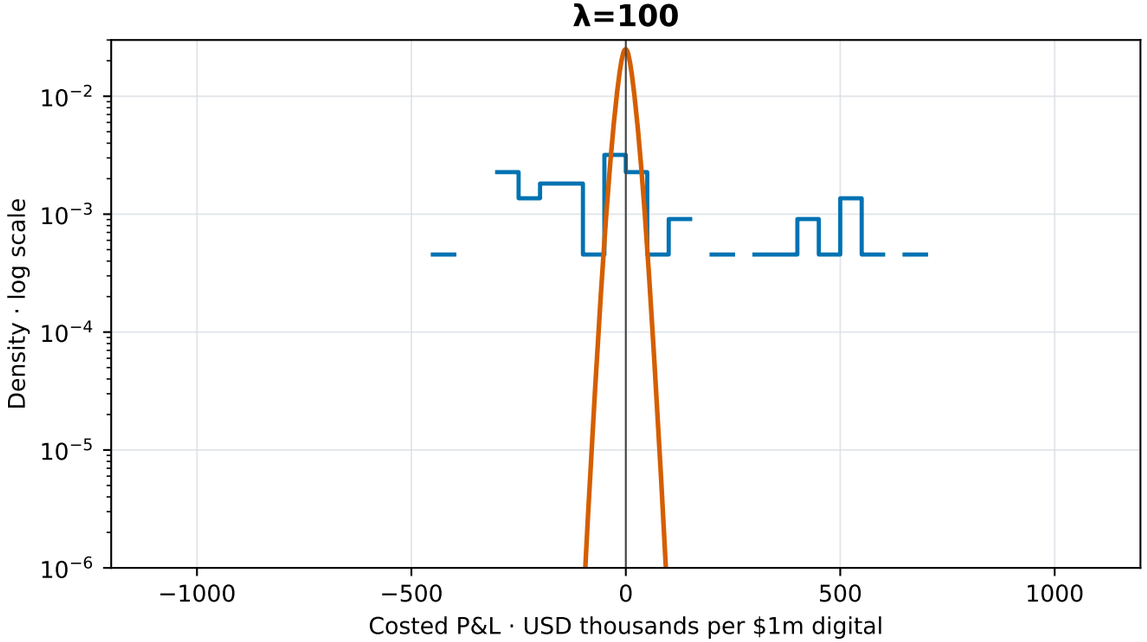
Realized, raw Fitted 252-day demeaned SANOS returns

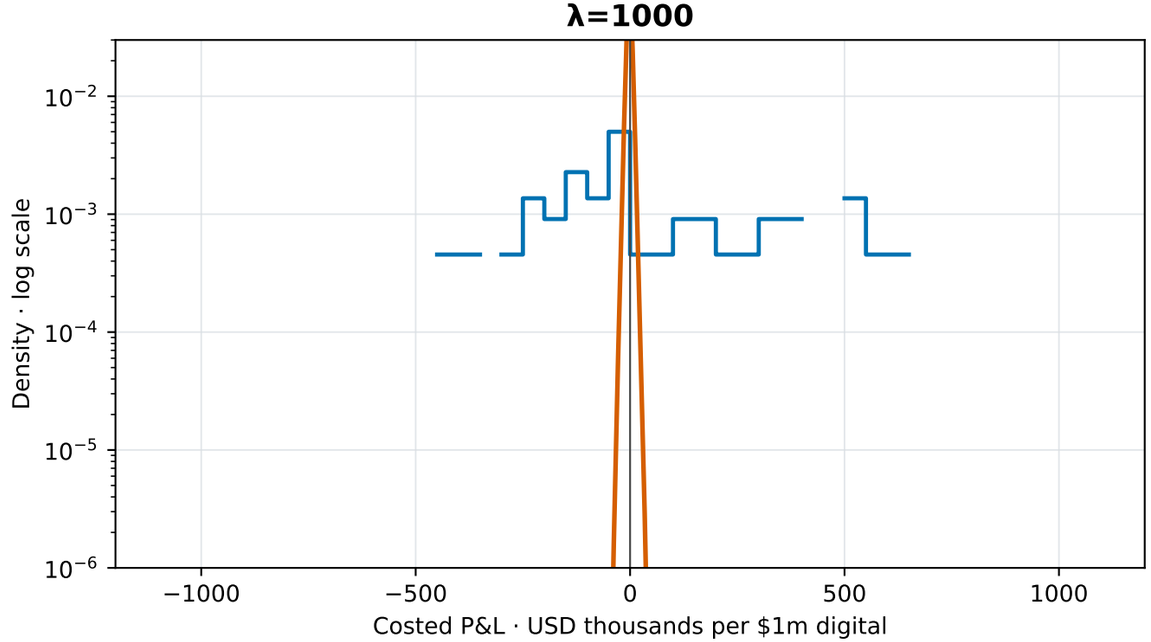
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-nms-0.25: SANOS parametric · Power-exponential



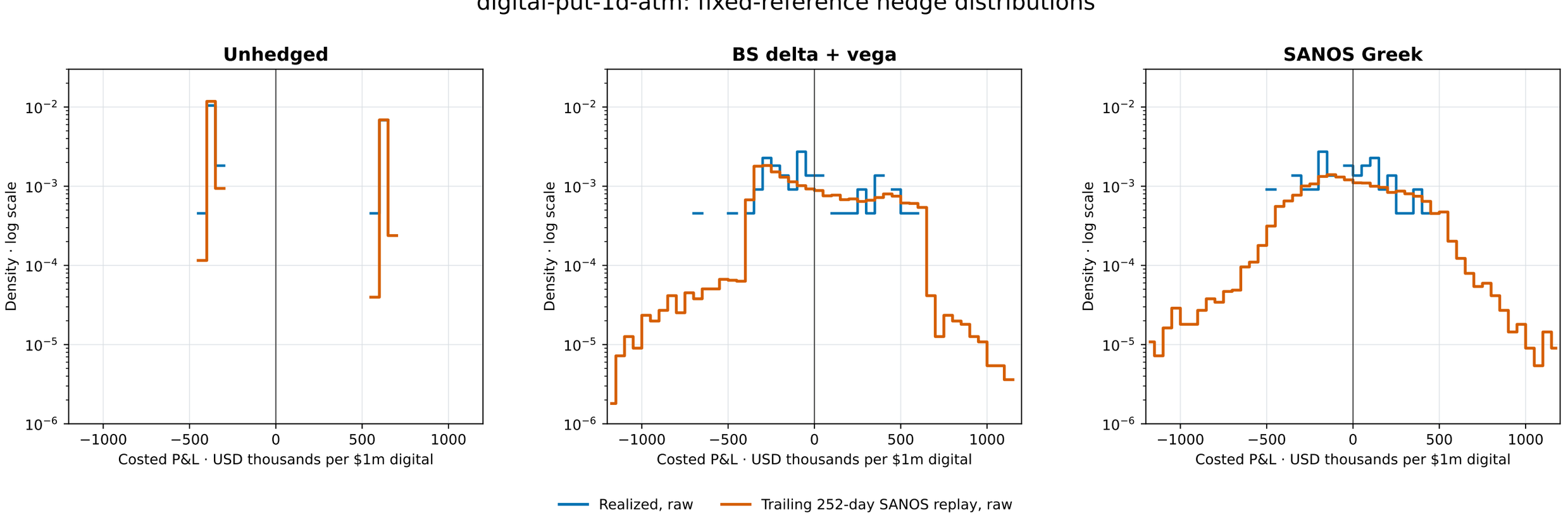




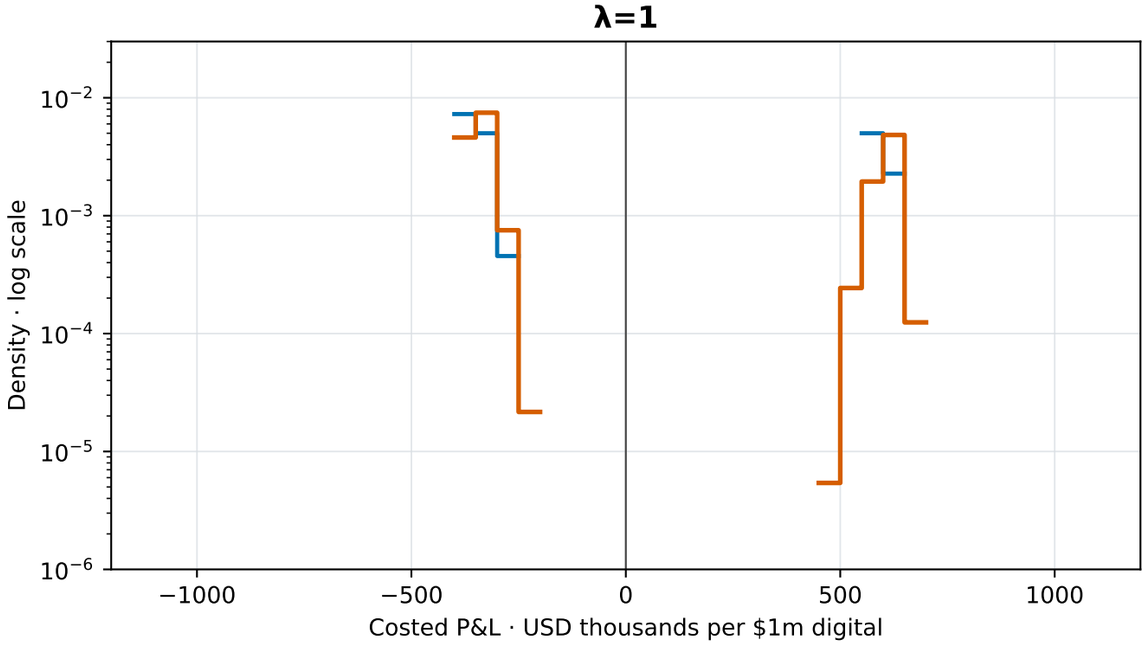


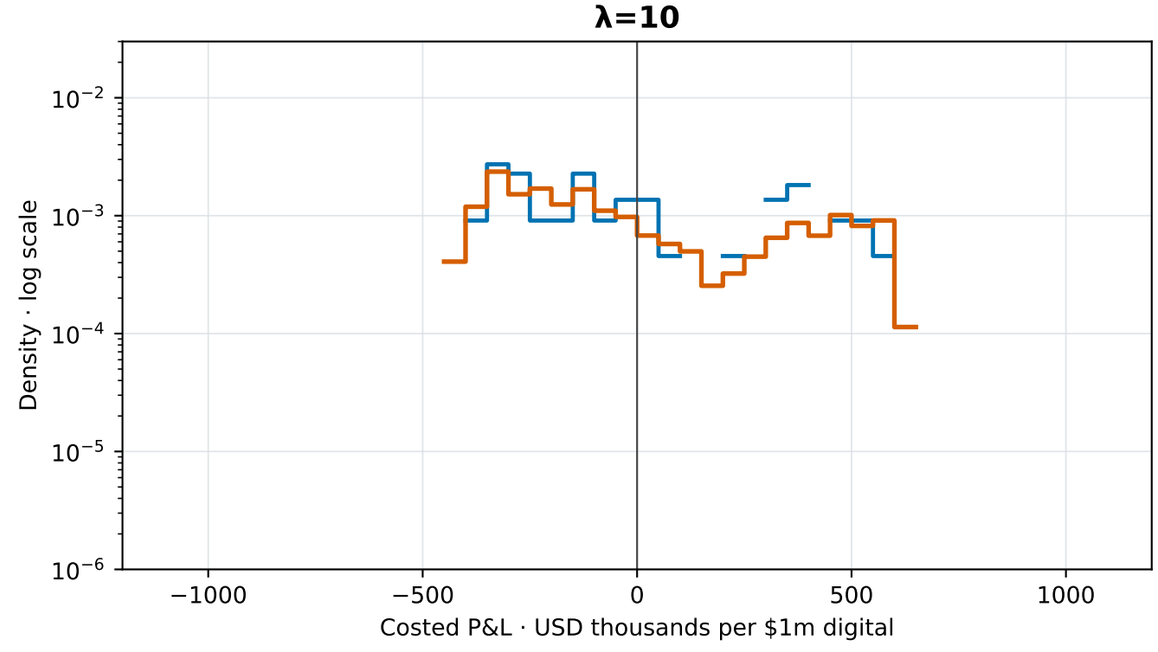
Realized, raw Fitted centered factor Gaussian

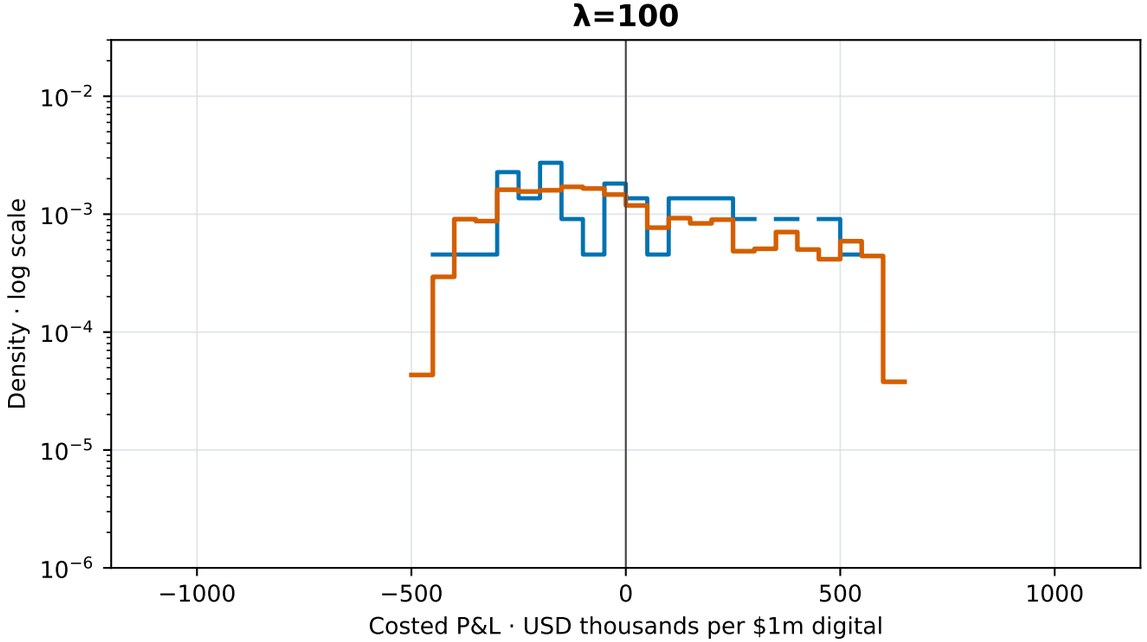
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

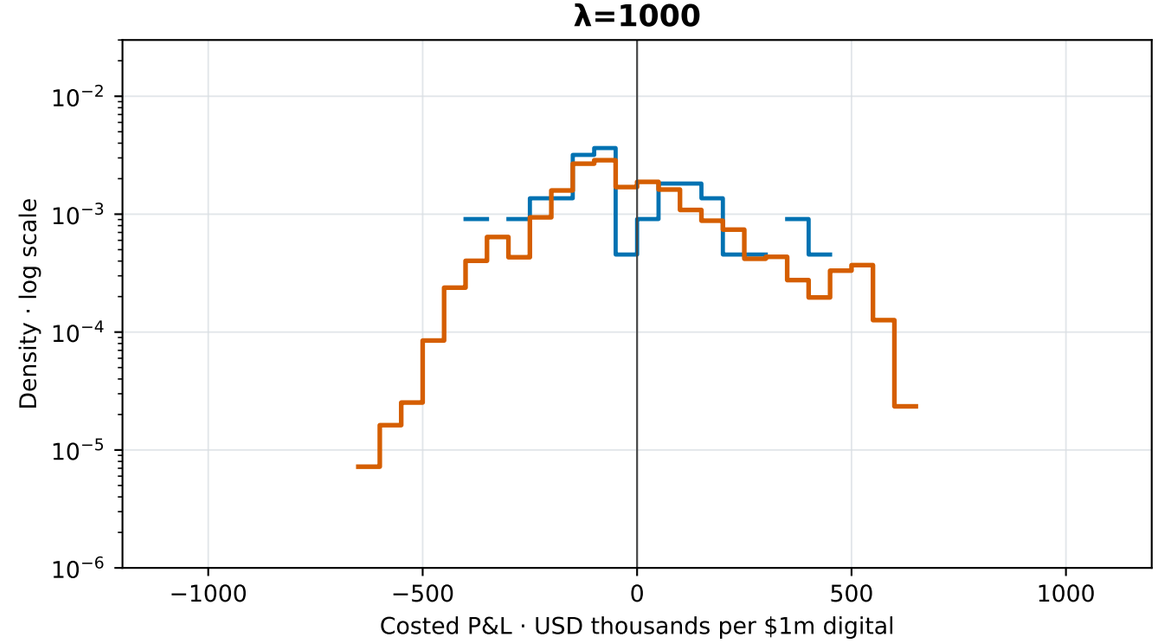

digital-put-1d-atm: fixed-reference hedge distributions
Unhedged
BS delta + vega
SANOS Greek
Density · log scale
Costed P&L · USD thousands per $1m digital
Realized, raw
Trailing 252-day SANOS replay, raw
No mean correction: 44 realized next-day outcomes versus preceding 252 raw returns. References have no fitted hedge-return law.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-atm: SANOS historical · Mean-variance





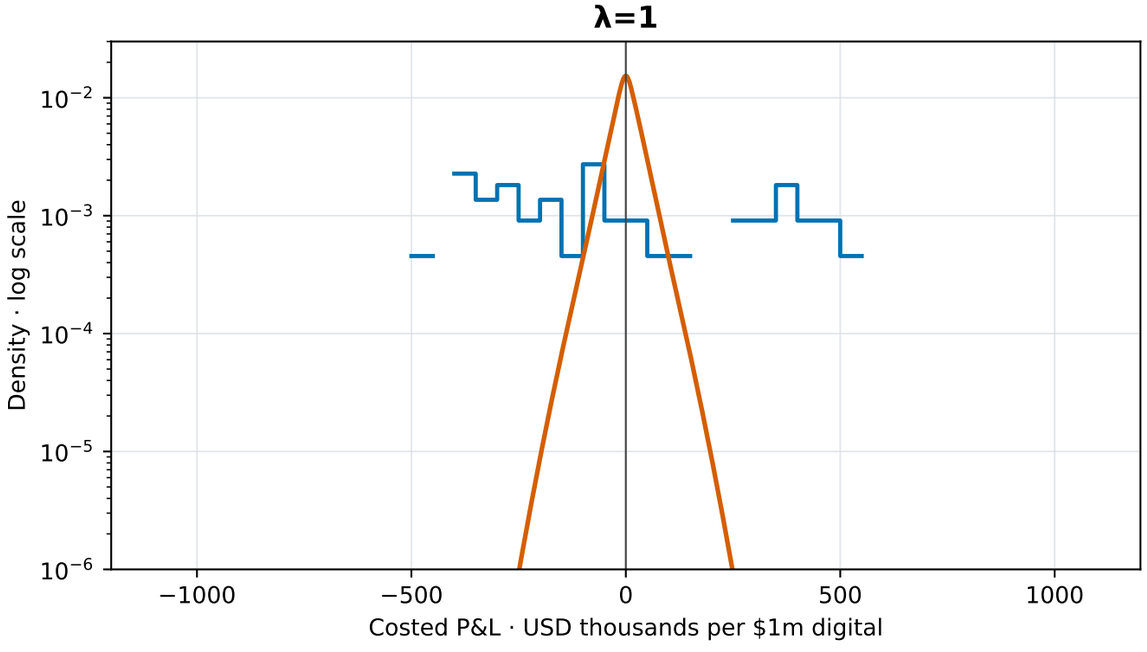


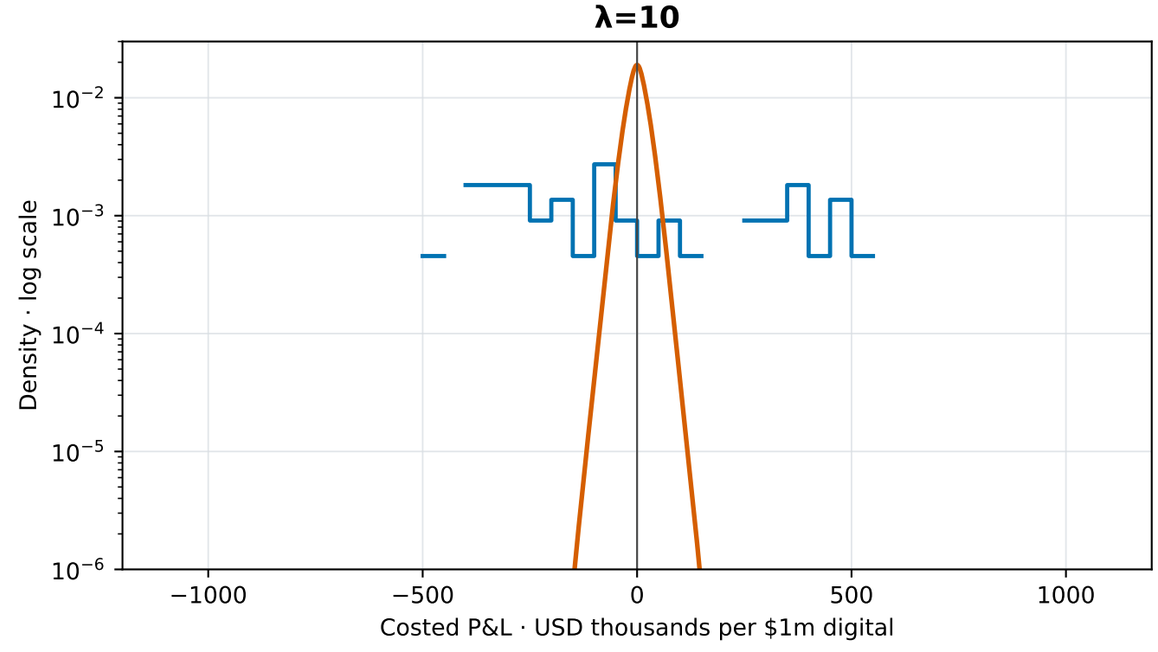


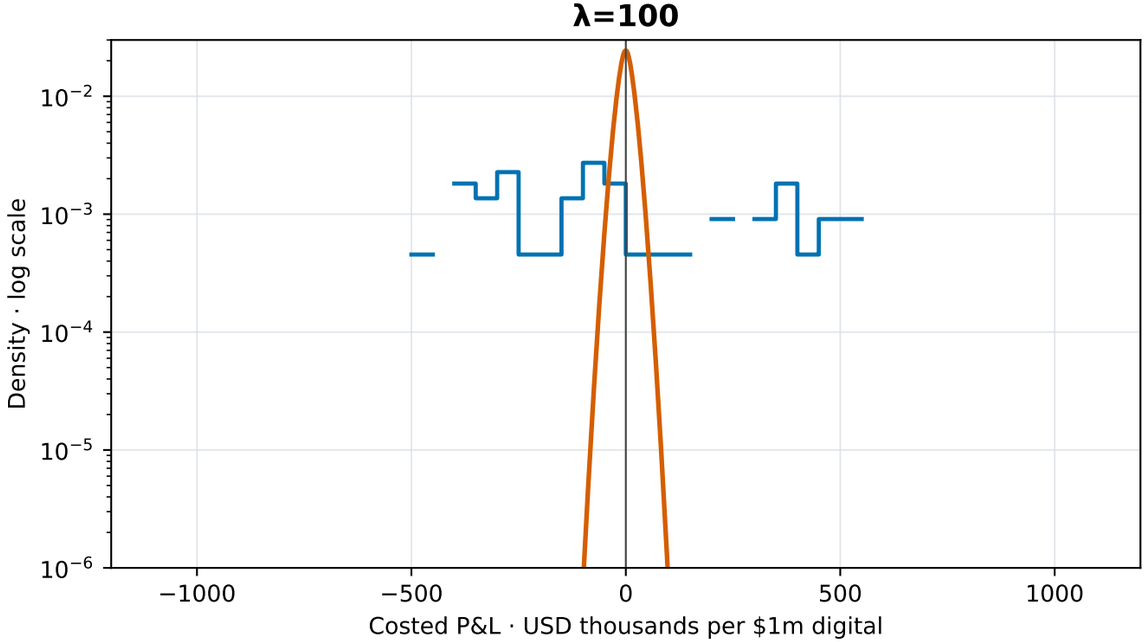
Realized, raw Fitted 252-day demeaned SANOS returns

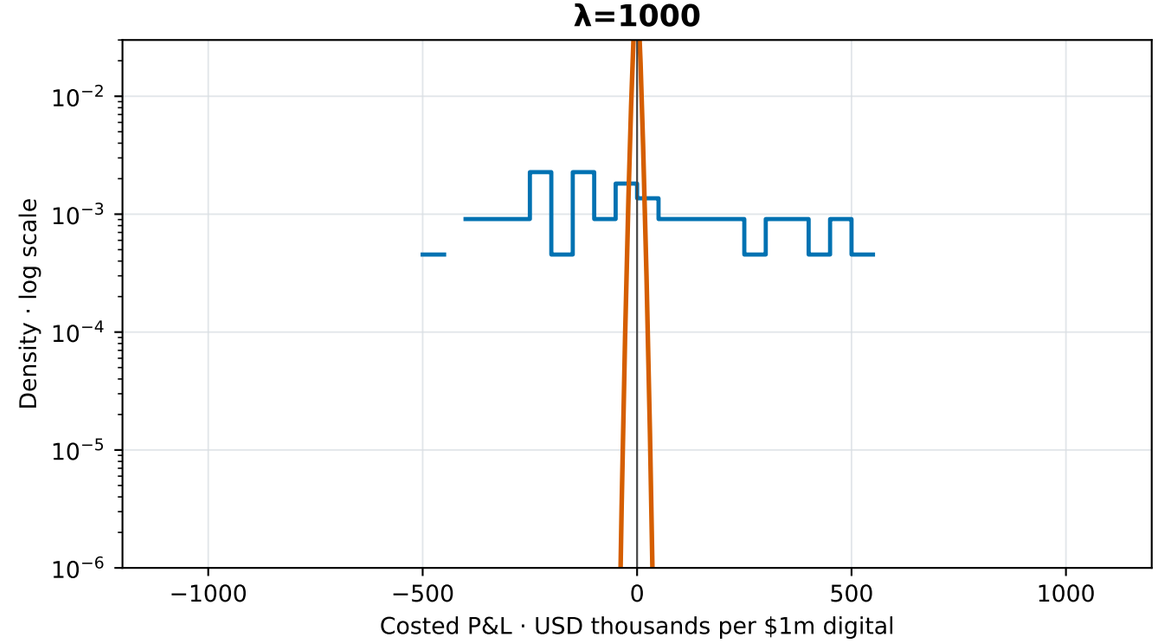
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-atm: SANOS parametric · Mean-variance





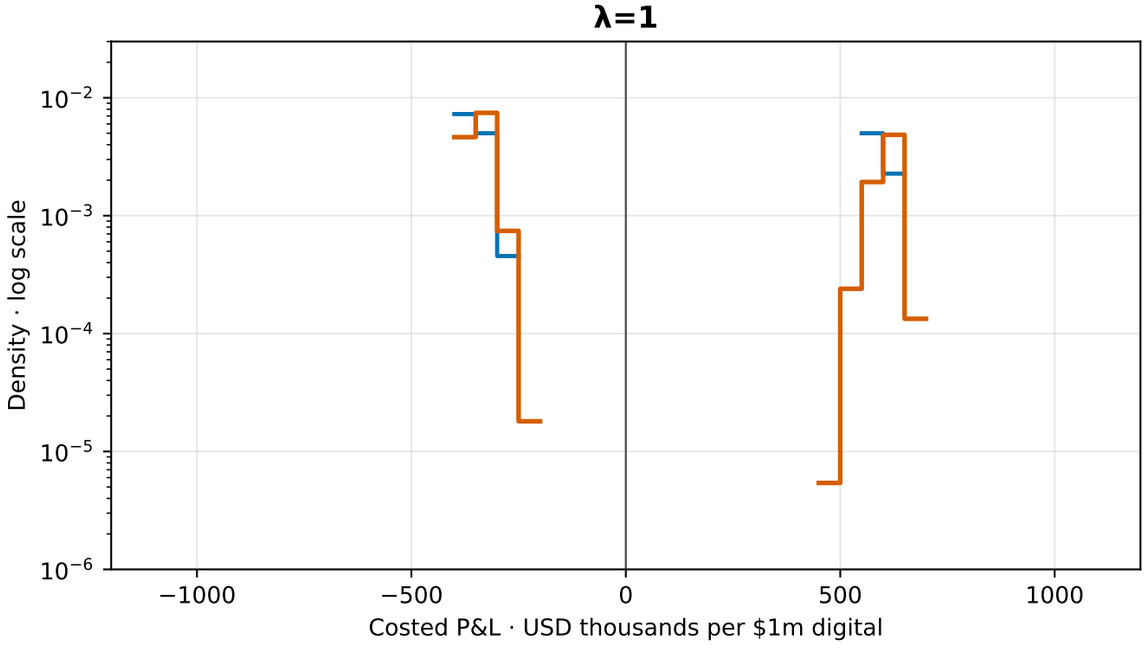


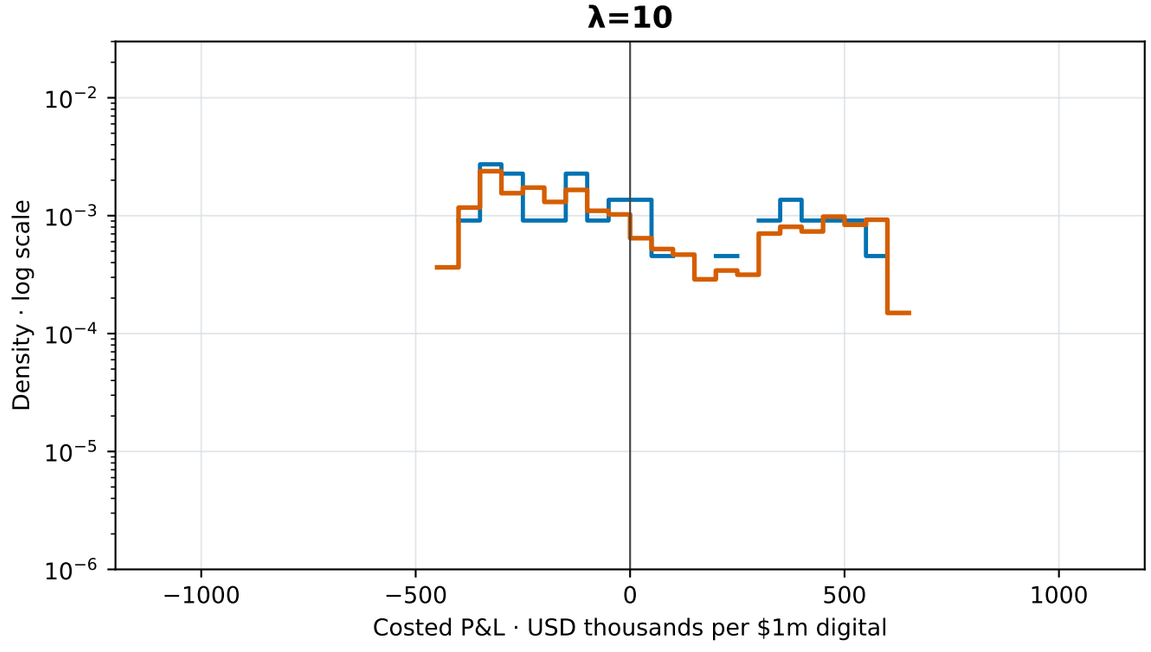


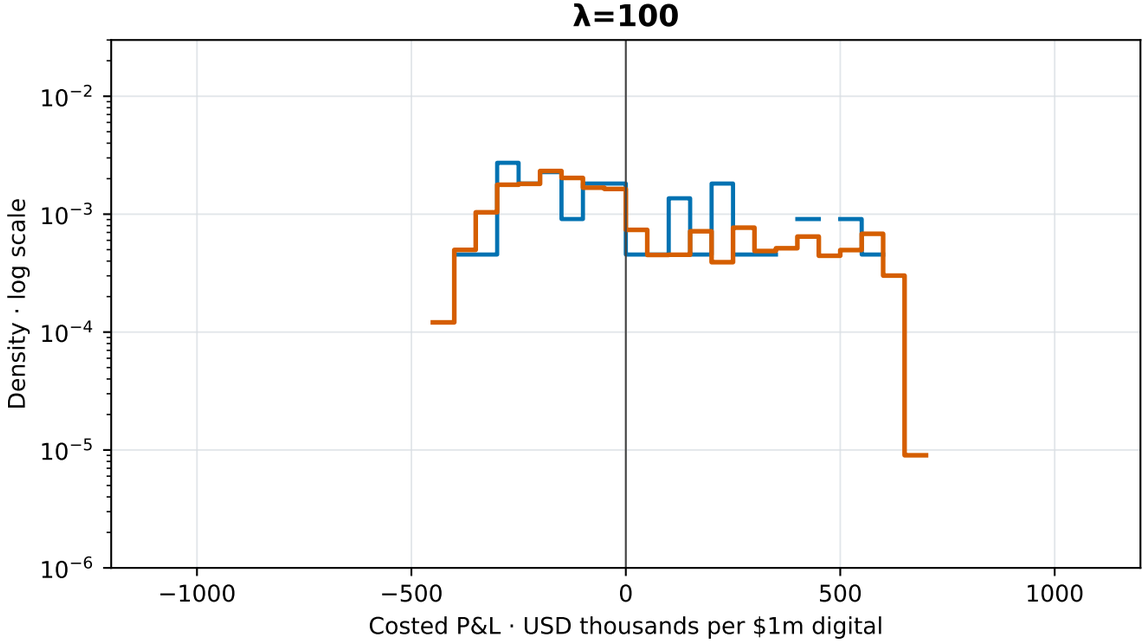
Realized, raw Fitted centered factor Gaussian

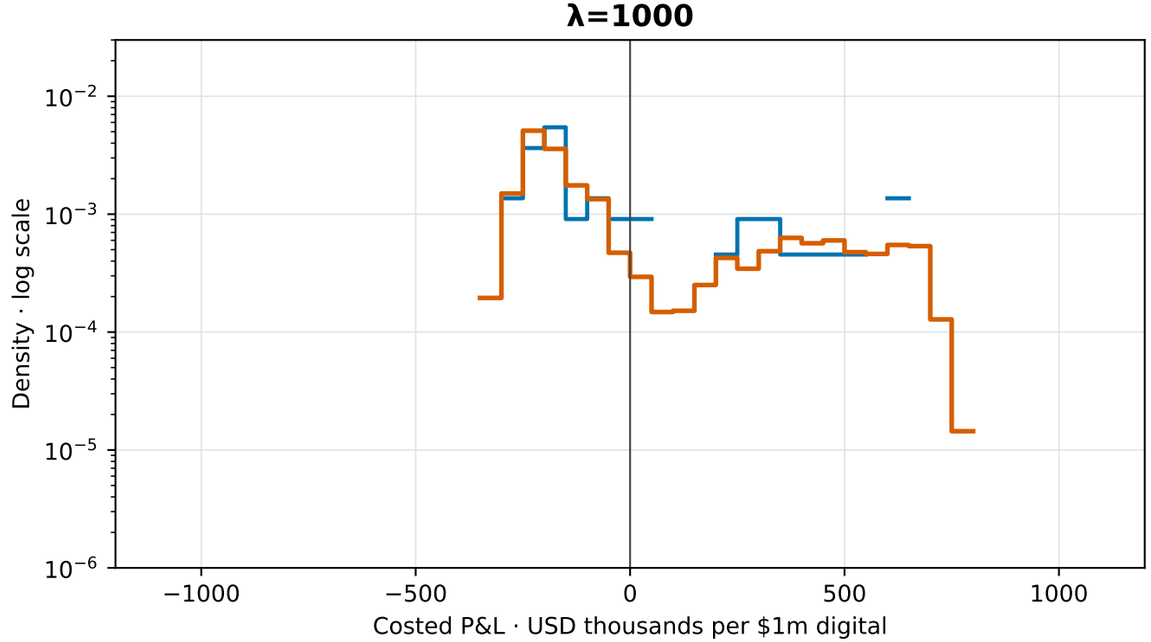
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-atm: SANOS historical · Entropy





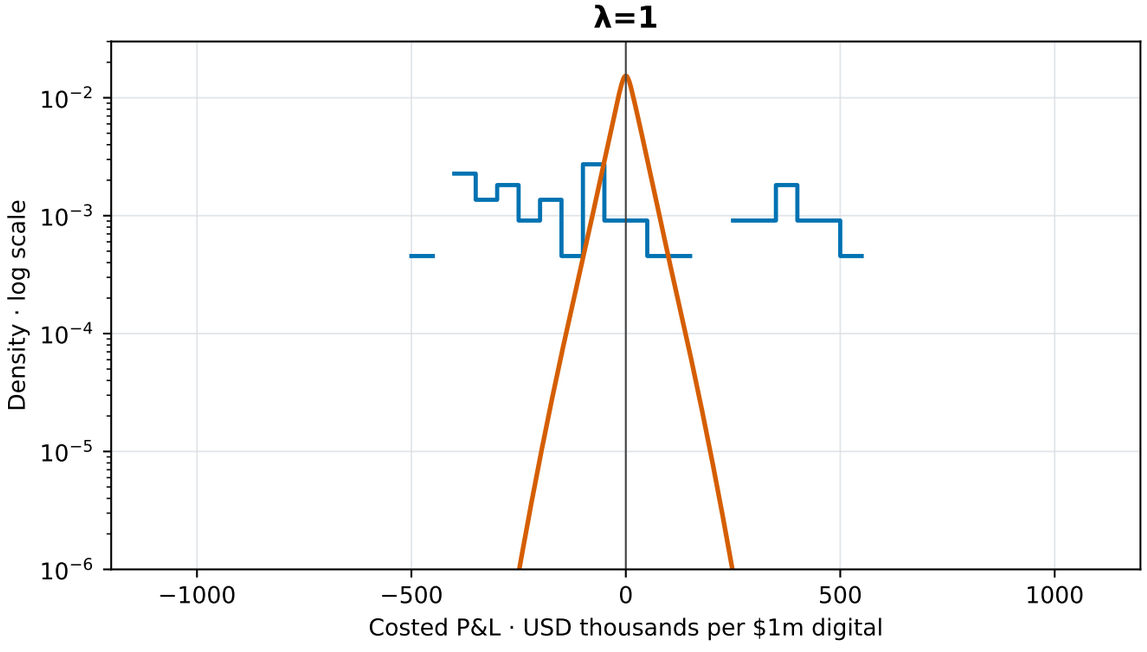


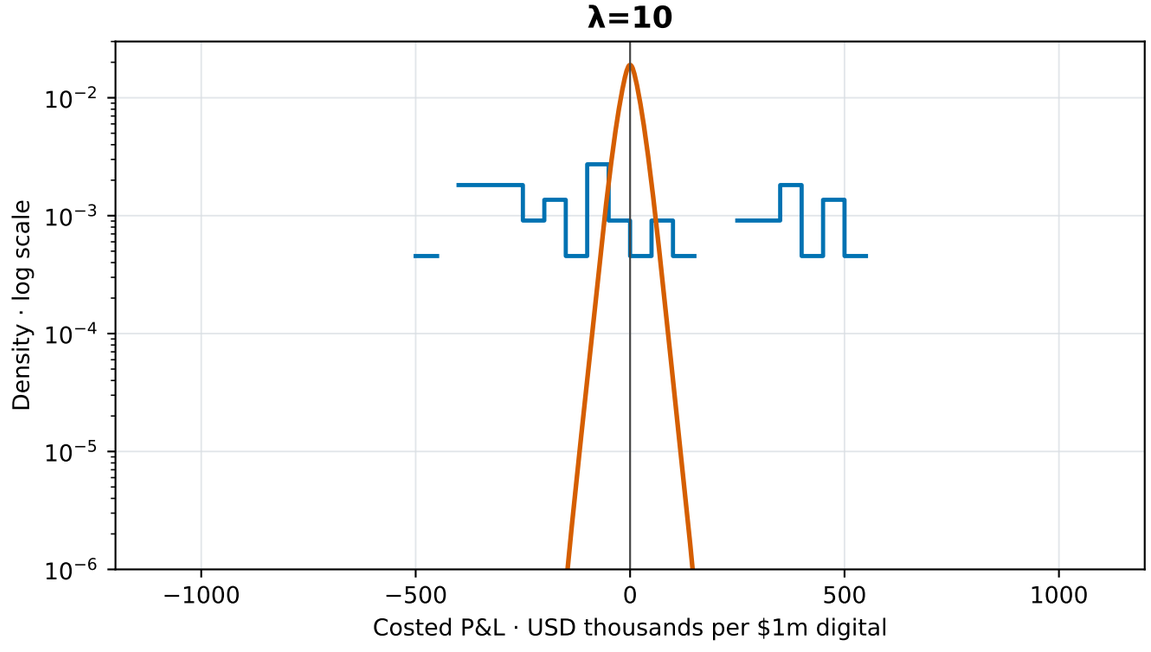


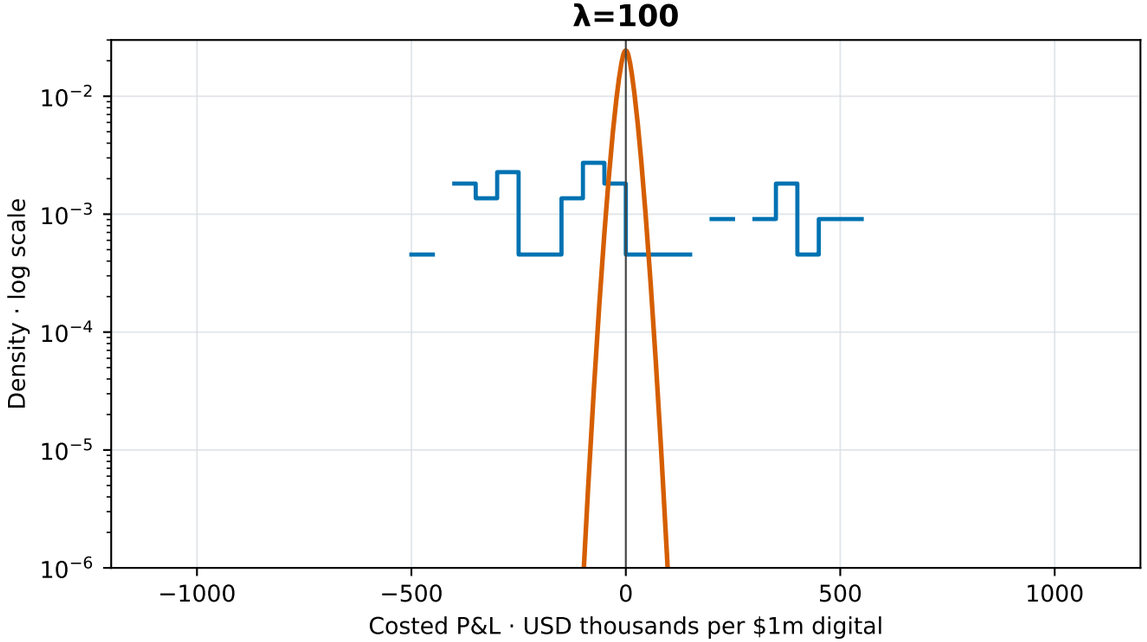
Realized, raw Fitted 252-day demeaned SANOS returns

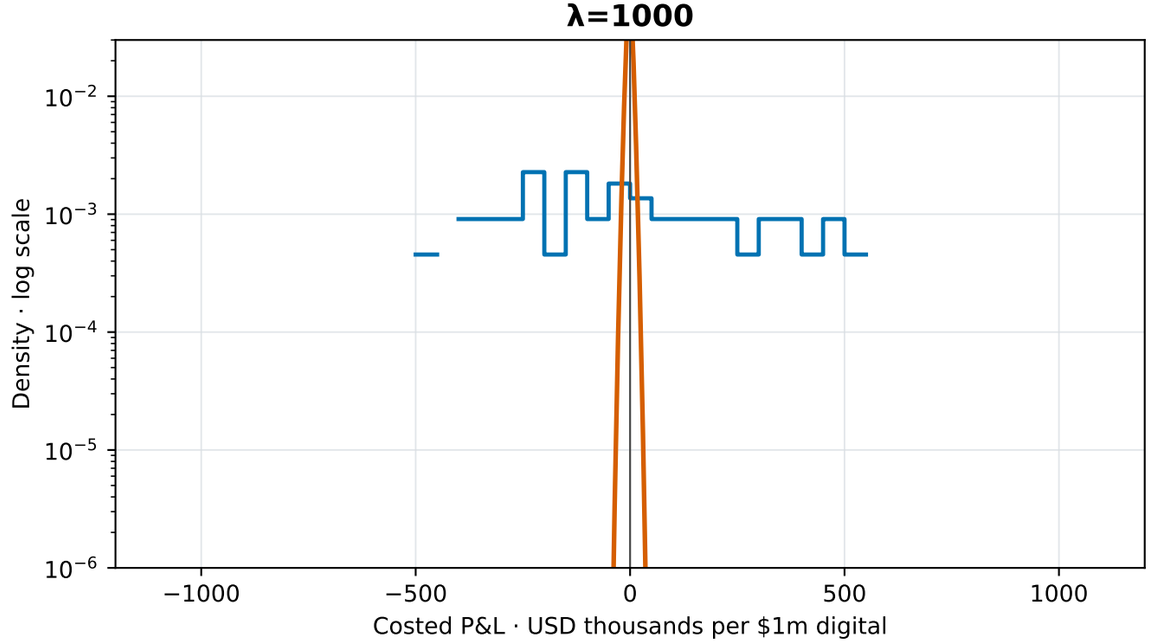
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-atm: SANOS parametric · Entropy





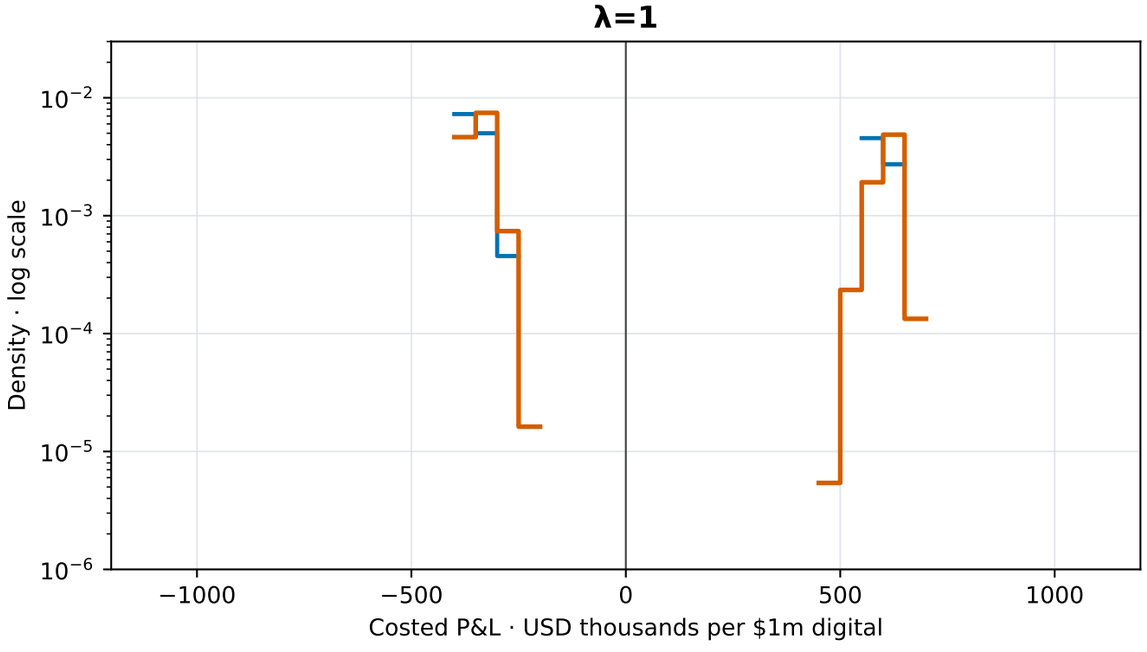


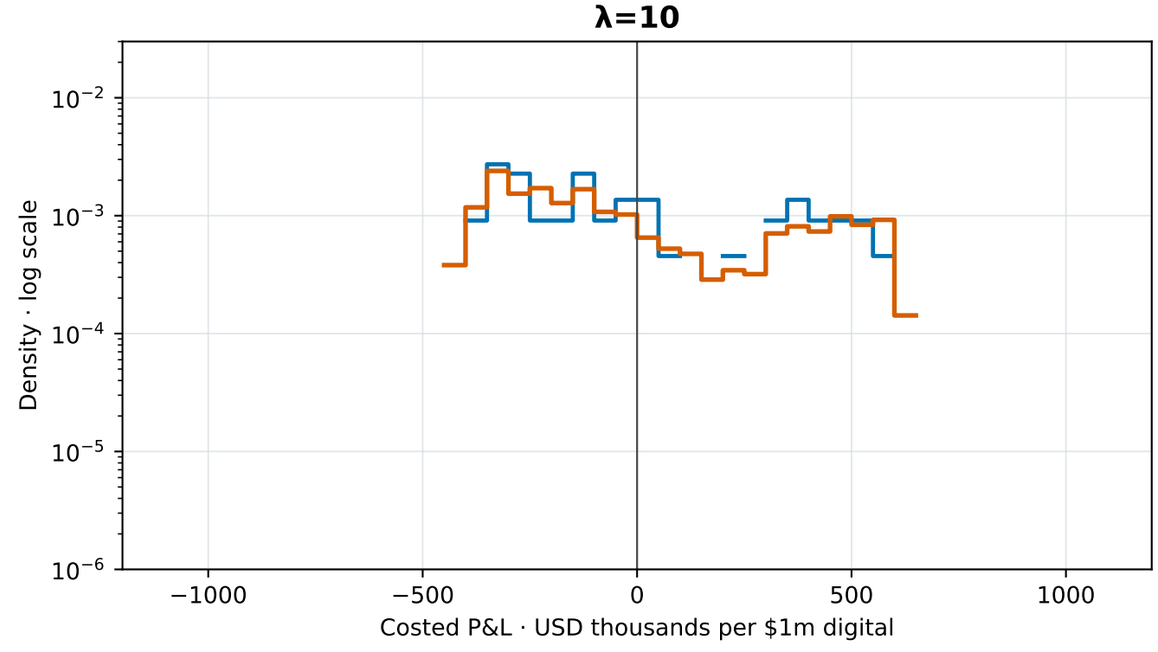


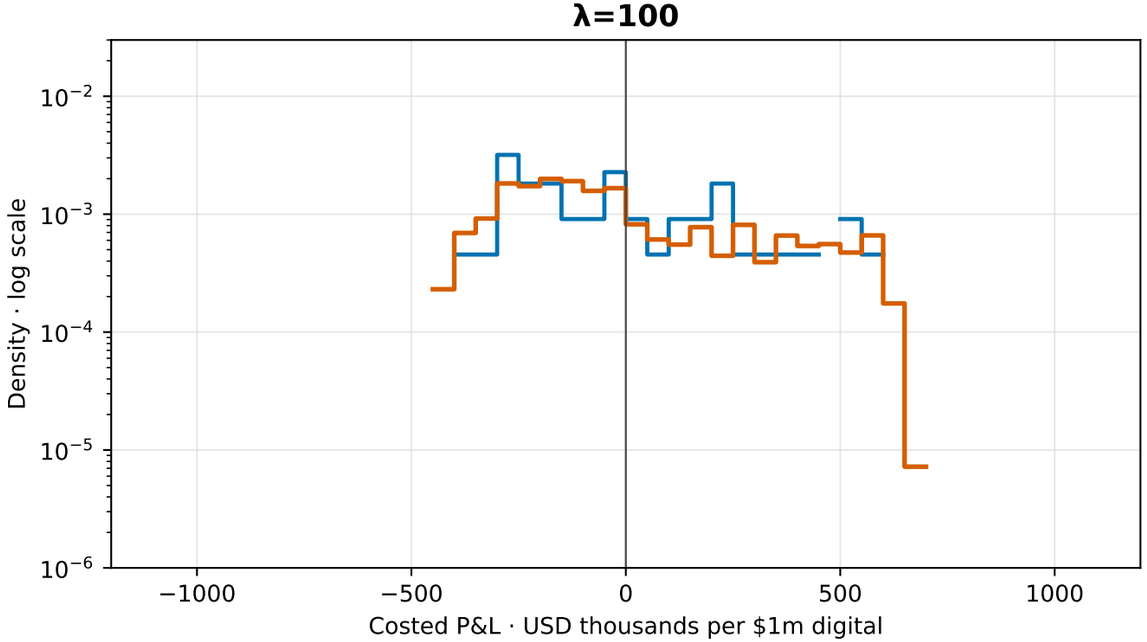
Realized, raw Fitted centered factor Gaussian

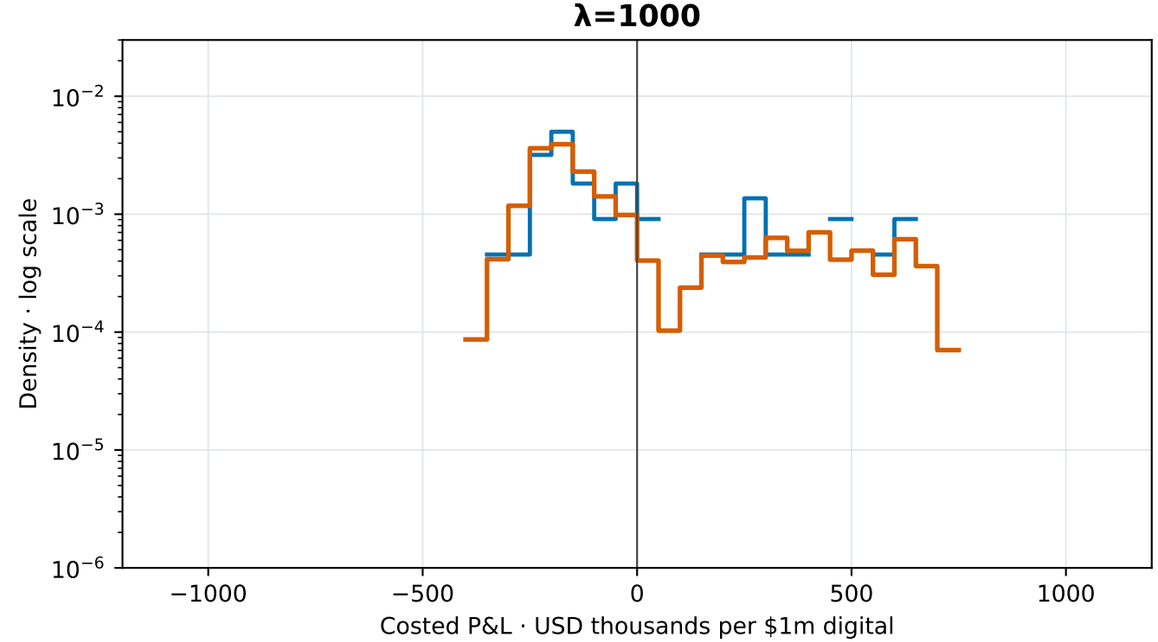
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-atm: SANOS historical · Power-exponential





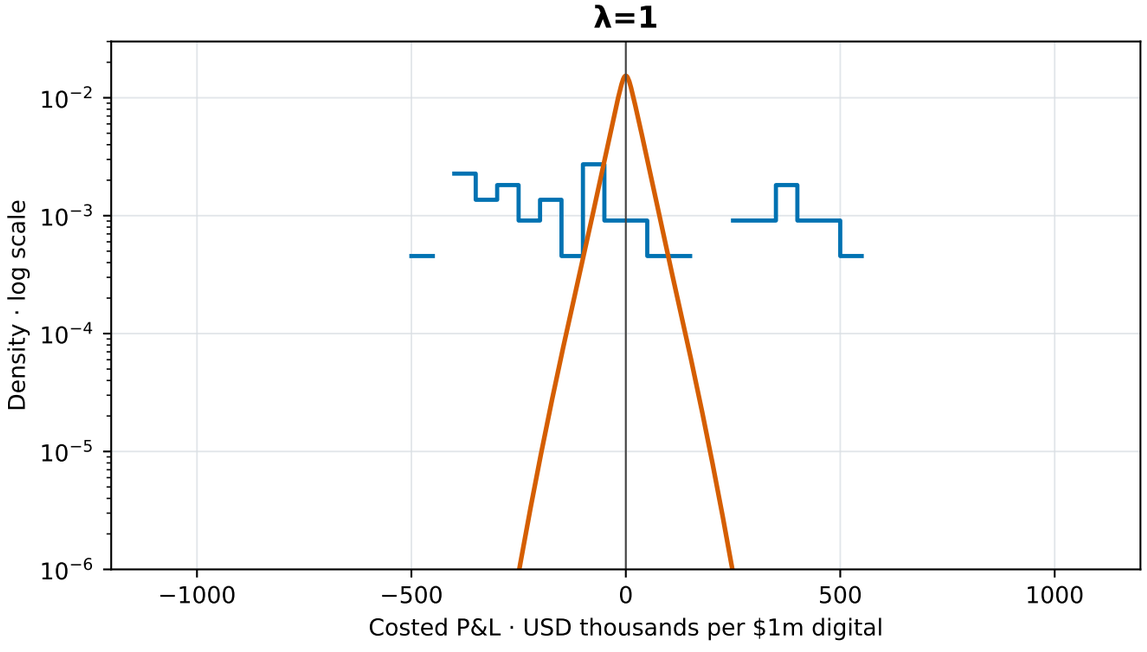


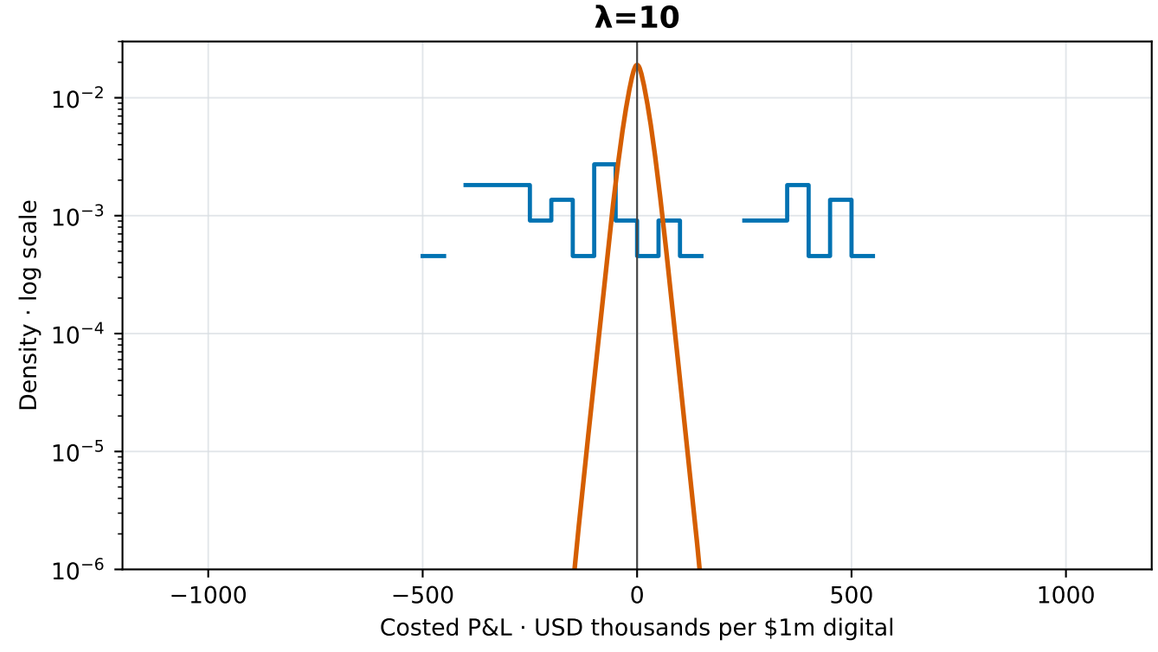


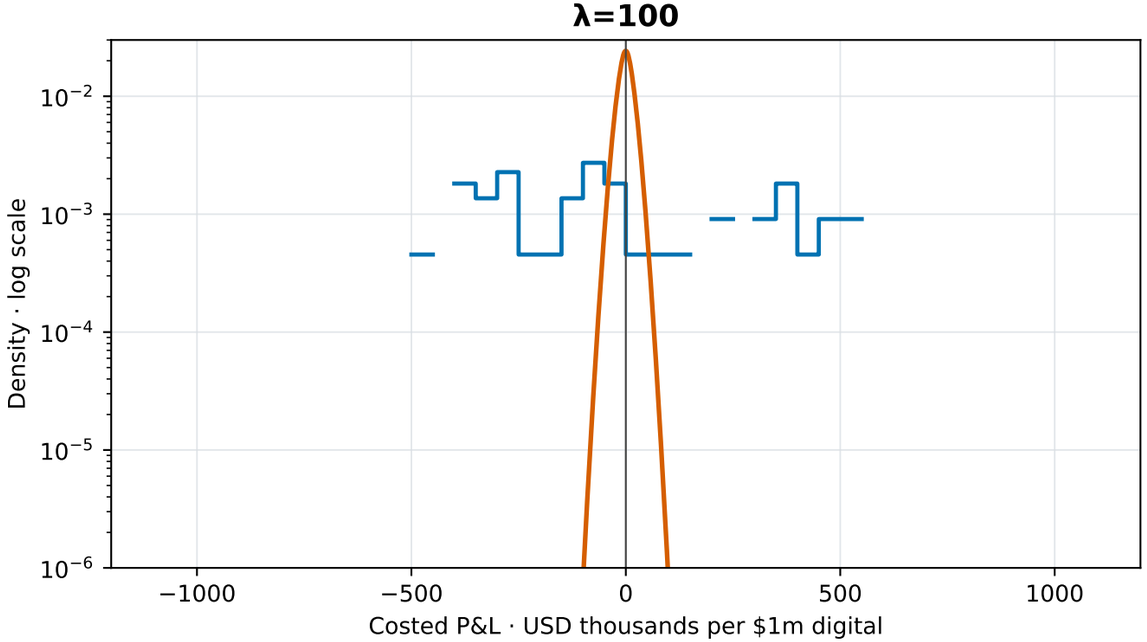
Realized, raw Fitted 252-day demeaned SANOS returns

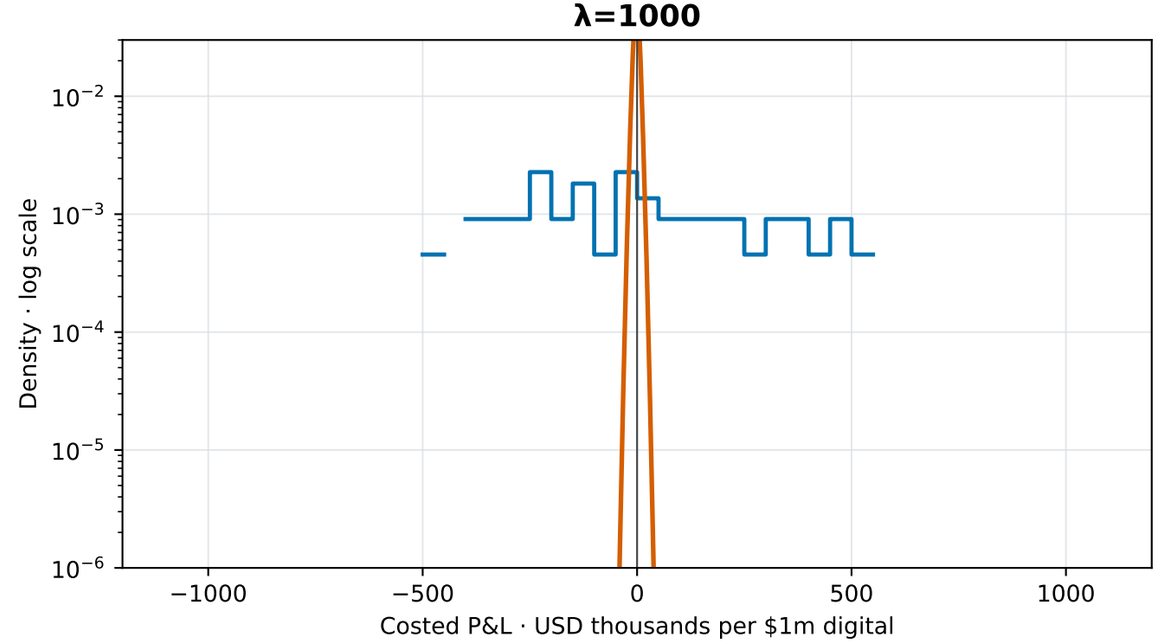
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-put-1d-atm: SANOS parametric · Power-exponential



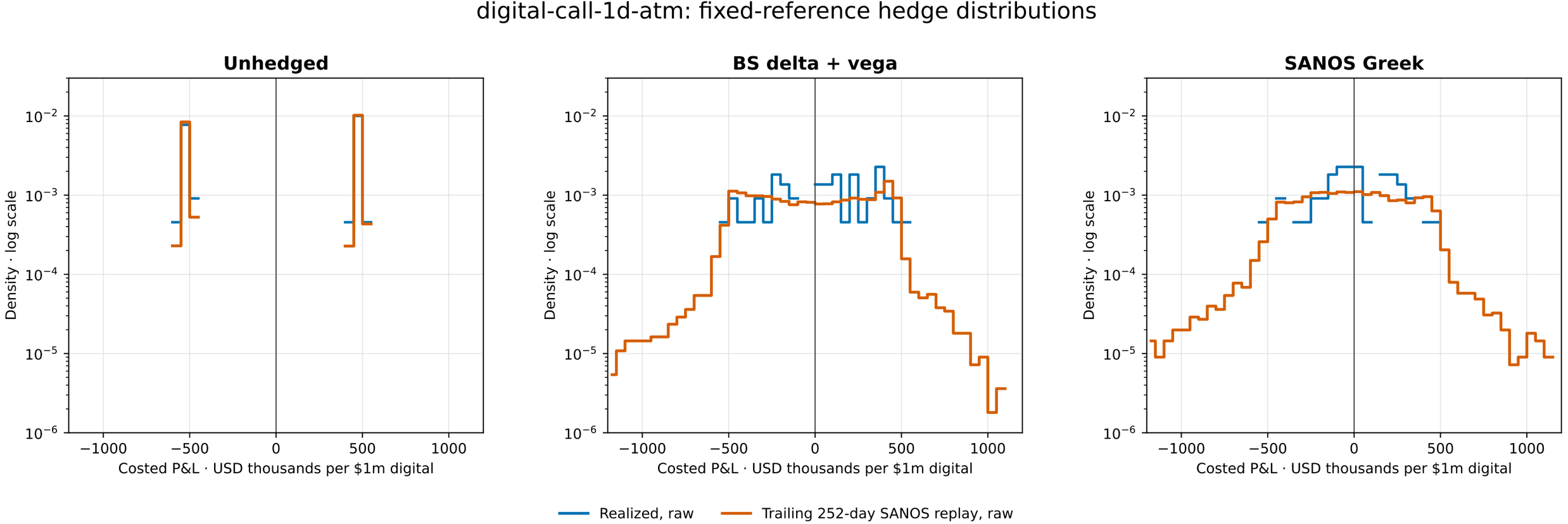




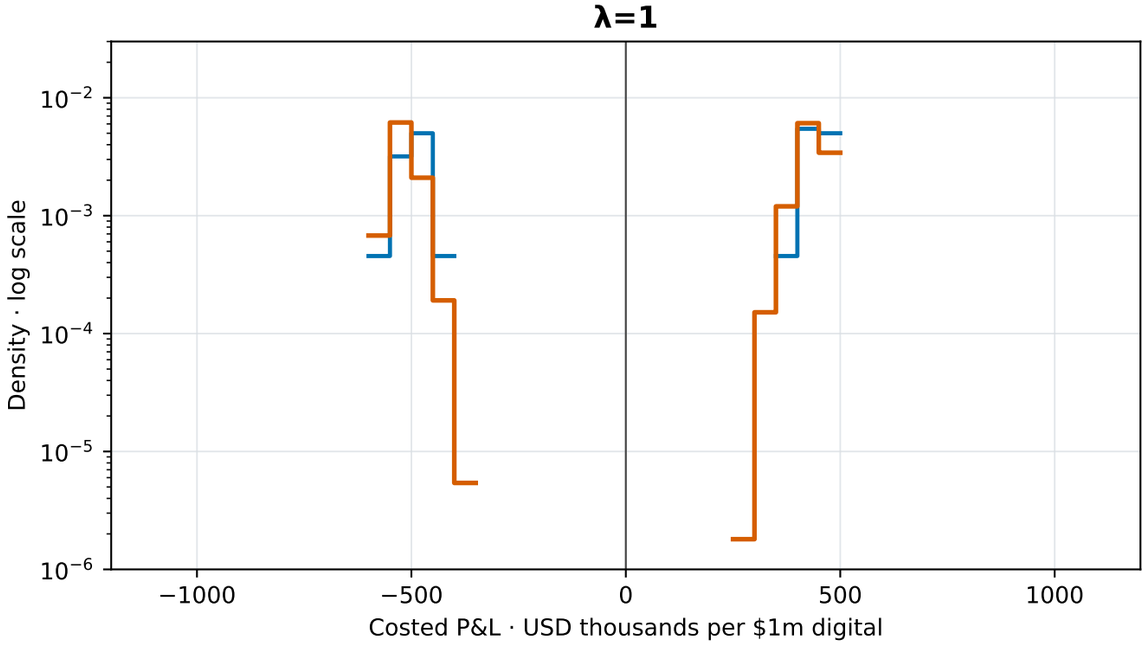


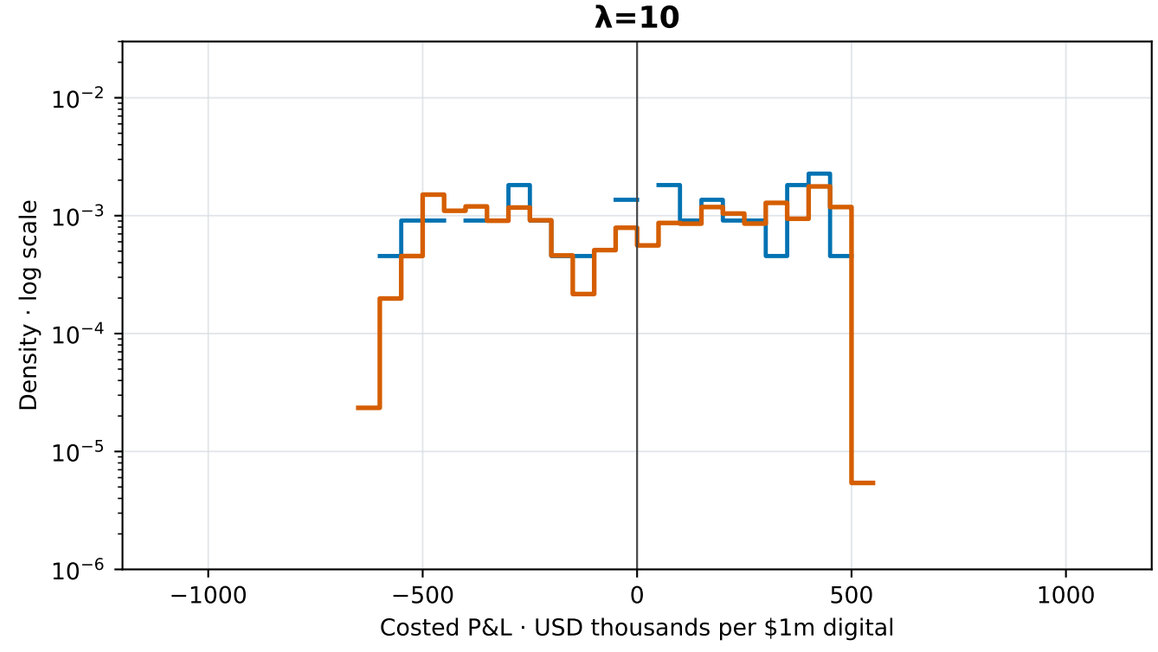
Realized, raw Fitted centered factor Gaussian

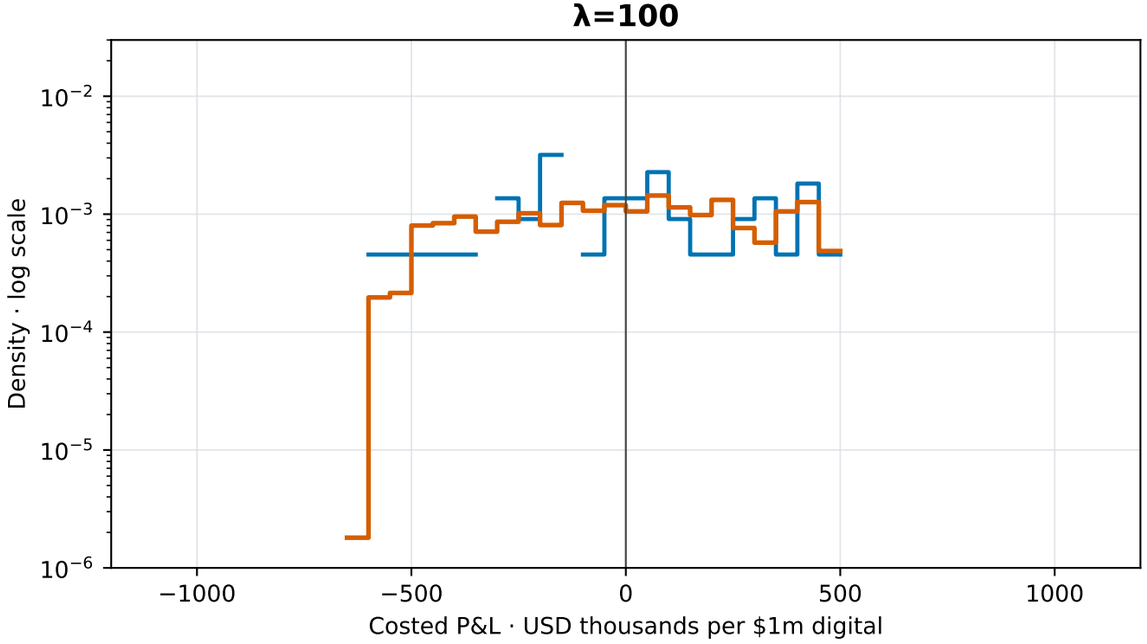
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

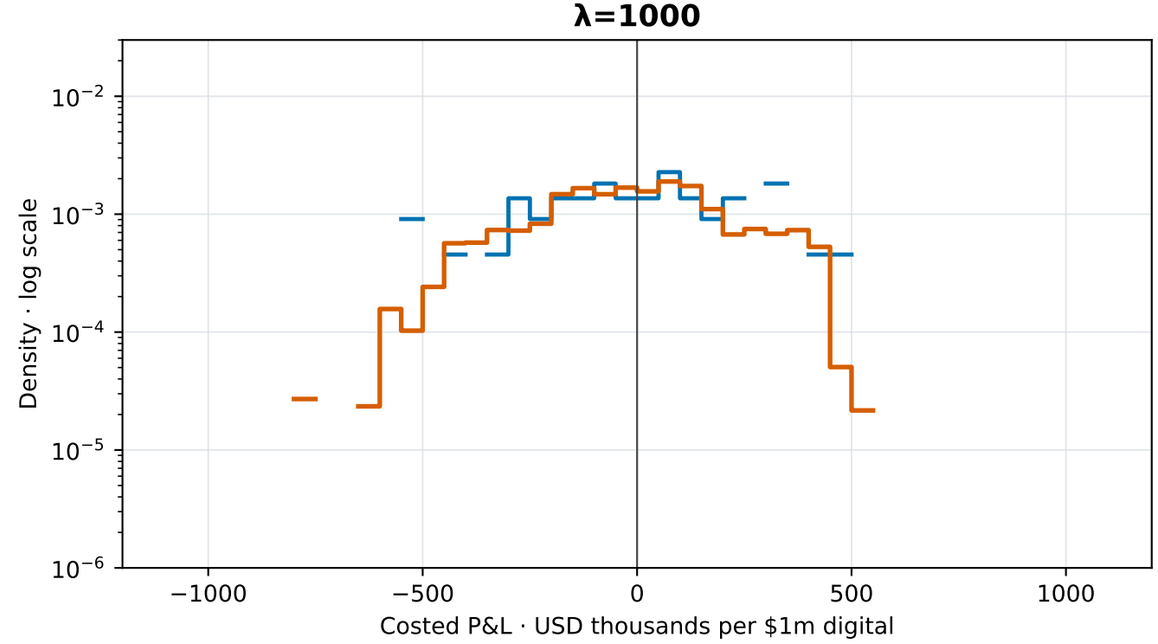

digital-call-1d-atm: fixed-reference hedge distributions
Unhedged
BS delta + vega
SANOS Greek
Density · log scale
Costed P&L · USD thousands per $1m digital
Realized, raw
Trailing 252-day SANOS replay, raw
No mean correction: 44 realized next-day outcomes versus preceding 252 raw returns. References have no fitted hedge-return law.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-atm: SANOS historical · Mean-variance





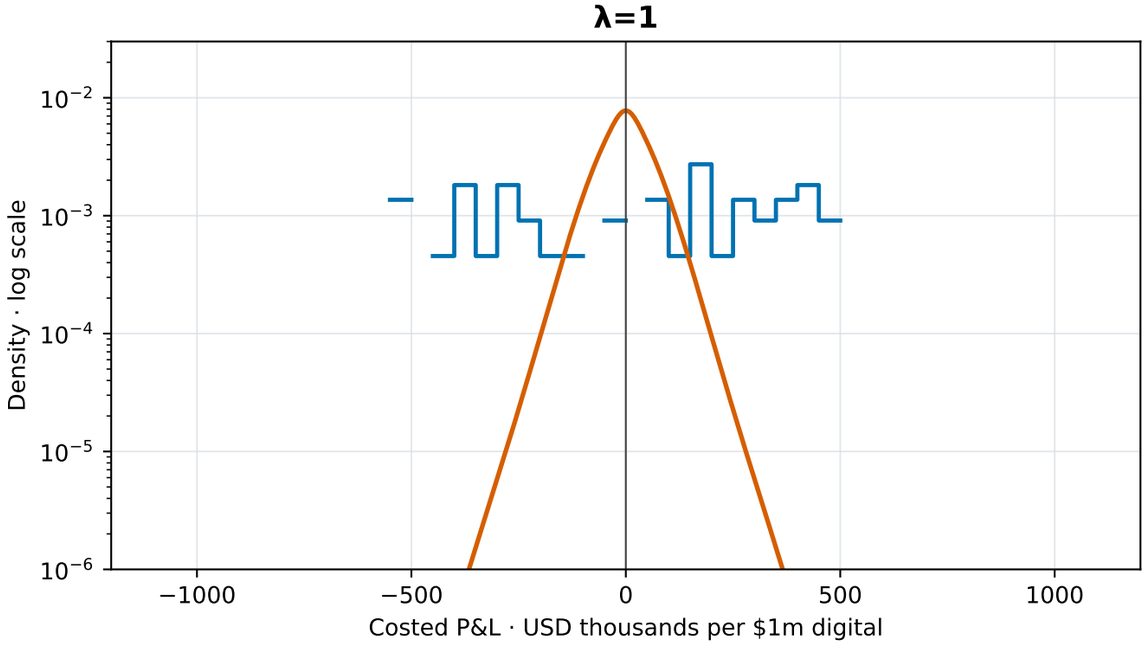


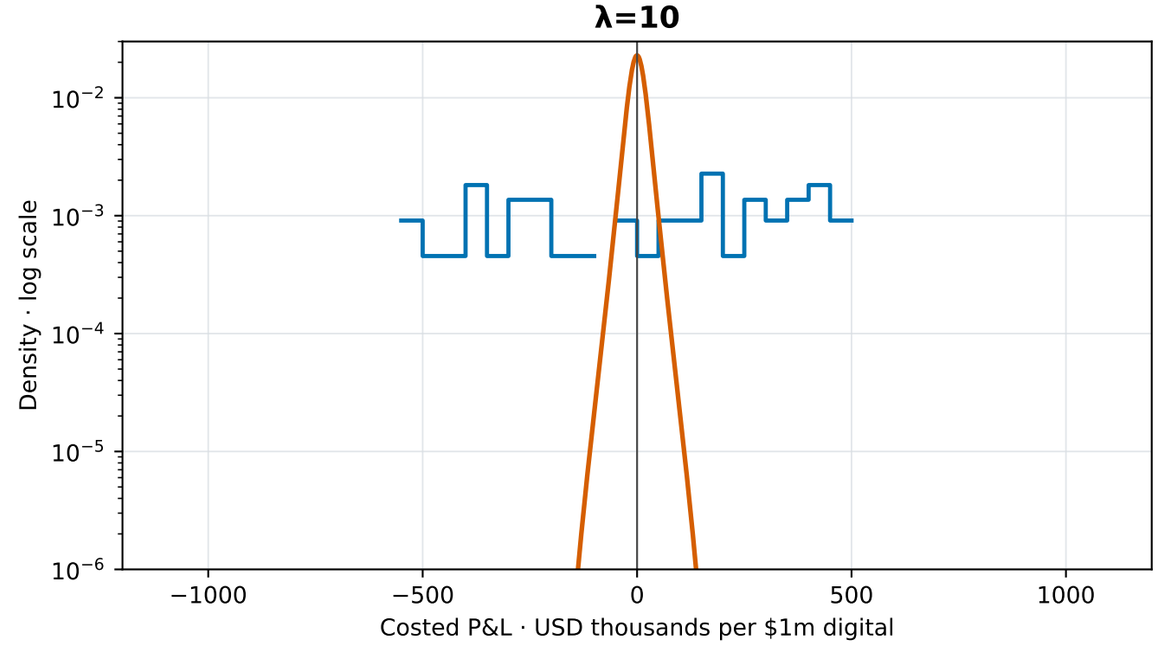


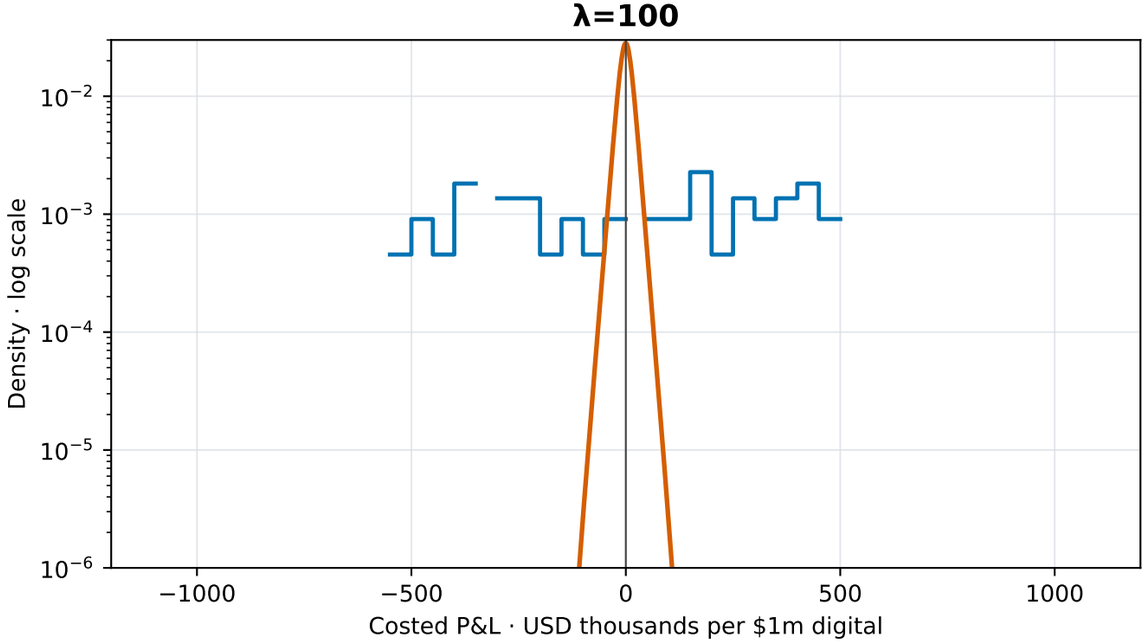
Realized, raw Fitted 252-day demeaned SANOS returns

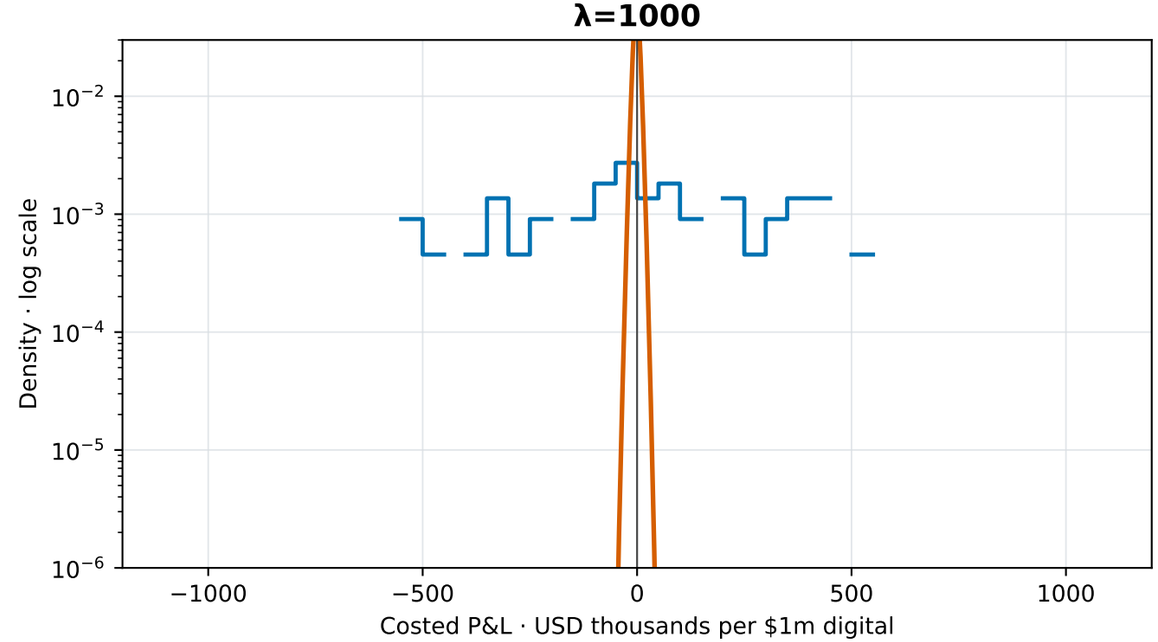
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-atm: SANOS parametric · Mean-variance





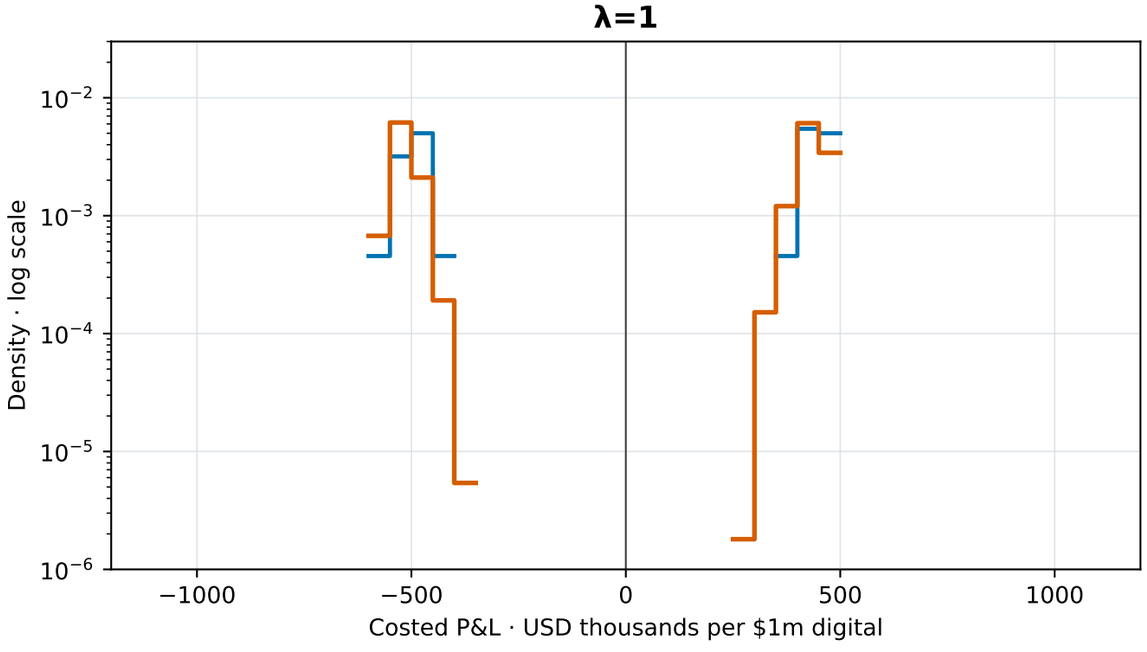


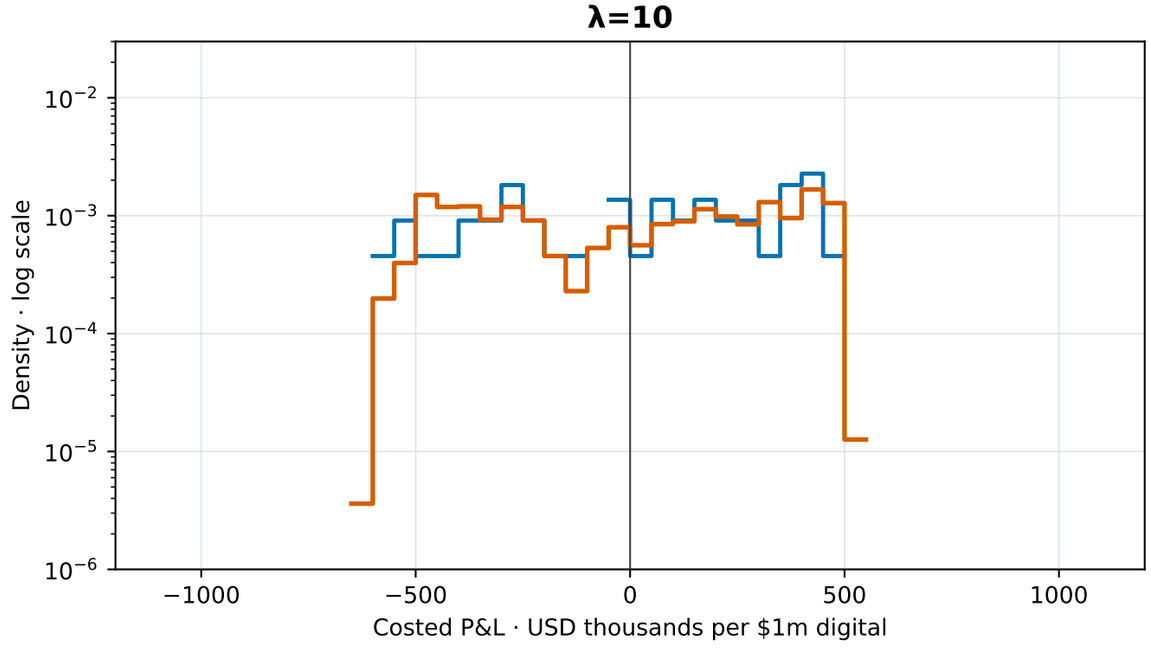


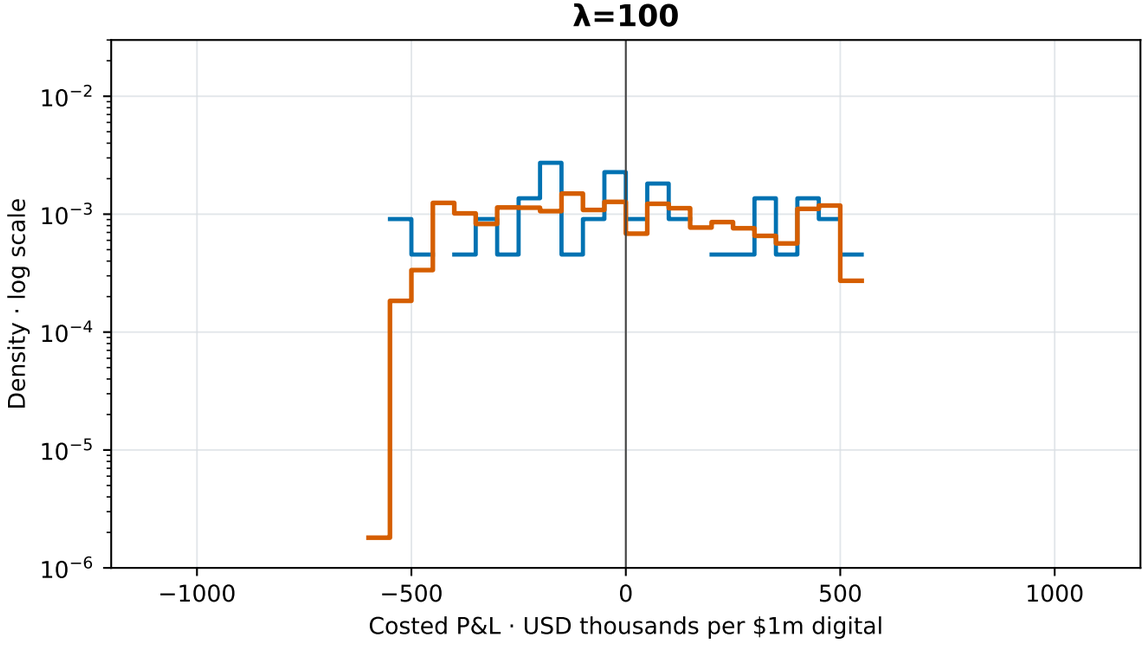
Realized, raw Fitted centered factor Gaussian

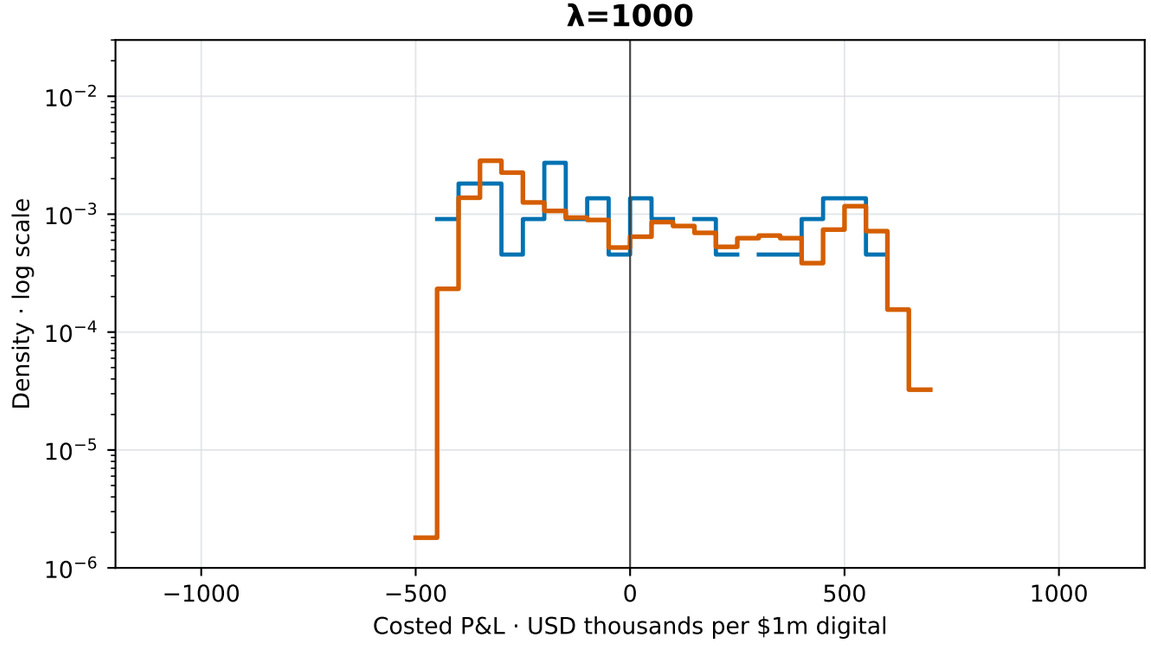
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-atm: SANOS historical · Entropy





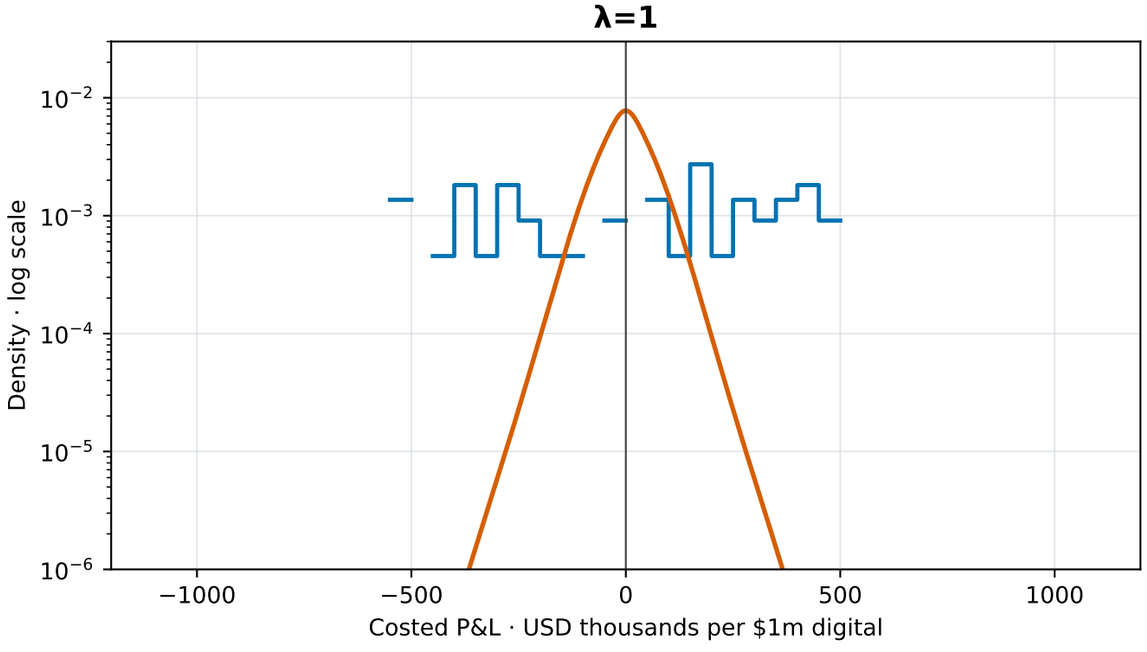


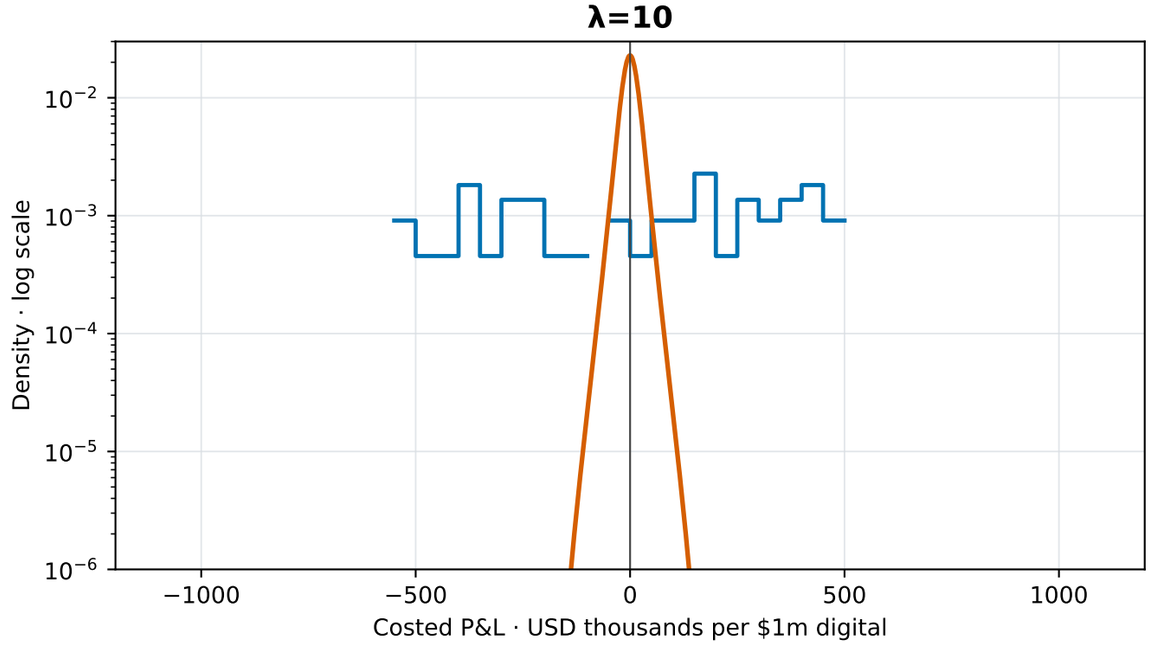


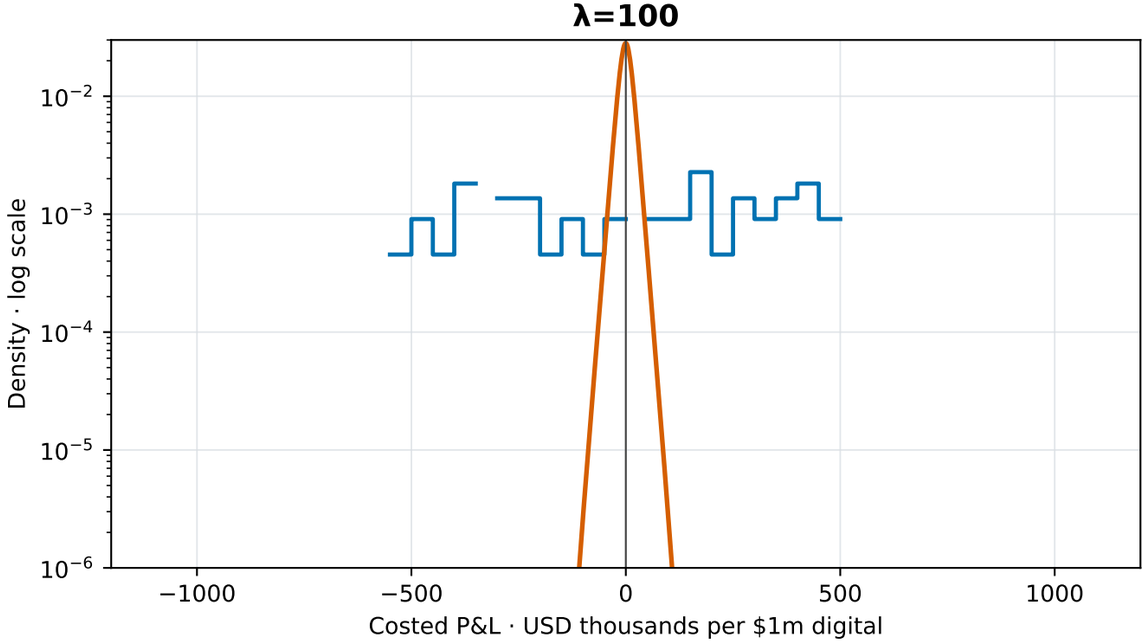
Realized, raw Fitted 252-day demeaned SANOS returns

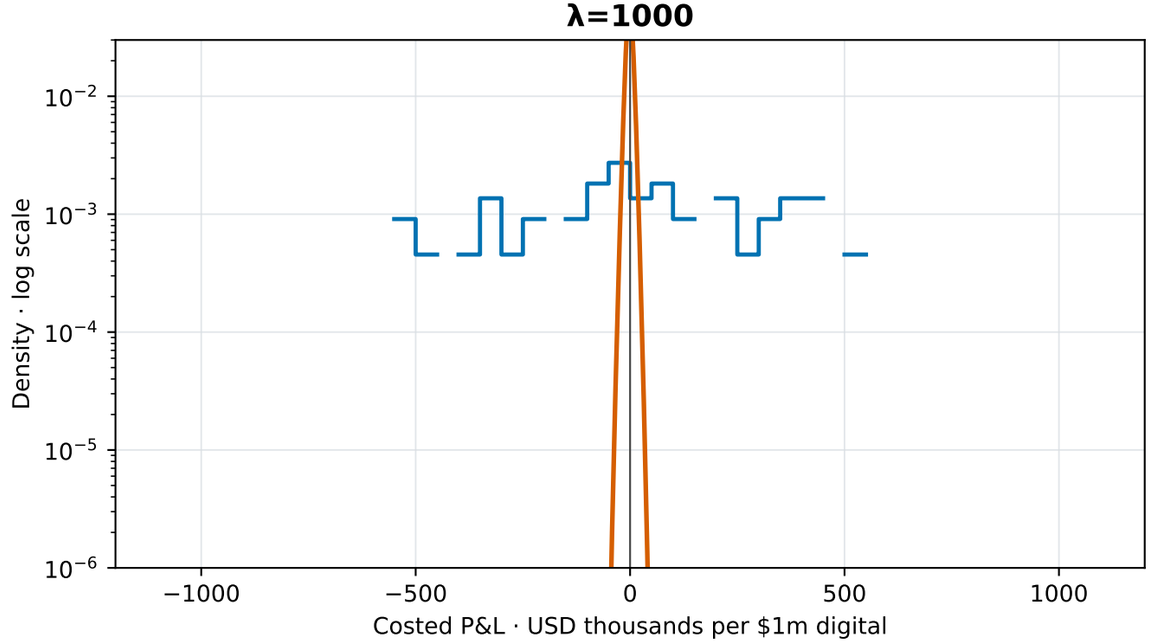
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-atm: SANOS parametric · Entropy





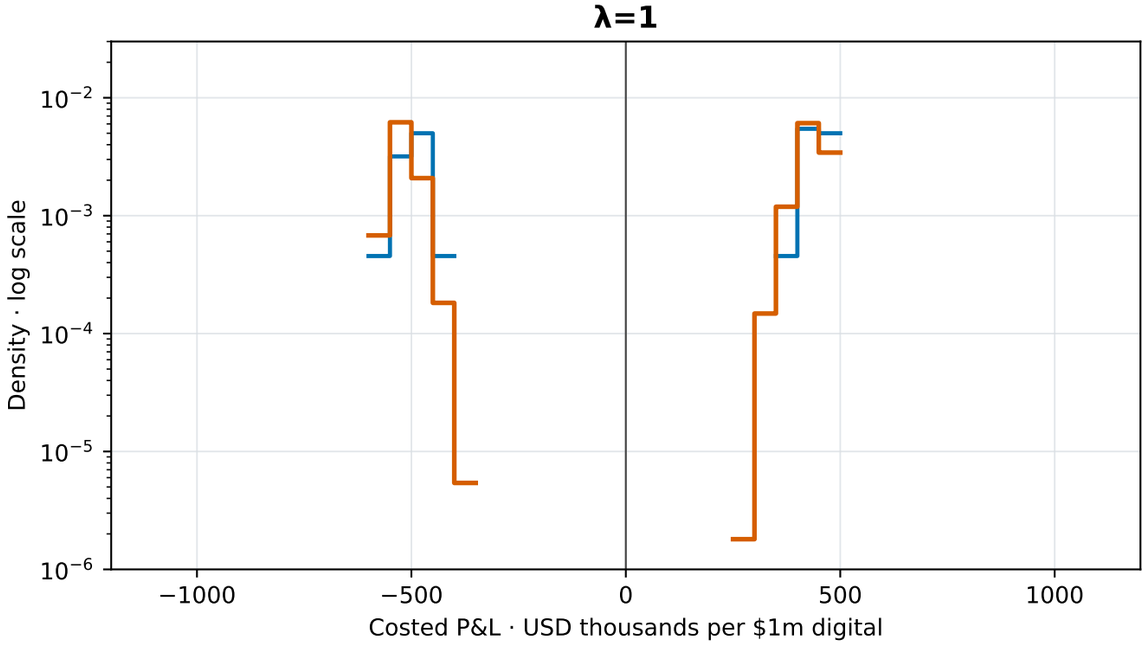


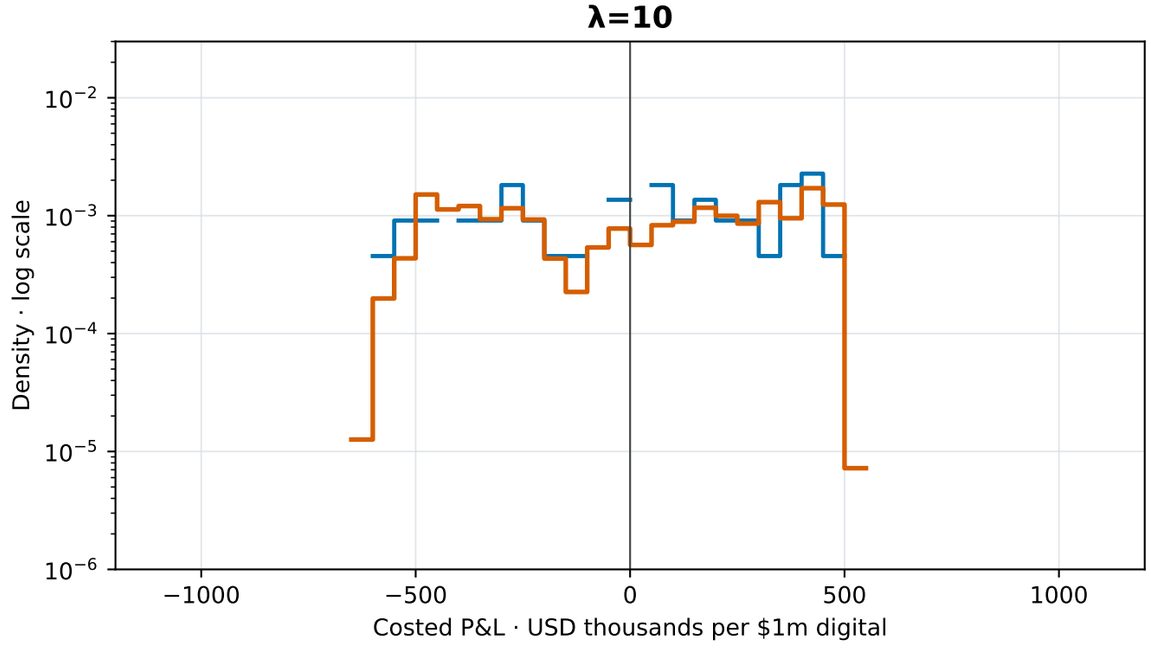


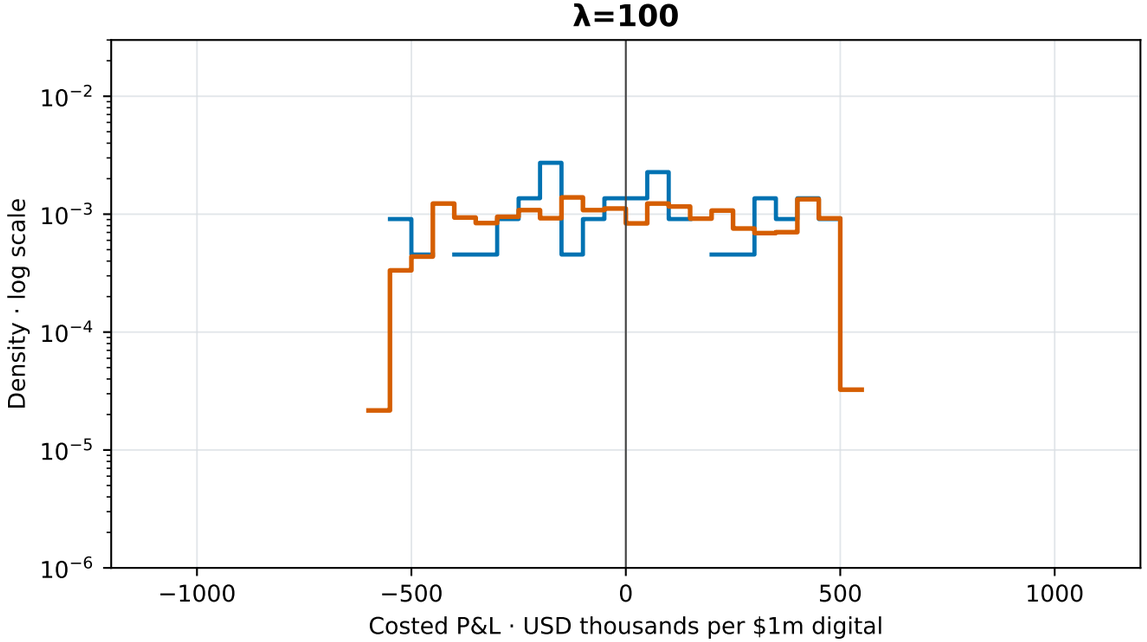
Realized, raw Fitted centered factor Gaussian

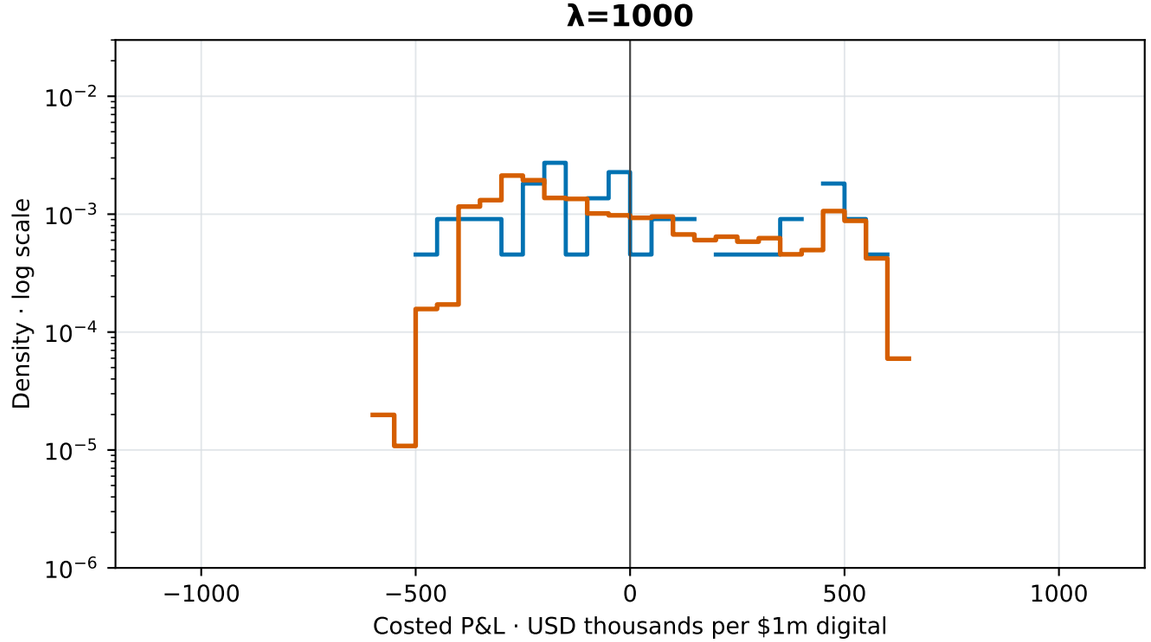
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-atm: SANOS historical · Power-exponential





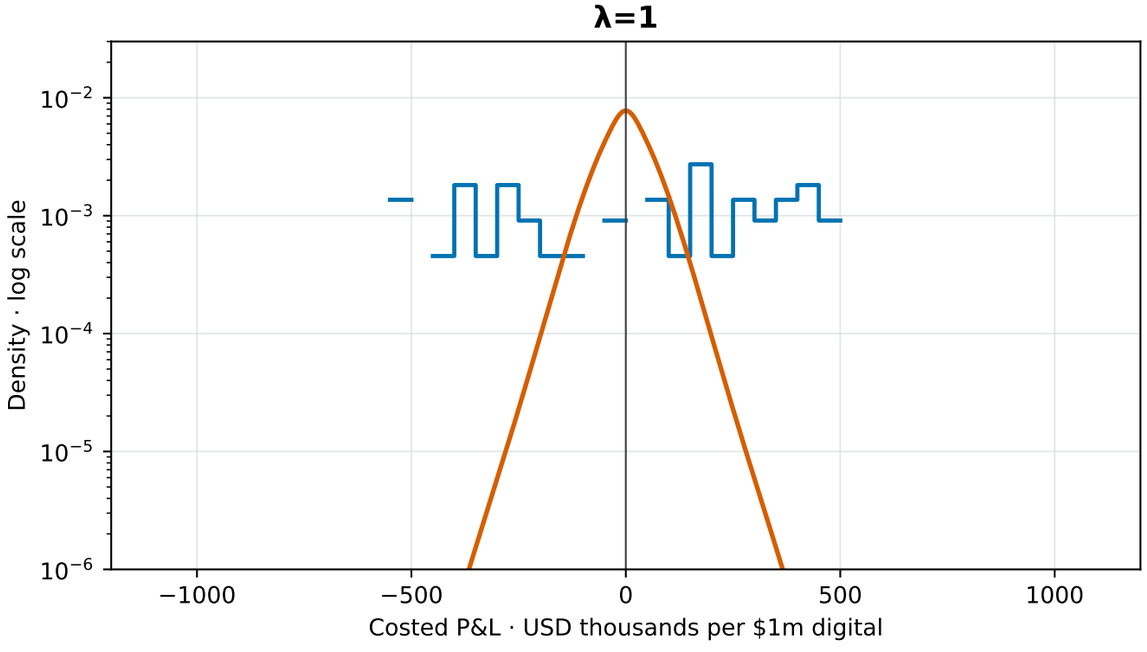


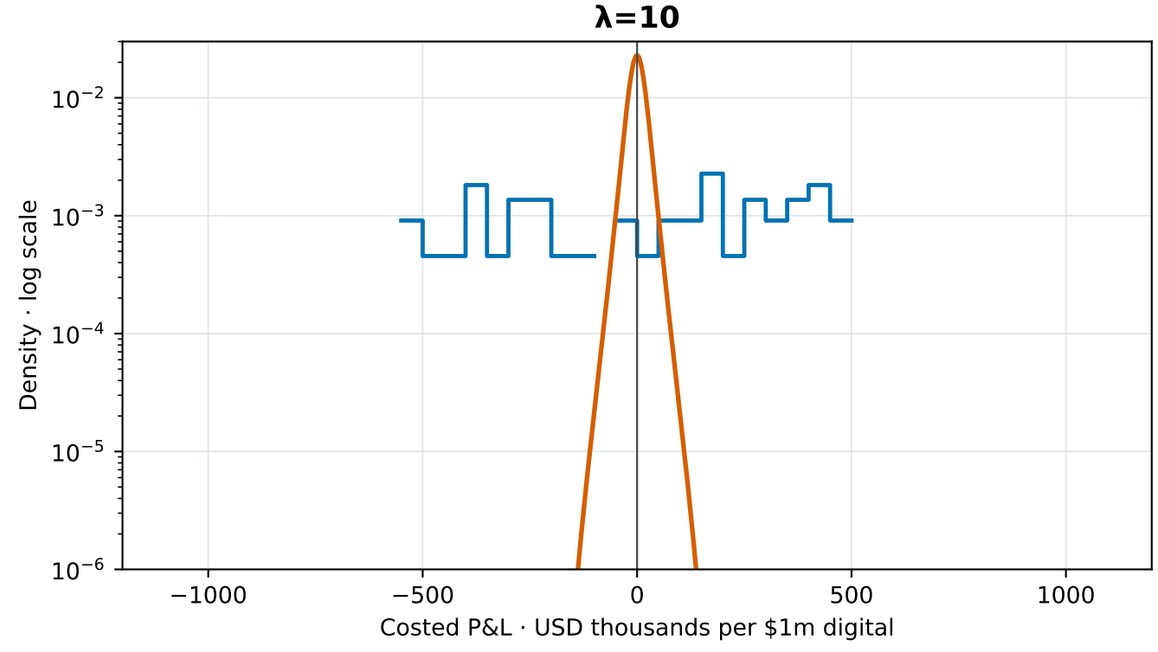


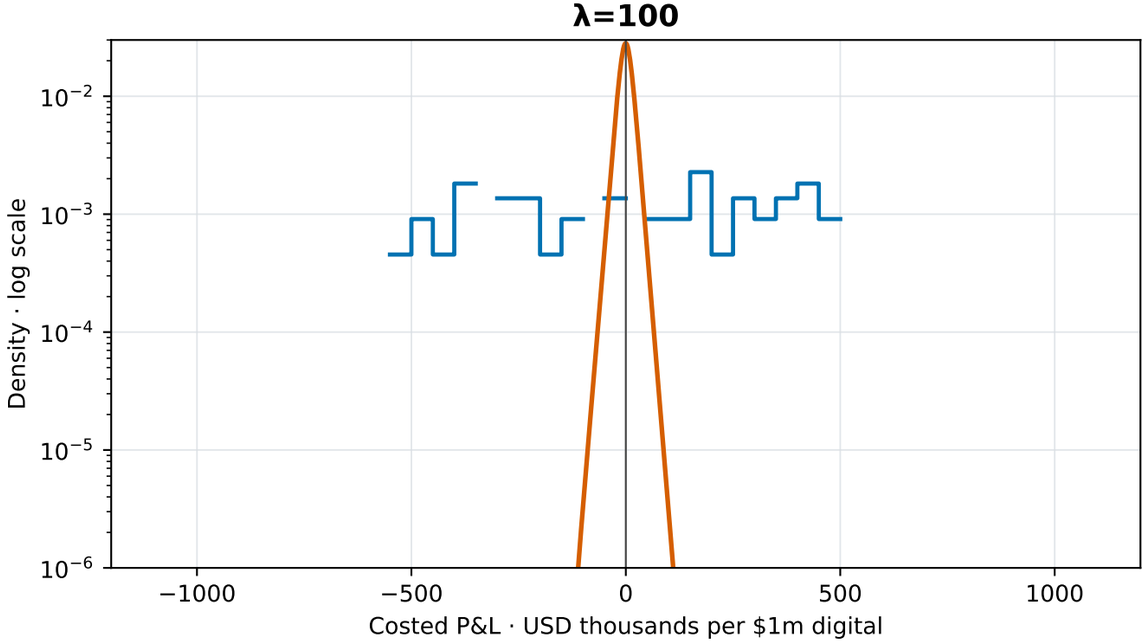
Realized, raw Fitted 252-day demeaned SANOS returns

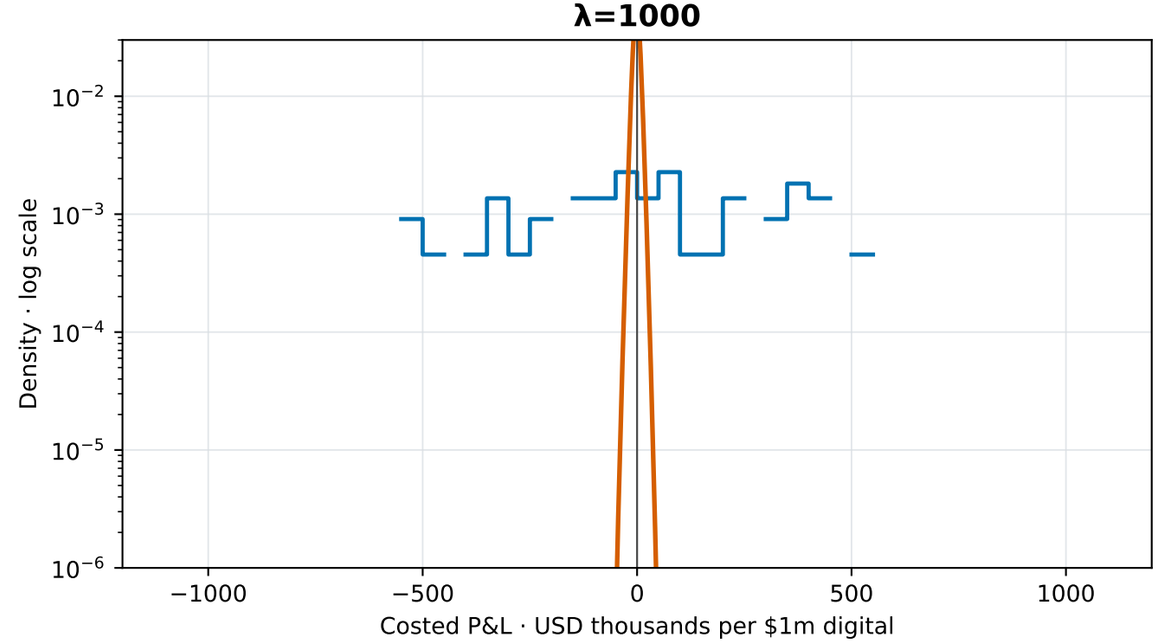
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-atm: SANOS parametric · Power-exponential



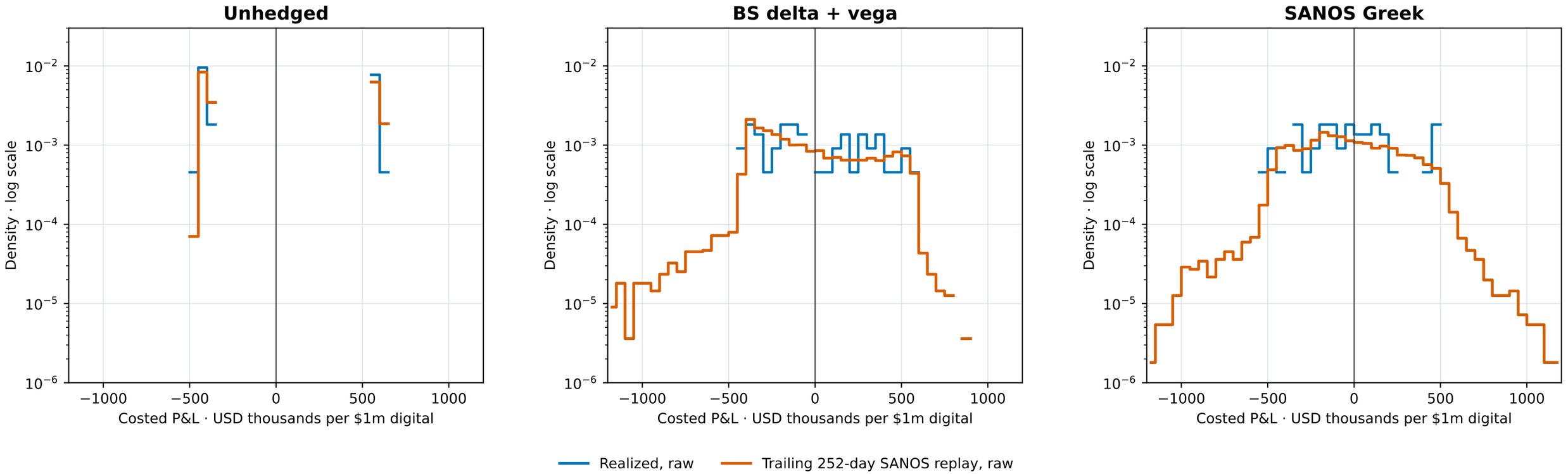




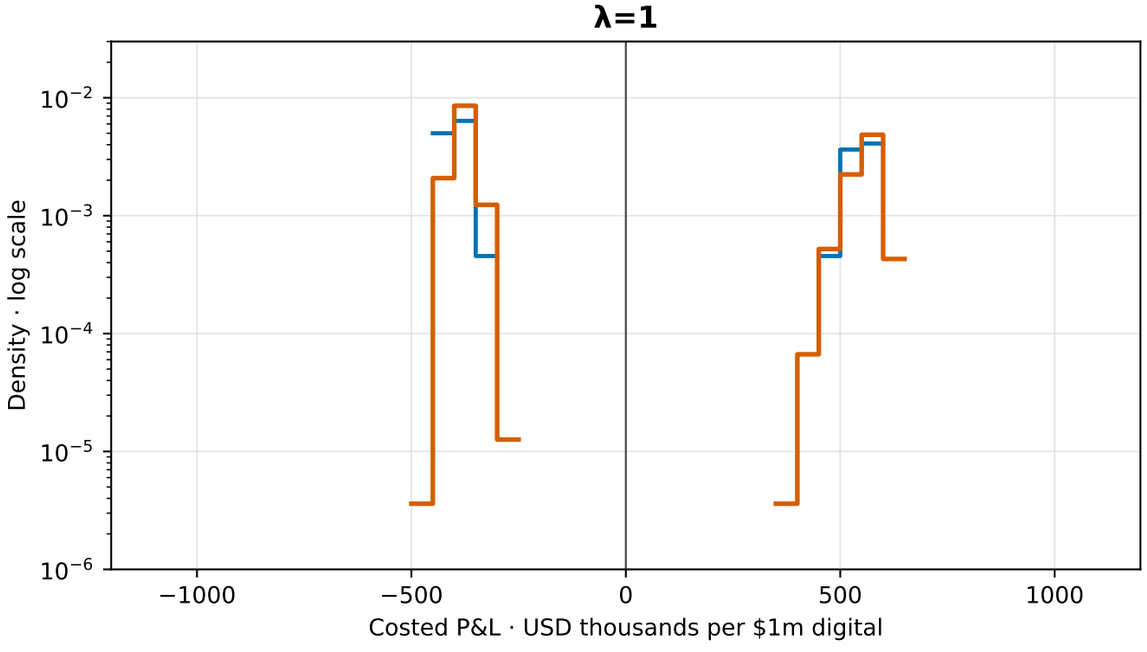


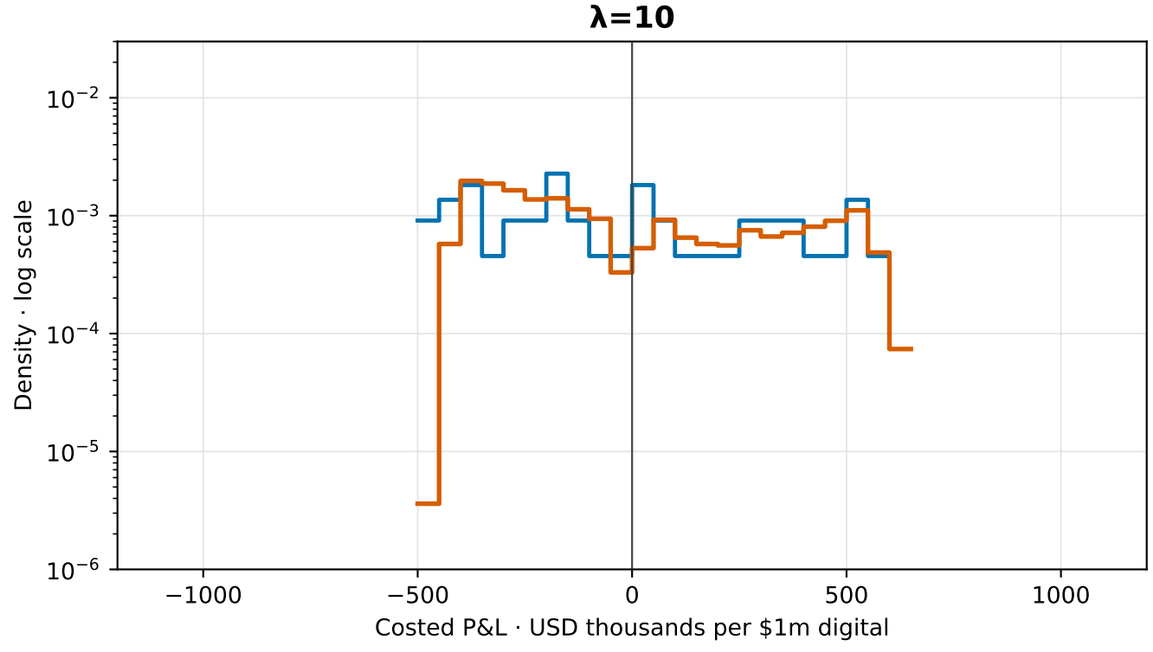
Realized, raw Fitted centered factor Gaussian

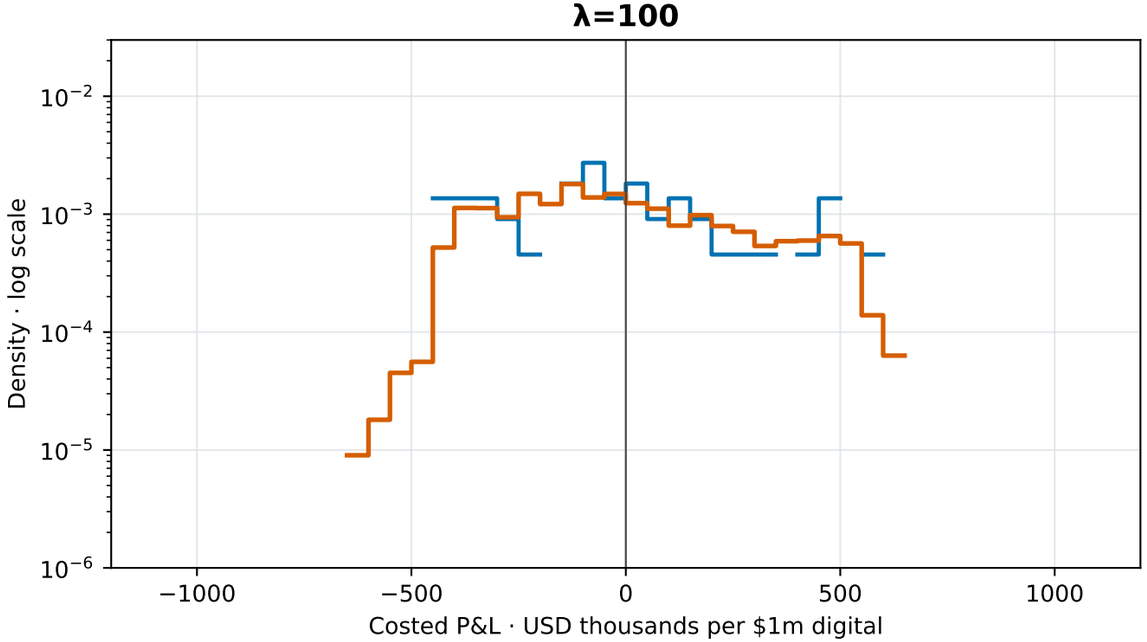
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

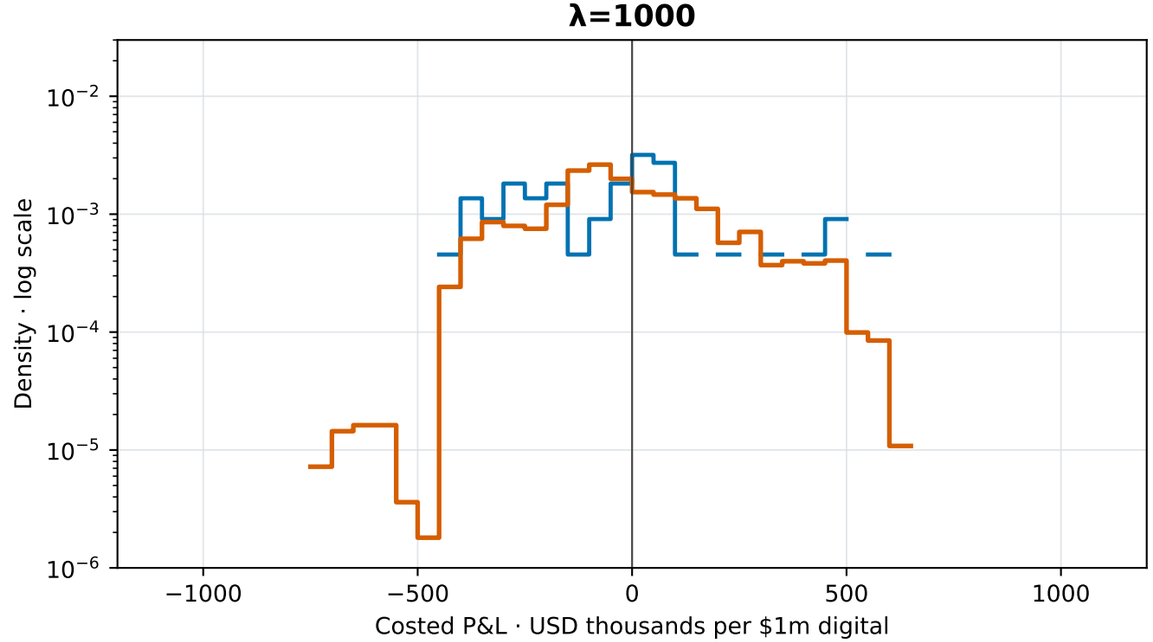

digital-call-1d-nms+0.25: fixed-reference hedge distributions
Unhedged
BS delta + vega
SANOS Greek
Density · log scale
Costed P&L · USD thousands per $1m digital
Realized, raw
Trailing 252-day SANOS replay, raw
No mean correction: 44 realized next-day outcomes versus preceding 252 raw returns. References have no fitted hedge-return law.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.25: SANOS historical · Mean-variance





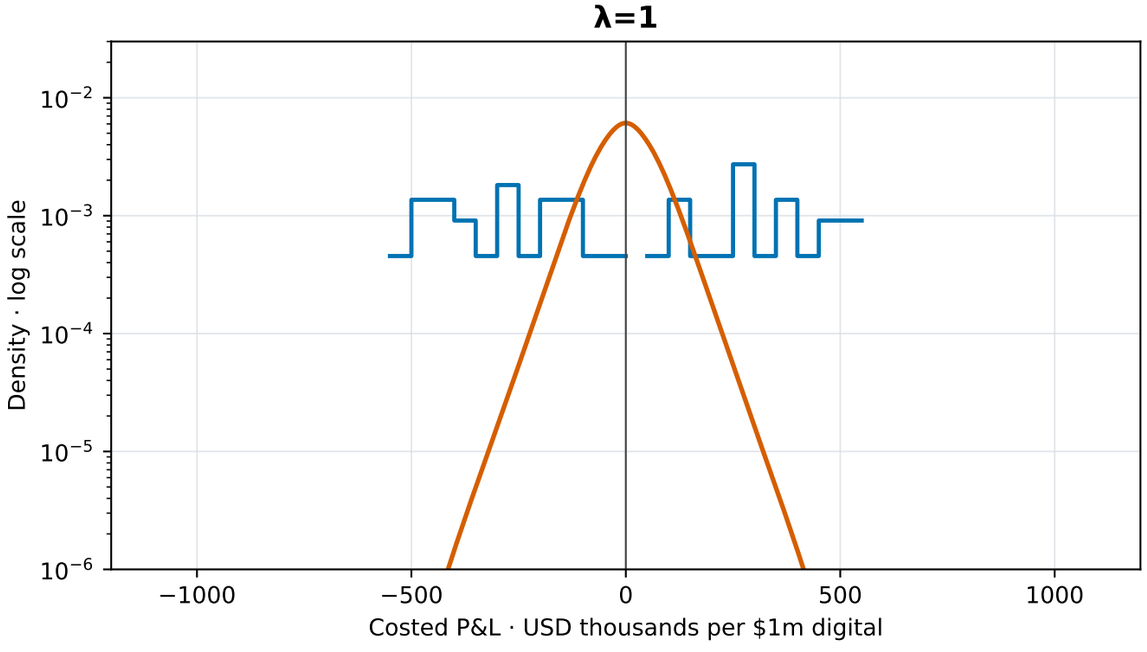


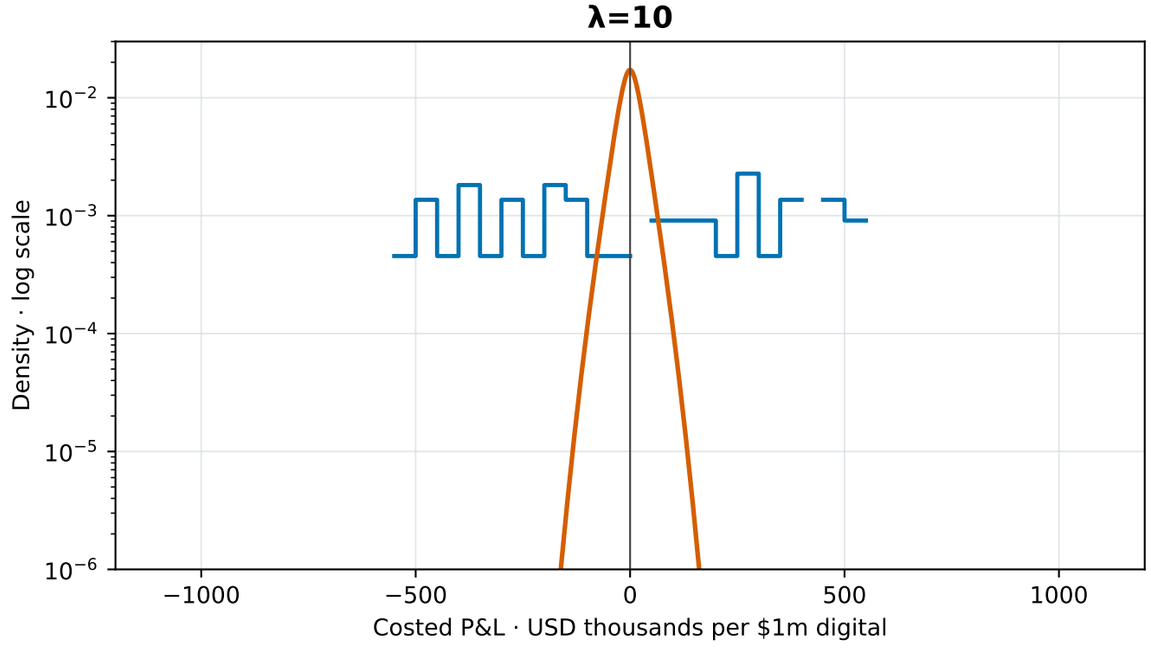


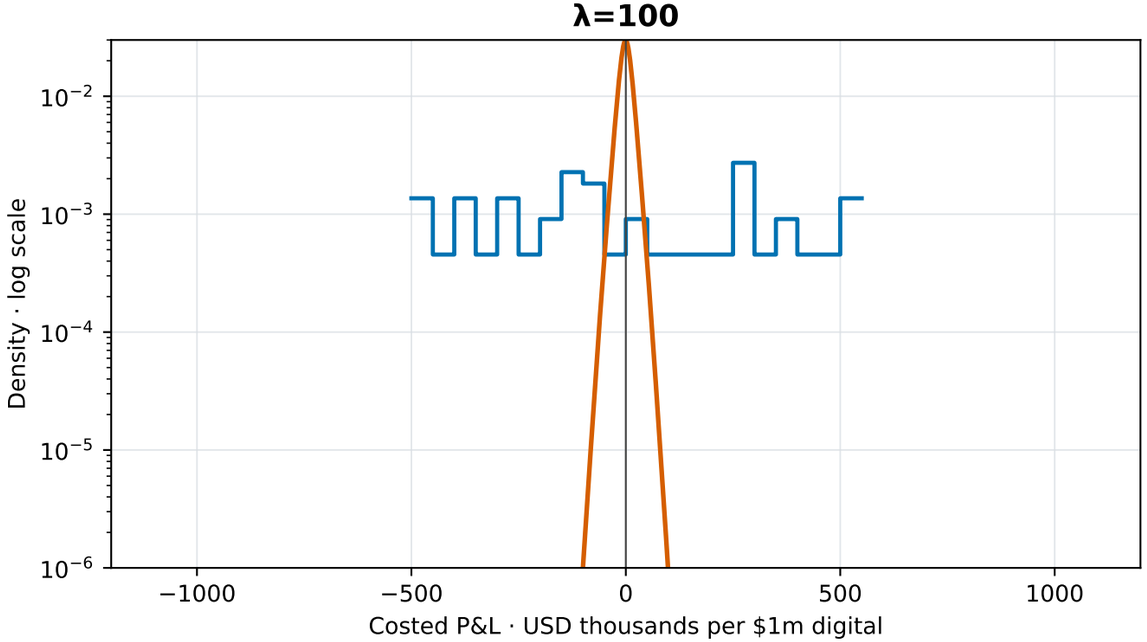
Realized, raw Fitted 252-day demeaned SANOS returns

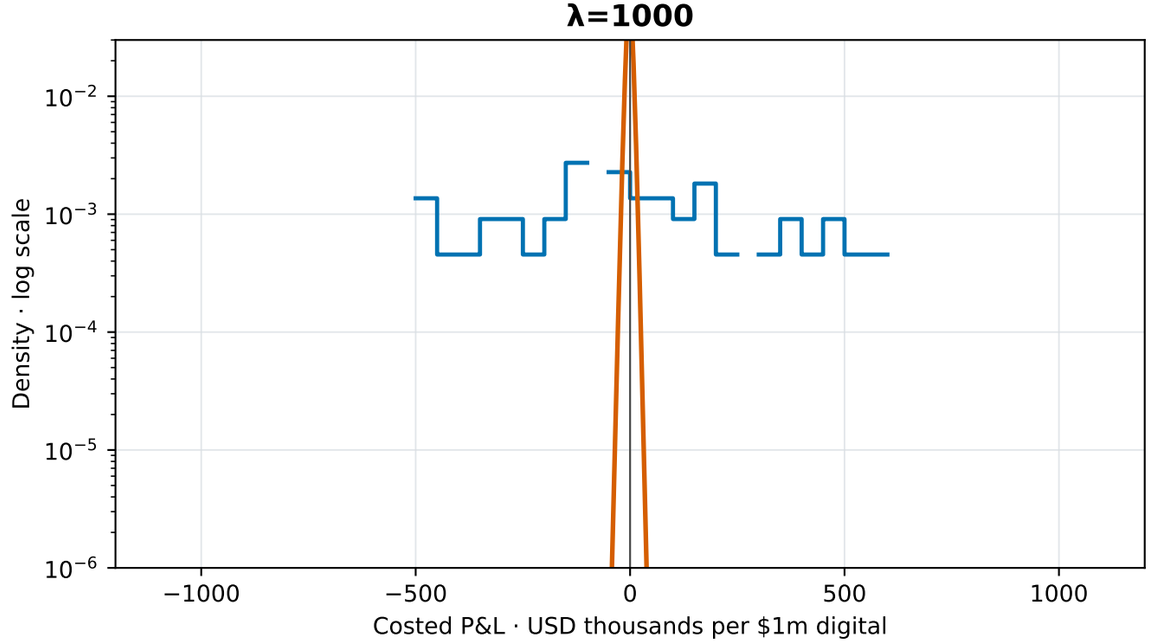
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.25: SANOS parametric · Mean-variance





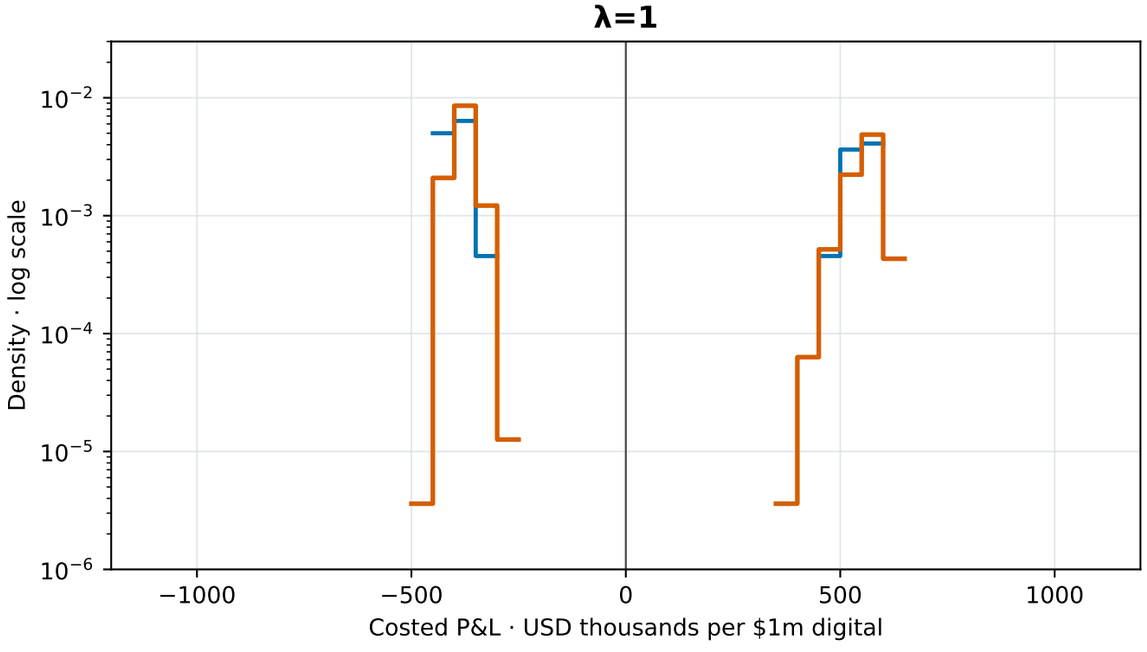


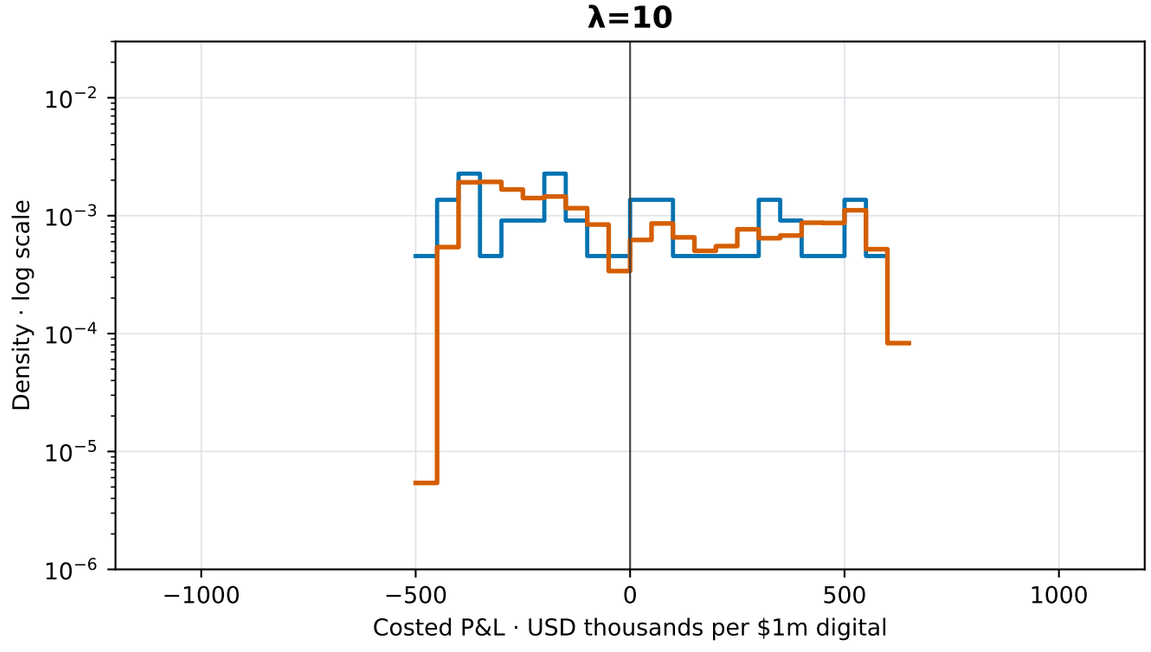


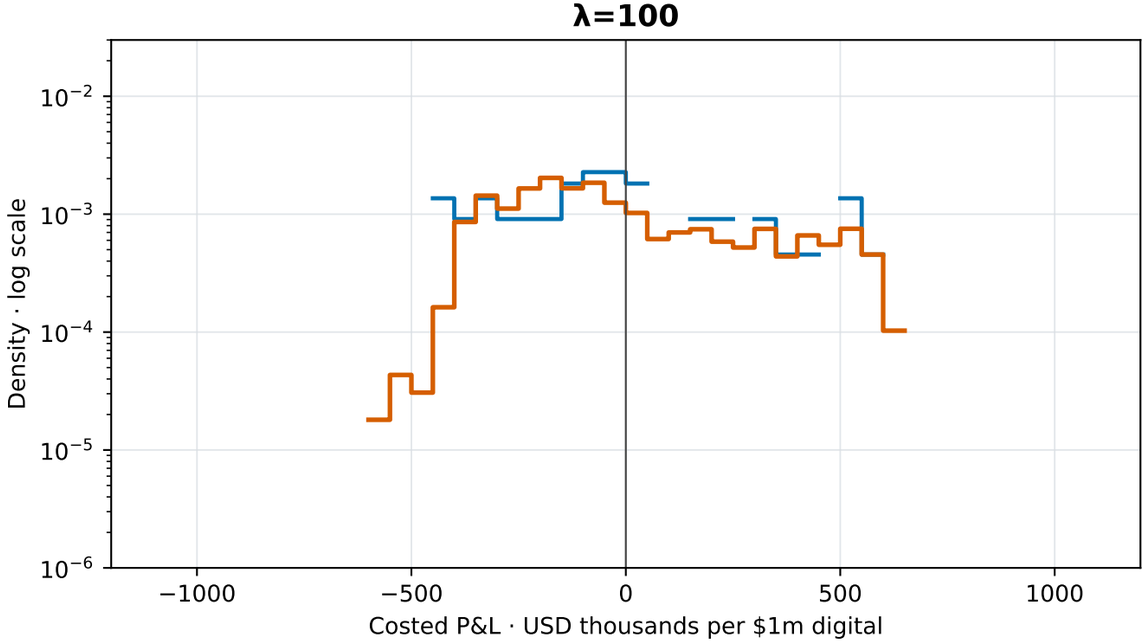
Realized, raw Fitted centered factor Gaussian

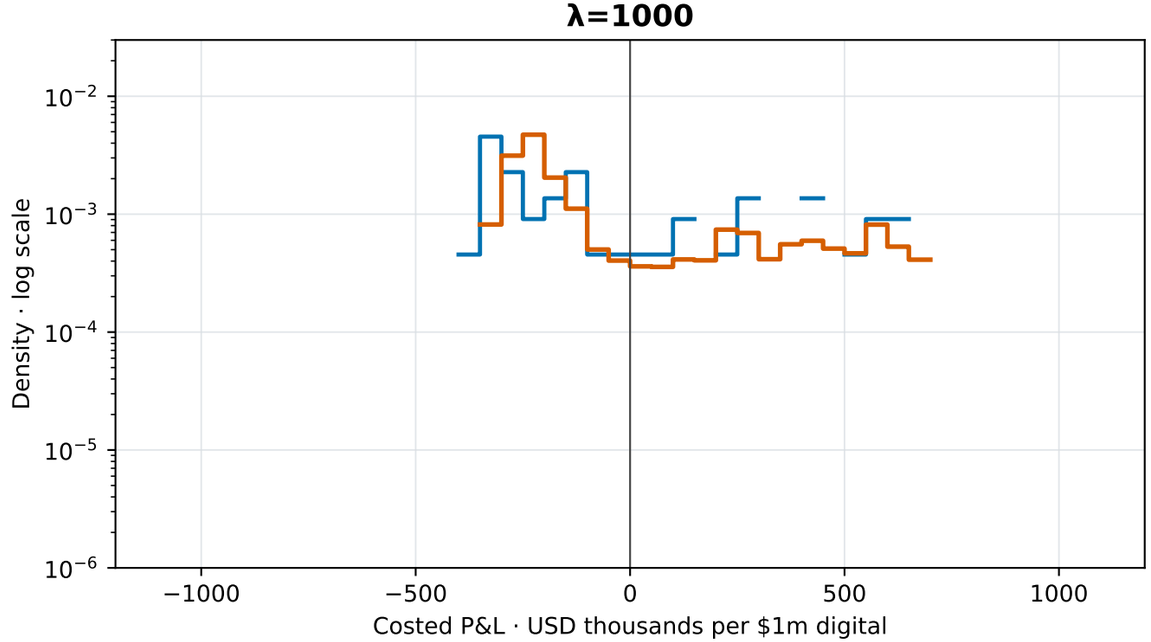
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.25: SANOS historical · Entropy





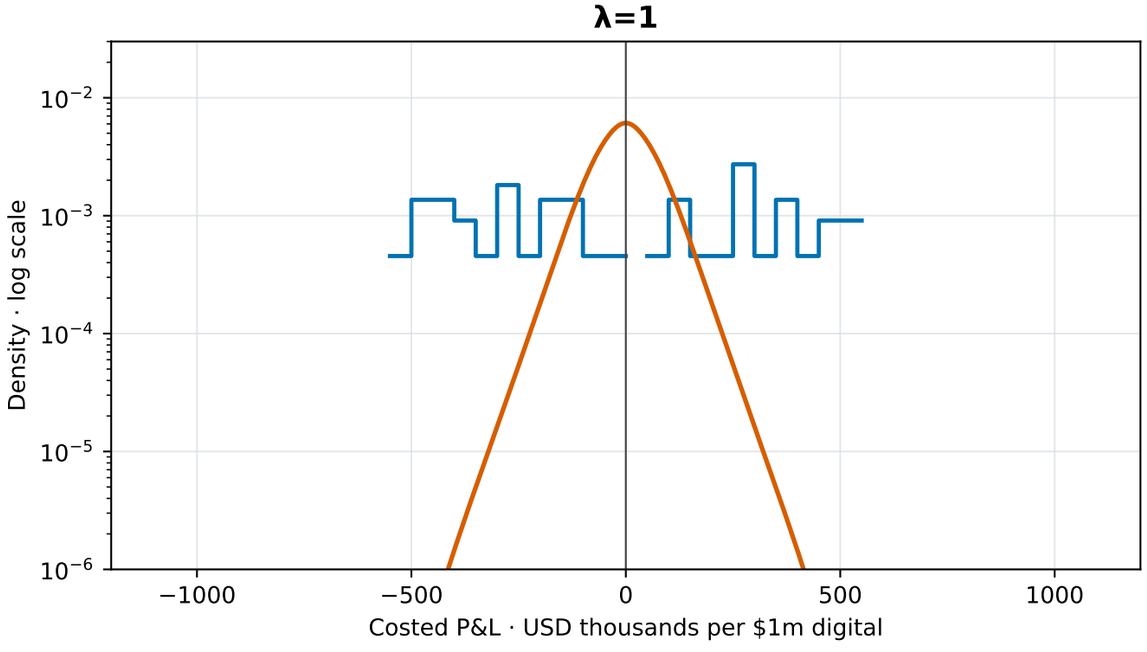


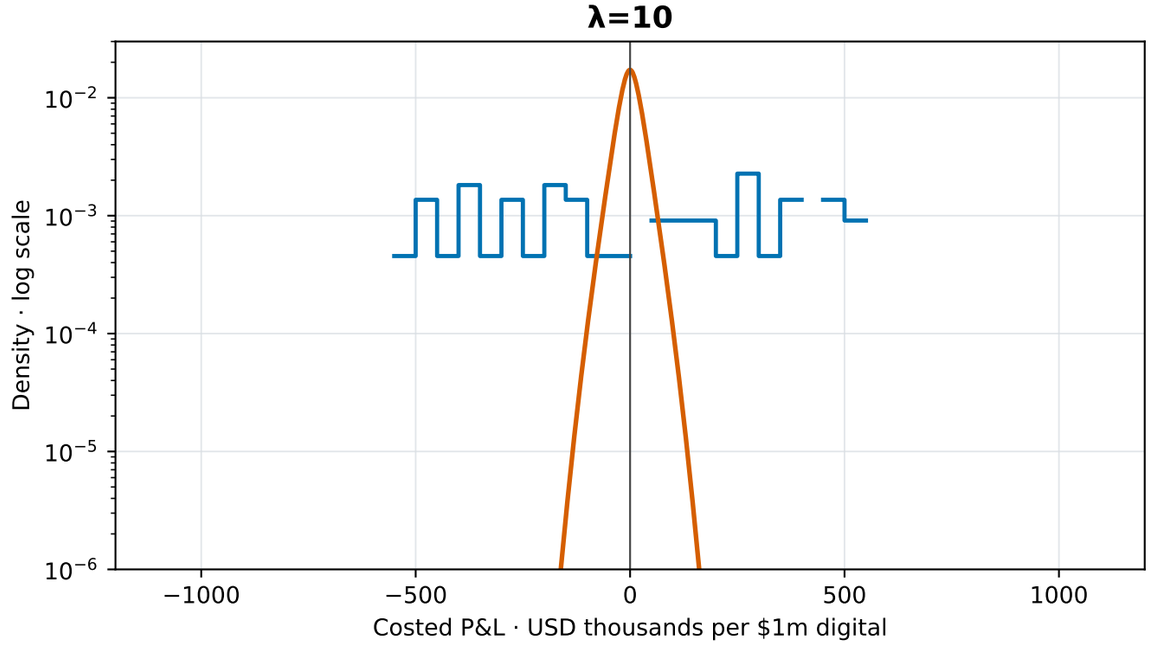


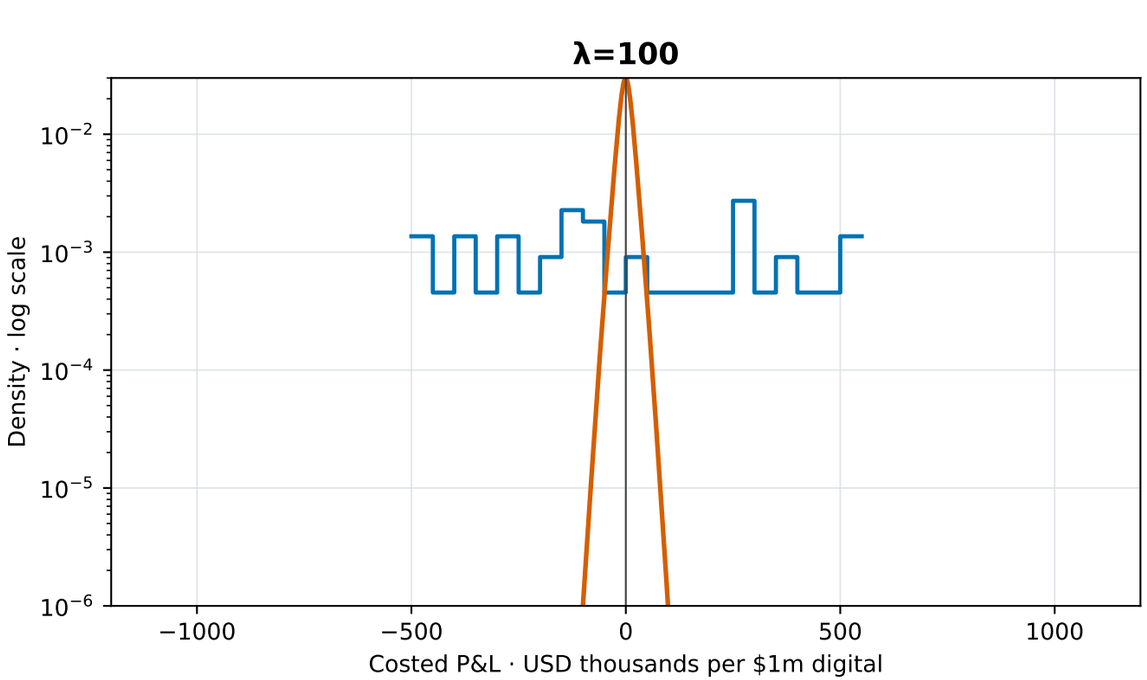
Realized, raw Fitted 252-day demeaned SANOS returns

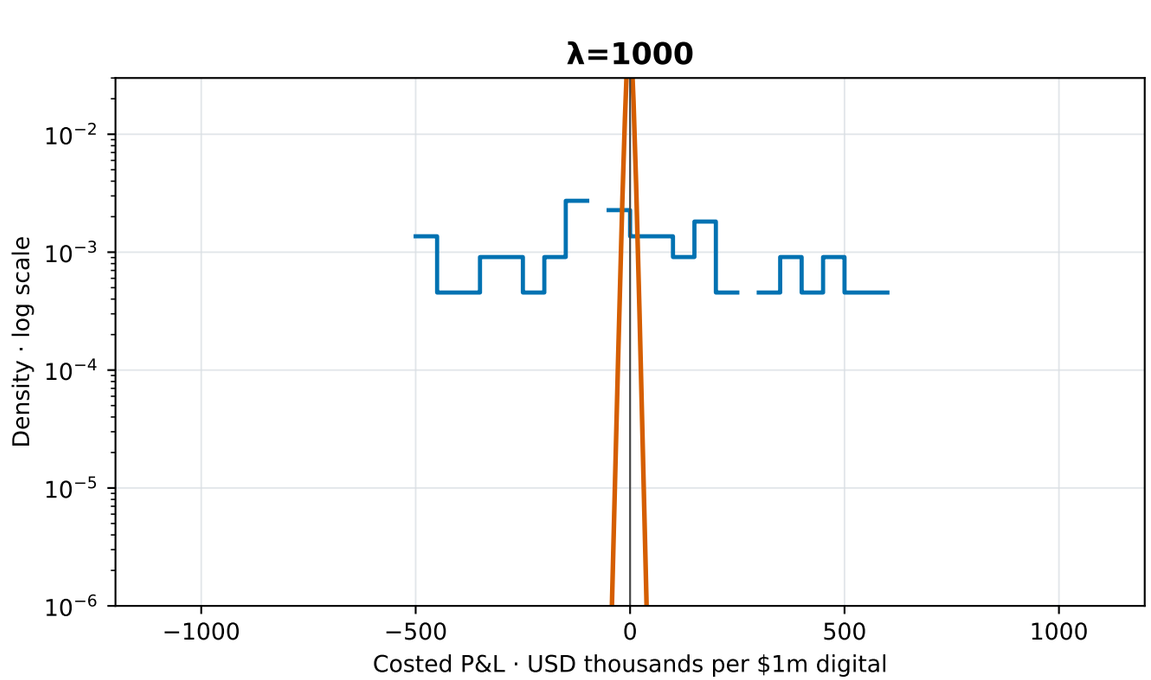
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.25: SANOS parametric · Entropy





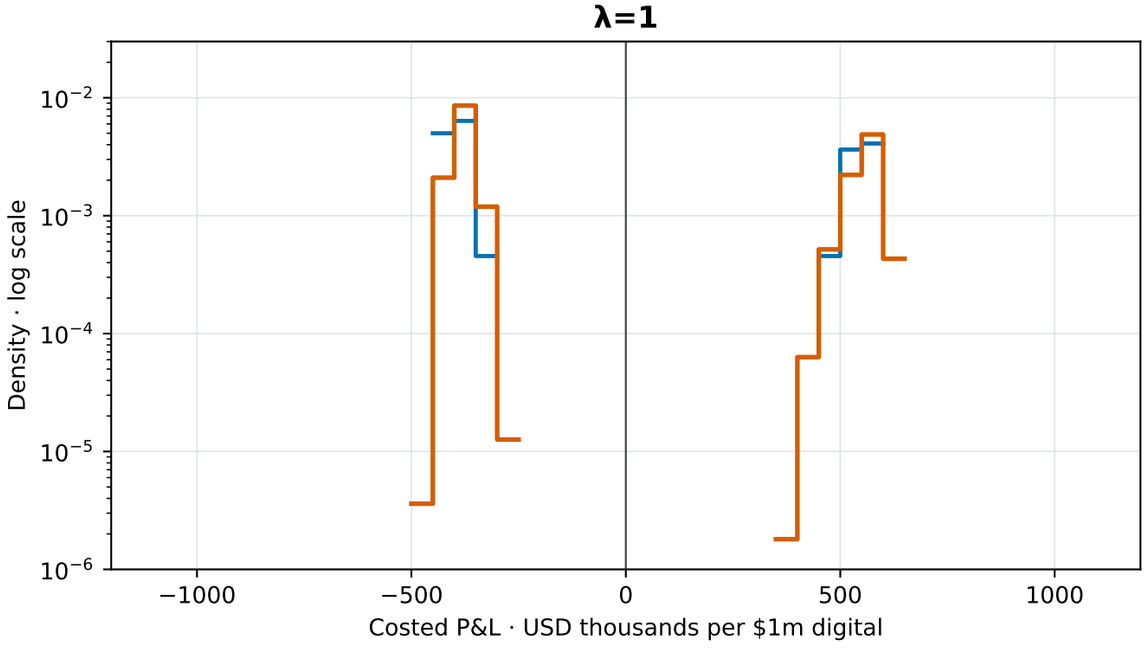


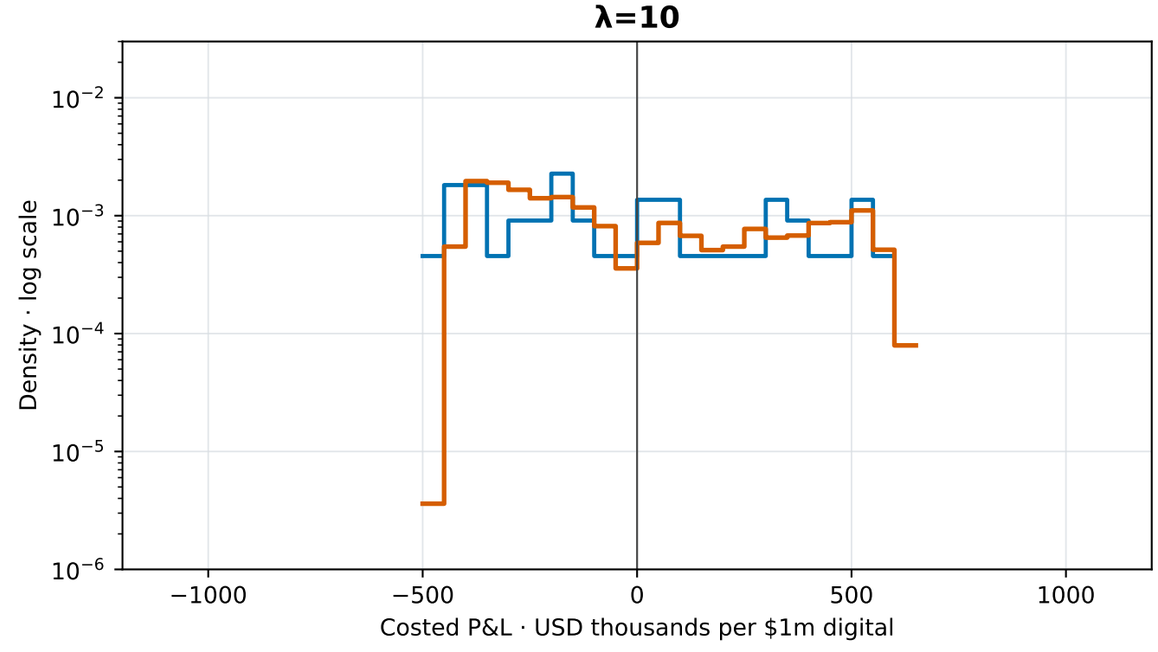


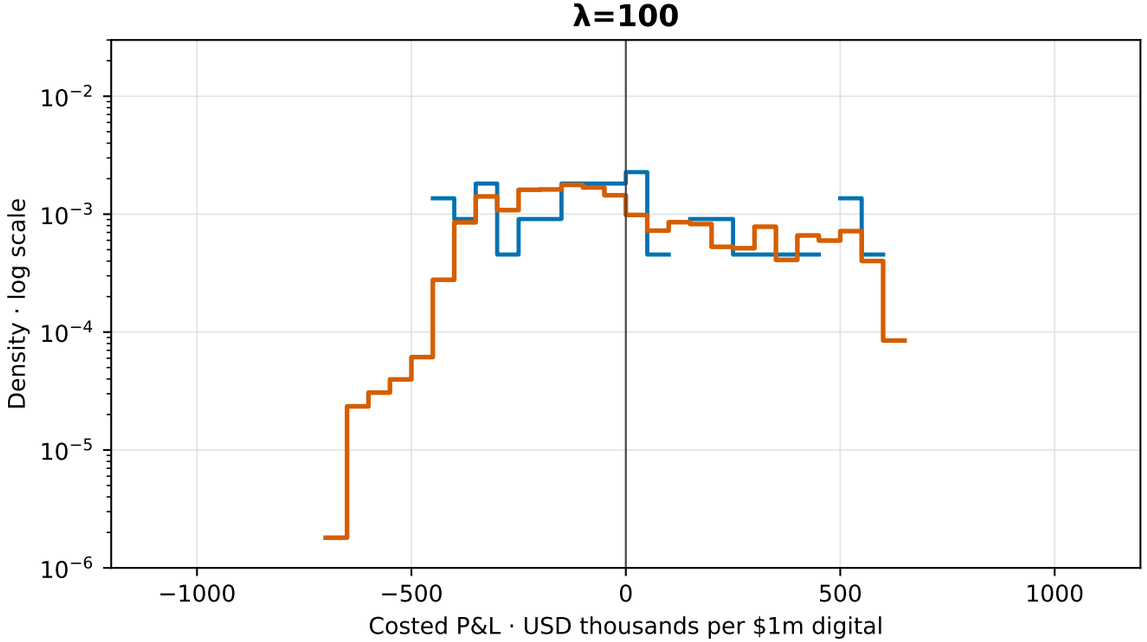
Realized, raw Fitted centered factor Gaussian

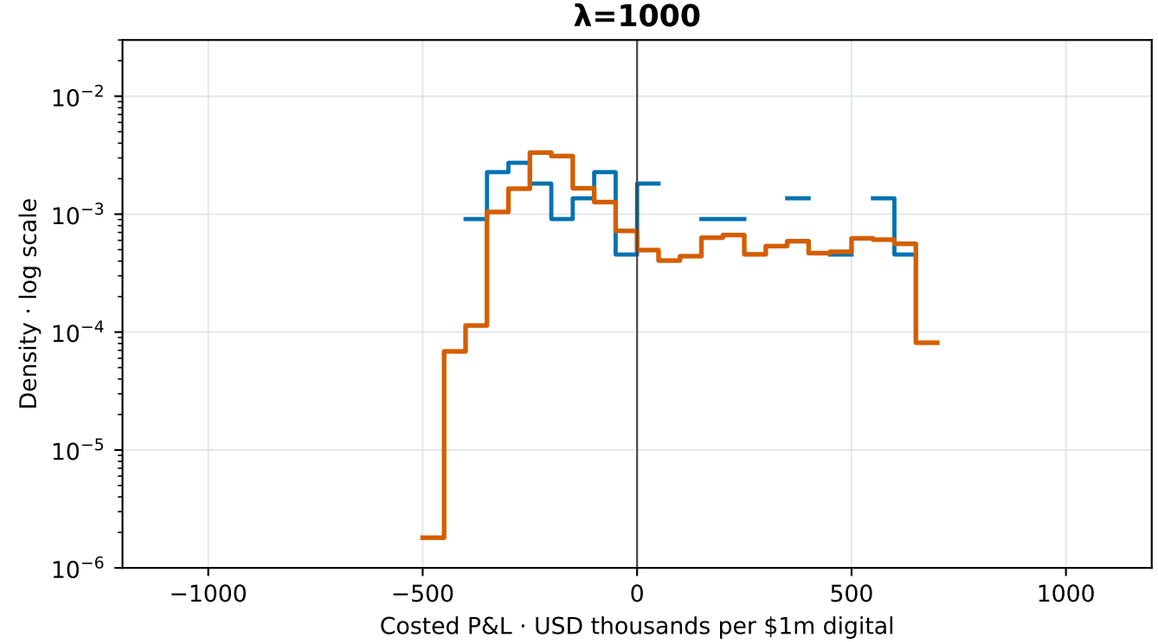
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.25: SANOS historical · Power-exponential





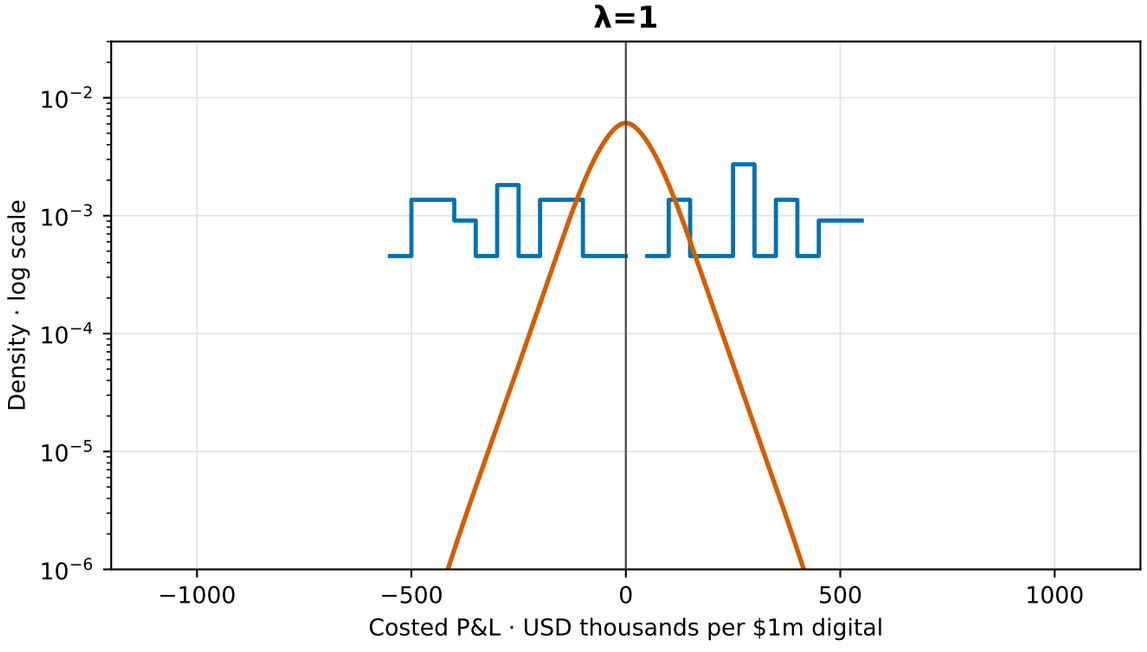


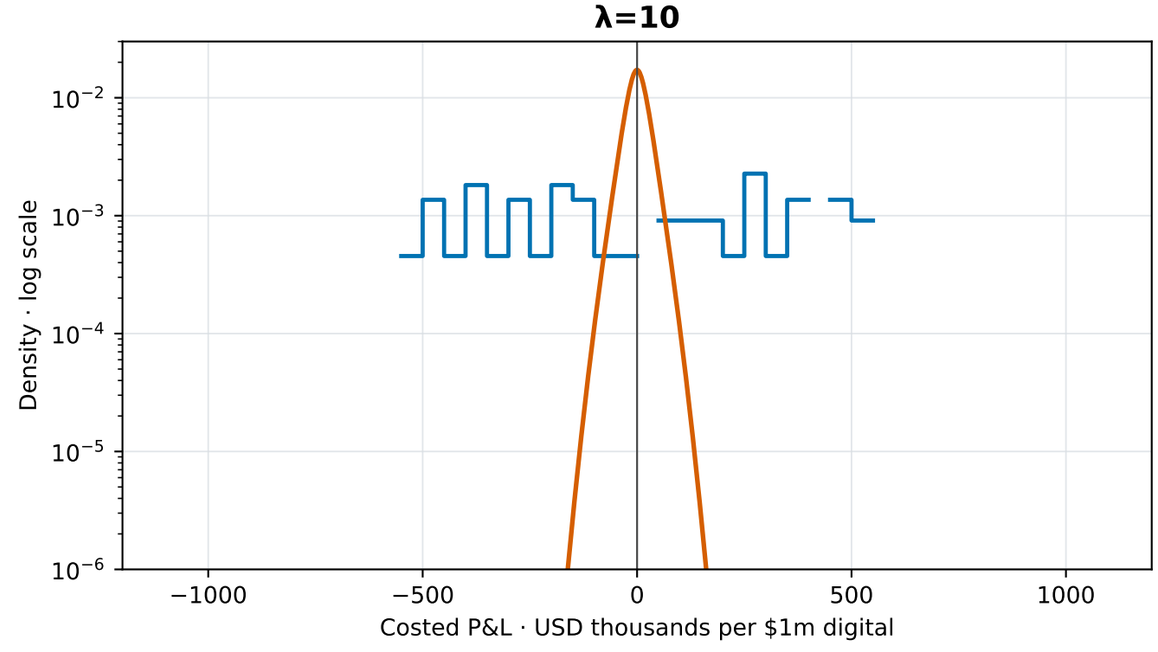


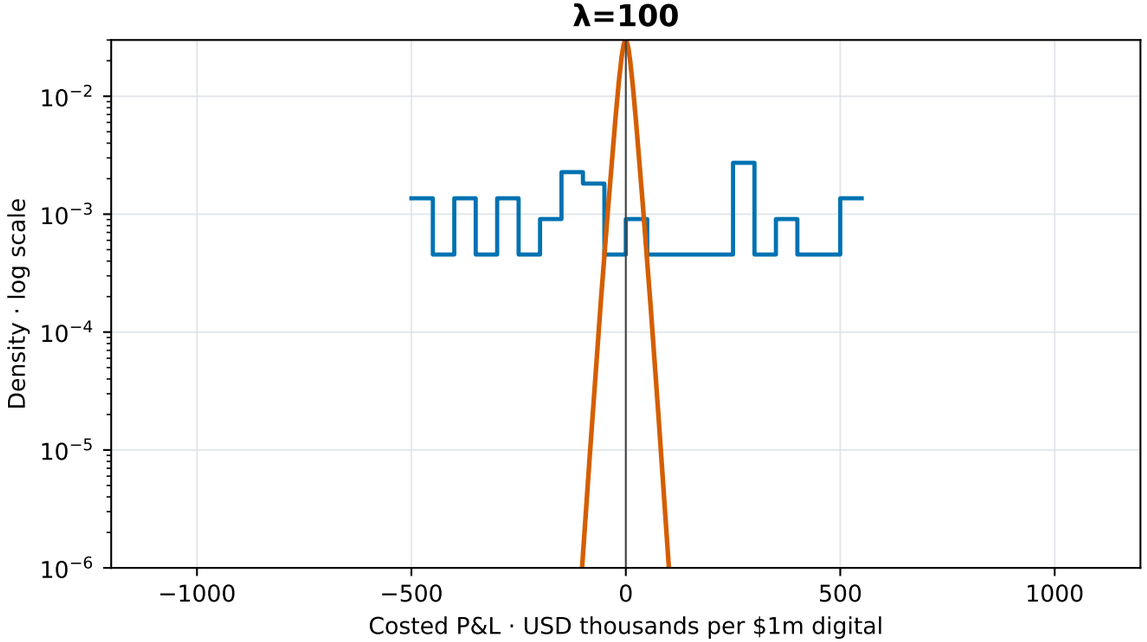
Realized, raw Fitted 252-day demeaned SANOS returns

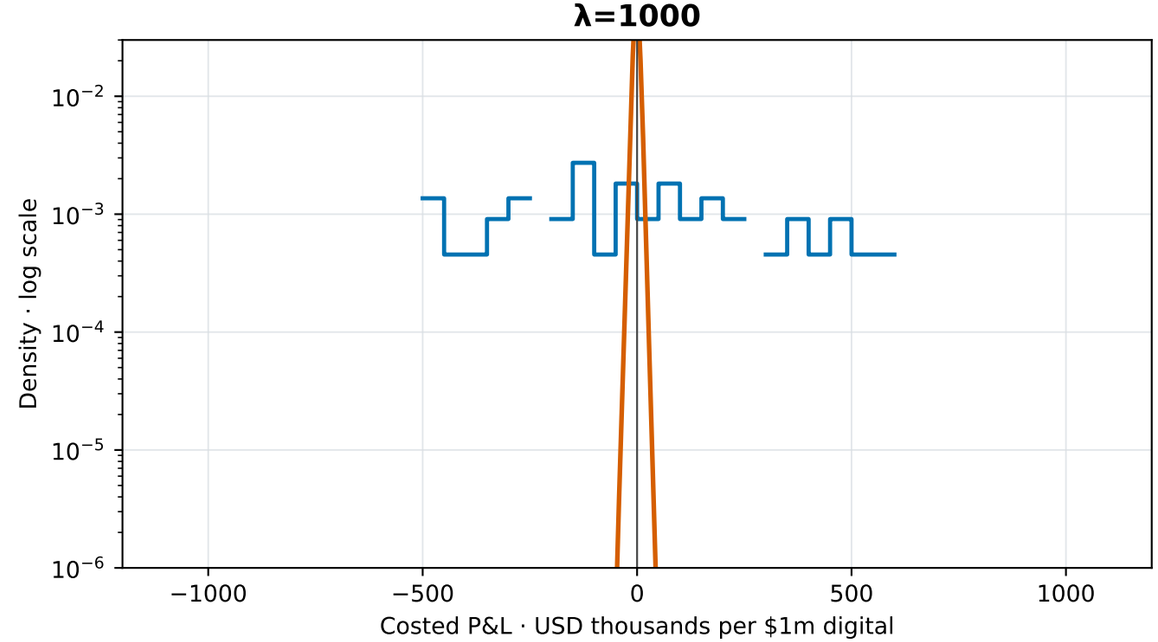
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.25: SANOS parametric · Power-exponential



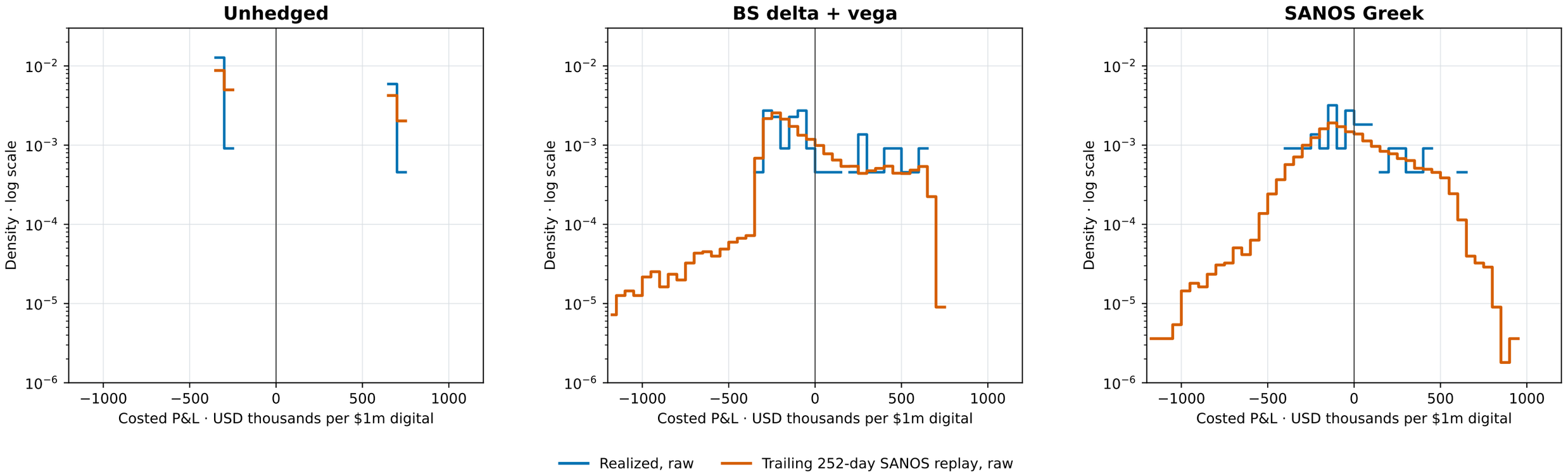




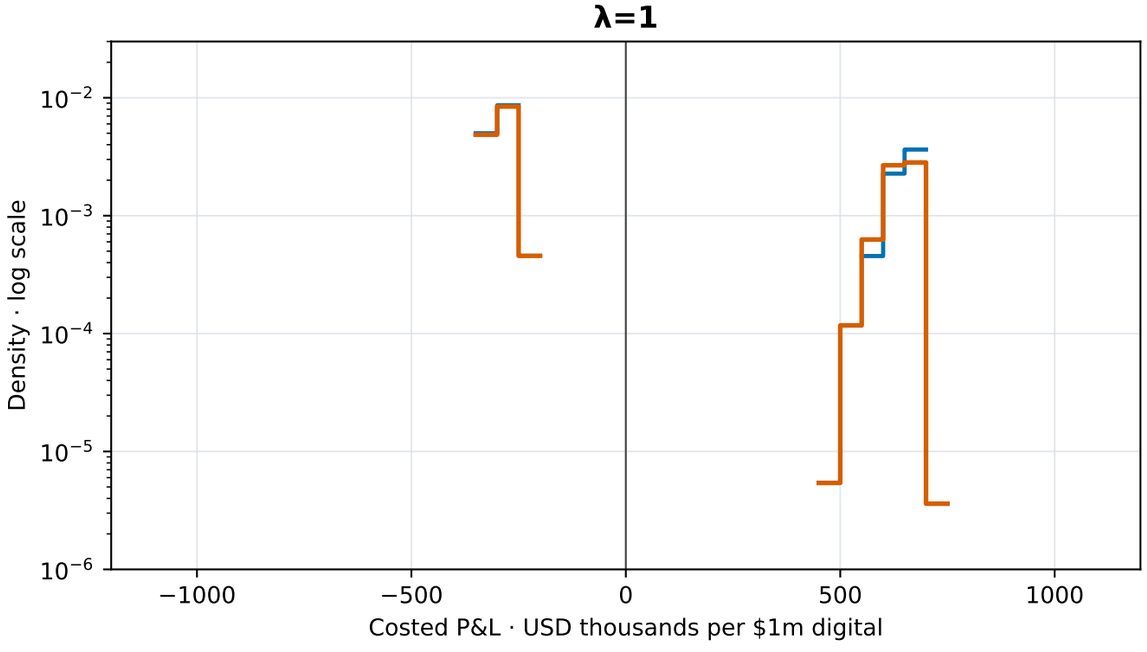


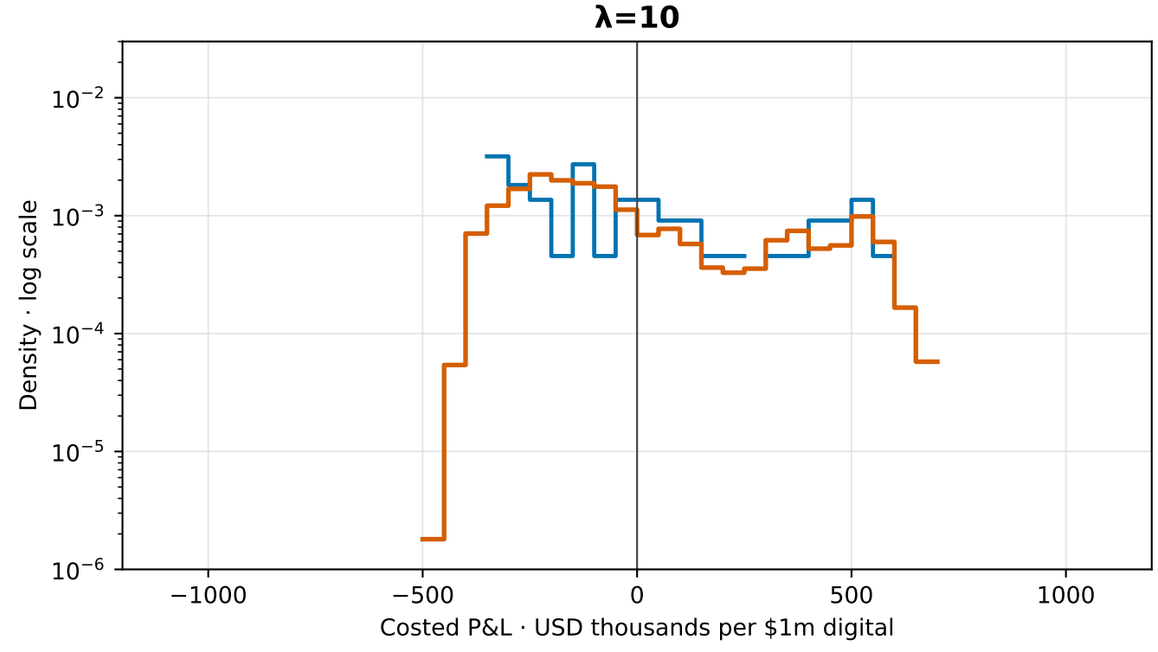
Realized, raw Fitted centered factor Gaussian

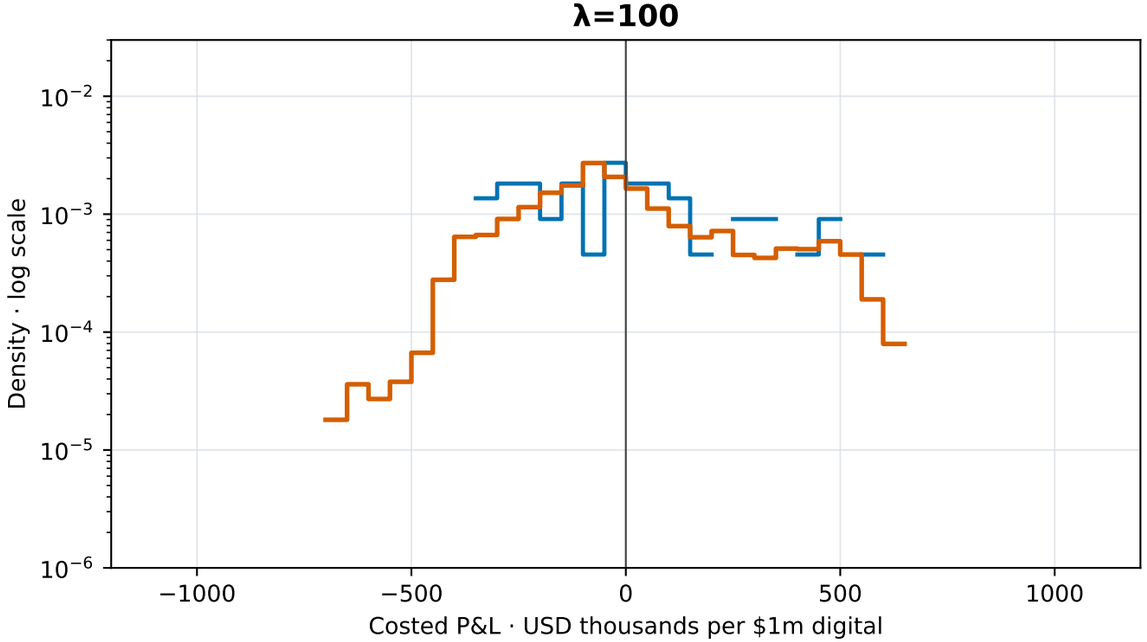
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

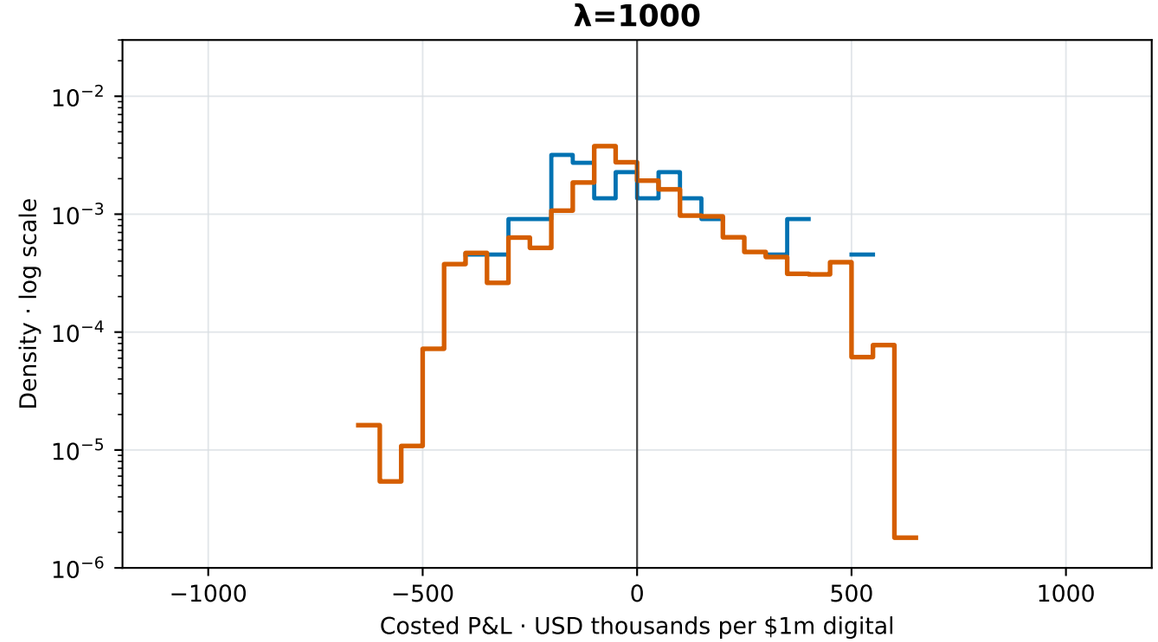

digital-call-1d-nms+0.50: fixed-reference hedge distributions
Unhedged
BS delta + vega
SANOS Greek
Density · log scale
Costed P&L · USD thousands per $1m digital
Realized, raw
Trailing 252-day SANOS replay, raw
No mean correction: 44 realized next-day outcomes versus preceding 252 raw returns. References have no fitted hedge-return law.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.50: SANOS historical · Mean-variance





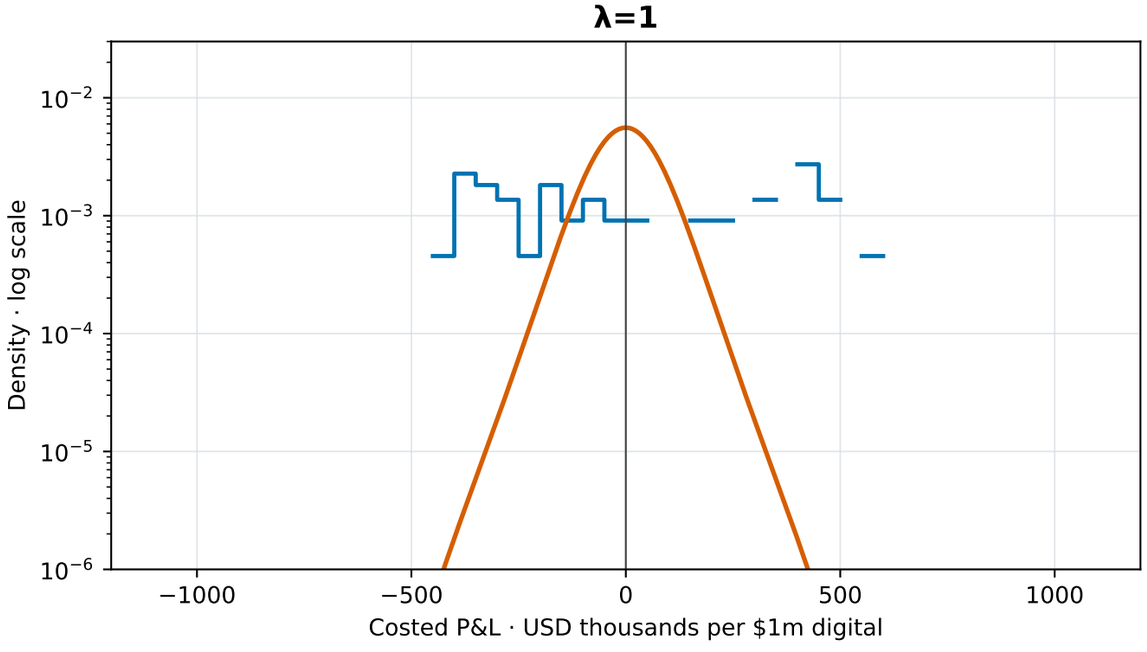


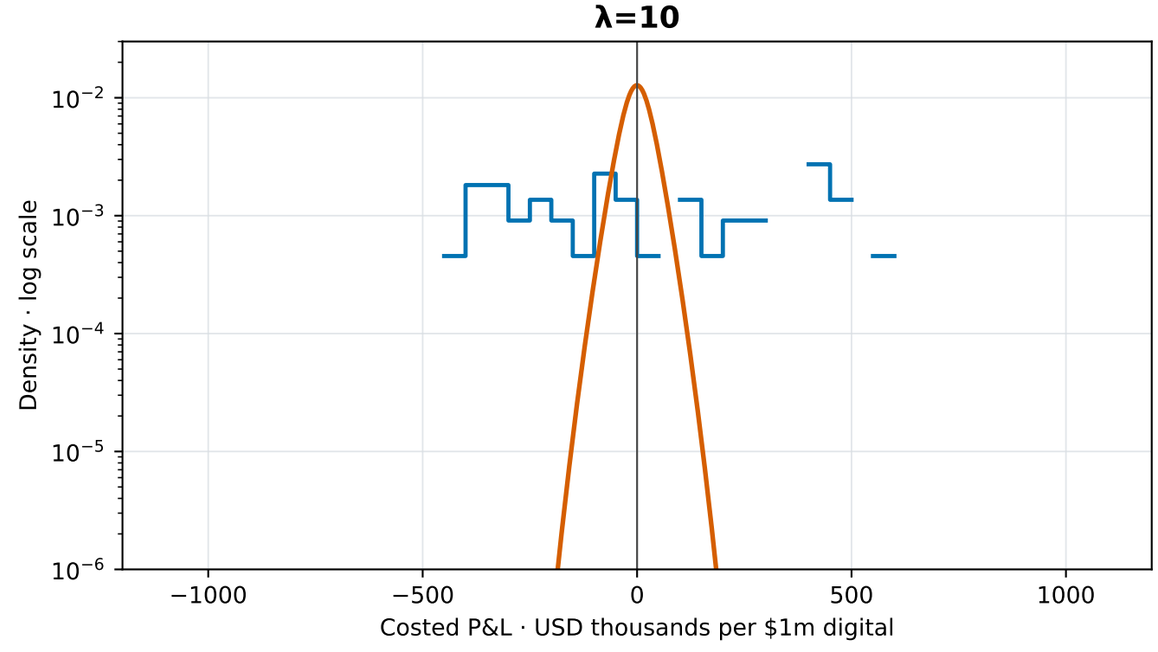


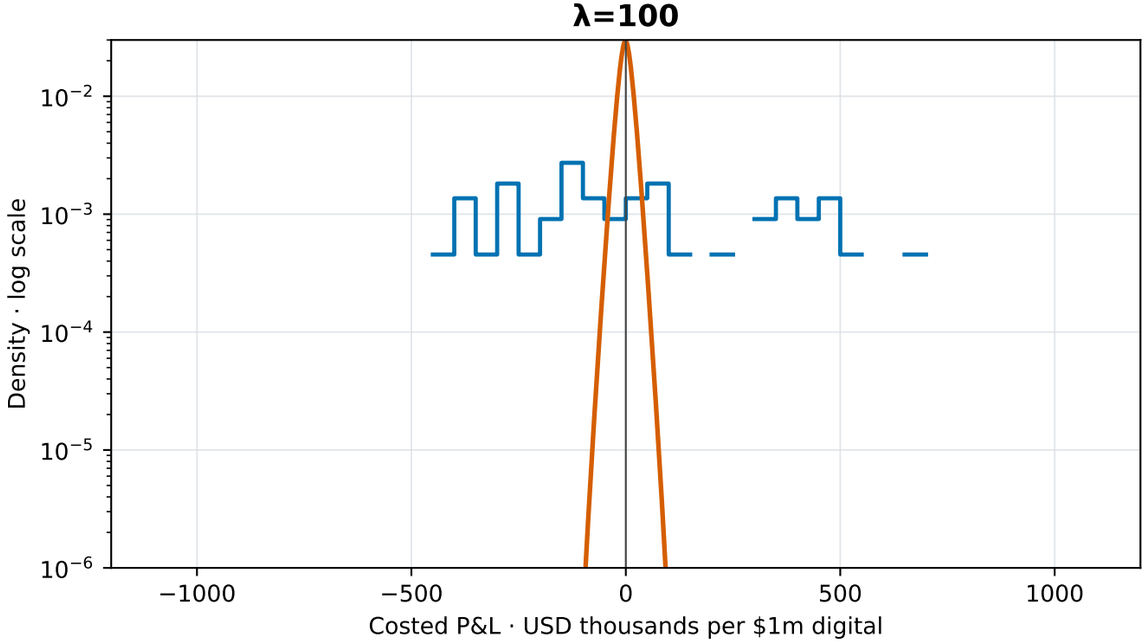
Realized, raw Fitted 252-day demeaned SANOS returns

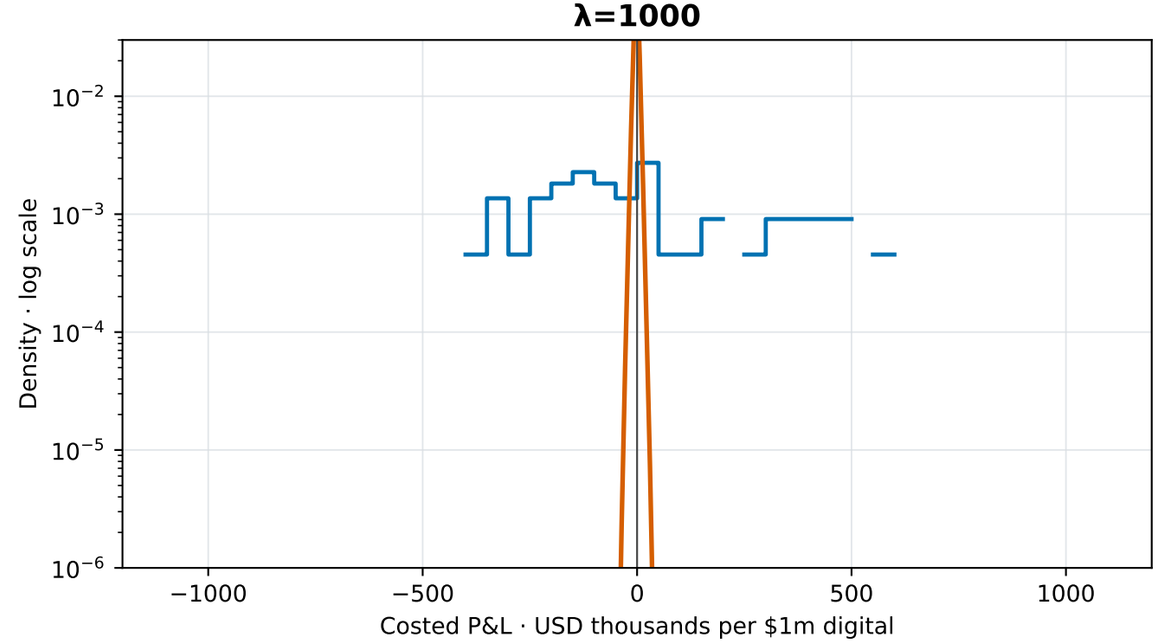
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.50: SANOS parametric · Mean-variance





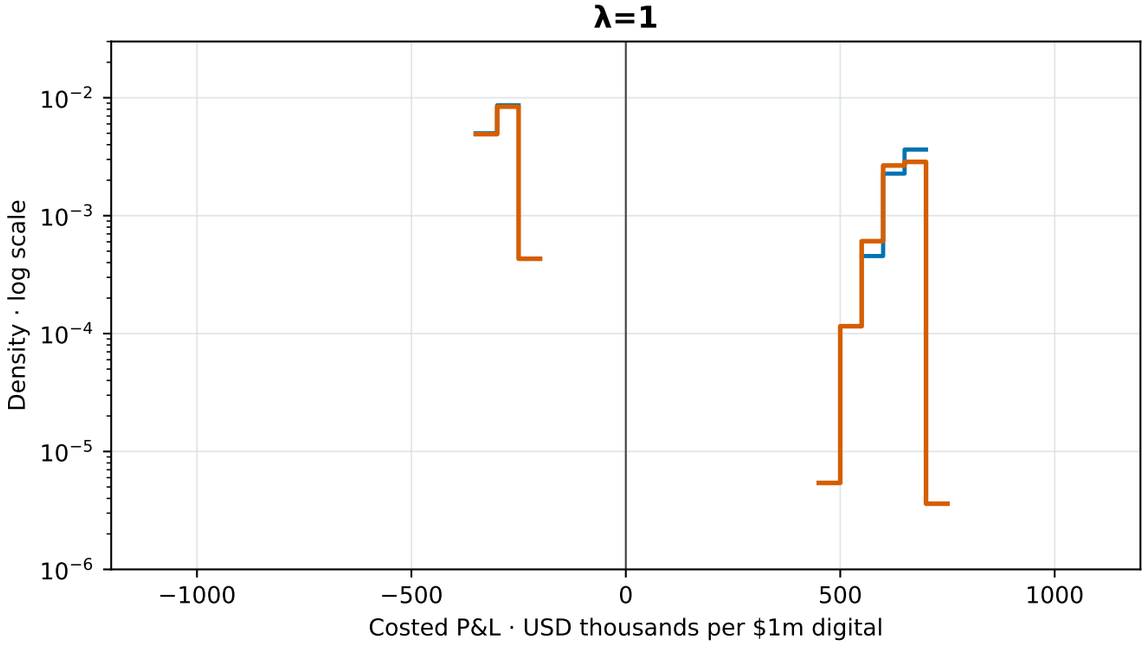


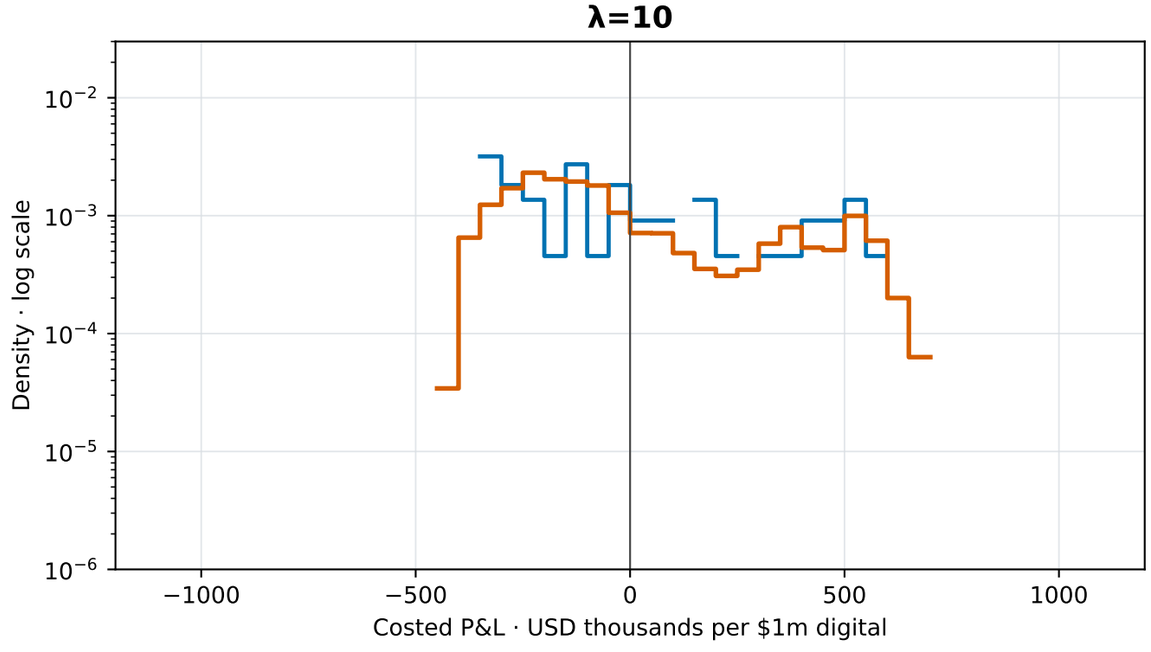


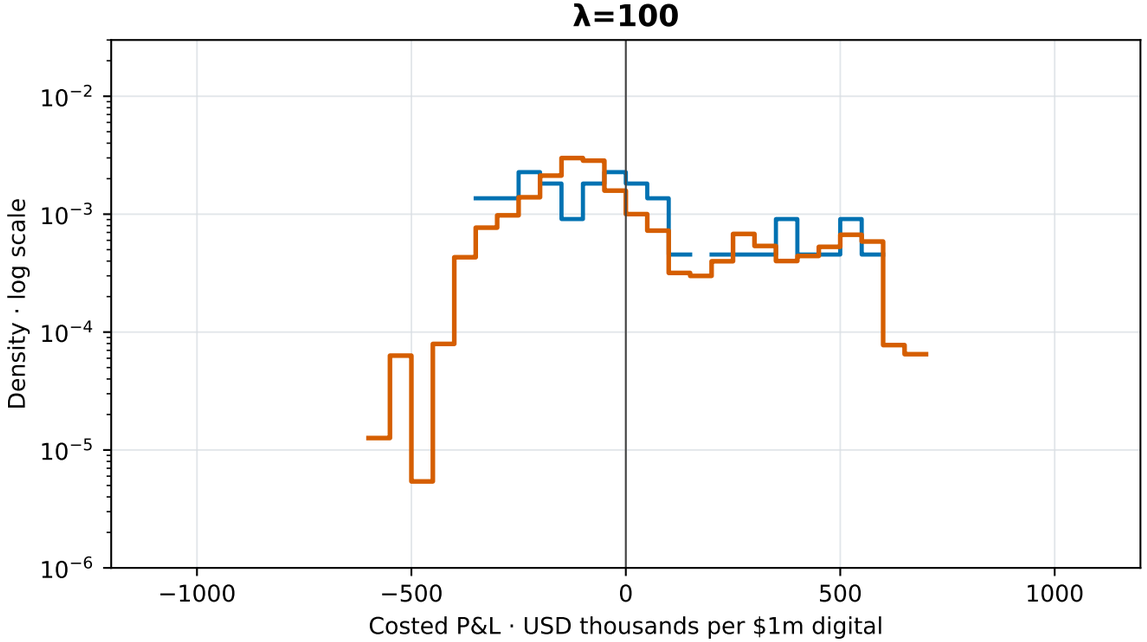
Realized, raw Fitted centered factor Gaussian

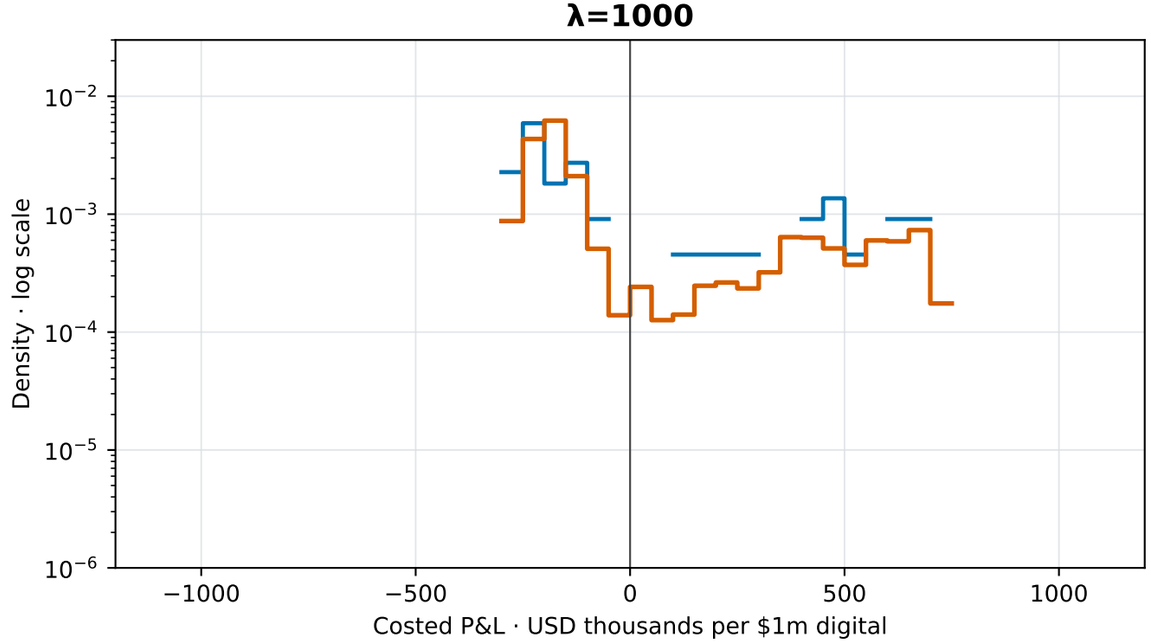
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.50: SANOS historical · Entropy





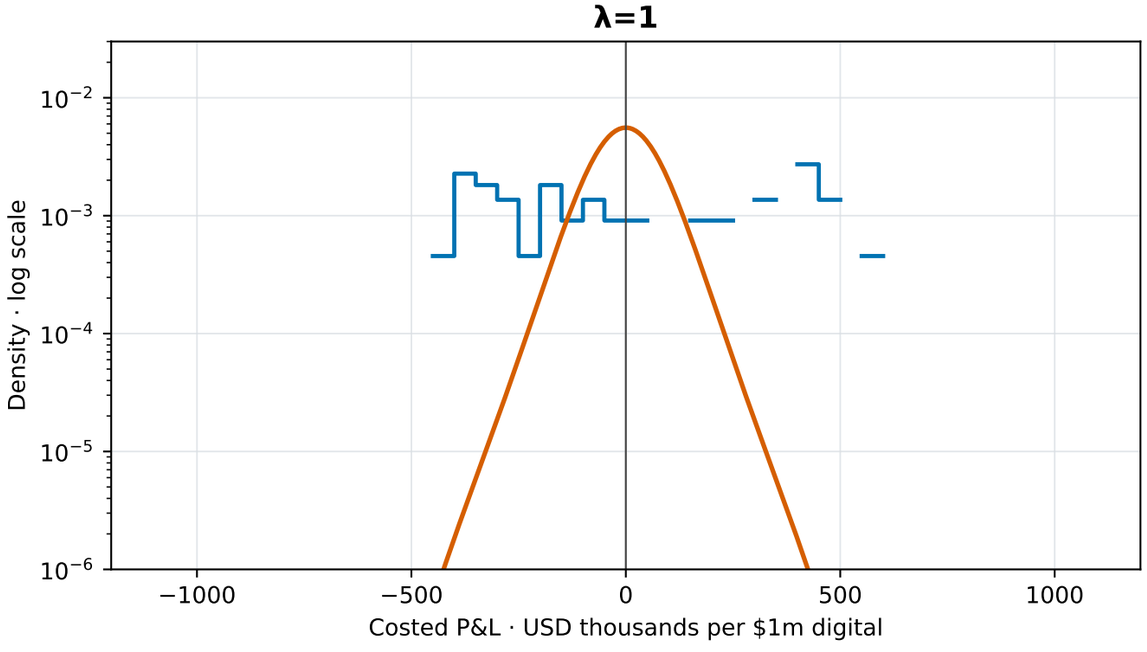


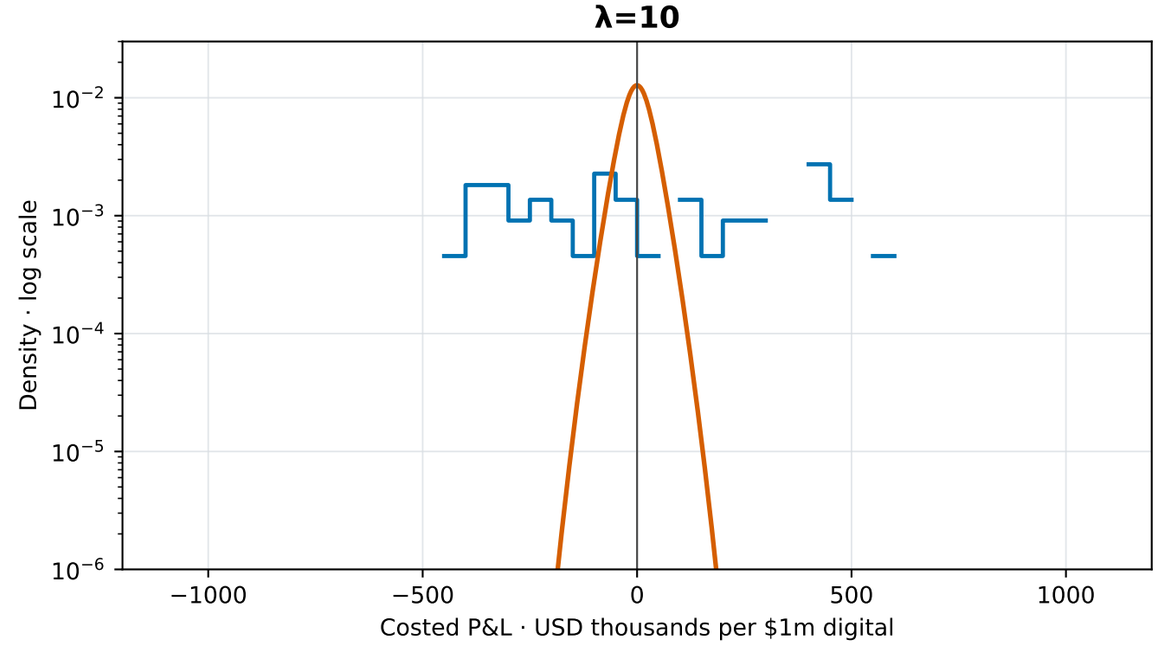


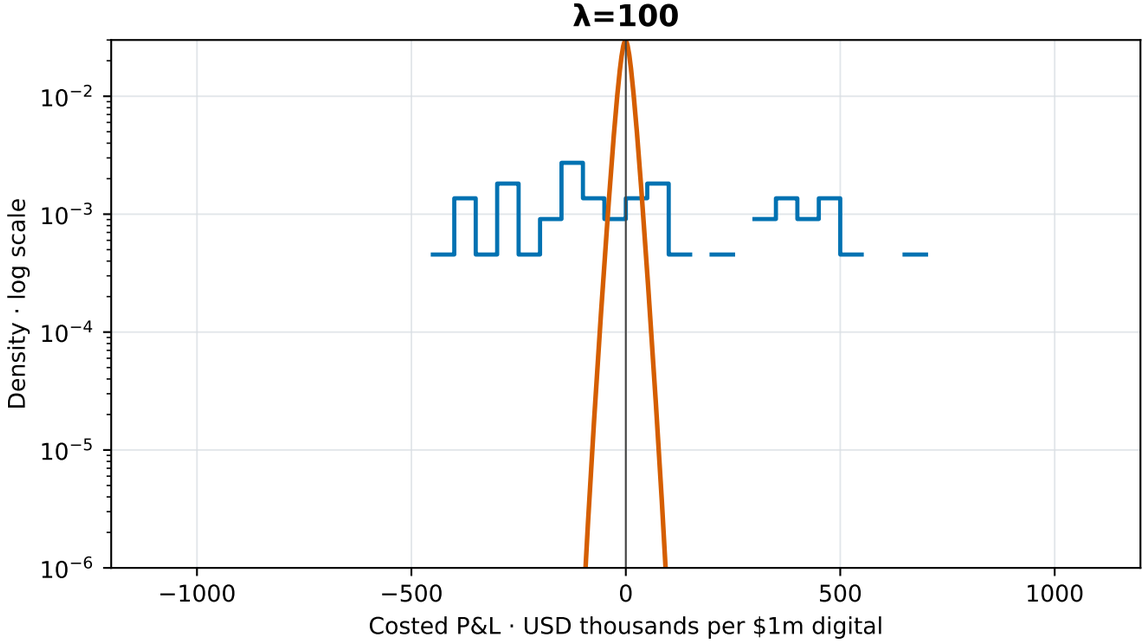
Realized, raw Fitted 252-day demeaned SANOS returns

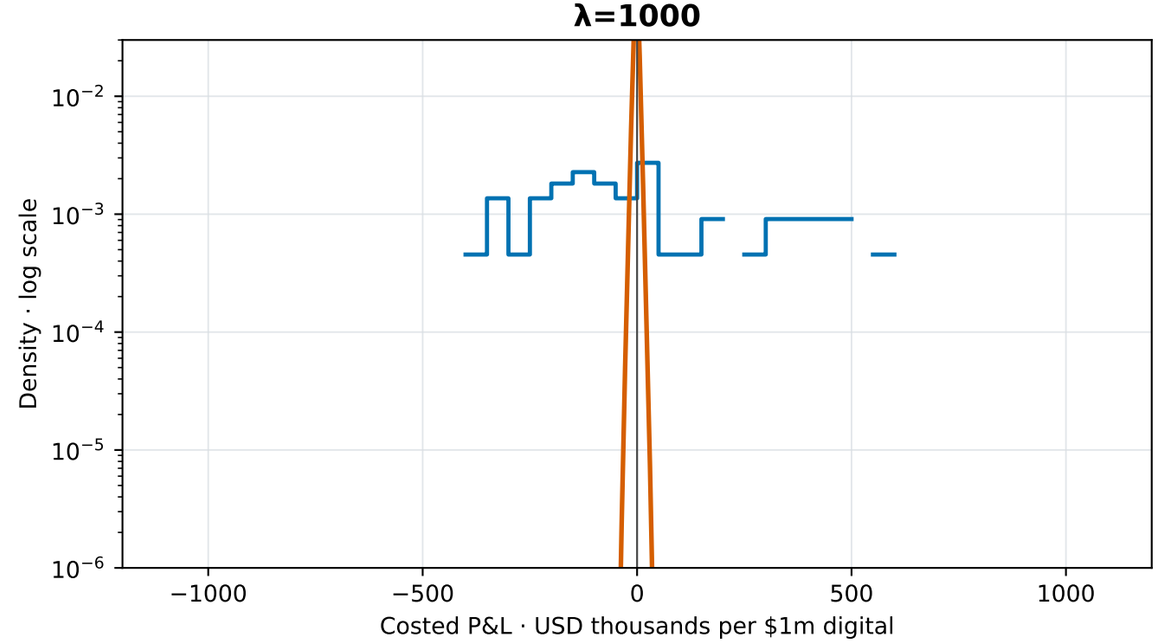
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.50: SANOS parametric · Entropy





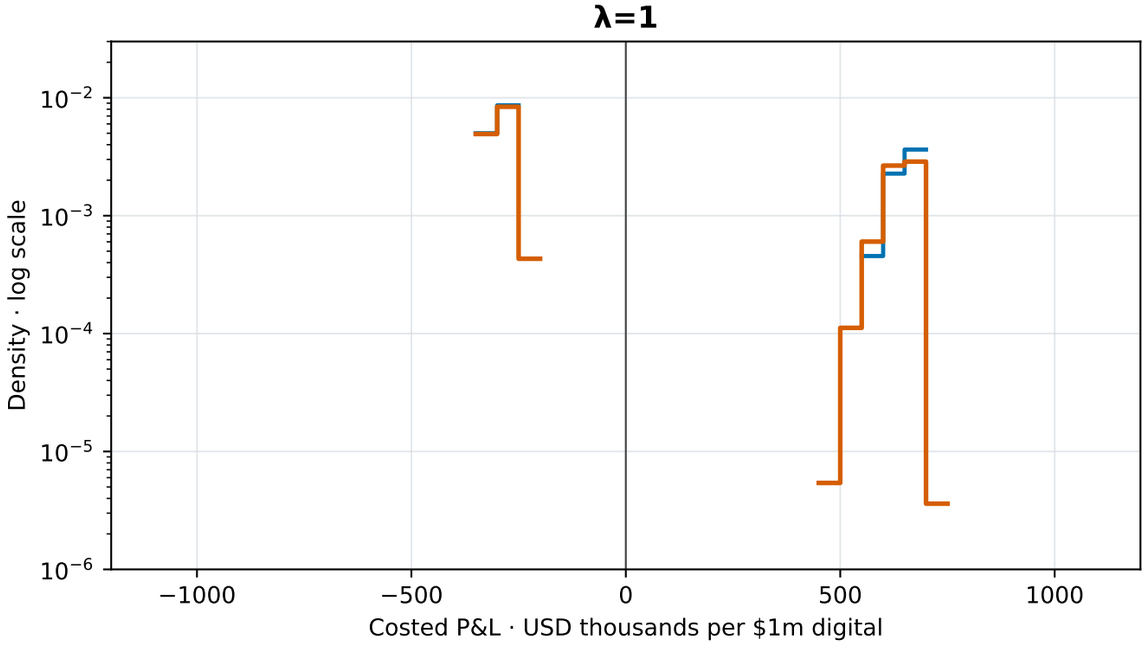


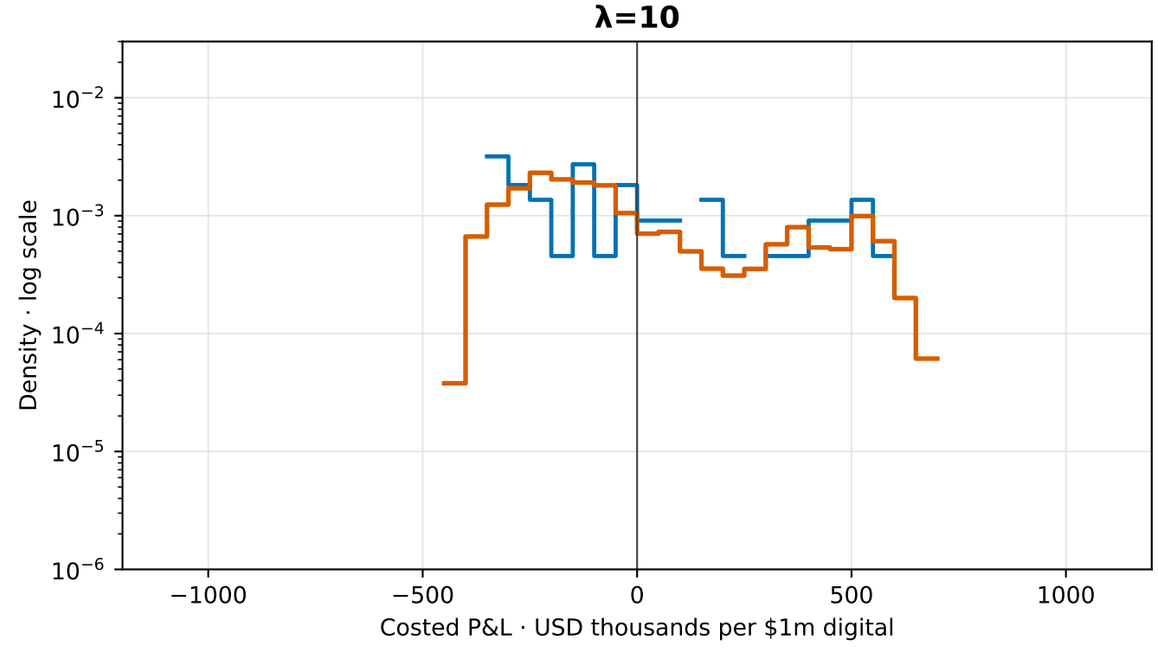


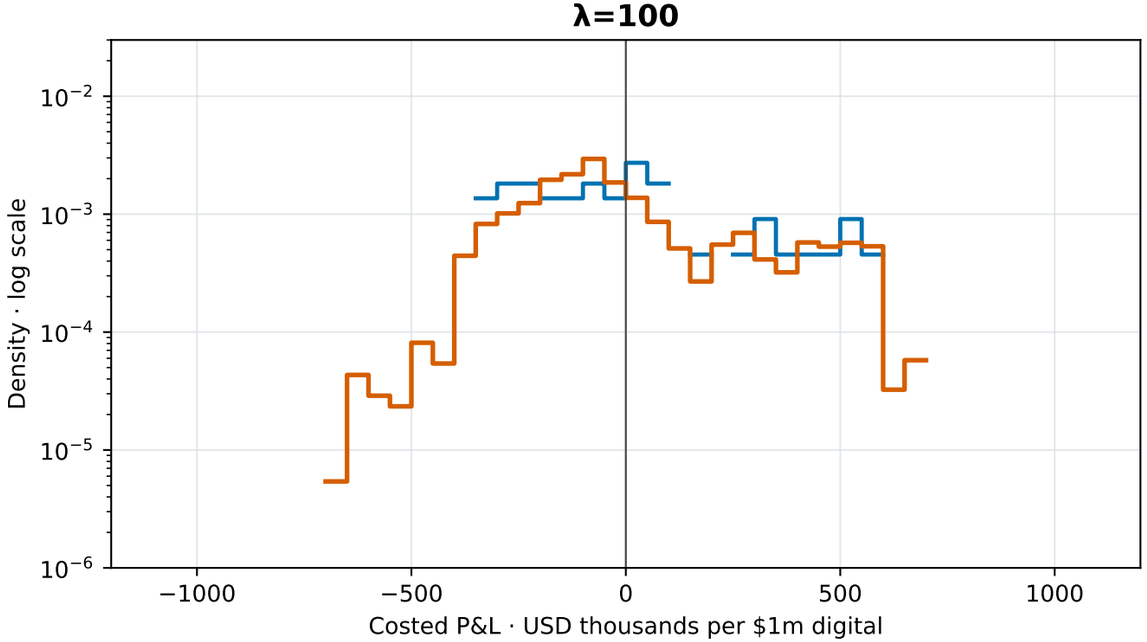
Realized, raw Fitted centered factor Gaussian

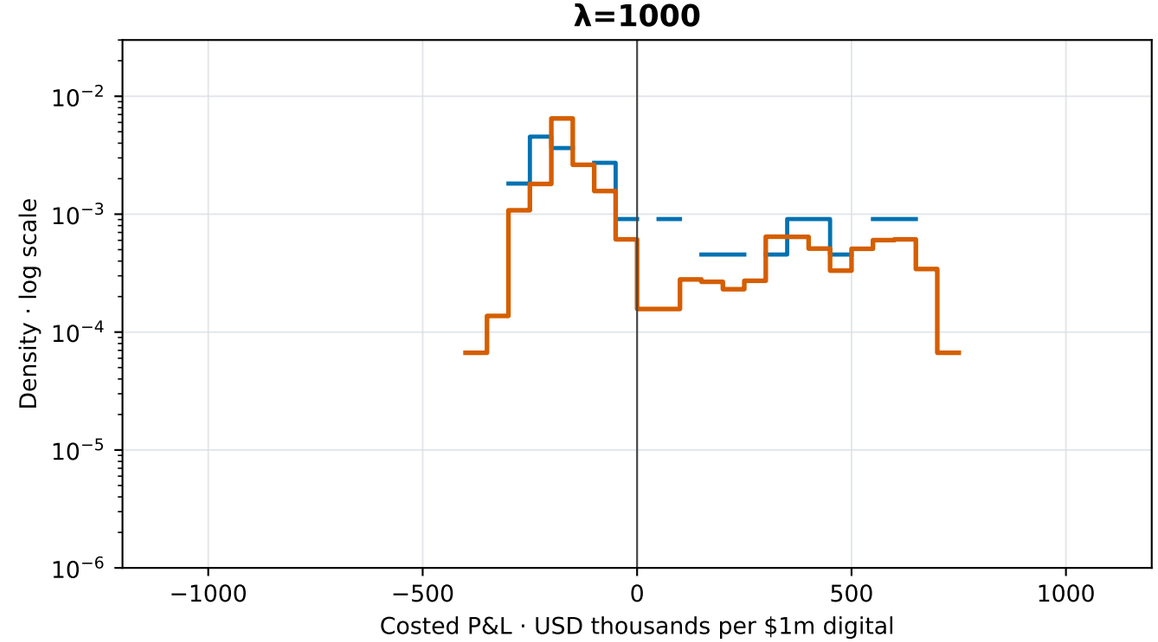
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.50: SANOS historical · Power-exponential





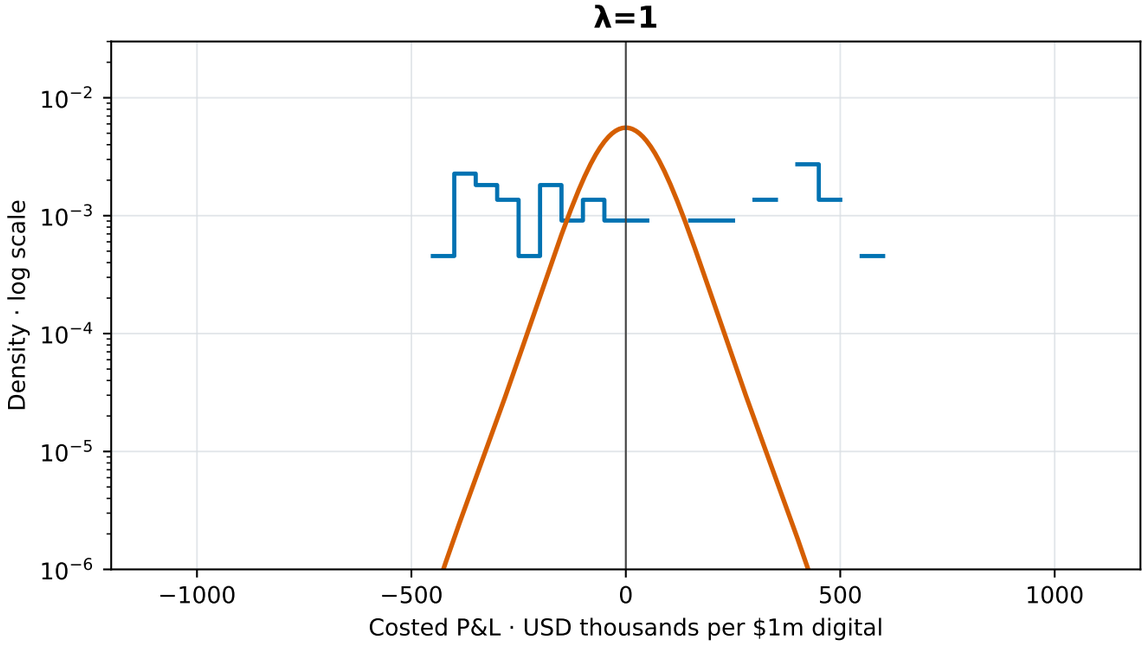


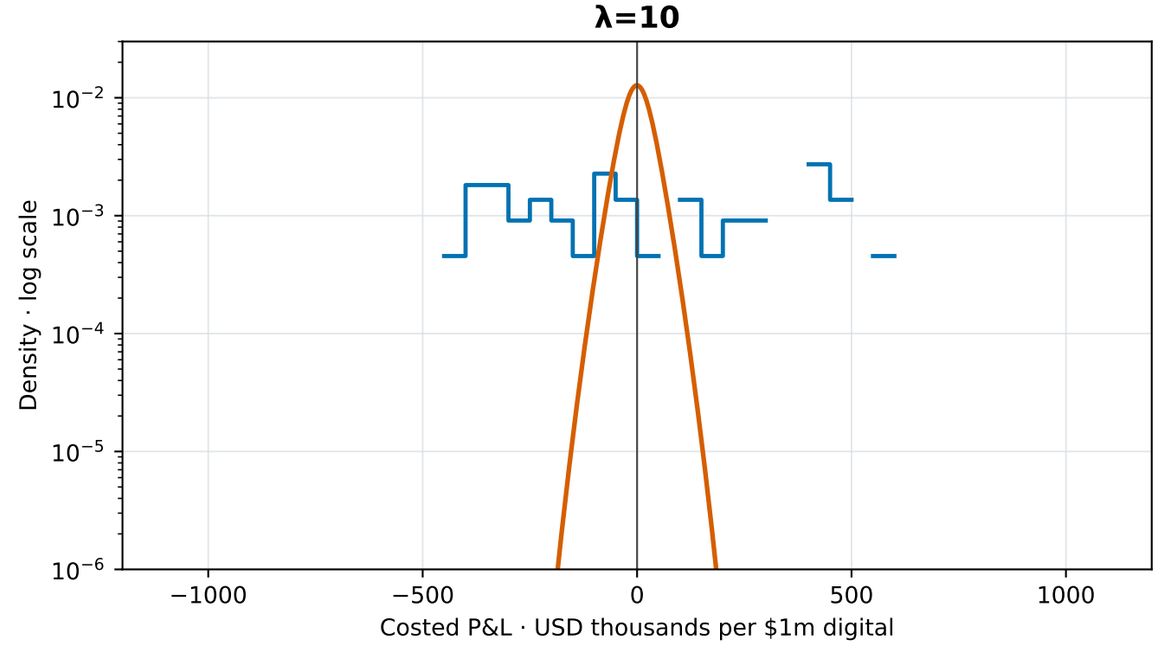


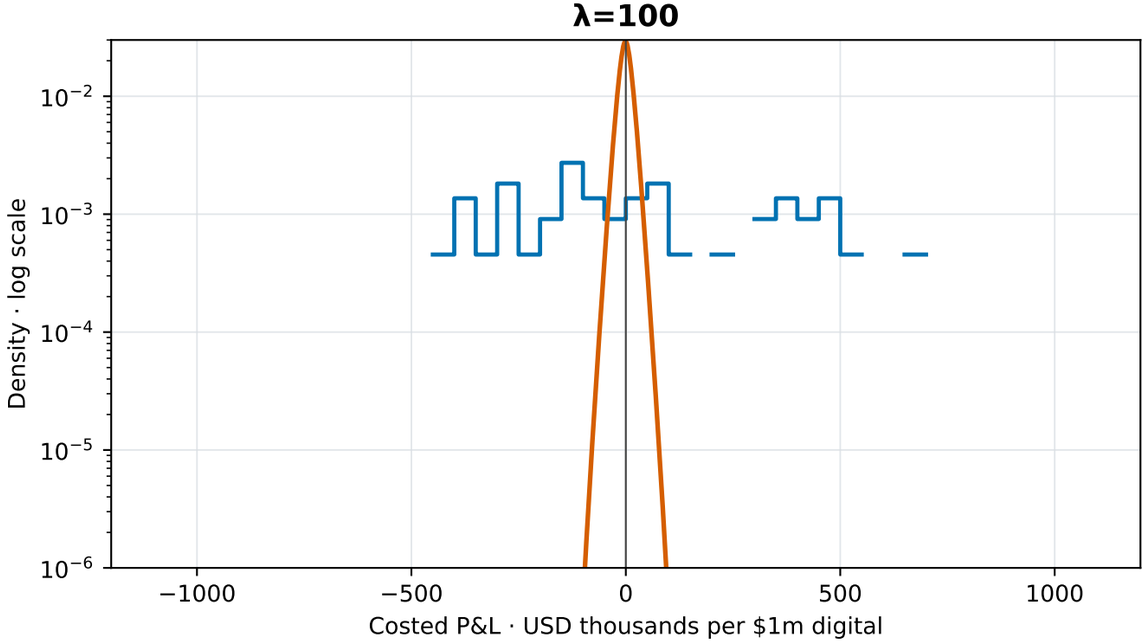
Realized, raw Fitted 252-day demeaned SANOS returns

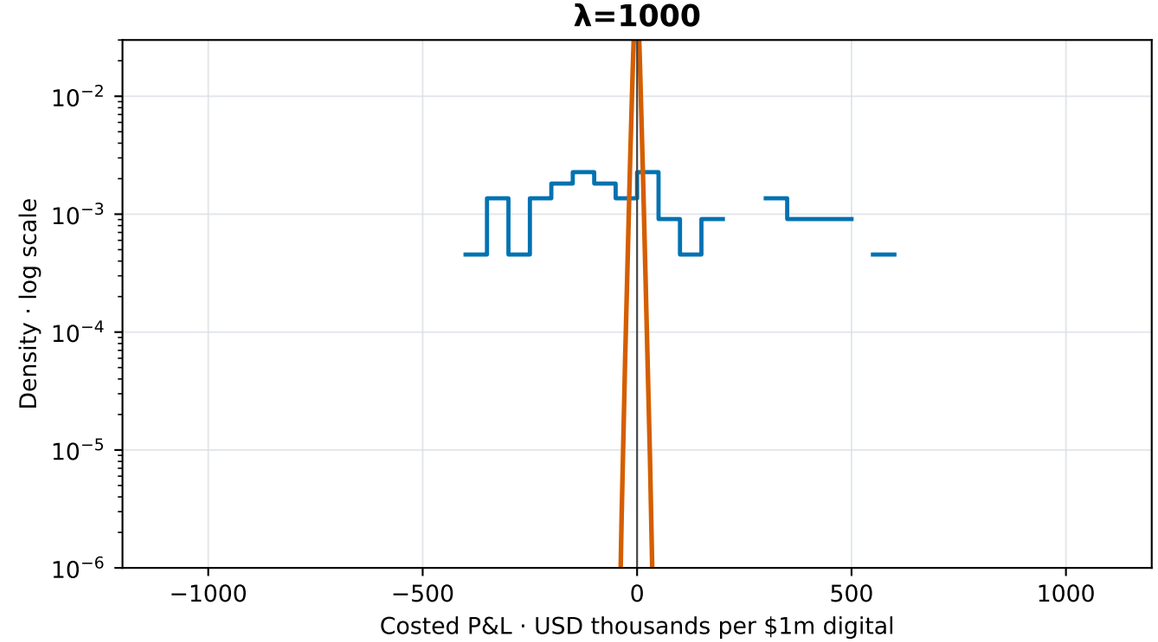
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.50: SANOS parametric · Power-exponential



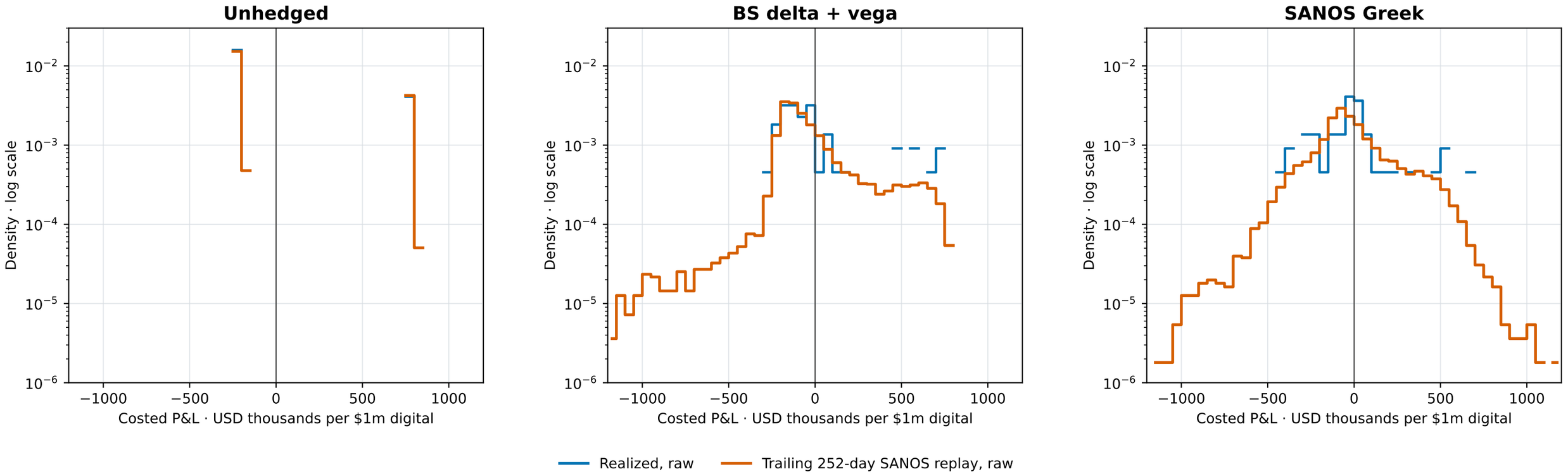




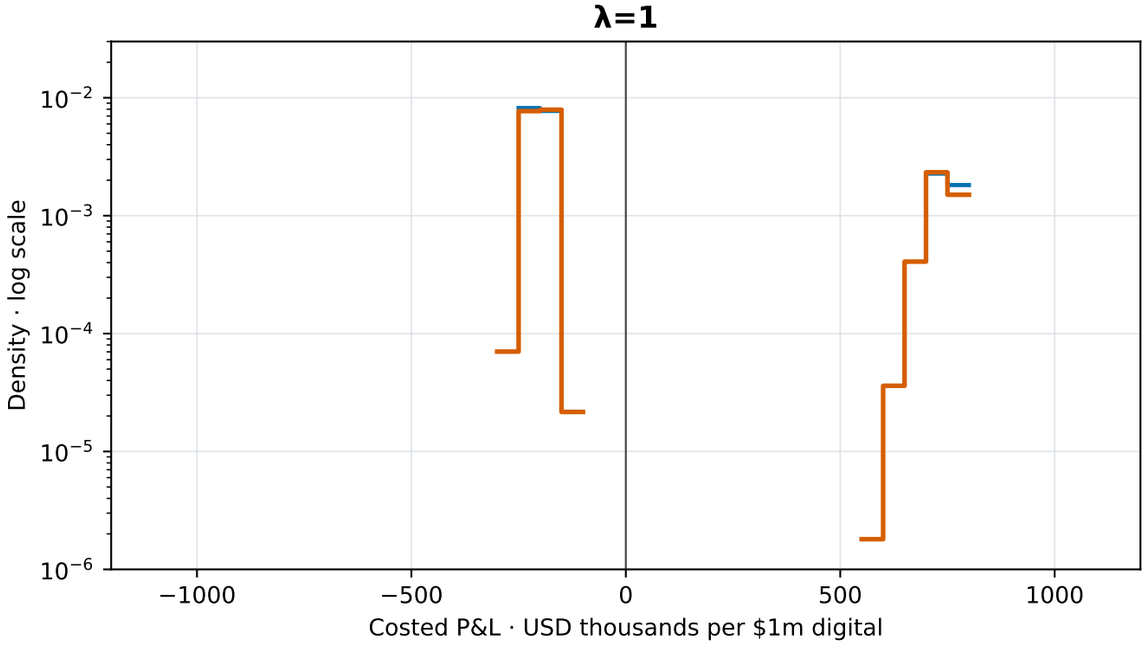


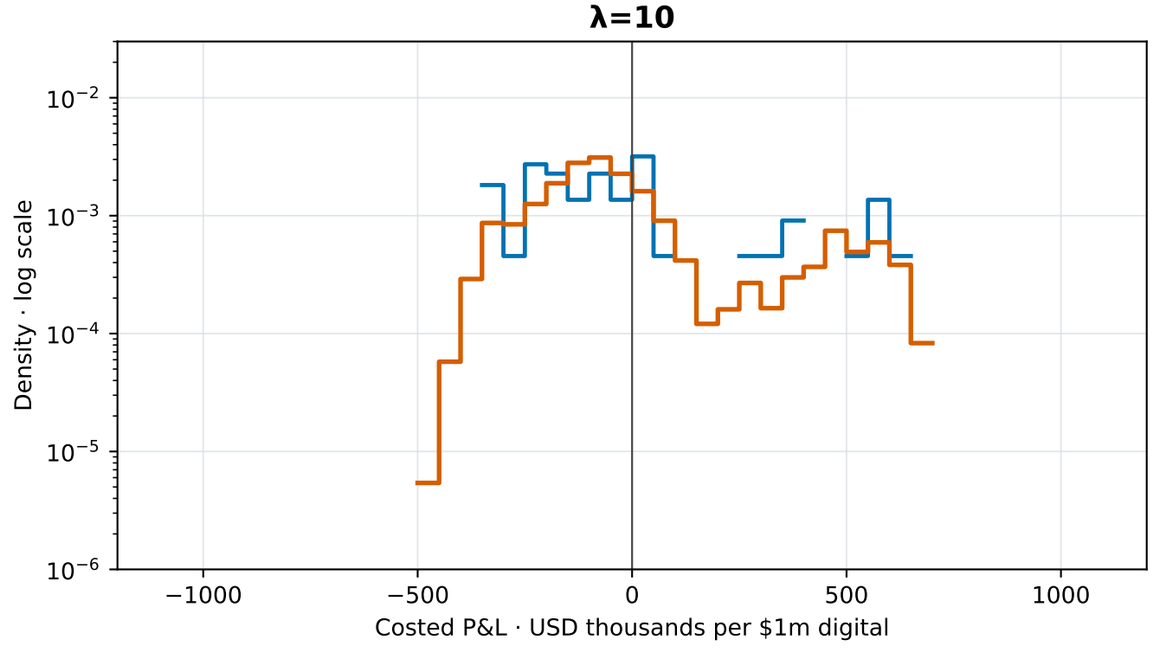
Realized, raw Fitted centered factor Gaussian

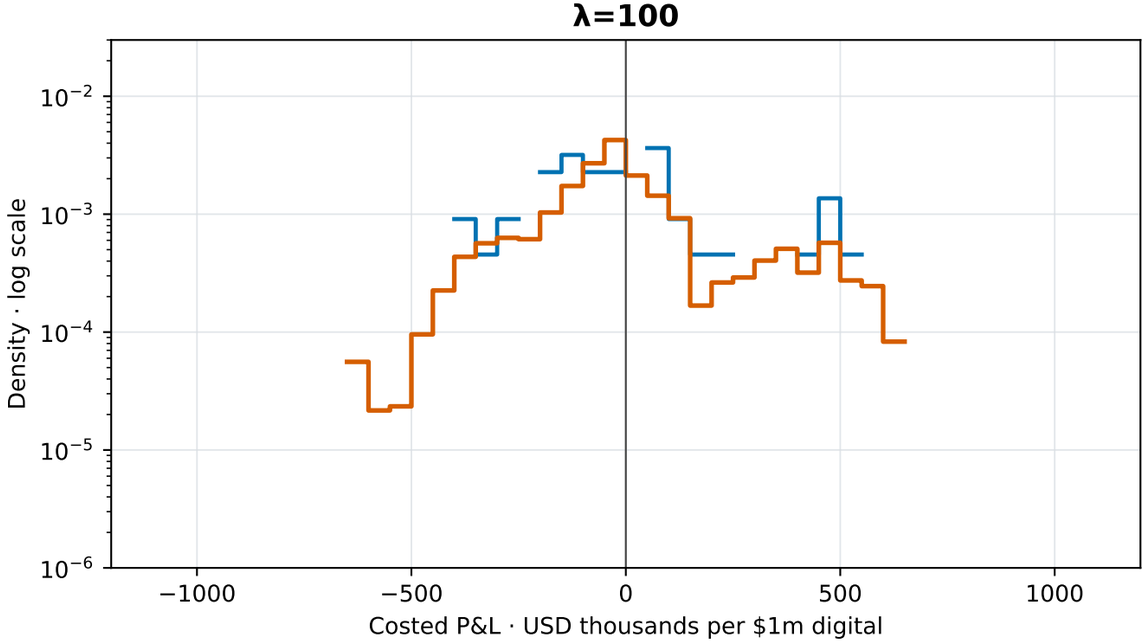
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

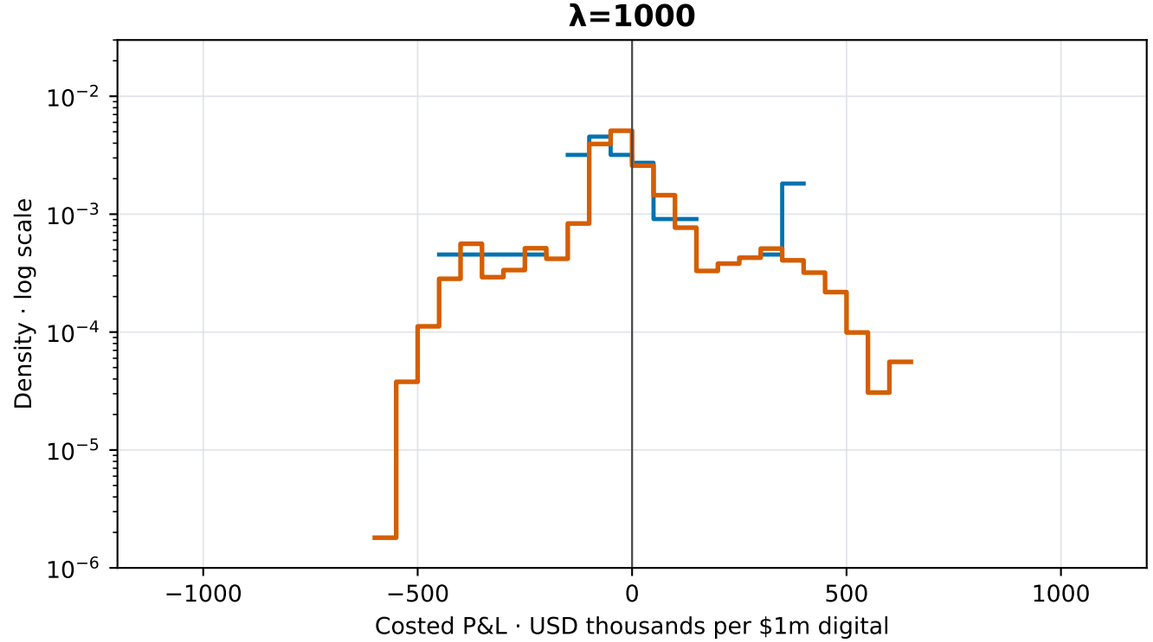

digital-call-1d-nms+0.75: fixed-reference hedge distributions
Unhedged
BS delta + vega
SANOS Greek
Density · log scale
Costed P&L · USD thousands per $1m digital
Realized, raw
Trailing 252-day SANOS replay, raw
No mean correction: 44 realized next-day outcomes versus preceding 252 raw returns. References have no fitted hedge-return law.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.75: SANOS historical · Mean-variance





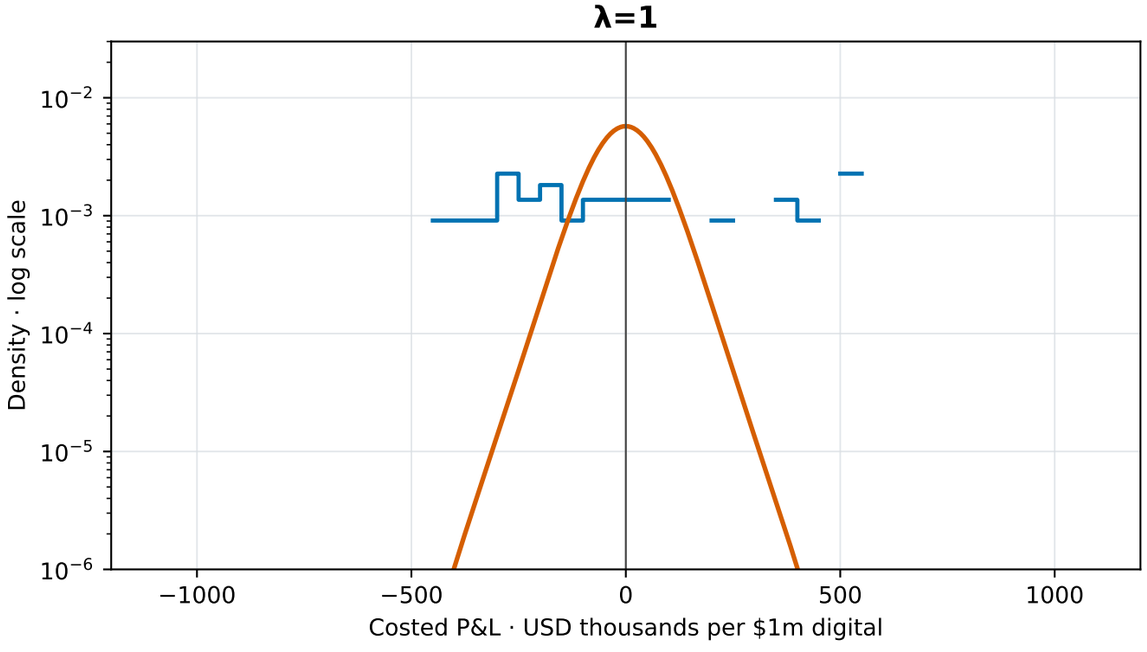


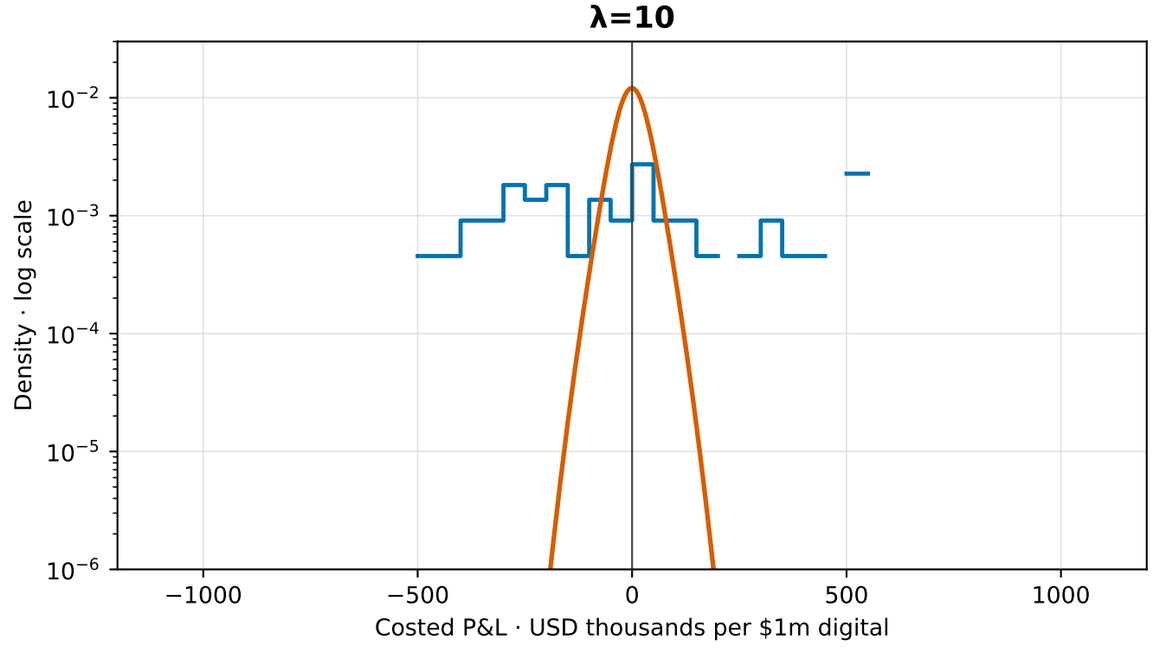


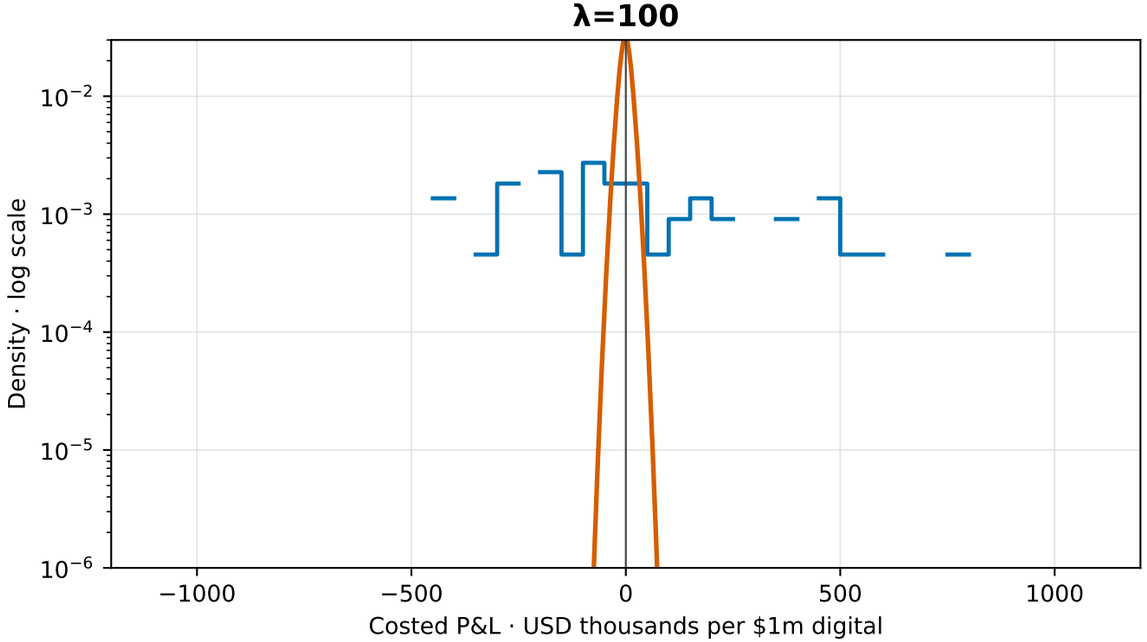
Realized, raw Fitted 252-day demeaned SANOS returns

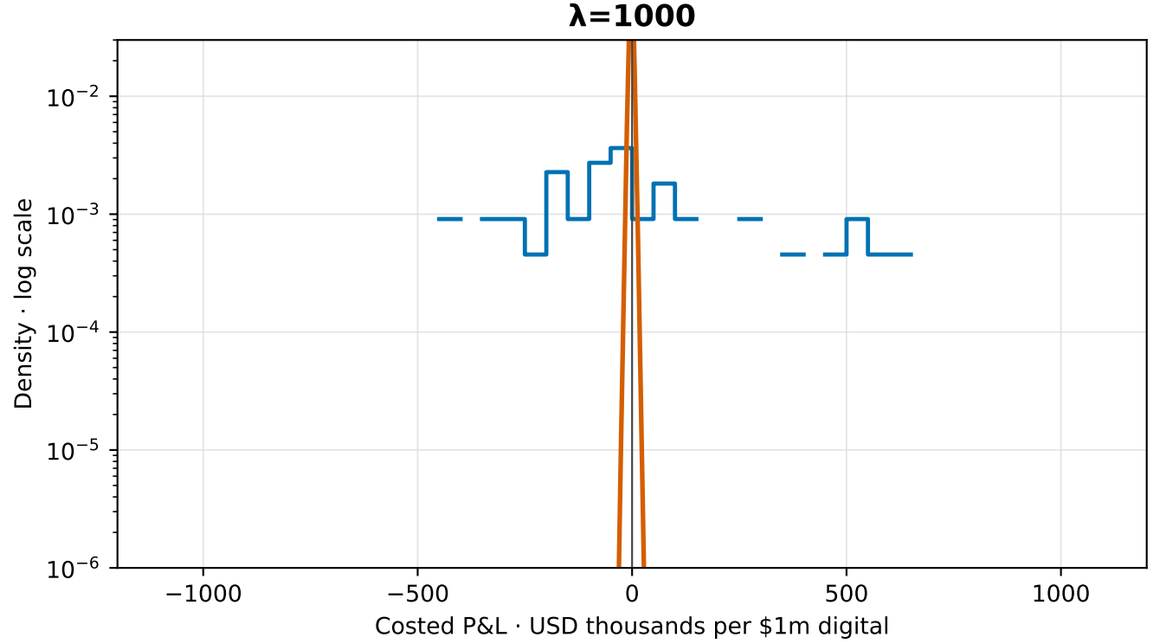
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.75: SANOS parametric · Mean-variance





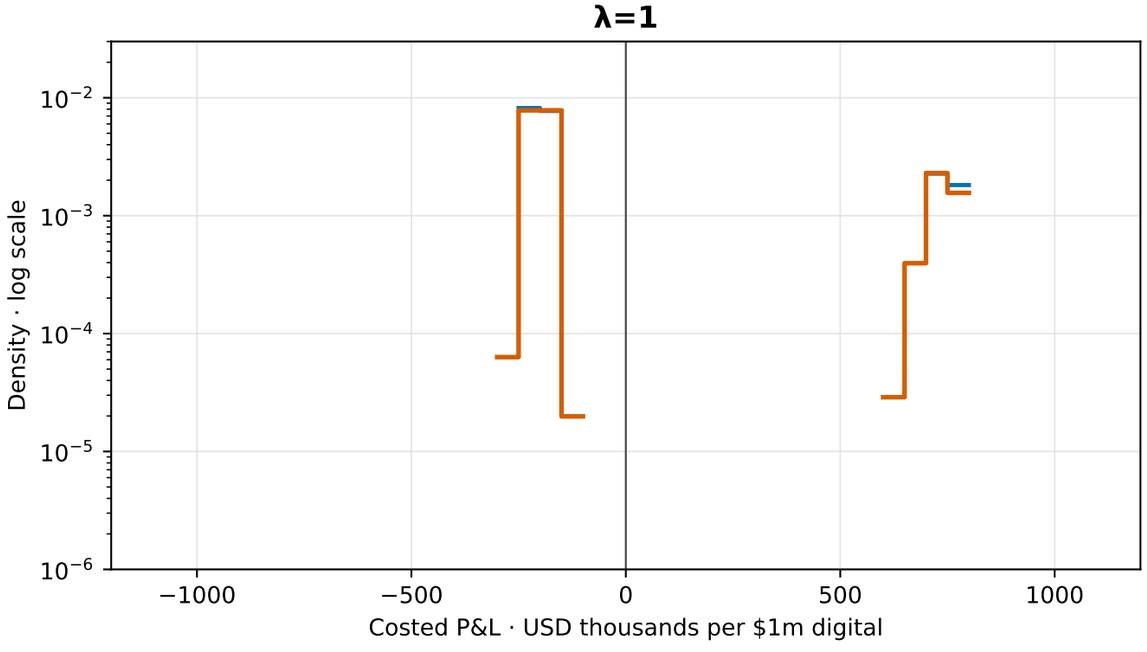


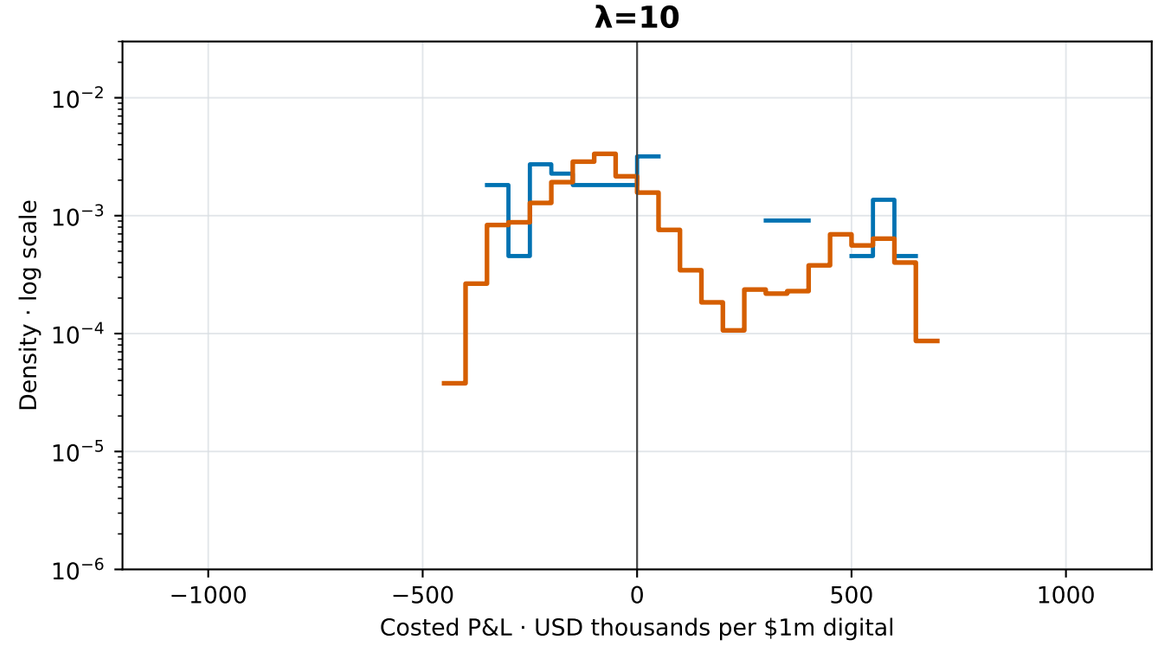


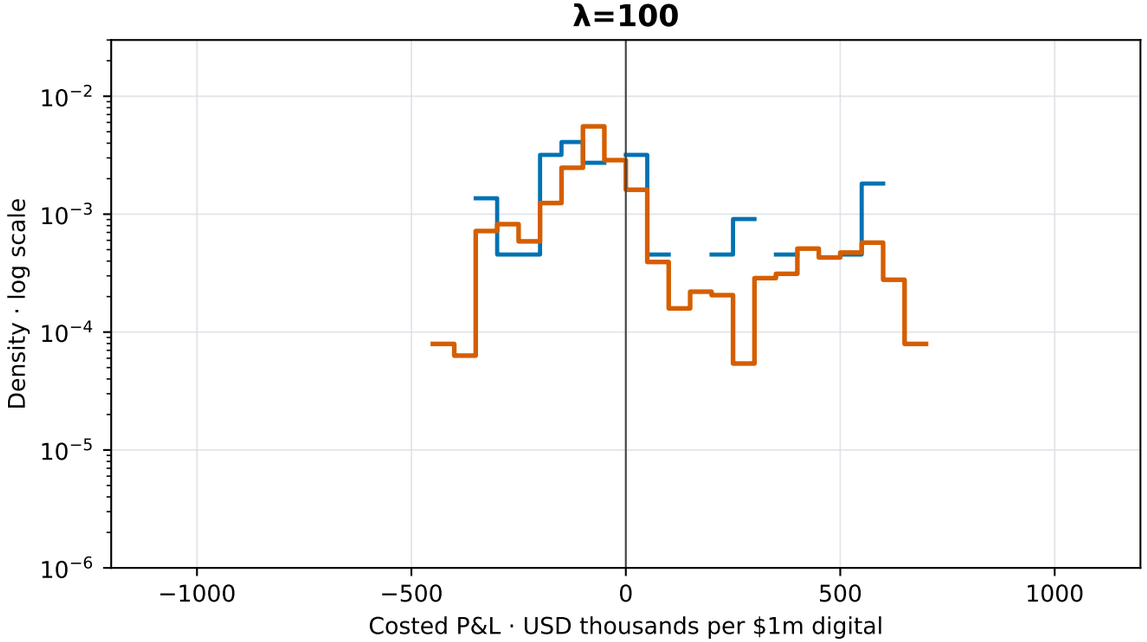
Realized, raw Fitted centered factor Gaussian

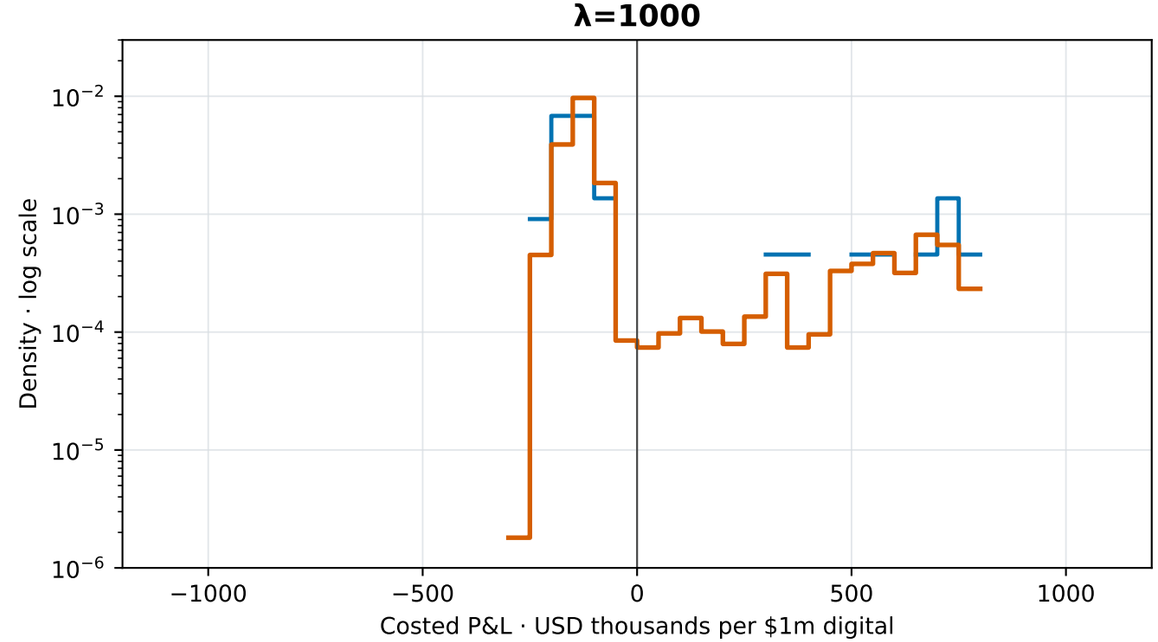
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.75: SANOS historical · Entropy





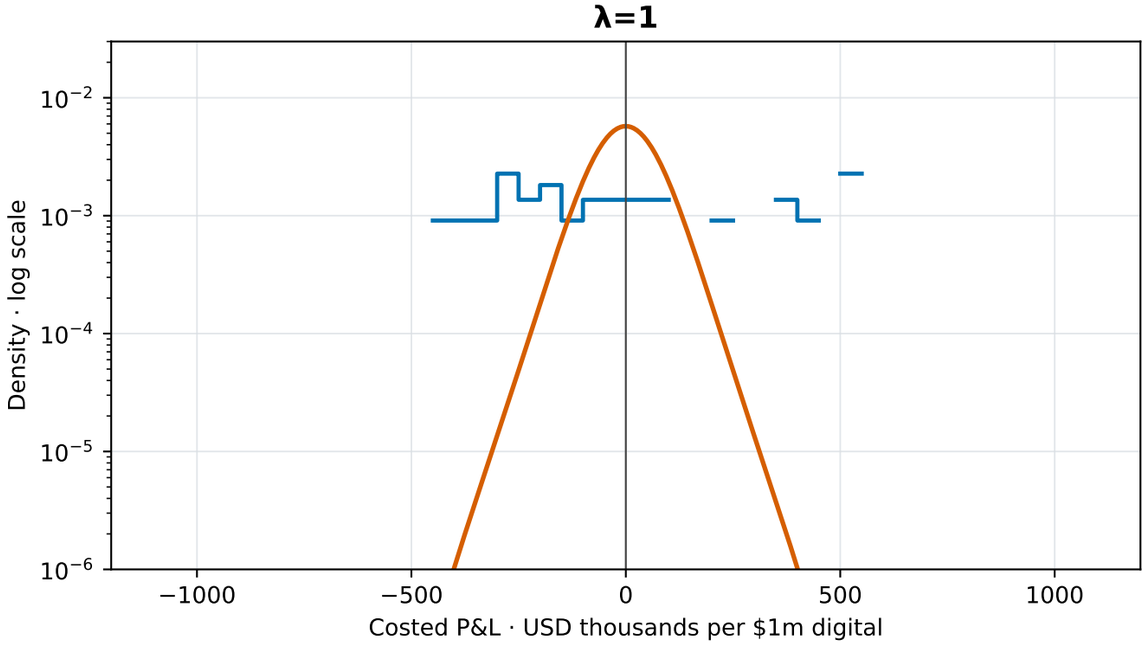


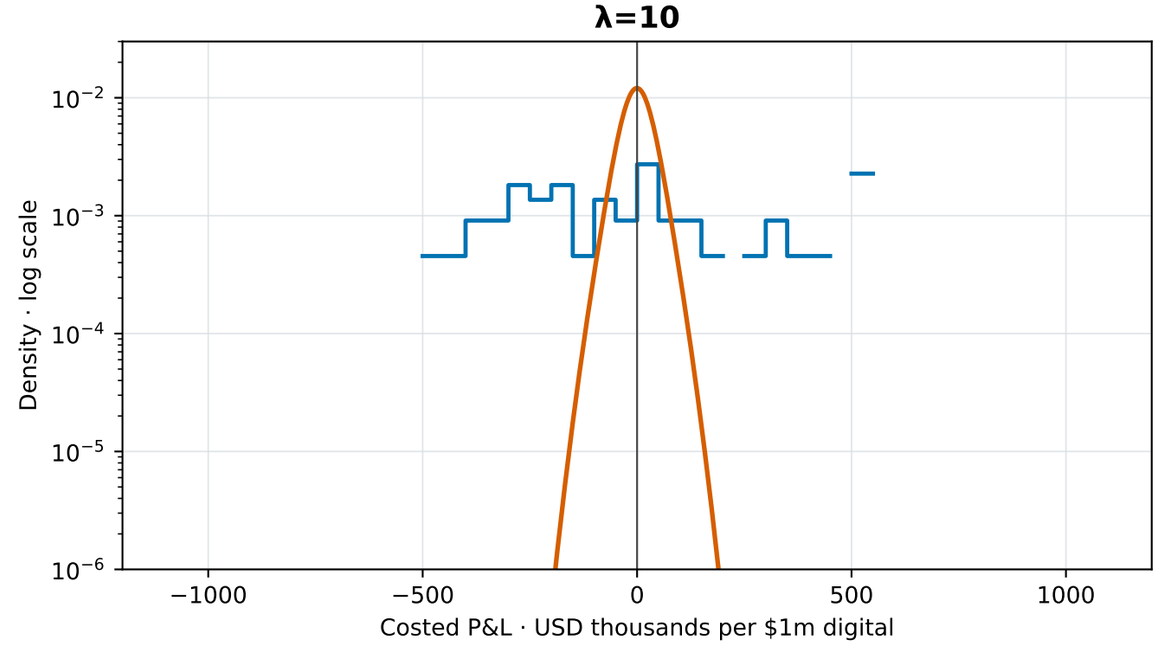


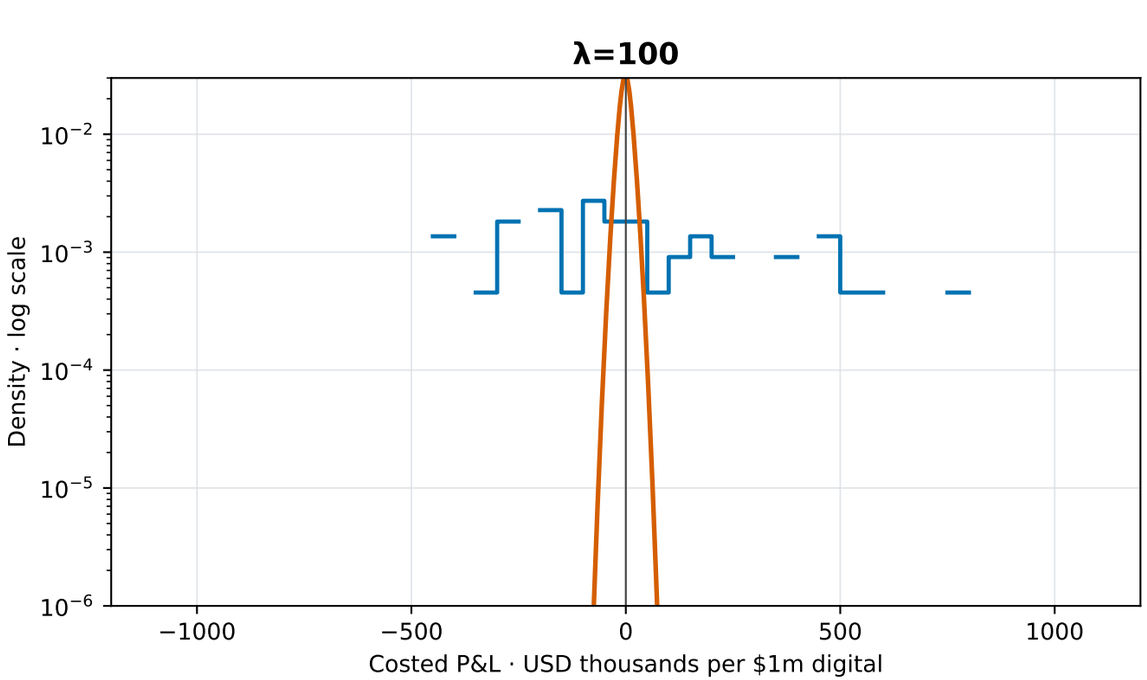
Realized, raw Fitted 252-day demeaned SANOS returns

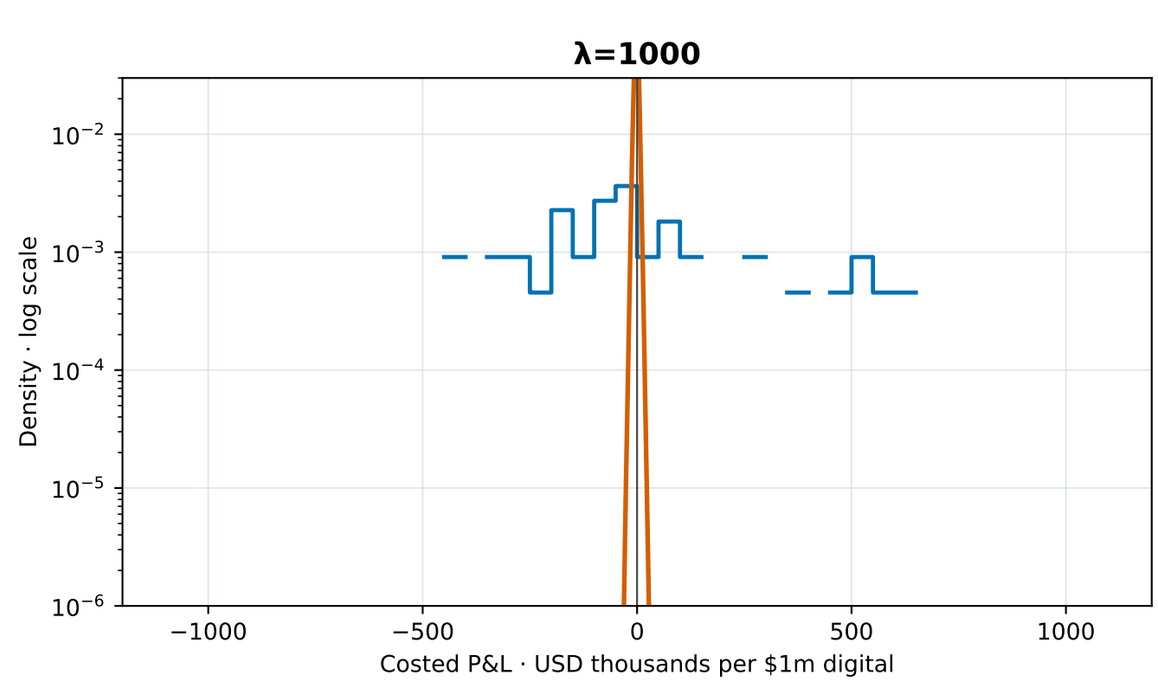
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.75: SANOS parametric · Entropy





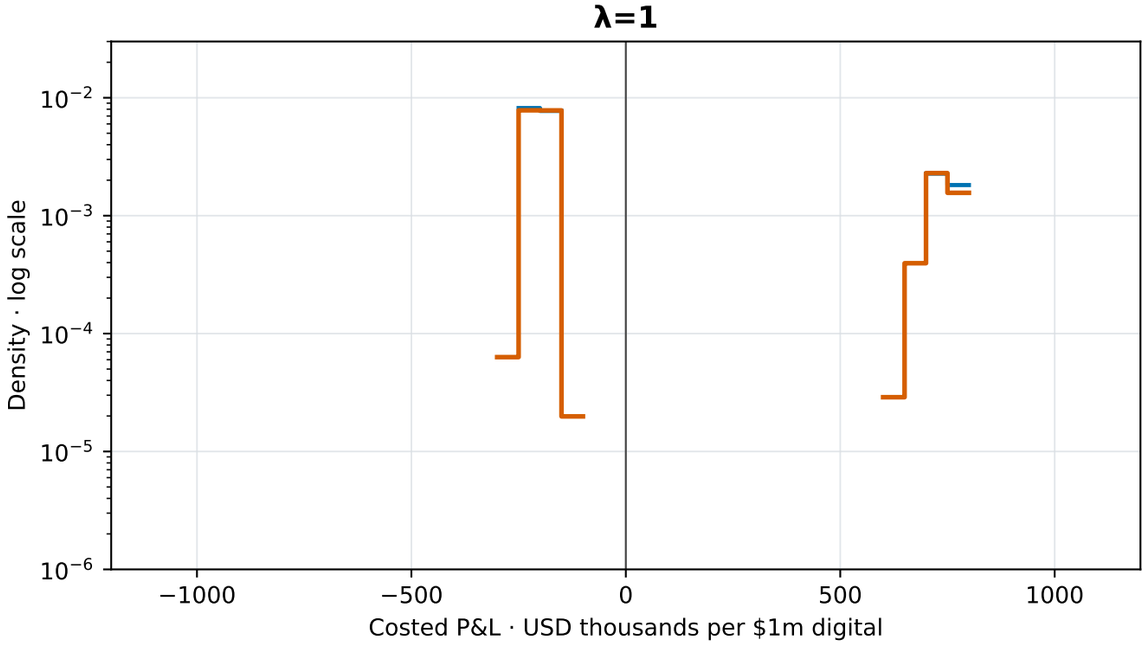


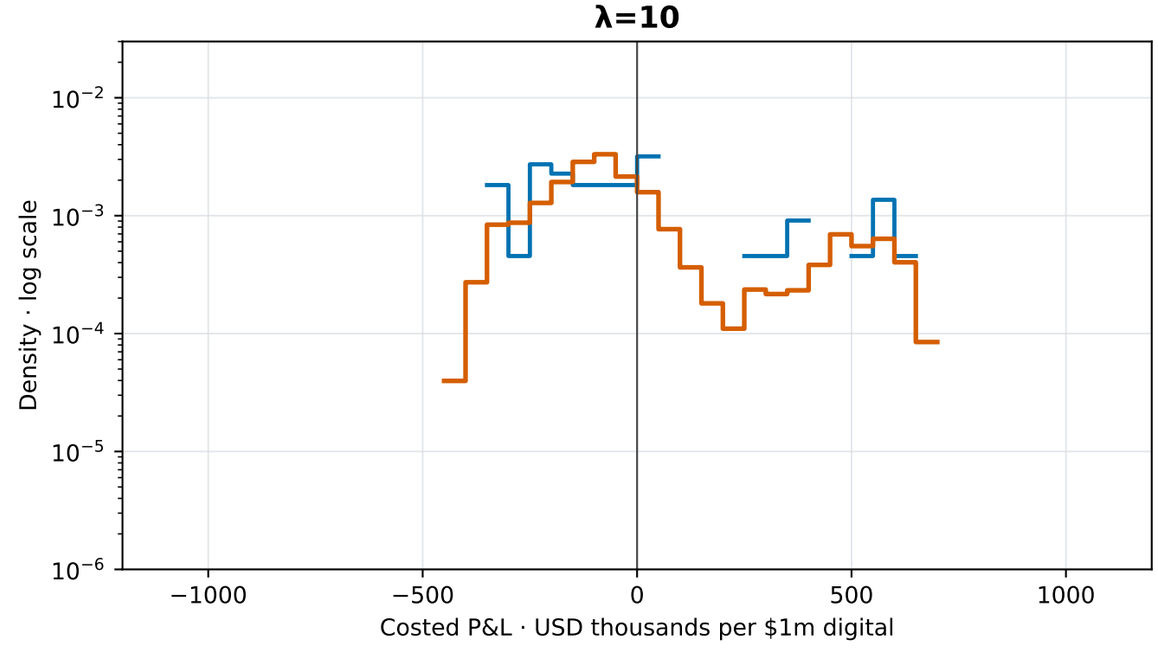


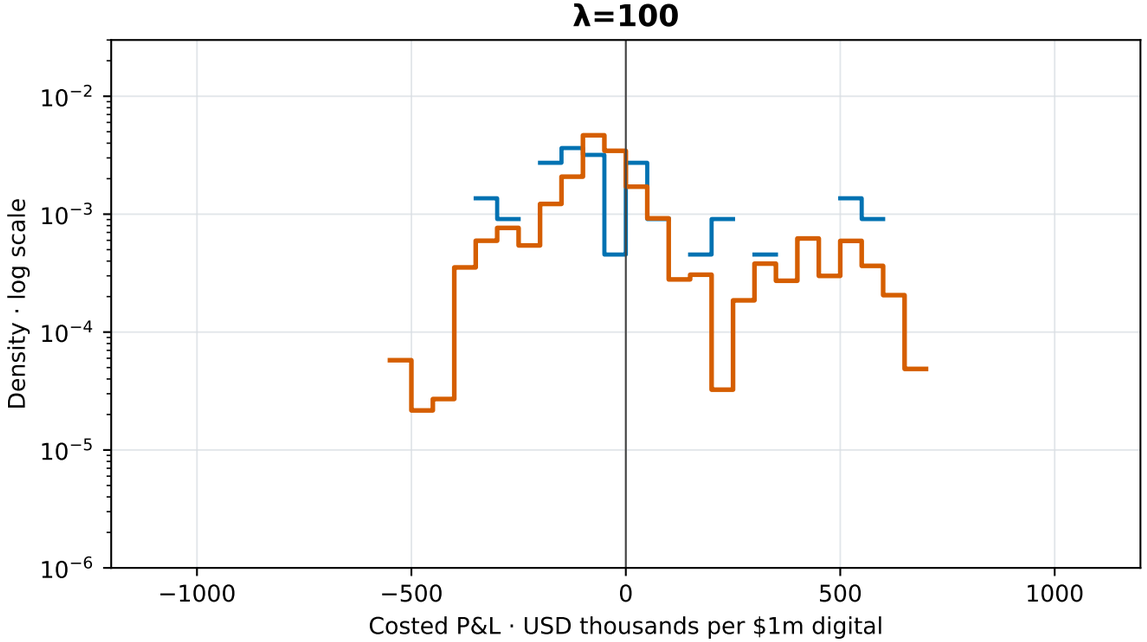
Realized, raw Fitted centered factor Gaussian

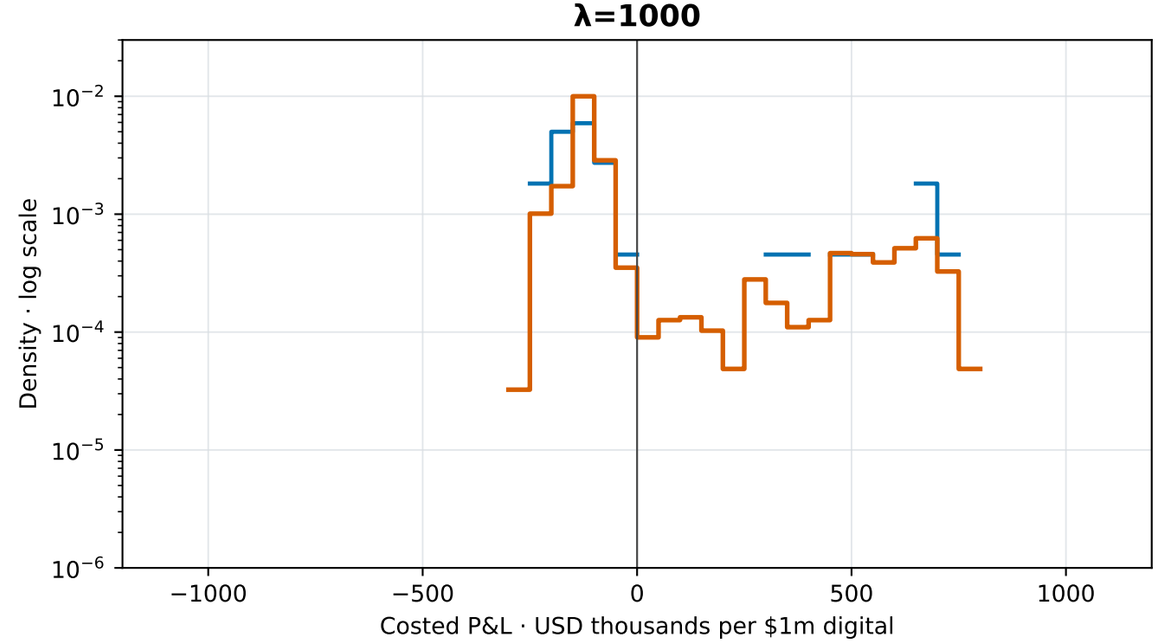
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.75: SANOS historical · Power-exponential





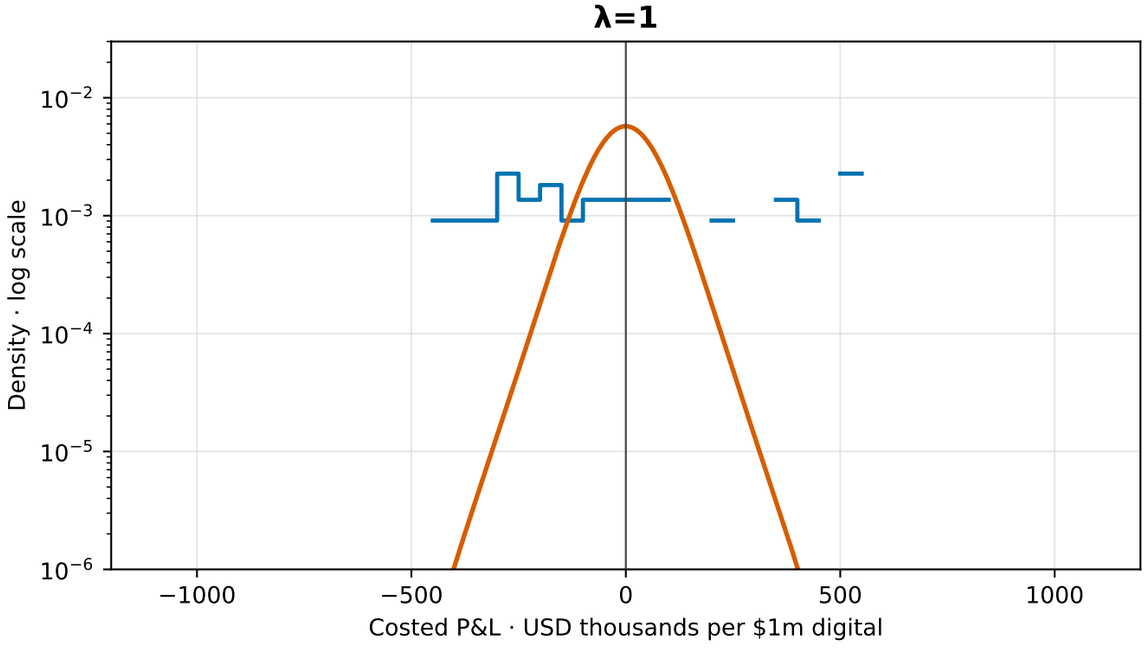


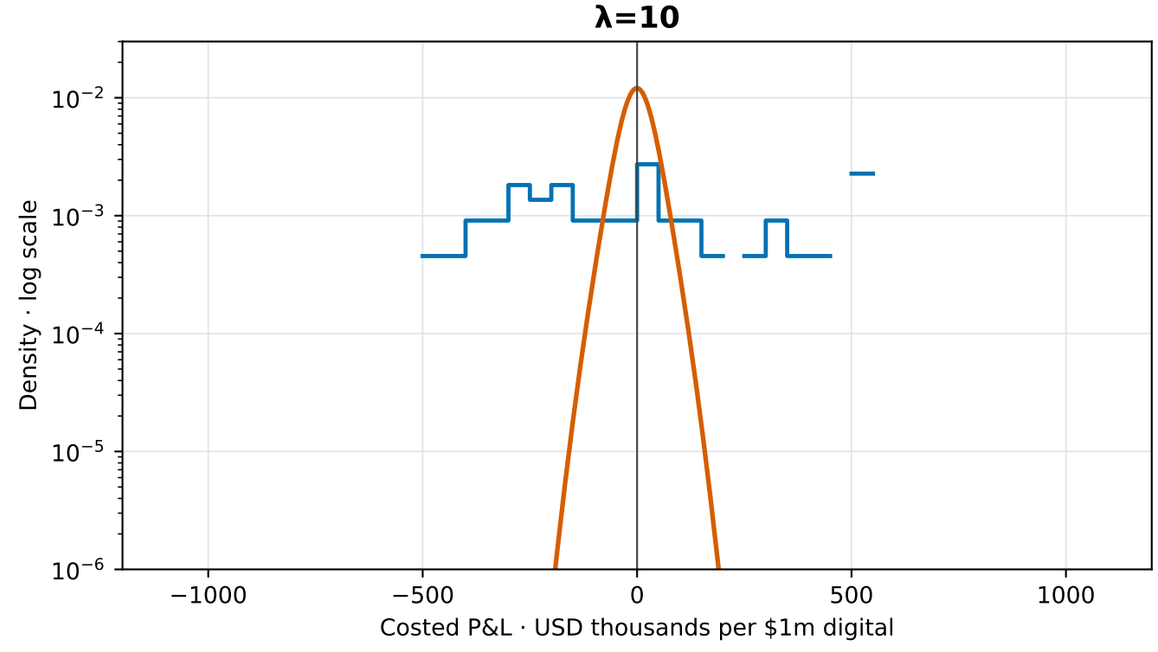


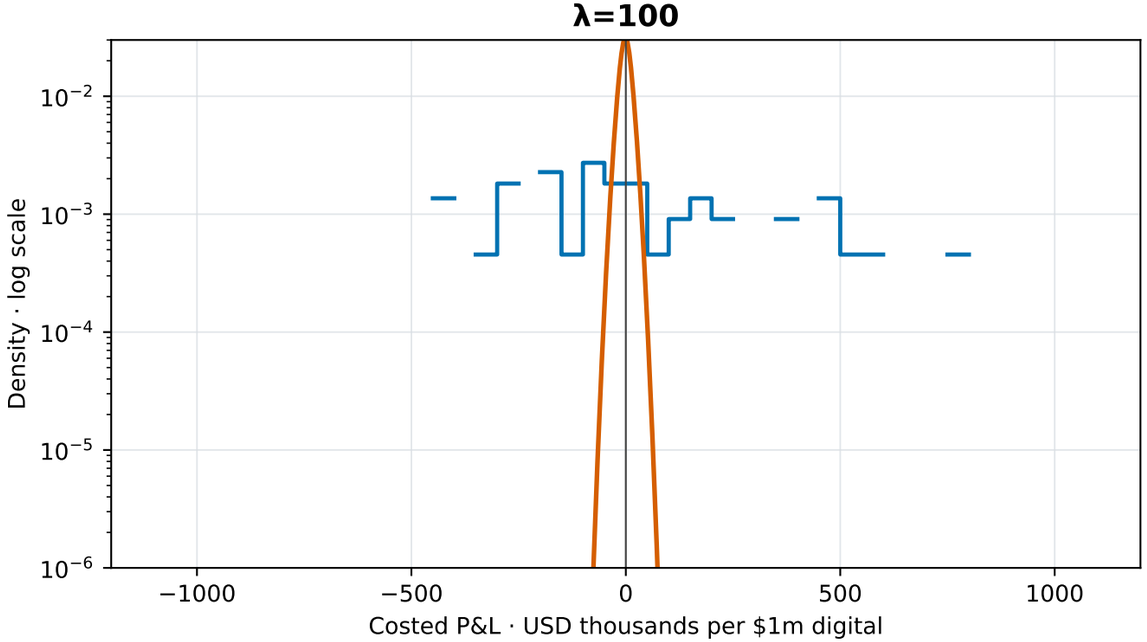
Realized, raw Fitted 252-day demeaned SANOS returns

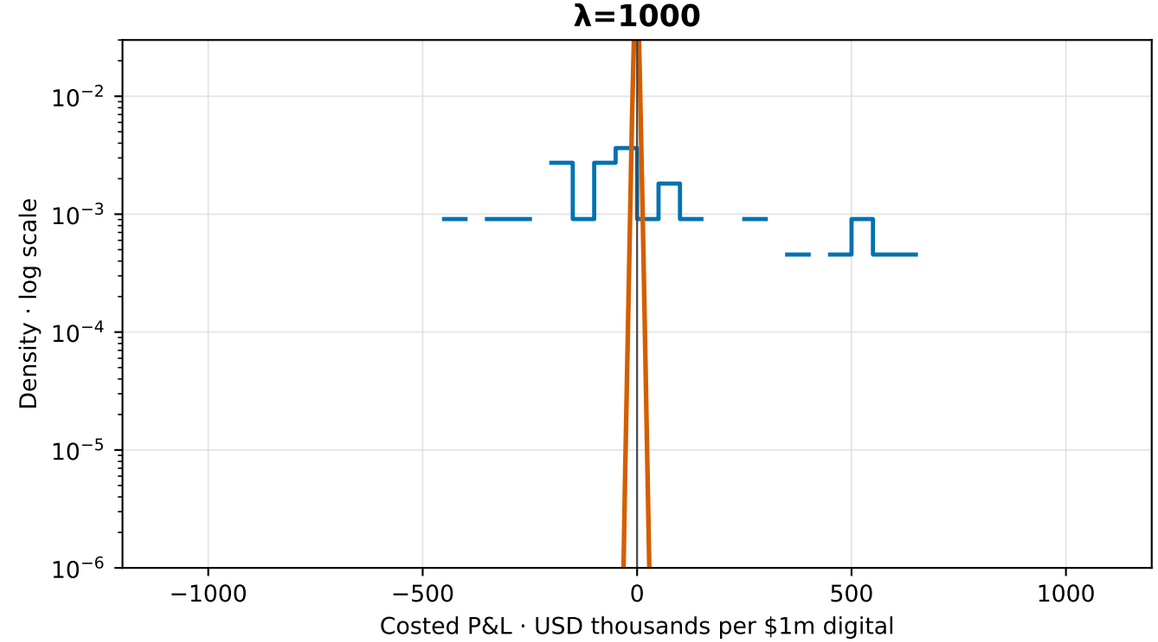
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+0.75: SANOS parametric · Power-exponential



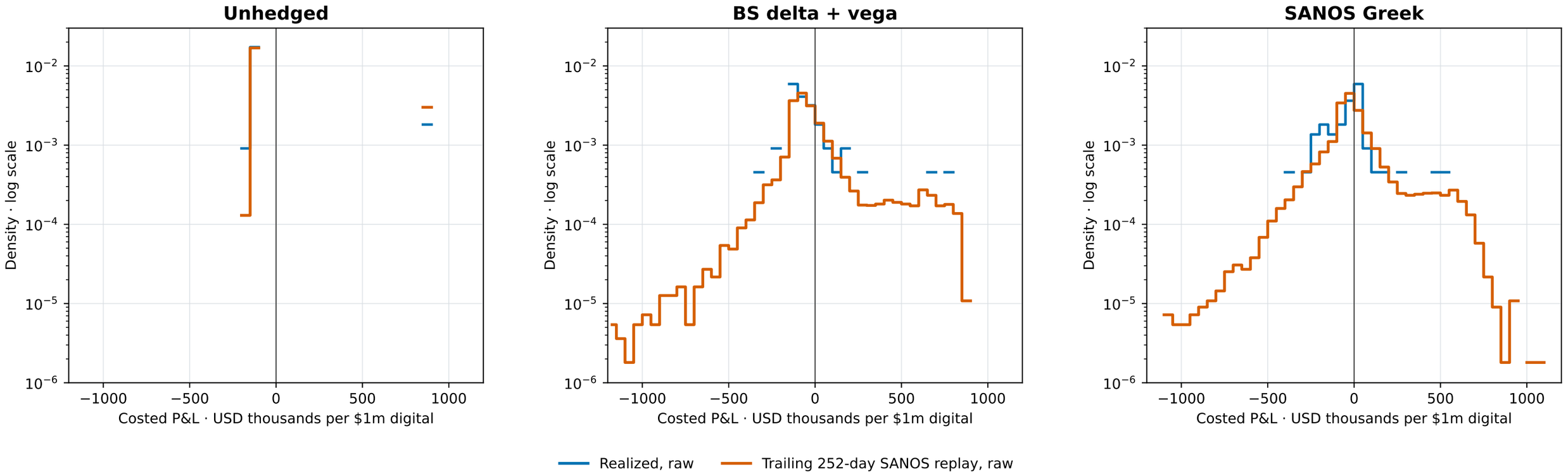




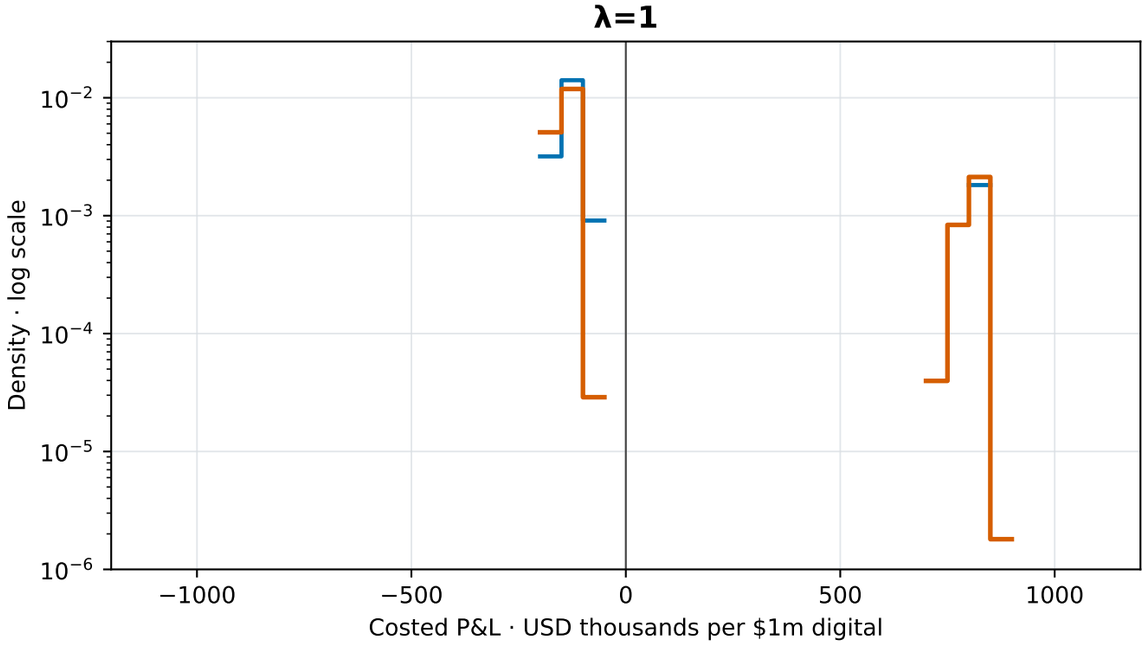


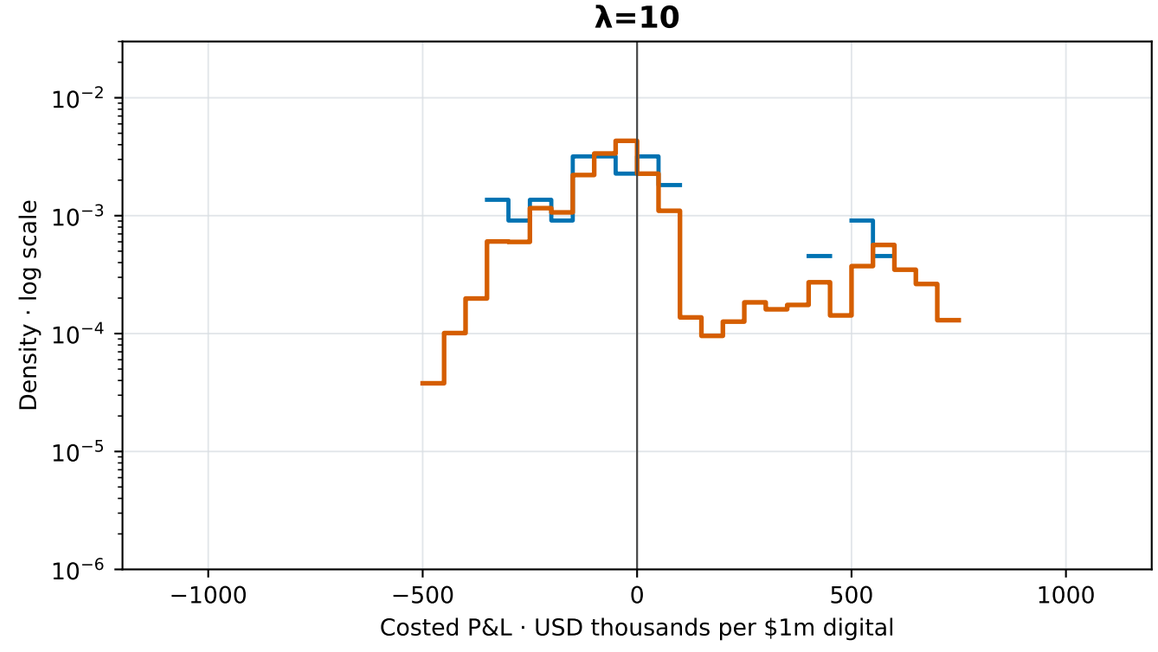
Realized, raw Fitted centered factor Gaussian

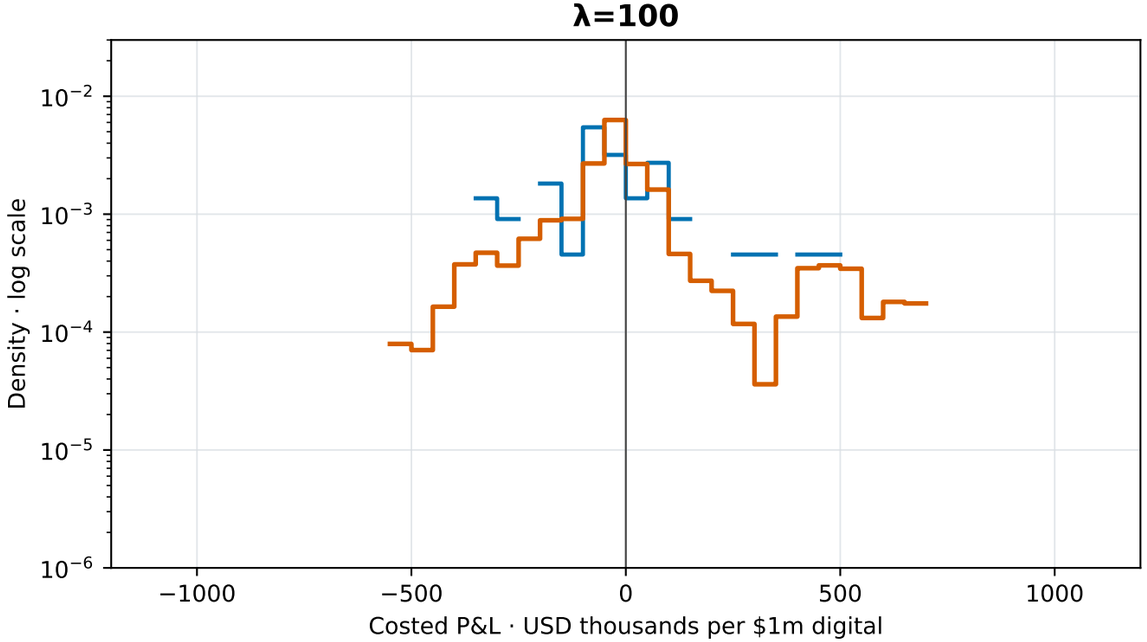
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

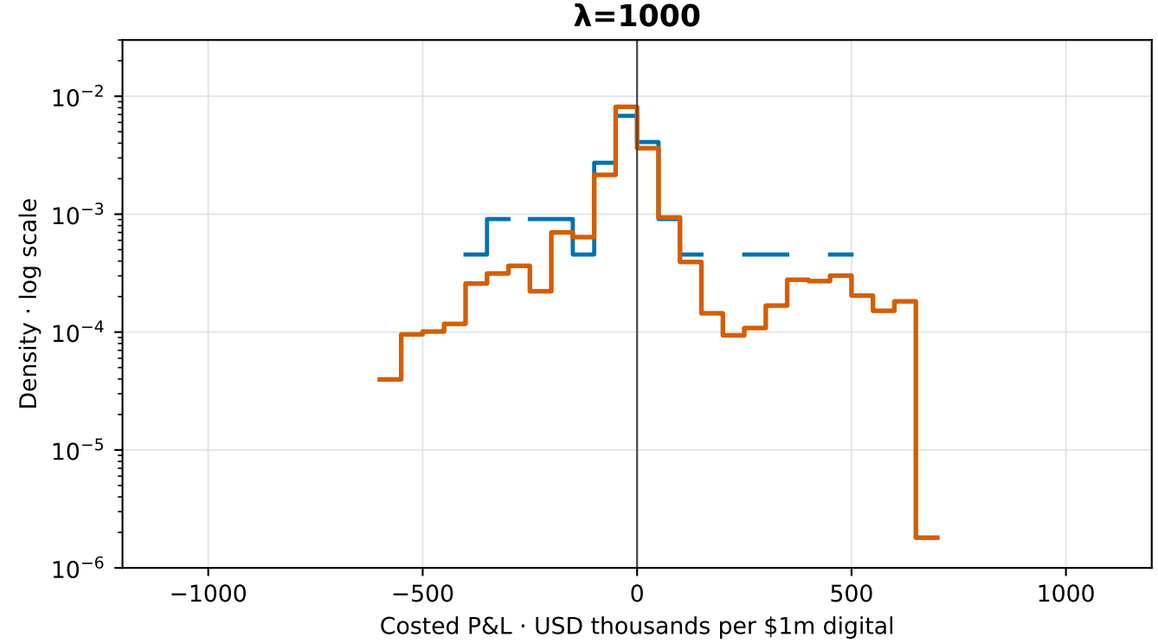

digital-call-1d-nms+1.00: fixed-reference hedge distributions
Unhedged
BS delta + vega
SANOS Greek
Density · log scale
Costed P&L · USD thousands per $1m digital
Realized, raw
Trailing 252-day SANOS replay, raw
No mean correction: 44 realized next-day outcomes versus preceding 252 raw returns. References have no fitted hedge-return law.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+1.00: SANOS historical · Mean-variance





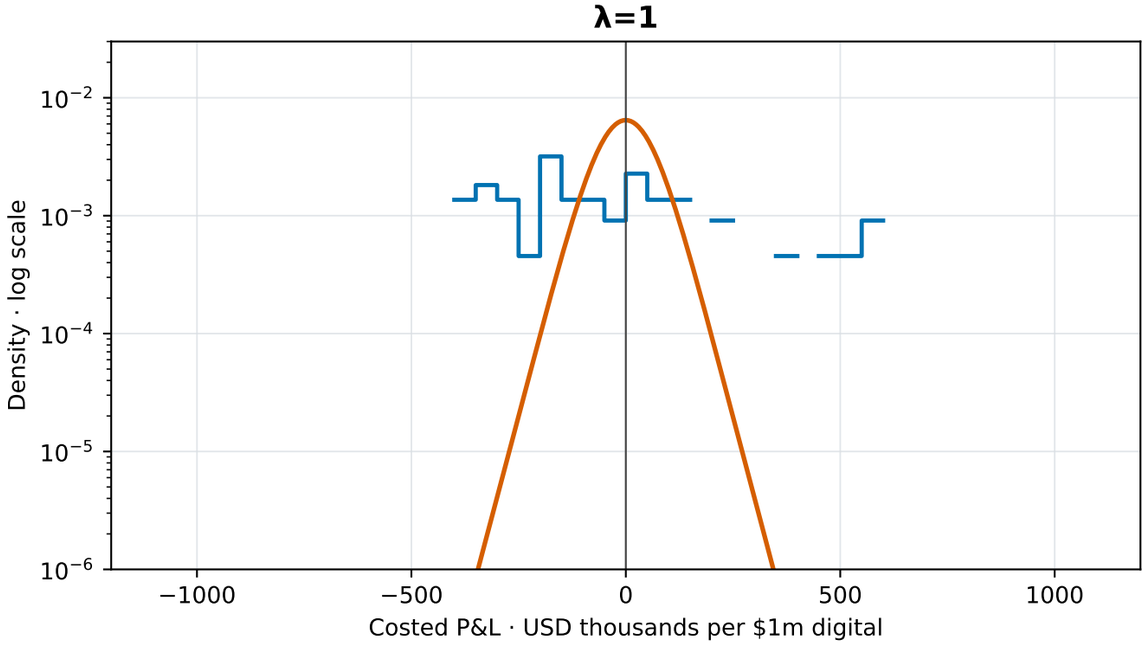


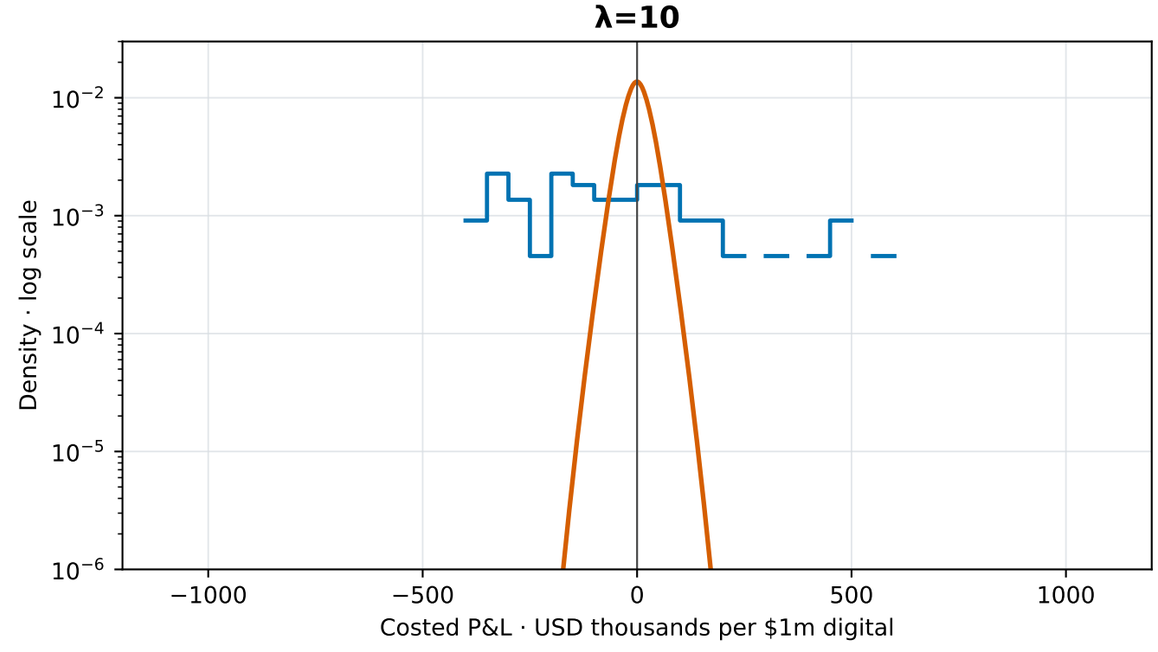


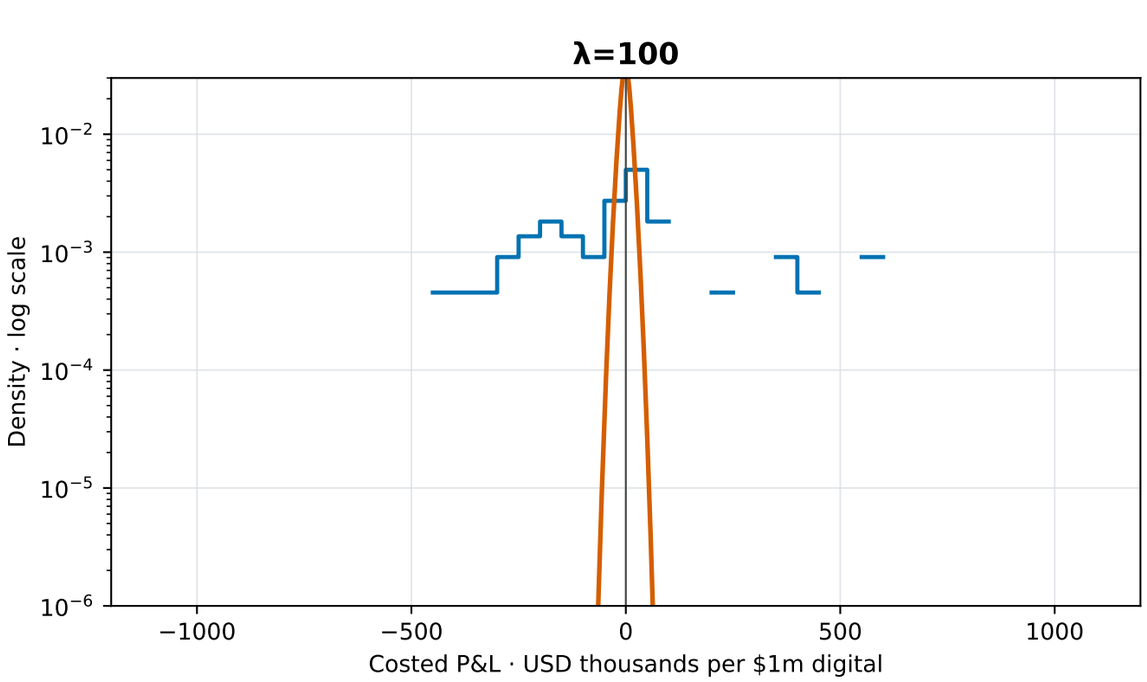
Realized, raw Fitted 252-day demeaned SANOS returns

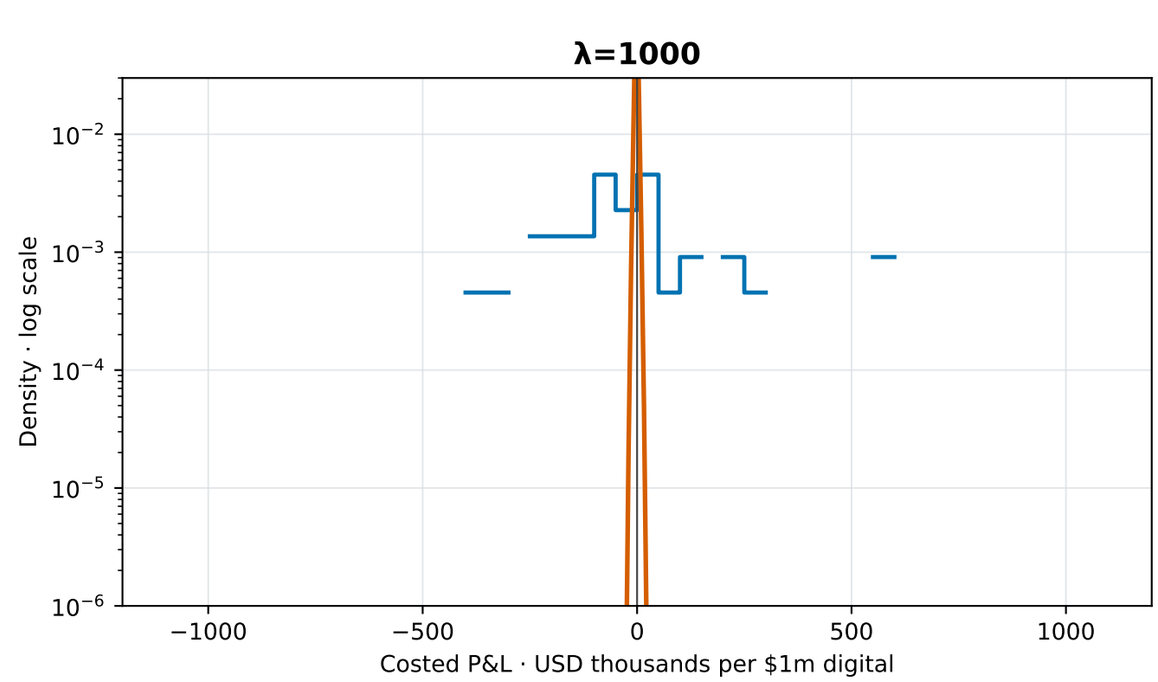
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+1.00: SANOS parametric · Mean-variance





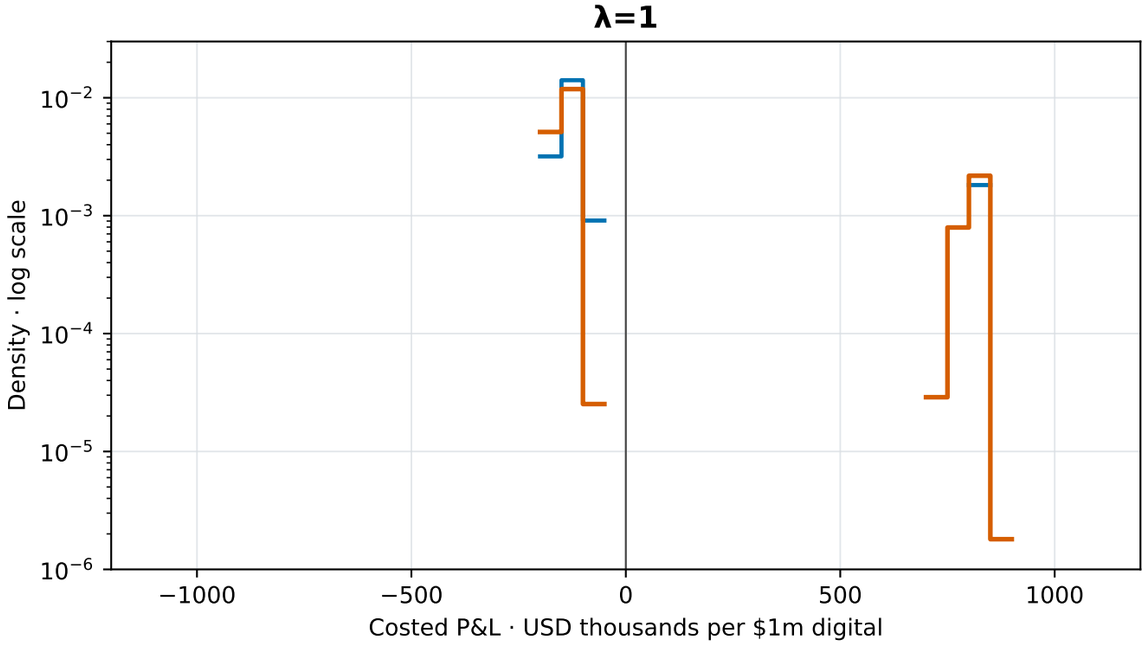


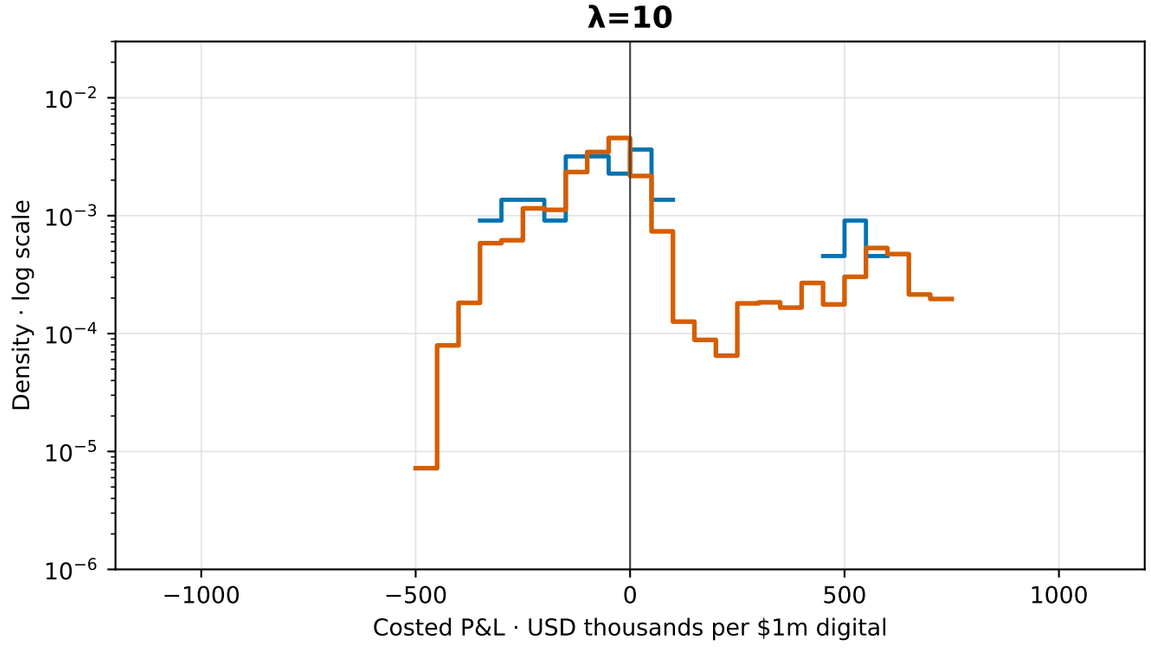


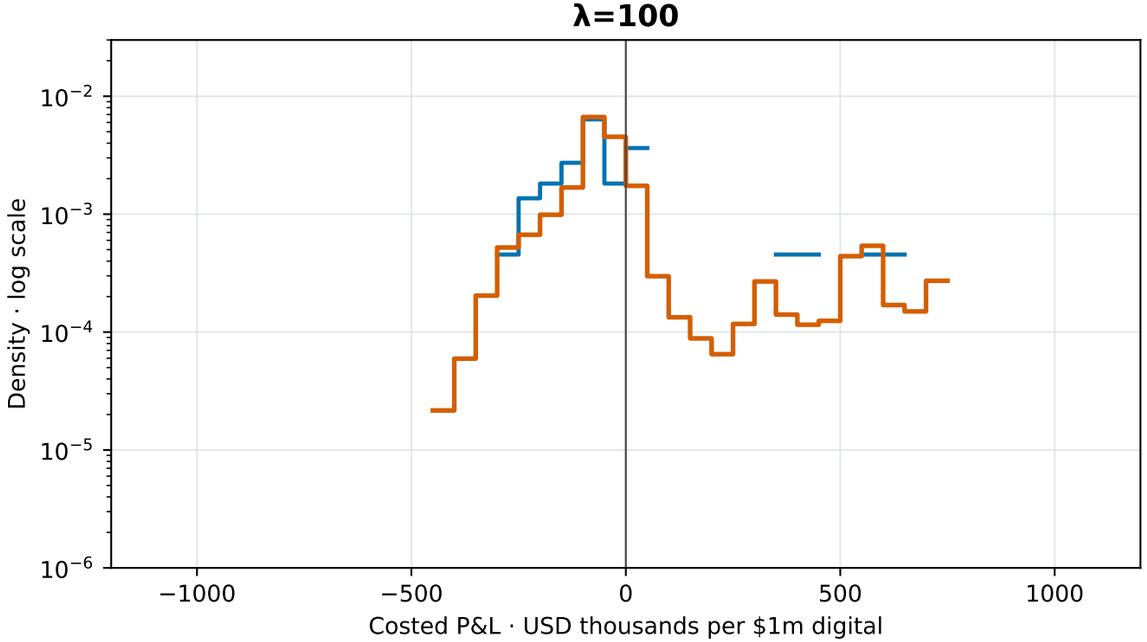
Realized, raw Fitted centered factor Gaussian

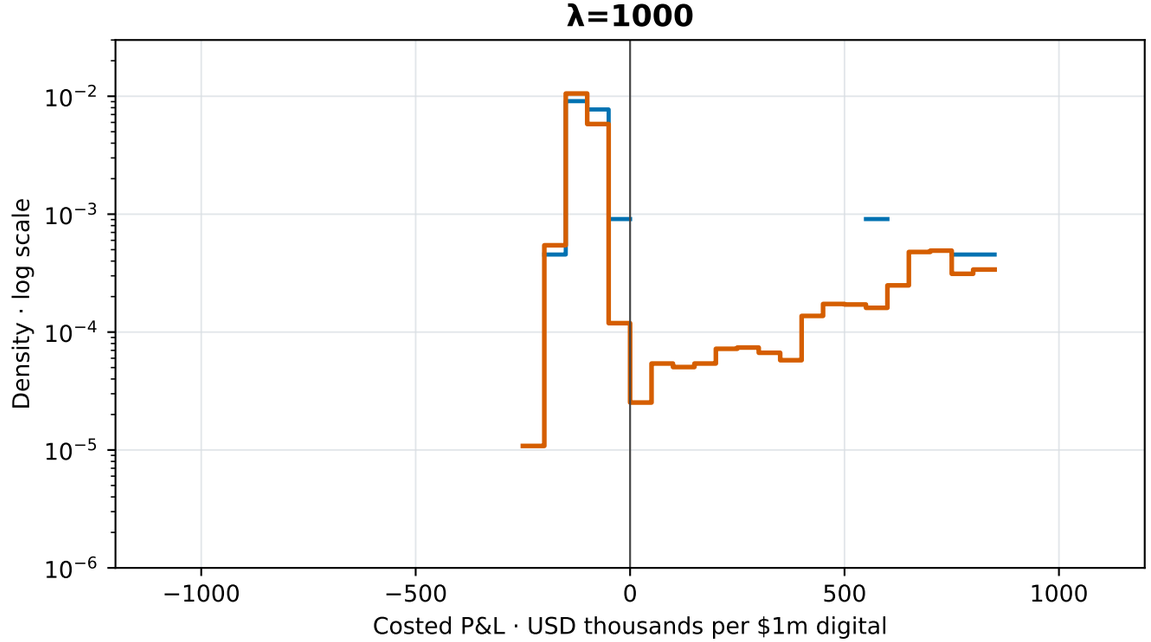
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+1.00: SANOS historical · Entropy





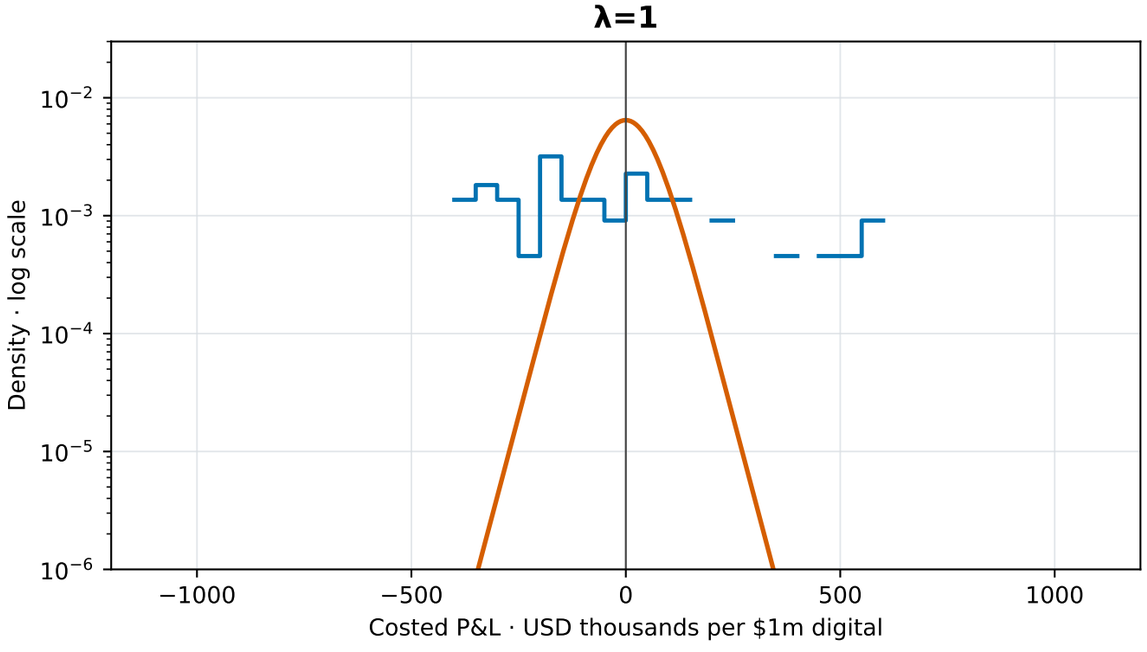


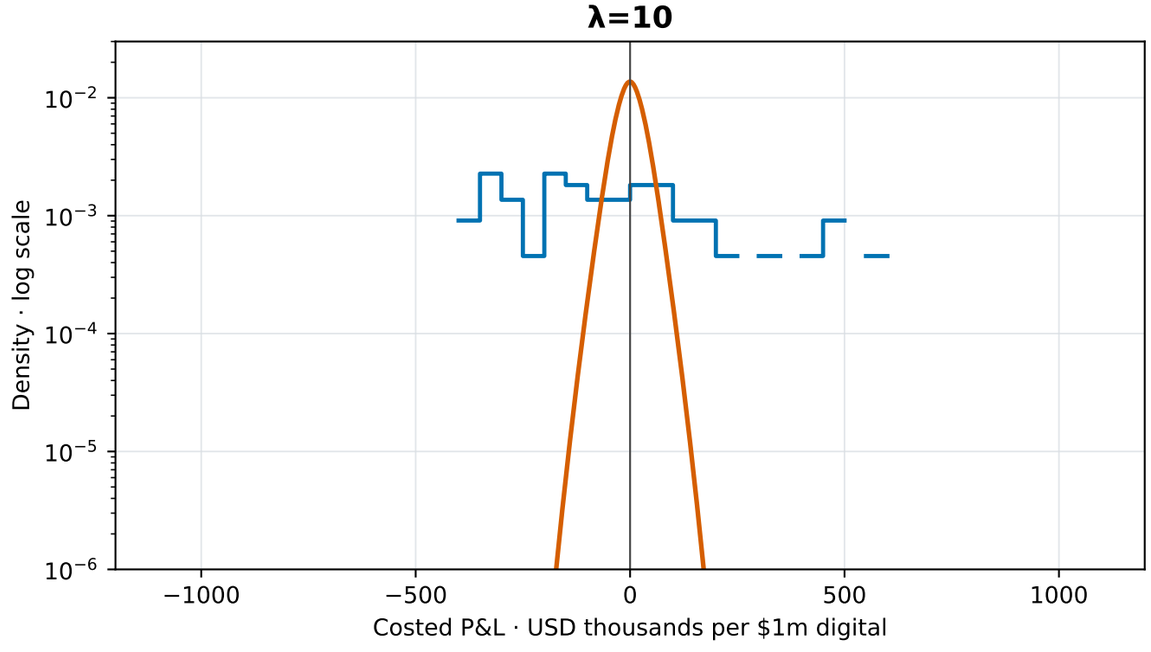


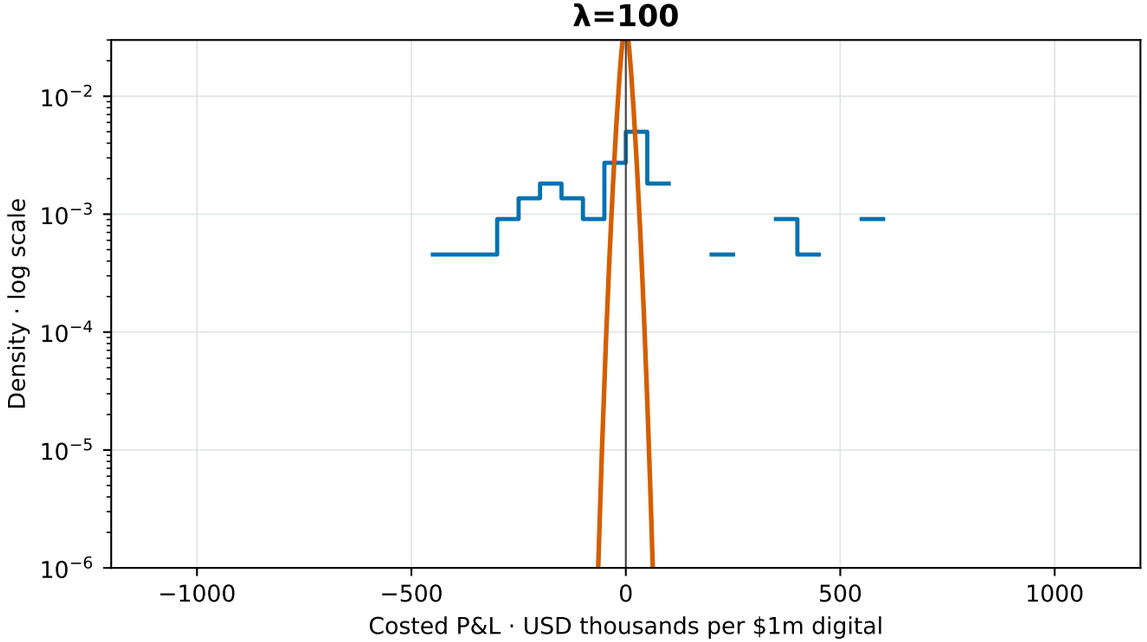
Realized, raw Fitted 252-day demeaned SANOS returns

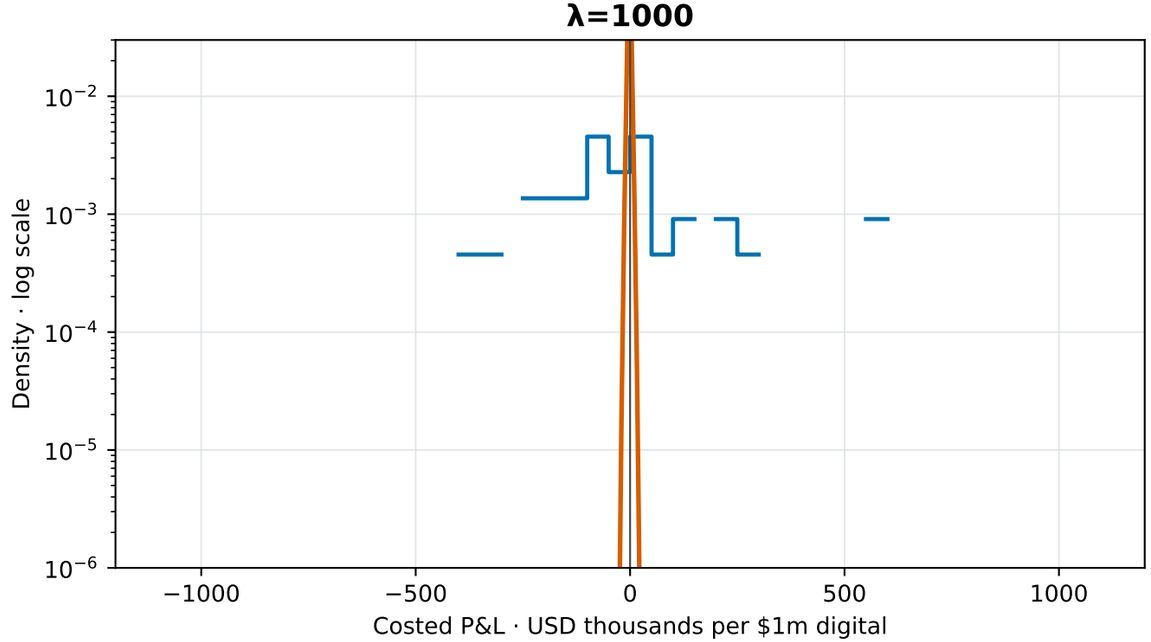
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+1.00: SANOS parametric · Entropy





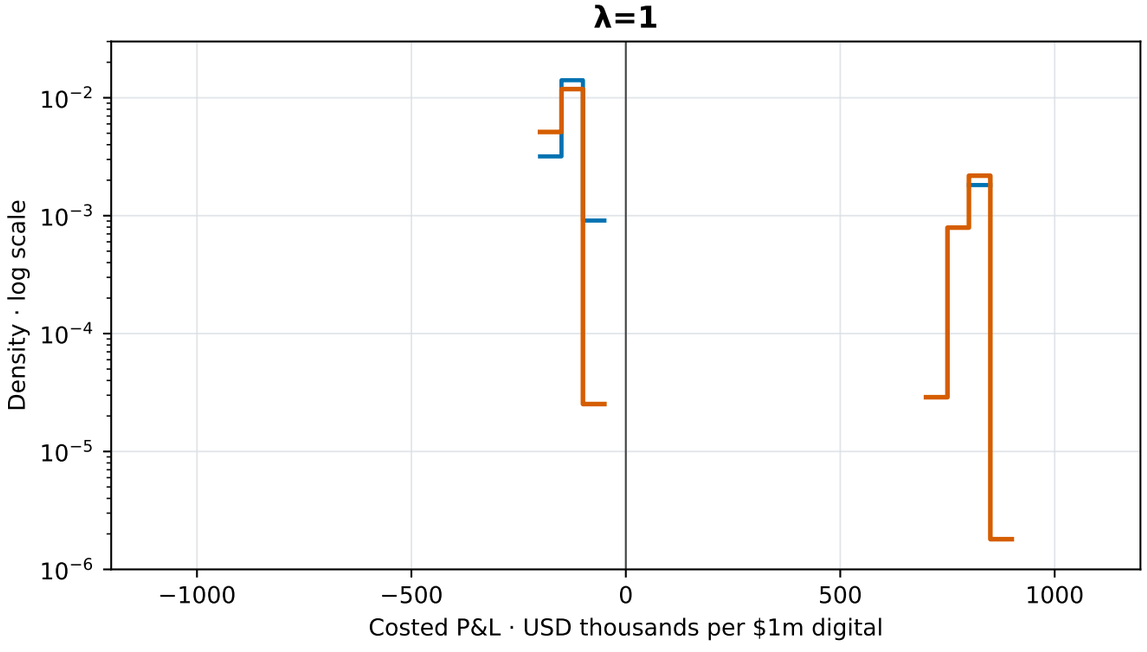


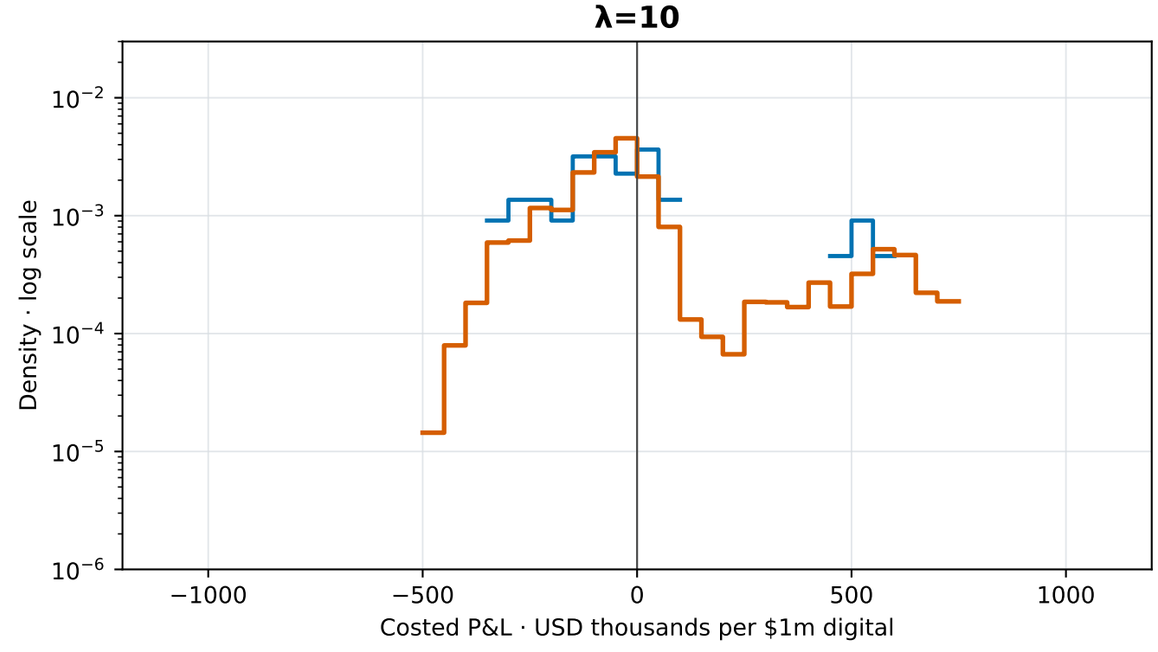


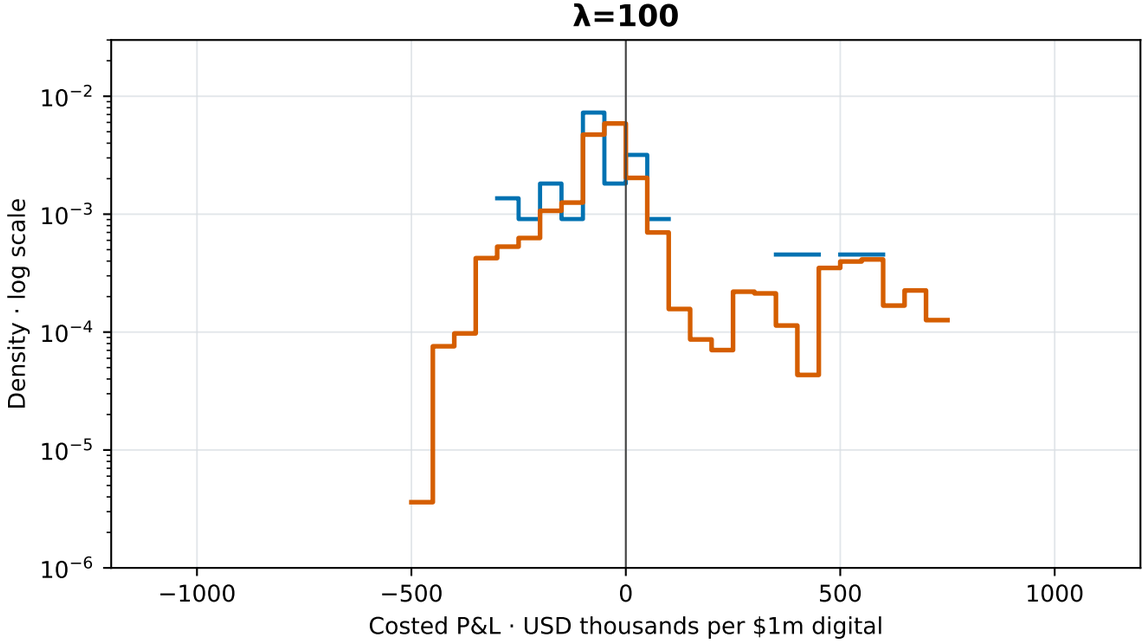
Realized, raw Fitted centered factor Gaussian

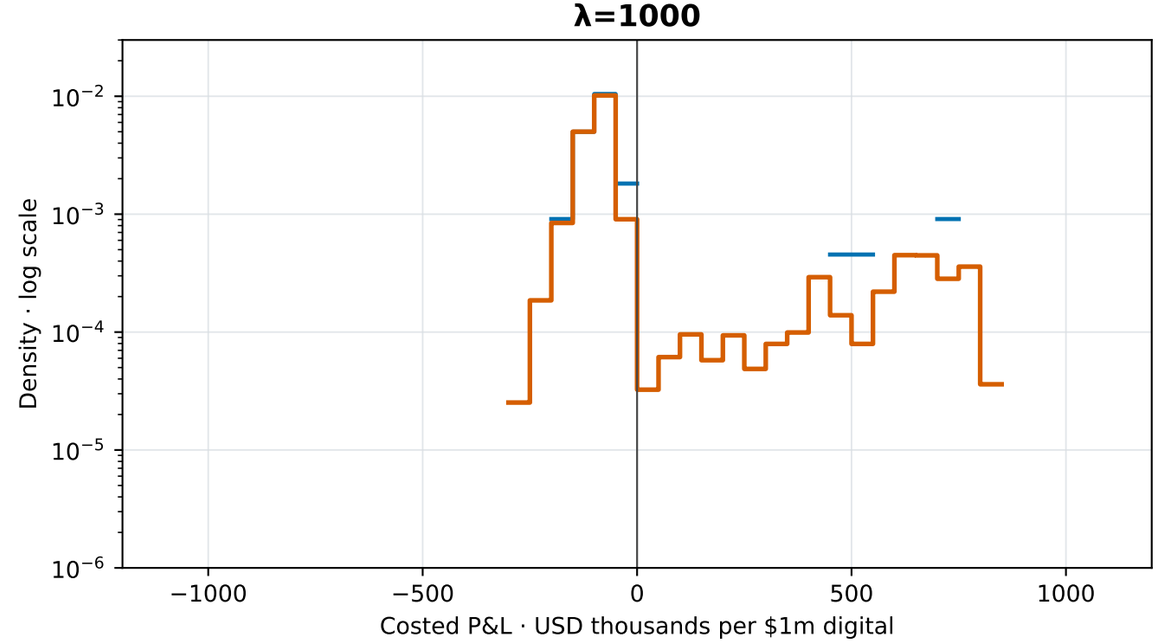
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+1.00: SANOS historical · Power-exponential





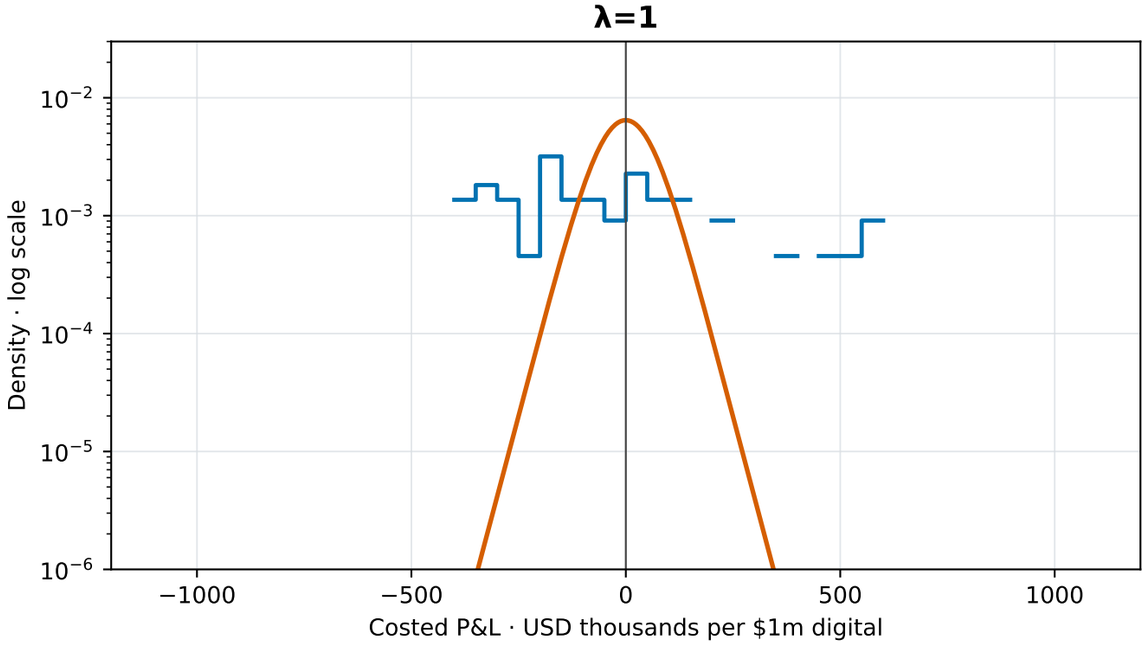


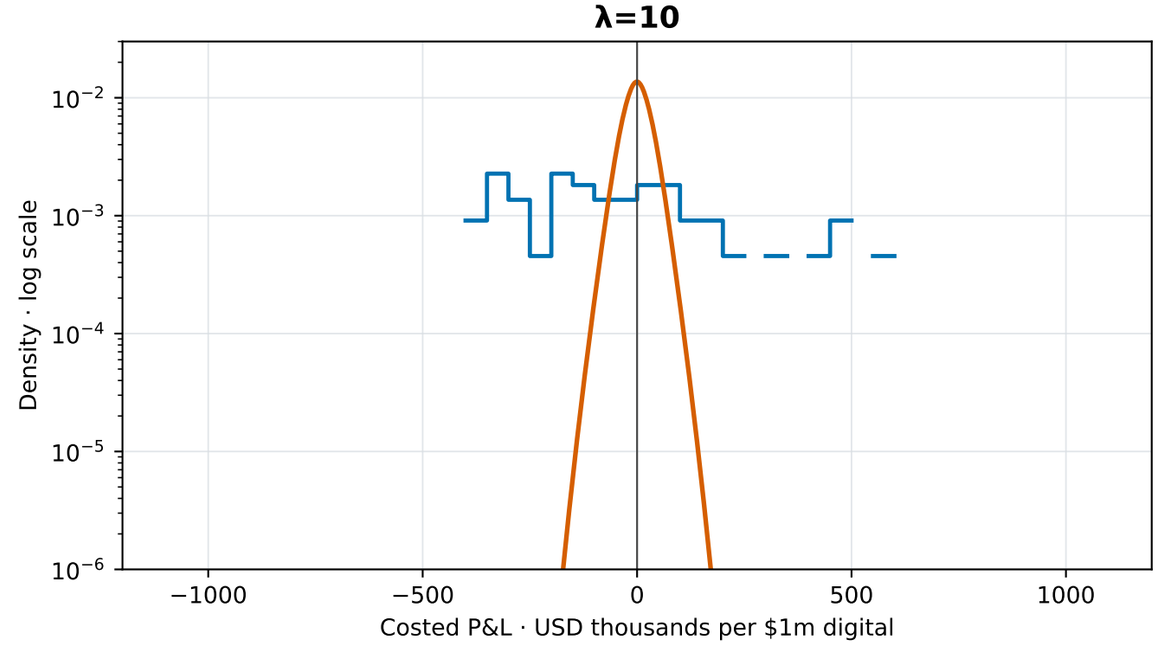


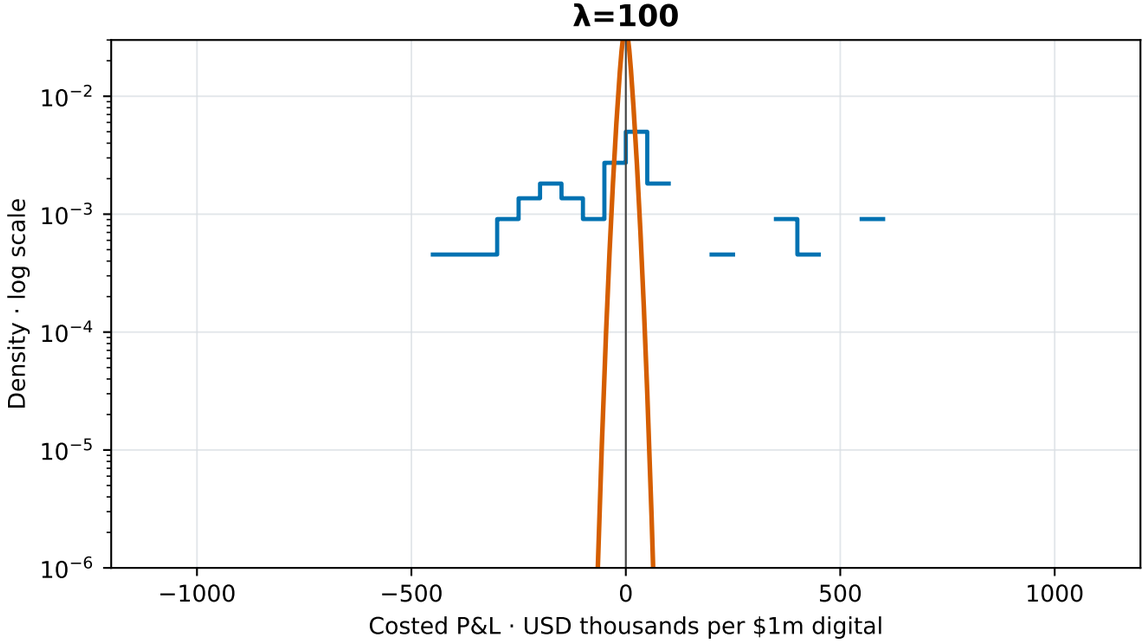
Realized, raw Fitted 252-day demeaned SANOS returns

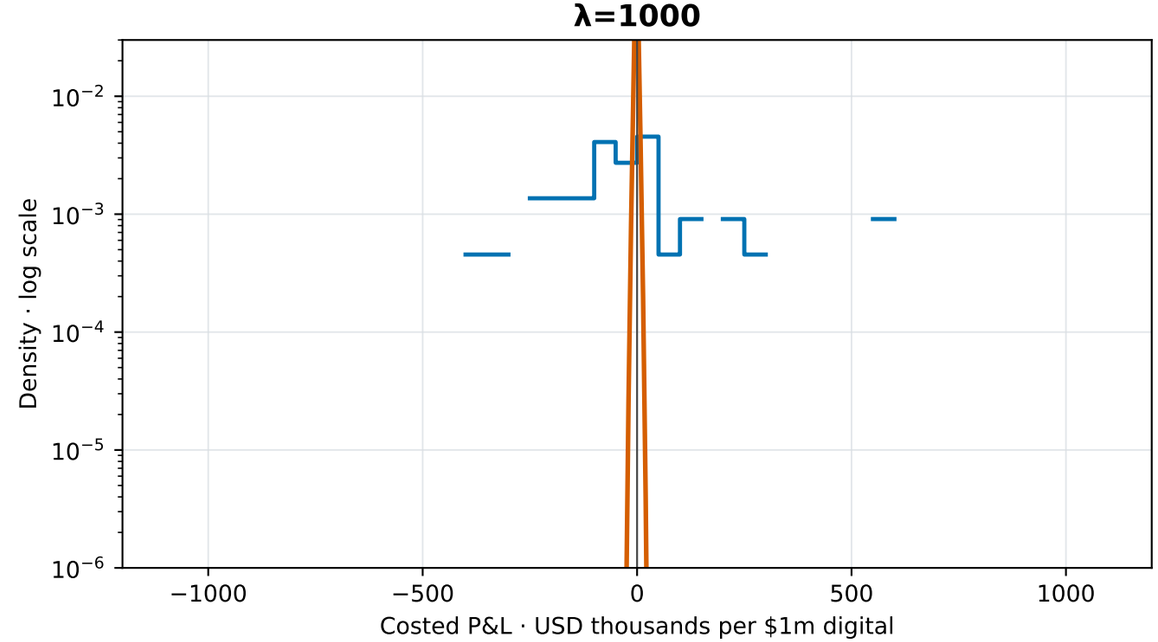
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

digital-call-1d-nms+1.00: SANOS parametric · Power-exponential





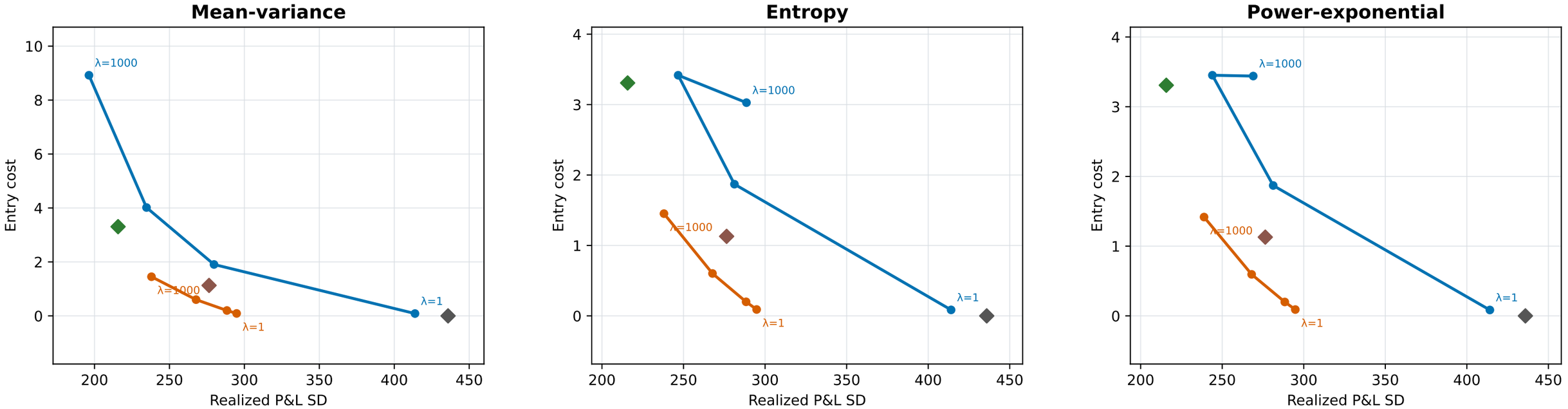




Realized, raw Fitted centered factor Gaussian

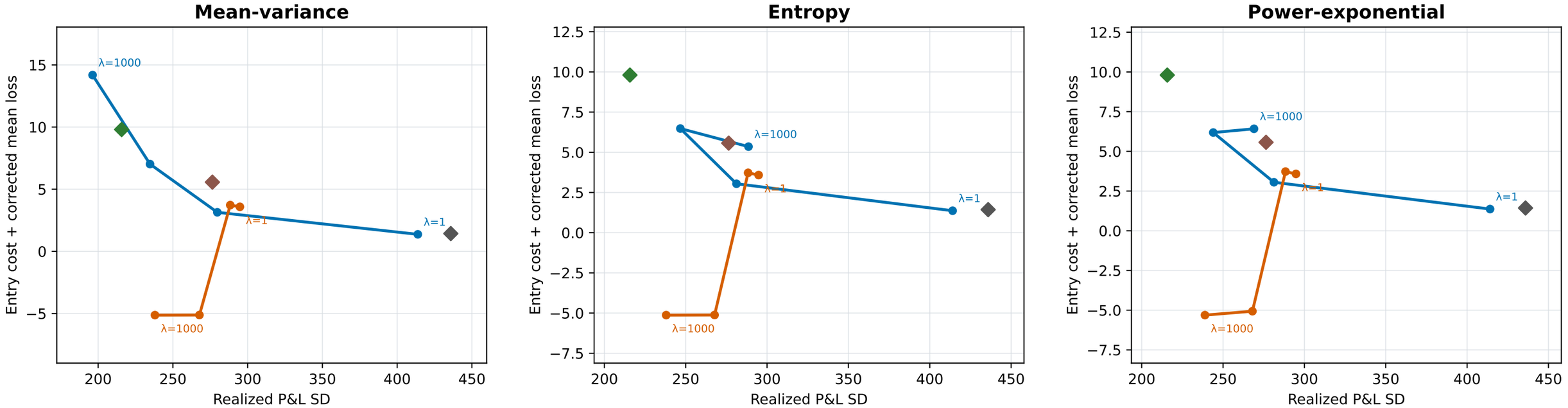
Realized P&L is not mean-adjusted; each fitted law precedes its next-day outcome across 44 dates.
Density peaks above 0.03 and outcomes beyond ±$1.2m are clipped in panels only.

Entry cost versus realized SD



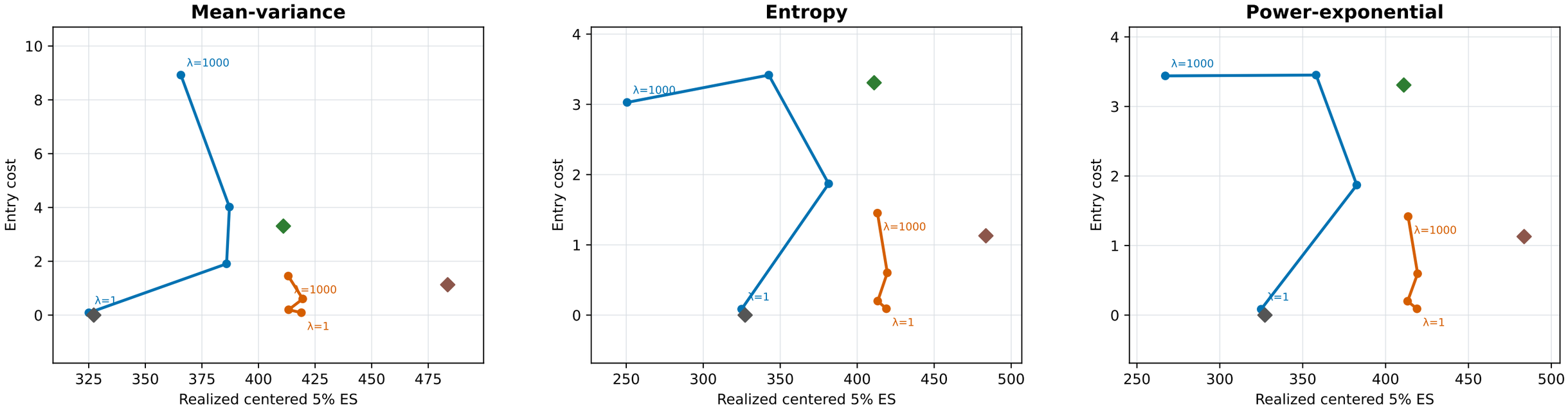
Corrected mean subtracts preceding 252-day target/instrument sample means; risk axes use raw next-day P&L.
Centered 5% ES uses each target's worst two dates plus 20% of its third-worst after removing its realized mean. Lines join λ=1, 10, 100, 1000.

Entry cost + corrected mean loss versus realized SD



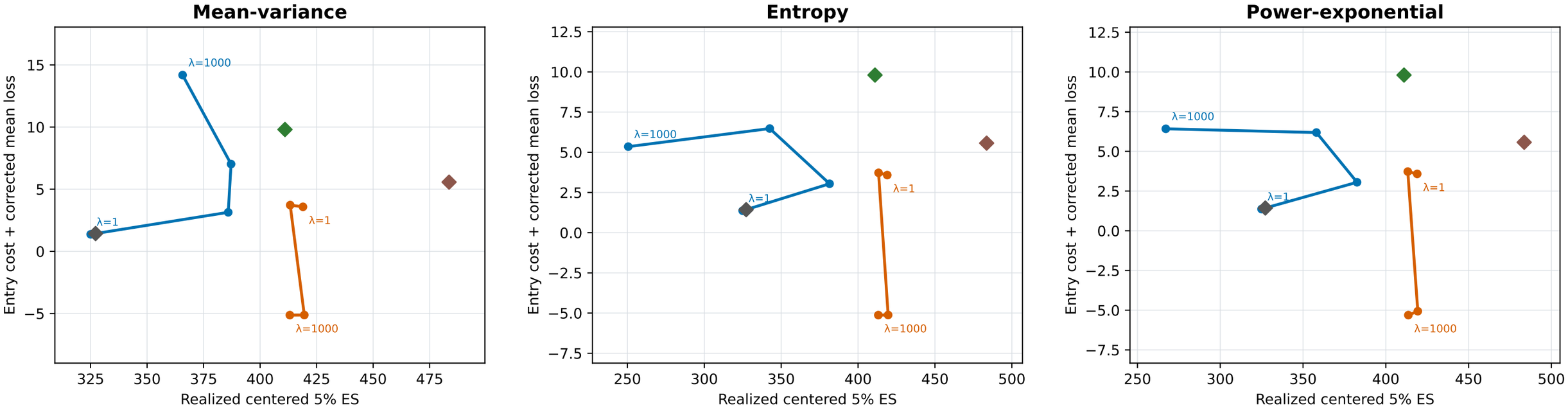
Corrected mean subtracts preceding 252-day target/instrument sample means; risk axes use raw next-day P&L.
Centered 5% ES uses each target's worst two dates plus 20% of its third-worst after removing its realized mean. Lines join λ=1, 10, 100, 1000.

Entry cost versus realized centered 5% ES



Corrected mean subtracts preceding 252-day target/instrument sample means; risk axes use raw next-day P&L.
Centered 5% ES uses each target's worst two dates plus 20% of its third-worst after removing its realized mean. Lines join λ=1, 10, 100, 1000.

Entry cost + corrected mean loss versus realized centered 5% ES



Corrected mean subtracts preceding 252-day target/instrument sample means; risk axes use raw next-day P&L.
Centered 5% ES uses each target's worst two dates plus 20% of its third-worst after removing its realized mean. Lines join λ=1, 10, 100, 1000.